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ACTIN-BINDING LIM PROTEIN 3 (ABLIM3) iRNA AGENT COMPOSITIONS AND METHODS OF USE THEREOF

Abstract

The disclosure relates to double stranded ribonucleic acid (dsRNAi) agents and compositions targeting a Actin binding LIM protein 3 (ABLIM3) gene, as well as methods of inhibiting expression of an ABLIM3 gene and methods of treating subjects having an ABLIM3-associated disease or disorder, e.g., PTSD or age-related memory loss, using such dsRNAi agents and compositions.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of priority to U.S. Provisional Application No. 63/338,645, filed on May 5, 2022, and claims the benefit of priority to U.S. Provisional Application No. 63/352,573, filed on Jun. 15, 2022. The entire contents of the foregoing applications are hereby incorporated herein by reference.

SEQUENCE LISTING

[0002] The instant application contains a Sequence Listing which has been submitted electronically in XML format and is

hereby incorporated by reference in its entirety. The XML copy, created on May 2, 2023, is named A108868_1570WO_SL.xml and is 4,029,028 bytes in size.

BACKGROUND OF THE INVENTION

[0003] The actin binding LIM protein 3 (ABLIM3) gene encoding the actin binding LIM protein 3 protein is located in the chromosomal region 5q32 on chromosome 5. ABLIM3 is part of the actin binding LIM family of proteins which are characterized by an N-terminal LIM domain and a C-terminal dematin-like domain.

[0004] ABLIM3 is expressed in the brain and also expressed in other tissues, such as in fat tissue, heart, placenta, and liver. ABLIM3 can bind to the actin cytoskeleton, localizes to adherens junctions, stabilizes branched F-actin, and also functions as a protein binding interface through its LIM domain.

[0005] ABLIM3 has recently been implicated in memory as a molecular brake in connectivity of dentate granule cells and stratum lucidum interneurons. Upon learning, the levels of ABLIM3 decrease resulting in improved remote memory precision and a decrease in generalization of remote fear memories. Accordingly, ABLIM3 may play a role in post-traumatic stress disorder (PTSD) and age-related memory loss.

[0006] The current treatments for PTSD and age-related memory loss are limited and have low efficacy. Accordingly, there is a need for agents that can treat, prevent, and/or inhibit the progression or the formation of PTSD and age-related memory loss.

BRIEF SUMMARY OF THE INVENTION

[0007] The present disclosure provides RNAi compositions, which effect the RNA-induced silencing complex (RISC)-mediated cleavage of RNA transcripts of an ABLIM3 gene. The ABLIM3 gene may be within a cell, e.g., a cell within a subject, such as a human. The use of these iRNAs enables the targeted degradation of mRNAs of the corresponding gene (ABLIM3 gene) in mammals.

[0008] The iRNAs of the invention have been designed to target an ABLIM3 gene, e.g., an ABLIM3 wild type gene and/or an ABLIM3 gene variant. The iRNAs of the invention inhibit the expression of the ABLIM3 gene by at least about 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, or 99%. Without intending to be limited by theory, it is believed that a combination or sub-combination of the foregoing properties and the specific target sites, or the specific modifications in these iRNAs confer to the iRNAs of the invention improved efficacy, stability, potency, durability, and safety. In one aspect, the present invention provides double stranded ribonucleic acid (dsRNA) agent for inhibiting expression of ABLIM3, wherein the dsRNA agent comprises a sense strand and an antisense strand forming a double stranded region, wherein the sense strand comprises at least 15 contiguous nucleotides differing by no more than 3 nucleotides from the nucleotide sequence of SEQ ID NO: 1 and the antisense strand comprises at least 15 contiguous nucleotides differing by no more than 3 nucleotides from the nucleotide sequence of SEQ ID NO: 2.

[0009] In another aspect, the present invention provides a double stranded ribonucleic acid (dsRNA) agent for inhibiting expression of ABLIM3, wherein the dsRNA agent comprises a sense strand and an antisense strand forming a double stranded region, wherein the antisense strand comprises a region of complementarity to an mRNA encoding ABLIM3, and wherein the region of complementarity comprises at least 15 contiguous nucleotides differing by no more than 3 nucleotides from the nucleotide sequence of SEQ ID NO:2.

[0010] In yet another aspect, the present invention provides a double stranded ribonucleic acid (dsRNA) agent for inhibiting expression of ABLIM3, wherein the dsRNA agent comprises a sense strand and an antisense strand forming a double stranded region, wherein the antisense strand comprises a region of complementarity to an mRNA encoding ABLIM3, and wherein the region of complementarity comprises at least 15 contiguous nucleotides differing by no more than 3 nucleotides from any one of the antisense nucleotide sequences in Tables 3 or 4.

[0011] In one embodiment, the sense strand comprises at least 15 contiguous nucleotides differing by no more than three nucleotides from any one of the nucleotide sequence of nucleotides 82-102, 134-154, 232-252, 276-296, 293-313, 326-346, 351-371, 366-386, 381-401, 414-434, 482-502, 500-520, 542-562, 558-578, 582-602, 622-642, 674-694, 692-712, 713-733, 728-748, 743-763, 779-799, 887-907, 942-962, 983-1003, 1001-1021, 1082-1102, 1102-1122, 1117-1137, 1177-1197, 1192-1212, 1242-1262, 1269-1289, 1313-1333, 1344-1364, 1359-1379, 1374-1394, 1408-1428, 1437-1457, 1550-1570, 1594-1614, 1624-1644, 1645-1665, 1674-1694, 1715-1735, 1746-1766, 1767-1787, 1812-1832, 1859-1879, 1884-1904, 1912-1932, 1935-1955, 1971-1991, 2039-2059, 2096-2116, 2127-2147, 2169-2189, 2184-2204, 2218-2238, 2245-2265, 2273-2293, 2325-2345, 2343-2363, 2385-2405, 2422-2442, 2459-2479, 2483-2503, 2520-2540, 2535-2555, 2561-2581, 2585-2605, 2667-2687, 2703-2723, 2729-2749, 2764-2784, 2788-2808, 2830-2850, 2861-2881, 2885-2905, 2937-2957, 2958-2978, 2974-2994, 3011-3031, 3044-3064, 3061-3081, 3079-3099, 3110-3130, 3146-3166, 3203-3223, 3220-3240, 3279-3299, 3303-3323, 3321-3341, 3338-3358, 3388-3408, 3407-3427, 3422-3442, 3437-3457, 3452-3472, 3532-3552, 3548-3568, 3579-3599, 3594-3614, 3626-3646, 3644-3664, 3682-3702, 3706-3726, 3721-3741, 3746-3766, 3763-3783, 3796-3816, 3814-3834, 3849-3869, 3866-3886, 3897-3917, 3919-3939, 3986-4006, 4018-4038, 4037-4057, 4058-4078, 4084-4104, 4099-4119, 4131-4151, 4159-4179, 4225-4245, 4241-4261, 4276-4296, 4291-4311, 4306-4326, 4336-4356, 4354-4374, 4369-4389, 4387-4407, 4404-4424, or 4421-4441 of SEQ ID NO: 1, and the antisense strand comprises at least 15 contiguous nucleotides from the corresponding nucleotide sequence of SEQ ID NO: 2.

[0012] In one embodiment, the antisense strand comprises at least 15 contiguous nucleotides differing by no more than three nucleotides from any one of the antisense strand nucleotide sequences of a duplex selected from the group consisting of AD-1807455, AD-1807481, AD-1807539, AD-1807561, AD-1807578, AD-1807591, AD-1807616, AD-1807631, AD-1807646, AD-1807659, AD-1807707, AD-1807725, AD-1807764, AD-1807780, AD-1807804, AD-1807844, AD-1807870, AD-1807888, AD-1807889, AD-1807904, AD-1807919, AD-1807955, AD-1808063, AD-1808098, AD-

1808139, AD-1808157, AD-1808238, AD-1808258, AD-1808273, AD-1808312, AD-1808327, AD-1808356, AD-1808383, AD-1808407, AD-1808418, AD-1808433, AD-1808448, AD-1808482, AD-1808511, AD-1808561, AD-1808598, AD-1808628, AD-1808649, AD-1808678, AD-1808698, AD-1808727, AD-1808748, AD-1808773, AD-1808789, AD-1808814, AD-1808842, AD-1808865, AD-1808901, AD-1808941, AD-1808978, AD-1809009, AD-1809051, AD-1809066, AD-1809080, AD-1809107, AD-1809135, AD-1809187, AD-1809205, AD-1809246, AD-1809283, AD-1809293, AD-1809317, AD-1809354, AD-1809369, AD-1809395, AD-1809414, AD-1809456, AD-1809472, AD-1809498, AD-1809511, AD-1809535, AD-1809550, AD-1809561, AD-1809585, AD-1809596, AD-1809617, AD-1809633, AD-1809670, AD-1809702, AD-1809719, AD-1809737, AD-1809757, AD-1809773, AD-1809822, AD-1809839, AD-1809895, AD-1809919, AD-1809937, AD-1809954, AD-1810004, AD-1810023, AD-1810038, AD-1810053, AD-1810068, AD-1810127, AD-1810143, AD-1810174, AD-1810189, AD-1810221, AD-1810239, AD-1810255, AD-1810277, AD-1810292, AD-1810296, AD-1810313, AD-1810345, AD-1810363, AD-1810398, AD-1810415, AD-1810446, AD-1810468, AD-1810494, AD-1810526, AD-1810545, AD-1810566, AD-1810591, AD-1810606, AD-1810638, AD-1810666, AD-1810676, AD-1810692, AD-1810727, AD-1810742, AD-1810757, AD-1810767, AD-1810785, AD-1810800, AD-1810818, AD-1810822, and AD-1810839.

[0013] In some embodiments, the nucleotide sequence of the sense and antisense strand comprises any one of the sense strand nucleotide sequences in Tables 3 or 4.

[0014] In one embodiment, the sense strand, the antisense strand, or both the sense strand and the antisense strand is conjugated to one or more lipophilic moieties.

[0015] In one embodiment, the lipophilic moiety is conjugated to one or more internal positions in the double stranded region of the dsRNA agent.

[0016] In one embodiment, the lipophilic moiety is conjugated via a linker or carrier.

[0017] In one embodiment, the lipophilicity of the lipophilic moiety, measured by log Kow, exceeds 0.

[0018] In one embodiment, the hydrophobicity of the double-stranded RNAi agent, measured by the unbound fraction in a plasma protein binding assay of the double-stranded RNAi agent, exceeds 0.2.

[0019] In one embodiment, the plasma protein binding assay is an electrophoretic mobility shift assay using human serum albumin protein.

[0020] In some embodiments, the dsRNA agent comprises at least one modified nucleotide.

[0021] In one embodiment, no more than five of the sense strand nucleotides and no more than five of the nucleotides of the antisense strand are unmodified nucleotides

[0022] In one embodiment, all of the nucleotides of the sense strand and all of the nucleotides of the antisense strand are modified nucleotides.

[0023] In one embodiment, at least one of the modified nucleotides is selected from the group a deoxy-nucleotide, a 3'-terminal deoxythymidine (dT) nucleotide, a 2'-O-methyl modified nucleotide, a 2'-fluoro modified nucleotide, a 2'-deoxy-modified nucleotide, a locked nucleotide, an unlocked nucleotide, a conformationally restricted nucleotide, a constrained ethyl nucleotide, an abasic nucleotide, a 2'-amino-modified nucleotide, a 2'-O-allyl-modified nucleotide, 2'-C-alkyl-modified nucleotide, 2'-hydroxyl-modified nucleotide, a 2'-methoxyethyl modified nucleotide, a 2'-O-alkyl-modified nucleotide, a morpholino nucleotide, a phosphoramidate, a non-natural base comprising nucleotide, a tetrahydropyran modified nucleotide, a 1,5-anhydrohexitol modified nucleotide, a cyclohexenyl modified nucleotide, a nucleotide comprising a 5'-phosphorothioate group, a nucleotide comprising a 5'-methylphosphonate group, a nucleotide comprising a 5' phosphate or 5' phosphate mimic, a nucleotide comprising vinyl phosphonate, a glycol nucleic acid (GNA) (e.g., a nucleotide comprising adenosine-glycol nucleic acid (GNA)), an S-glycol nucleic acid (S-GNA) (e.g., a nucleotide comprising thymidine-glycol nucleic acid (GNA) S-Isomer), a nucleotide comprising 2-hydroxymethyl-tetrahydrofuran-5-phosphate, a nucleotide comprising 2'-deoxythymidine-3'phosphate, a nucleotide comprising 2'-deoxyguanosine-3'-phosphate, a nucleotide comprising a 2' phosphate ("3'-RNA", e.g., G2p, C2p, A2p, U2p), and a terminal nucleotide linked to a cholesteryl derivative and a dodecanoic acid bisdecylamide group; and combinations thereof.

[0024] In one embodiment, the modified nucleotide is selected from the group consisting of a 2'-deoxy-2'-fluoro modified nucleotide, a 2'-deoxy-modified nucleotide, 3'-terminal deoxythymidine nucleotides (dT), a locked nucleotide, an abasic nucleotide, a 2'-amino-modified nucleotide, a 2'-alkyl-modified nucleotide, a morpholino nucleotide, a phosphoramidate, and a non-natural base comprising nucleotide.

[0025] In one embodiment, the modified nucleotide comprises a short sequence of 3'-terminal deoxythymidine nucleotides (dT).

[0026] In one embodiment, the modifications on the nucleotides are 2'-O-methyl, GNA and 2'fluoro modifications.

[0027] In some embodiments, the dsRNA agent further comprises at least one phosphorothioate internucleotide linkage.

[0028] In one embodiment, the dsRNA agent comprises 6-8 phosphorothioate internucleotide linkages.

[0029] In one embodiment, each strand is no more than 30 nucleotides in length.

[0030] In one embodiment, at least one strand comprises a 3' overhang of at least 1 nucleotide. In another embodiment, at least one strand comprises a 3' overhang of at least 2 nucleotides.

[0031] The double stranded region may be 15-30 nucleotide pairs in length; 17-23 nucleotide pairs in length; 17-25 nucleotide pairs in length; 23-27 nucleotide pairs in length; 19-21 nucleotide pairs in length; or 21-23 nucleotide pairs in length.

[0032] Each strand may have 19-30 nucleotides; 19-23 nucleotides; or 21-23 nucleotides.

[0033] In one embodiment, one or more lipophilic moieties are conjugated to one or more internal positions on at least one

strand, or via a linker or carrier.

[0034] In one embodiment, the internal positions include all positions except the terminal two positions from each end of the at least one strand.

[0035] In another embodiment, the internal positions include all positions except the terminal three positions from each end of the at least one strand.

[0036] In one embodiment, the internal positions exclude a cleavage site region of the sense strand.

[0037] In one embodiment, the internal positions include all positions except positions 9-12, counting from the 5'-end of the sense strand.

[0038] In another embodiment, the internal positions include all positions except positions 11-13, counting from the 3'-end of the sense strand.

[0039] In one embodiment, the internal positions exclude a cleavage site region of the antisense strand.

[0040] In one embodiment, the internal positions include all positions except positions 12-14, counting from the 5'-end of the antisense strand.

[0041] In one embodiment, the internal positions include all positions except positions 11-13 on the sense strand, counting from the 3'-end, and positions 12-14 on the antisense strand, counting from the 5'-end.

[0042] In one embodiment, the one or more lipophilic moieties are conjugated to one or more of the internal positions selected from the group consisting of positions 4-8 and 13-18 on the sense strand, and positions 6-10 and 15-18 on the antisense strand, counting from the 5' end of each strand.

[0043] In another embodiment, the one or more lipophilic moieties are conjugated to one or more of the internal positions selected from the group consisting of positions 5, 6, 7, 15, and 17 on the sense strand, and positions 15 and 17 on the antisense strand, counting from the 5'-end of each strand.

[0044] In one embodiment, the internal positions in the double stranded region exclude a cleavage site region of the sense strand.

[0045] In one embodiment, the sense strand is 21 nucleotides in length, the antisense strand is 23 nucleotides in length, and the lipophilic moiety is conjugated to position 21, position 20, position 15, position 1, position 7, position 6, or position 2 of the sense strand or position 16 of the antisense strand.

[0046] In one embodiment, the lipophilic moiety is conjugated to position 21, position 20, position 15, position 1, or position 7 of the sense strand.

[0047] In another embodiment, the lipophilic moiety is conjugated to position 21, position 20, or position 15 of the sense strand.

[0048] In yet another embodiment, the lipophilic moiety is conjugated to position 20 or position 15 of the sense strand.

[0049] In one embodiment, the lipophilic moiety is conjugated to position 16 of the antisense strand.

[0050] In one embodiment, the lipophilic moiety is an aliphatic, alicyclic, or polyalicyclic compound.

[0051] In one embodiment, the lipophilic moiety is selected from the group consisting of lipid, cholesterol, retinoic acid, cholic acid, adamantane acetic acid, 1-pyrene butyric acid, dihydrotestosterone, 1,3-bis-O(hexadecyl)glycerol, geranyloxyhexanol, hexadecylglycerol, borneol, menthol, 1,3-propanediol, heptadecyl group, palmitic acid, myristic acid, O3-(oleoyl) lithocholic acid, O3-(oleoyl)cholenic acid, dimethoxytrityl, or phenoxazine.

[0052] In one embodiment, the lipophilic moiety contains a saturated or unsaturated C4-C30 hydrocarbon chain, and an optional functional group selected from the group consisting of hydroxyl, amine, carboxylic acid, sulfonate, phosphate, thiol, azide, and alkyne.

[0053] In one embodiment, the lipophilic moiety contains a saturated or unsaturated C6-C18 hydrocarbon chain.

[0054] In one embodiment, the lipophilic moiety contains a saturated or unsaturated C16 hydrocarbon chain.

[0055] In one embodiment, the ligand is conjugated at the 2'-position of a nucleotide or modified nucleotide within the sense or antisense strand. For example, a C16 ligand may be conjugated as shown in the following structure:

##STR00001##

where * denotes a bond to an adjacent nucleotide, and B is a nucleobase or a nucleobase analog, optionally where B is adenine, guanine, cytosine, thymine or uracil.

[0056] In one embodiment, the saturated or unsaturated C16 hydrocarbon chain is conjugated to position 6, counting from the 5'-end of the strand.

[0057] In one embodiment, the lipophilic moiety is conjugated via a carrier that replaces one or more nucleotide(s) in the internal position(s) or the double stranded region.

[0058] In one embodiment, the carrier is a cyclic group selected from the group consisting of pyrrolidinyl, pyrazolinyl, pyrazolidinyl, imidazolinyl, imidazolidinyl, piperidinyl, piperazinyl, [1,3]dioxolanyl, oxazolidinyl, isoxazolidinyl, morpholinyl, thiazolidinyl, isothiazolidinyl, quinoxalinyl, pyridazinonyl, tetrahydrofuranlyl, and decalinyl; or is an acyclic moiety based on a serinol backbone or a diethanolamine backbone.

[0059] In one embodiment, the lipophilic moiety is conjugated to the double-stranded iRNA agent via a linker containing an ether, thioether, urea, carbonate, amine, amide, malcicide-thioether, disulfide, phosphodiester, sulfonamide linkage, a product of a click reaction, or carbamate.

[0060] In one embodiment, the lipophilic moiety is conjugated to a nucleobase, sugar moiety, or internucleosidic linkage.

[0061] In one embodiment, the lipophilic moiety or targeting ligand is conjugated via a bio-cleavable linker selected from the group consisting of DNA, RNA, disulfide, amide, functionalized monosaccharides or oligosaccharides of galactosamine, glucosamine, glucose, galactose, mannose, and combinations thereof.

[0062] In one embodiment, the 3' end of the sense strand is protected via an end cap which is a cyclic group having an amine, said cyclic group being selected from the group consisting of pyrrolidinyl, pyrazolinyl, pyrazolidinyl, imidazolinyl, imidazolidinyl, piperidinyl, piperazinyl, [1,3]dioxolanyl, oxazolidinyl, isoxazolidinyl, morpholinyl, thiazolidinyl, isothiazolidinyl, quinoxalinyl, pyridazinonyl, tetrahydrofuranyl, and decalinyl.

[0063] In one embodiment, the targeting ligand is a GalNAc conjugate.

[0064] In one embodiment, the dsRNA agent further comprises a terminal, chiral modification occurring at the first internucleotide linkage at the 3' end of the antisense strand, having the linkage phosphorus atom in Sp configuration, a terminal, chiral modification occurring at the first internucleotide linkage at the 5' end of the antisense strand, having the linkage phosphorus atom in Rp configuration, and a terminal, chiral modification occurring at the first internucleotide linkage at the 5' end of the sense strand, having the linkage phosphorus atom in either Rp configuration or Sp configuration.

[0065] In another embodiment, the dsRNA agent further comprises a terminal, chiral modification occurring at the first and second internucleotide linkages at the 3' end of the antisense strand, having the linkage phosphorus atom in Sp configuration, a terminal, chiral modification occurring at the first internucleotide linkage at the 5' end of the antisense strand, having the linkage phosphorus atom in Rp configuration, and a terminal, chiral modification occurring at the first internucleotide linkage at the 5' end of the sense strand, having the linkage phosphorus atom in either Rp or Sp configuration.

[0066] In yet another embodiment, the dsRNA agent further comprises a terminal, chiral modification occurring at the first, second and third internucleotide linkages at the 3' end of the antisense strand, having the linkage phosphorus atom in Sp configuration, a terminal, chiral modification occurring at the first internucleotide linkage at the 5' end of the antisense strand, having the linkage phosphorus atom in Rp configuration, and a terminal, chiral modification occurring at the first internucleotide linkage at the 5' end of the sense strand, having the linkage phosphorus atom in either Rp or Sp configuration.

[0067] In another embodiment, the dsRNA agent further comprises a terminal, chiral modification occurring at the first, and second internucleotide linkages at the 3' end of the antisense strand, having the linkage phosphorus atom in Sp configuration, a terminal, chiral modification occurring at the third internucleotide linkages at the 3' end of the antisense strand, having the linkage phosphorus atom in Rp configuration, a terminal, chiral modification occurring at the first internucleotide linkage at the 5' end of the antisense strand, having the linkage phosphorus atom in Rp configuration, and a terminal, chiral modification occurring at the first internucleotide linkage at the 5' end of the sense strand, having the linkage phosphorus atom in either Rp or Sp configuration.

[0068] In another embodiment, the dsRNA agent further comprises a terminal, chiral modification occurring at the first, and second internucleotide linkages at the 3' end of the antisense strand, having the linkage phosphorus atom in Sp configuration, a terminal, chiral modification occurring at the first, and second internucleotide linkages at the 5' end of the antisense strand, having the linkage phosphorus atom in Rp configuration, and a terminal, chiral modification occurring at the first internucleotide linkage at the 5' end of the sense strand, having the linkage phosphorus atom in either Rp or Sp configuration.

[0069] In one embodiment, the dsRNA agent further comprises a phosphate or phosphate mimic at the 5'-end of the antisense strand.

[0070] In one embodiment, the phosphate mimic is a 5'-vinyl phosphonate (VP).

[0071] In one embodiment, the base pair at the 1 position of the 5'-end of the antisense strand of the duplex is an AU base pair.

[0072] In one embodiment, the sense strand has a total of 21 nucleotides and the antisense strand has a total of 23 nucleotides.

[0073] In another aspect, the present invention provides a double stranded ribonucleic acid (dsRNA) agent for inhibiting expression of actin binding LIM protein 3 (ABLIM3) in a cell, wherein the dsRNA agent comprises a sense strand and an antisense strand forming a double stranded region. The sense strand comprises a nucleotide sequence of any one of the agents in Table 3 or 4 and the antisense strand comprises a nucleotide sequence of any one of the agents in Table 3 or 4. Substantially all of the nucleotides of the sense strand and substantially all of the nucleotides of the antisense strand are modified nucleotides, and the dsRNA agent is conjugated to a ligand.

[0074] In various embodiments of the aforementioned dsRNA agents, the dsRNA agent targets a hotspot region of an mRNA encoding ABLIM3. In one embodiment, the hotspot region comprises nucleotides 3042-3081, 3420-3457, 4223-4261, 726-763, 1080-1137, 1342-1379, 1910-1955, 2167-2204, 480-520, 1080-1122, 1622-1665, 1744-1787, 2323-2363, 540-578, 1175-1212, 274-313, 349-401, or 2518-2555 of SEQ ID NO: 1. The dsRNA agent may be selected from the group consisting of AD-1809702, AD-1809719, AD-1810038, AD-1810053, AD-1810676, AD-1810692, AD-1807904, AD-1807919, AD-1808238, AD-1808258, AD-1808273, AD-1808418, AD-1808433, AD-1808842, AD-1808865, AD-1809051, AD-1809066, AD-1807707, AD-1807725, AD-1808628, AD-1808649, AD-1808727, AD-1808748, AD-1809187, AD-1809205, AD-1807764, AD-1807780, AD-1808312, AD-1808327, AD-1807561, AD-1807578, AD-1807616, AD-1807631, AD-1807646, AD-1809354, and AD-1809369.

[0075] In another aspect, the present invention provides a dsRNA agent that targets a hotspot region of an actin binding LIM protein 3 (ABLIM3) mRNA.

[0076] The present invention also provides cells and pharmaceutical compositions for inhibiting expression of a gene encoding ABLIM3 comprising the dsRNA agents of the invention.

[0077] In one embodiment, the dsRNA agent is in an unbuffered solution, such as saline or water.

[0078] In another embodiment, the dsRNA agent is in a buffer solution, such as a buffer solution comprising acetate, citrate, prolamine, carbonate, or phosphate or any combination thereof; or phosphate buffered saline (PBS).

[0079] In one aspect, the present invention provides a method of inhibiting expression of an ABLIM3 gene in a cell, the method comprising contacting the cell with a dsRNA agent of the invention, or a pharmaceutical composition of the invention, thereby inhibiting expression of the ABLIM3 gene in the cell.

[0080] In one embodiment, the cell is within a subject.

[0081] In one embodiment, the cell is a central nervous system (CNS) cell. In another embodiment, the cell is a brain cell. In certain embodiments, the cell is an astrocyte, neuron, or oligodendrocyte.

[0082] In one embodiment, the subject is a human.

[0083] In one embodiment, the subject has an ABLIM3-associated disorder.

[0084] In one embodiment, the subject has post-traumatic stress disorder (PTSD). In one embodiment, the subject has age-related memory loss.

[0085] In one embodiment, contacting the cell with the dsRNA agent inhibits the expression of ABLIM3 by at least 30%.

[0086] In one embodiment, inhibiting expression of ABLIM3 decreases ABLIM3 protein level in serum of the subject by at least 30%.

[0087] In one aspect, the present invention provides method of treating a subject having a disorder that would benefit from reduction in ABLIM3 expression, comprising administering to the subject a therapeutically effective amount of a dsRNA agent of the invention, or a pharmaceutical composition of the invention, thereby treating the subject having the disorder that would benefit from reduction in ABLIM3 expression.

[0088] In another aspect, the present invention provides a method of preventing at least one symptom or sign in a subject having a disorder that would benefit from reduction in ABLIM3 expression, comprising administering to the subject a prophylactically effective amount of a dsRNA agent of the invention, or a pharmaceutical composition of the invention, thereby preventing at least one symptom or sign in the subject having the disorder that would benefit from reduction in ABLIM3 expression.

[0089] In another aspect, the present invention provides a method of preventing a subject from having a disorder that would benefit from reduction in ABLIM3 expression, comprising administering to the subject a prophylactically effective amount of the dsRNA agent of the invention, or the pharmaceutical composition of the invention, thereby preventing the subject from having the disorder that would benefit from reduction in ABLIM3 expression.

[0090] In one embodiment, the disorder is an ABLIM3-associated disorder.

[0091] In one embodiment, the disorder is post-traumatic stress disorder (PTSD).

[0092] In one embodiment, the disorder is age-related memory loss.

[0093] In one embodiment, the subject is human.

[0094] In one embodiment, the administration of the agent to the subject causes improved memory, or a decrease in anxiety, depression, fear, agitation, hostility, and/or distress.

[0095] In one embodiment, the dsRNA agent is administered to the subject at a dose of about 0.01 mg/kg to about 50 mg/kg.

[0096] In one embodiment, the dsRNA agent is administered to the subject subcutaneously.

[0097] In another embodiment, the dsRNA agent is administered to the subject intrathecally.

[0098] In one embodiment, the methods of the invention further comprise determining the level of ABLIM3 in a sample(s) from the subject.

[0099] In one embodiment, the level of ABLIM3 in the subject sample(s) is an ABLIM3 protein level in a blood, serum, or cerebrospinal fluid sample(s).

[0100] In one embodiment, the methods of the invention further comprise administering to the subject an additional therapeutic agent.

[0101] In one aspect, the present invention provides a kit comprising a dsRNA agent of the invention, or a pharmaceutical composition of the invention.

[0102] In another aspect, the present invention provides a vial comprising a dsRNA agent of the invention, or a pharmaceutical composition of the invention.

[0103] In yet another aspect, the present invention provides a syringe comprising a dsRNA agent of the invention, or a pharmaceutical composition of the invention.

[0104] In another aspect, the present invention provides an intrathecal pump comprising a dsRNA agent of the invention, or a pharmaceutical composition of the invention.

Description

DETAILED DESCRIPTION OF THE INVENTION

[0105] The present disclosure provides RNAi compositions, which effect the RNA-induced silencing complex (RISC)-mediated cleavage of RNA transcripts of an ABLIM3 gene. The ABLIM3 gene may be within a cell, e.g., a cell within a subject, such as a human. The use of these iRNAs enables the targeted degradation of mRNAs of the corresponding gene (ABLIM3 gene) in mammals.

[0106] The iRNAs of the invention have been designed to target an ABLIM3 gene, e.g., an ABLIM3 gene either with or without nucleotide modifications. The iRNAs of the invention inhibit the expression of the ABLIM3 gene by at least about 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, or 99%. Without intending to be limited by theory, it is believed that a combination or sub-combination of the foregoing properties and the specific target sites, or the specific modifications in these iRNAs confer to the iRNAs of the invention improved efficacy, stability, potency, durability, and safety.

[0107] Accordingly, the present disclosure also provides methods of using the RNAi compositions of the disclosure for inhibiting the expression of an ABLIM3 gene or for treating a subject having a disorder that would benefit from inhibiting or reducing the expression of an ABLIM3 gene, e.g., PTSD or age-related memory loss.

[0108] The RNAi agents of the disclosure include an RNA strand (the antisense strand) having a region which is about 30 nucleotides or less in length, e.g., 15-30, 15-29, 15-28, 15-27, 15-26, 15-25, 15-24, 15-23, 15-22, 15-21, 15-20, 15-19, 15-18, 15-17, 18-30, 18-29, 18-28, 18-27, 18-26, 18-25, 18-24, 18-23, 18-22, 18-21, 18-20, 19-30, 19-29, 19-28, 19-27, 19-26, 19-25, 19-24, 19-23, 19-22, 19-21, 19-20, 20-30, 20-29, 20-28, 20-27, 20-26, 20-25, 20-24, 20-23, 20-22, 20-21, 21-30, 21-29, 21-28, 21-27, 21-26, 21-25, 21-24, 21-23, or 21-22 nucleotides in length, which region is substantially complementary to at least part of an mRNA transcript of an ABLIM3 gene, e.g., an ABLIM3 exon. In certain embodiments, the RNAi agents of the disclosure include an RNA strand (the antisense strand) having a region which is about 21-23 nucleotides in length, which region is substantially complementary to at least part of an mRNA transcript of an ABLIM3 gene.

[0109] In certain embodiments, the RNAi agents of the disclosure include an RNA strand (the antisense strand) which can include longer lengths, for example up to 66 nucleotides, e.g., 36-66, 26-36, 25-36, 31-60, 22-43, 27-53 nucleotides in length with a region of at least 19 contiguous nucleotides that is substantially complementary to at least a part of an mRNA transcript of an ABLIM3 gene. These RNAi agents with the longer length antisense strands preferably include a second RNA strand (the sense strand) of 20-60 nucleotides in length wherein the sense and antisense strands form a duplex of 18-30 contiguous nucleotides.

[0110] The use of these RNAi agents enables the targeted degradation and/or inhibition of mRNAs of an ABLIM3 gene in mammals. Thus, methods and compositions including these RNAi agents are useful for treating a subject who would benefit by a reduction in the levels or activity of an ABLIM3 protein, such as a subject having PTSD or age-related memory loss.

[0111] The following detailed description discloses how to make and use compositions containing RNAi agents to inhibit the expression of an ABLIM3 gene, as well as compositions and methods for treating subjects having diseases and disorders that would benefit from inhibition or reduction of the expression of the genes.

I. Definitions

[0112] In order that the present disclosure may be more readily understood, certain terms are first defined. In addition, it should be noted that whenever a value or range of values of a parameter are recited, it is intended that values and ranges intermediate to the recited values are also intended to be part of this disclosure.

[0113] The articles “a” and “an” are used herein to refer to one or to more than one (i.e., to at least one) of the grammatical object of the article. By way of example, “an element” means one element or more than one element, e.g., a plurality of elements.

[0114] The term “including” is used herein to mean, and is used interchangeably with, the phrase “including but not limited to”. The term “or” is used herein to mean, and is used interchangeably with, the term “and/or,” unless context clearly indicates otherwise.

[0115] The term “about” is used herein to mean within the typical ranges of tolerances in the art. For example, “about” can be understood as about 2 standard deviations from the mean. In certain embodiments, about means $\pm 10\%$. In certain embodiments, about means $\pm 5\%$. When about is present before a series of numbers or a range, it is understood that “about” can modify each of the numbers in the series or range.

[0116] The term “at least” prior to a number or series of numbers is understood to include the number adjacent to the term “at least”, and all subsequent numbers or integers that could logically be included, as clear from context. For example, the number of nucleotides in a nucleic acid molecule must be an integer. For example, “at least 18 nucleotides of a 21 nucleotide nucleic acid molecule” means that 18, 19, 20, or 21 nucleotides have the indicated property. When at least is present before a series of numbers or a range, it is understood that “at least” can modify each of the numbers in the series or range.

[0117] As used herein, “no more than” or “or less” is understood as the value adjacent to the phrase and logical lower values or integers, as logical from context, to zero. For example, a duplex with an overhang of “no more than 2 nucleotides” has a 2, 1, or 0 nucleotide overhang. When “no more than” is present before a series of numbers or a range, it is understood that “no more than” can modify each of the numbers in the series or range.

[0118] As used herein, methods of detection can include determination that the amount of analyte present is below the level of detection of the method.

[0119] In the event of a conflict between an indicated target site and the nucleotide sequence for a sense or antisense strand, the indicated sequence takes precedence.

[0120] In the event of a conflict between a chemical structure and a chemical name, the chemical structure takes precedence.

[0121] As used herein, the term “Actin binding LIM protein 3” used interchangeably with the term “ABLIM3,” refers to

the well-known gene and polypeptide encoded by that gene, also known in the art as “actin binding LIM protein family member 3,” “KIAA0843,” and “HMFN1661”. The ABLIM3 gene is expressed and active in the brain and is expressed in other tissues, including fat tissue, heart, placenta, and liver. ABLIM3 is expressed in some cells of the central nervous system, including dentate granule cells. RNAi therapy can specifically target brain ABLIM3, thereby avoiding peripheral side effects.

[0122] ABLIM3 has recently been implicated in memory as a molecular brake in connectivity of dentate granule cells and stratum lucidum interneurons (Guo, et al., 2018, Nat Med. 24(4):438-449). Using an animal model of learning, the levels of ABLIM3 were shown to decrease upon learning. The study found that downregulation of ABLIM3 in dentate granule cells decreased remote memory generalization and increased remote memory precision, which could help alleviate memory impairments and PTSD.

[0123] Exemplary nucleotide and amino acid sequences of ABLIM3 can be found, for example, at GenBank Accession No. NM_001301015.3 (*Homo sapiens* ABLIM3. SEQ ID NO: 1, reverse complement, SEQ ID NO: 2); and GenBank Accession No.: NM_001164491.1 (*Mus musculus* ABLIM3).

[0124] The nucleotide sequence of the genomic region of human chromosome harboring the ABLIM3 gene may be found in, for example, the Genome Reference Consortium Human Build 38 (also referred to as Human Genome build 38 or GRCh38) available at GenBank. The nucleotide sequence of the genomic region of human chromosome 5 harboring the ABLIM3 gene may also be found at, for example, GenBank Accession No. NC_000005.10, corresponding to nucleotides 149,141,193-14,9260,542 of human chromosome 5. The nucleotide sequence of the human ABLIM3 gene may be found in, for example, GenBank Accession No. NG_051566.1.

[0125] Further examples of ABLIM3 sequences can be found in publically available databases, for example, GenBank, OMIM, and UniProt.

[0126] Additional information on ABLIM3 can be found, for example, at <https://www.ncbi.nlm.nih.gov/gene/22885>. The term ABLIM3 as used herein also refers to variations of the ABLIM3 gene including variants provided in the clinical variant database, for example, at [https://www.ncbi.nlm.nih.gov/clinvar/?term=ABLIM3\[gene\]](https://www.ncbi.nlm.nih.gov/clinvar/?term=ABLIM3[gene]).

[0127] The entire contents of each of the foregoing GenBank Accession numbers and the Gene database numbers are incorporated herein by reference as of the date of filing this application.

[0128] As used herein, “target sequence” refers to a contiguous portion of the nucleotide sequence of an mRNA molecule formed during the transcription of an ABLIM3 gene, including mRNA that is a product of RNA processing of a primary transcription product. In one embodiment, the target portion of the sequence will be at least long enough to serve as a substrate for RNAi-directed cleavage at or near that portion of the nucleotide sequence of an mRNA molecule formed during the transcription of an ABLIM3 gene.

[0129] The target sequence is about 15-30 nucleotides in length. For example, the target sequence can be from about 15-30 nucleotides, 15-29, 15-28, 15-27, 15-26, 15-25, 15-24, 15-23, 15-22, 15-21, 15-20, 15-19, 15-18, 15-17, 18-30, 18-29, 18-28, 18-27, 18-26, 18-25, 18-24, 18-23, 18-22, 18-21, 18-20, 19-30, 19-29, 19-28, 19-27, 19-26, 19-25, 19-24, 19-23, 19-22, 19-21, 19-20, 20-30, 20-29, 20-28, 20-27, 20-26, 20-25, 20-24, 20-23, 20-22, 20-21, 21-30, 21-29, 21-28, 21-27, 21-26, 21-25, 21-24, 21-23, or 21-22 nucleotides in length. In certain embodiments, the target sequence is 19-23 nucleotides in length, optionally 21-23 nucleotides in length. Ranges and lengths intermediate to the above recited ranges and lengths are also contemplated to be part of the disclosure.

[0130] As used herein, the term “strand comprising a sequence” refers to an oligonucleotide comprising a chain of nucleotides that is described by the sequence referred to using the standard nucleotide nomenclature. “G,” “C,” “A,” “T,” and “U” each generally stand for a nucleotide that contains guanine, cytosine, adenine, thymidine, and uracil as a base, respectively in the context of a modified or unmodified nucleotide. However, it will be understood that the term “ribonucleotide” or “nucleotide” can also refer to a modified nucleotide, as further detailed below, or a surrogate replacement moiety (see, e.g., Table 2). The skilled person is well aware that guanine, cytosine, adenine, thymidine, and uracil can be replaced by other moieties without substantially altering the base pairing properties of an oligonucleotide comprising a nucleotide bearing such replacement moiety. For example, without limitation, a nucleotide comprising inosine as its base can base pair with nucleotides containing adenine, cytosine, or uracil. Hence, nucleotides containing uracil, guanine, or adenine can be replaced in the nucleotide sequences of dsRNA featured in the disclosure by a nucleotide containing, for example, inosine. In another example, adenine and cytosine anywhere in the oligonucleotide can be replaced with guanine and uracil, respectively to form G-U Wobble base pairing with the target mRNA. Sequences containing such replacement moieties are suitable for the compositions and methods featured in the disclosure.

[0131] The terms “iRNA,” “RNAi agent,” “iRNA agent,” “RNA interference agent” as used interchangeably herein, refer to an agent that contains RNA as that term is defined herein, and which mediates the targeted cleavage of an RNA transcript via an RNA-induced silencing complex (RISC) pathway. RNA interference (RNAi) is a process that directs the sequence-specific degradation of mRNA. RNAi modulates, e.g., inhibits, the expression of ABLIM3 in a cell, e.g., a cell within a subject, such as a mammalian subject.

[0132] In one embodiment, an RNAi agent of the disclosure includes a single stranded RNAi that interacts with a target RNA sequence, e.g., an ABLIM3 target mRNA sequence, to direct the cleavage of the target RNA. Without wishing to be bound by theory it is believed that long double stranded RNA introduced into cells is broken down into double-stranded short interfering RNAs (siRNAs) comprising a sense strand and an antisense strand by a Type III endonuclease known as Dicer (Sharp et al. (2001) *Genes Dev.* 15:485). Dicer, a ribonuclease-III-like enzyme, processes this dsRNA into 19-23 base pair short interfering RNAs with characteristic two base 3' overhangs (Bernstein, et al., (2001) *Nature* 409:363).

These siRNAs are then incorporated into an RNA-induced silencing complex (RISC) where one or more helicases unwind the siRNA duplex, enabling the complementary antisense strand to guide target recognition (Nykanen, et al., (2001) *Cell* 107:309). Upon binding to the appropriate target mRNA, one or more endonucleases within the RISC cleave the target to induce silencing (Elbashir, et al., (2001) *Genes Dev.* 15:188). Thus, in one aspect the disclosure relates to a single stranded RNA (ssRNA) (the antisense strand of a siRNA duplex) generated within a cell and which promotes the formation of a RISC complex to effect silencing of the target gene, i.e., an ABLIM3 gene. Accordingly, the term “siRNA” is also used herein to refer to an RNAi as described above.

[0133] In another embodiment, the RNAi agent may be a single-stranded RNA that is introduced into a cell or organism to inhibit a target mRNA. Single-stranded RNAi agents bind to the RISC endonuclease, Argonaute 2, which then cleaves the target mRNA. The single-stranded siRNAs are generally 15-30 nucleotides and are chemically modified. The design and testing of single-stranded RNAs are described in U.S. Pat. No. 8,101,348 and in Lima et al., (2012) *Cell* 150:883-894, the entire contents of each of which are hereby incorporated herein by reference. Any of the antisense nucleotide sequences described herein may be used as a single-stranded siRNA as described herein or as chemically modified by the methods described in Lima et al., (2012) *Cell* 150:883-894.

[0134] In another embodiment, a “RNAi agent” for use in the compositions and methods of the disclosure is a double stranded RNA and is referred to herein as a “double stranded RNAi agent,” “double stranded RNA (dsRNA) molecule,” “dsRNA agent,” or “dsRNA”. The term “dsRNA” refers to a complex of ribonucleic acid molecules, having a duplex structure comprising two anti-parallel and substantially complementary nucleic acid strands, referred to as having “sense” and “antisense” orientations with respect to a target RNA, i.e., an ABLIM3 gene. In some embodiments of the disclosure, a double stranded RNA (dsRNA) triggers the degradation of a target RNA, e.g., an mRNA, through a post-transcriptional gene-silencing mechanism referred to herein as RNA interference or RNAi.

[0135] In general, a dsRNA molecule can include ribonucleotides, but as described in detail herein, each or both strands can also include one or more non-ribonucleotides, e.g., a deoxyribonucleotide, a modified nucleotide. In addition, as used in this specification, an “RNAi agent” may include ribonucleotides with chemical modifications; an RNAi agent may include substantial modifications at multiple nucleotides. As used herein, the term “modified nucleotide” refers to a nucleotide having, independently, a modified sugar moiety, a modified internucleotide linkage, or a modified nucleobase. Thus, the term modified nucleotide encompasses substitutions, additions or removal of, e.g., a functional group or atom, to internucleoside linkages, sugar moieties, or nucleobases. The modifications suitable for use in the agents of the disclosure include all types of modifications disclosed herein or known in the art. Any such modifications, as used in a siRNA type molecule, are encompassed by “RNAi agent” for the purposes of this specification and claims.

[0136] In certain embodiments of the instant disclosure, inclusion of a deoxy-nucleotide if present within an RNAi agent can be considered to constitute a modified nucleotide.

[0137] The duplex region may be of any length that permits specific degradation of a desired target RNA through a RISC pathway, and may range from about 15-36 base pairs in length, for example, about 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, or 36 base pairs in length, such as about 15-30, 15-29, 15-28, 15-27, 15-26, 15-25, 15-24, 15-23, 15-22, 15-21, 15-20, 15-19, 15-18, 15-17, 18-30, 18-29, 18-28, 18-27, 18-26, 18-25, 18-24, 18-23, 18-22, 18-21, 18-20, 19-30, 19-29, 19-28, 19-27, 19-26, 19-25, 19-24, 19-23, 19-22, 19-21, 19-20, 20-30, 20-29, 20-28, 20-27, 20-26, 20-25, 20-24, 20-23, 20-22, 20-21, 21-30, 21-29, 21-28, 21-27, 21-26, 21-25, 21-24, 21-23, or 21-22 base pairs in length. In certain embodiments, the duplex region is 19-21 base pairs in length, e.g., 21 base pairs in length. Ranges and lengths intermediate to the above recited ranges and lengths are also contemplated to be part of the disclosure.

[0138] The two strands forming the duplex structure may be different portions of one larger RNA molecule, or they may be separate RNA molecules. Where the two strands are part of one larger molecule, and they may be connected by an uninterrupted chain of nucleotides between the 3'-end of one strand and the 5'-end of the respective other strand forming the duplex structure, the connecting RNA chain is referred to as a “hairpin loop.” A hairpin loop can comprise at least one unpaired nucleotide. In some embodiments, the hairpin loop can comprise at least 4, at least 5, at least 6, at least 7, at least 8, at least 9, at least 10, at least 20, at least 23 or more unpaired nucleotides or nucleotides not directed to the target site of the dsRNA. In some embodiments, the hairpin loop can be 10 or fewer nucleotides. In some embodiments, the hairpin loop can be 8 or fewer unpaired nucleotides. In some embodiments, the hairpin loop can be 4-10 unpaired nucleotides. In some embodiments, the hairpin loop can be 4-8 nucleotides.

[0139] Where the two substantially complementary strands of a dsRNA are comprised by separate RNA molecules, those molecules need not, but can be covalently connected. In certain embodiments where the two strands are connected covalently by means other than an uninterrupted chain of nucleotides between the 3'-end of one strand and the 5'-end of the respective other strand forming the duplex structure, the connecting structure is referred to as a “linker” (though it is noted that certain other structures defined elsewhere herein can also be referred to as a “linker”). The RNA strands may have the same or a different number of nucleotides. The maximum number of base pairs is the number of nucleotides in the shortest strand of the dsRNA minus any overhangs that are present in the duplex. In addition to the duplex structure, an RNAi may comprise one or more nucleotide overhangs. In one embodiment of the RNAi agent, at least one strand comprises a 3' overhang of at least 1 nucleotide. In another embodiment, at least one strand comprises a 3' overhang of at least 2 nucleotides, e.g., 2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14, or 15 nucleotides. In other embodiments, at least one strand of the RNAi agent comprises a 5' overhang of at least 1 nucleotide. In certain embodiments, at least one strand comprises a 5' overhang of at least 2 nucleotides, e.g., 2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14, or 15 nucleotides. In still other embodiments, both the 3' and the 5' end of one strand of the RNAi agent comprise an overhang of at least 1 nucleotide.

[0140] In one embodiment, an RNAi agent of the disclosure is a dsRNA, each strand of which independently comprises 19-23 nucleotides, that interacts with a target RNA sequence, e.g., an ABLIM3 target mRNA sequence, to direct the cleavage of the target RNA.

[0141] In some embodiments, an iRNA of the invention is a dsRNA of 24-30 nucleotides that interacts with a target RNA sequence, e.g., an ABLIM3 target mRNA sequence, to direct the cleavage of the target RNA.

[0142] As used herein, the term “nucleotide overhang” refers to at least one unpaired nucleotide that protrudes from the duplex structure of an RNAi agent, e.g., a dsRNA. For example, when a 3'-end of one strand of a dsRNA extends beyond the 5'-end of the other strand, or vice versa, there is a nucleotide overhang. A dsRNA can comprise an overhang of at least one nucleotide; alternatively, the overhang can comprise at least two nucleotides, at least three nucleotides, at least four nucleotides, at least five nucleotides or more. A nucleotide overhang can comprise or consist of a nucleotide/nucleoside analog, including a deoxynucleotide/nucleoside. The overhang(s) can be on the sense strand, the antisense strand or any combination thereof. Furthermore, the nucleotide(s) of an overhang can be present on the 5'-end, 3'-end or both ends of either an antisense or sense strand of a dsRNA.

[0143] In one embodiment, the antisense strand of a dsRNA has a 1-10 nucleotide, e.g., a 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 nucleotide, overhang at the 3'-end or the 5'-end. In one embodiment, the sense strand of a dsRNA has a 1-10 nucleotide, e.g., a 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 nucleotide, overhang at the 3'-end or the 5'-end. In another embodiment, one or more of the nucleotides in the overhang is replaced with a nucleoside thiophosphate.

[0144] In certain embodiments, the antisense strand of a dsRNA has a 1-10 nucleotide, e.g., 0-3, 1-3, 2-4, 2-5, 4-10, 5-10, e.g., a 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 nucleotide, overhang at the 3'-end or the 5'-end. In one embodiment, the sense strand of a dsRNA has a 1-10 nucleotide, e.g., a 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 nucleotide, overhang at the 3'-end or the 5'-end. In another embodiment, one or more of the nucleotides in the overhang is replaced with a nucleoside thiophosphate.

[0145] In certain embodiments, the antisense strand of a dsRNA has a 1-15 nucleotide, e.g., 0-3, 1-3, 2-4, 2-5, 4-10, 5-10, 6-12 or e.g., a 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, or 15 nucleotide, overhang at the 3'-end. In another embodiment, one or more of the nucleotides in the overhang is replaced with a nucleoside thiophosphate.

[0146] In certain embodiments, the overhang on the sense strand or the antisense strand, can include extended lengths longer than 10 or 15 nucleotides, e.g., 1-30 nucleotides, 2-30 nucleotides, 10-30 nucleotides, or 10-15 nucleotides in length. In certain embodiments, an extended overhang is on the sense strand of the duplex. In certain embodiments, an extended overhang is present on the 3' end of the sense strand of the duplex. In certain embodiments, an extended overhang is present on the 5' end of the sense strand of the duplex. In certain embodiments, an extended overhang is on the antisense strand of the duplex. In certain embodiments, an extended overhang is present on the 3' end of the antisense strand of the duplex. In certain embodiments, an extended overhang is present on the 5' end of the antisense strand of the duplex. In certain embodiments, one or more of the nucleotides in the overhang is replaced with a nucleoside thiophosphate. In certain embodiments, the overhang includes a self-complementary portion such that the overhang is capable of forming a hairpin structure that is stable under physiological conditions.

[0147] The terms “blunt” or “blunt ended” as used herein in reference to a dsRNA mean that there are no unpaired nucleotides or nucleotide analogs at a given terminal end of a dsRNA, i.e., no nucleotide overhang. One or both ends of a dsRNA can be blunt. Where both ends of a dsRNA are blunt, the dsRNA is said to be blunt ended. To be clear, a “blunt ended” dsRNA is a dsRNA that is blunt at both ends, i.e., no nucleotide overhang at either end of the molecule. Most often such a molecule will be double stranded over its entire length.

[0148] The term “antisense strand” or “guide strand” refers to the strand of an RNAi agent, e.g., a dsRNA, which includes a region that is substantially complementary to a target sequence, e.g., an ABLIM3 mRNA.

[0149] As used herein, the term “region of complementarity” refers to the region on the antisense strand that is substantially complementary to a sequence, for example a target sequence, e.g., an ABLIM3 nucleotide sequence, as defined herein. Where the region of complementarity is not fully complementary to the target sequence, the mismatches can be in the internal or terminal regions of the molecule. Generally, the most tolerated mismatches are in the terminal regions, e.g., within 5, 4, 3, or 2 nucleotides of the 5'- or 3'-terminus of the RNAi agent. In some embodiments, a double stranded RNA agent of the invention includes a nucleotide mismatch in the antisense strand. In some embodiments, the antisense strand of the double stranded RNA agent of the invention includes no more than 4 mismatches with the target mRNA, e.g., the antisense strand includes 4, 3, 2, 1, or 0 mismatches with the target mRNA. In some embodiments, the antisense strand double stranded RNA agent of the invention includes no more than 4 mismatches with the sense strand, e.g., the antisense strand includes 4, 3, 2, 1, or 0 mismatches with the sense strand. In some embodiments, a double stranded RNA agent of the invention includes a nucleotide mismatch in the sense strand. In some embodiments, the sense strand of the double stranded RNA agent of the invention includes no more than 4 mismatches with the antisense strand, e.g., the sense strand includes 4, 3, 2, 1, or 0 mismatches with the antisense strand. In some embodiments, the nucleotide mismatch is, for example, within 5, 4, 3 nucleotides from the 3'-end of the iRNA. In another embodiment, the nucleotide mismatch is, for example, in the 3'-terminal nucleotide of the iRNA agent. In some embodiments, the mismatch(s) is not in the seed region.

[0150] Thus, an RNAi agent as described herein can contain one or more mismatches to the target sequence. In one embodiment, an RNAi agent as described herein contains no more than 3 mismatches (i.e., 3, 2, 1, or 0 mismatches). In one embodiment, an RNAi agent as described herein contains no more than 2 mismatches. In one embodiment, an RNAi agent as described herein contains no more than 1 mismatch. In one embodiment, an RNAi agent as described herein contains 0 mismatches. In certain embodiments, when the antisense strand of the RNAi agent contains mismatches to the

target sequence, then the mismatch can be restricted to be within the last 5 nucleotides from either the 5'- or 3'-end of the region of complementarity. For example, in such embodiments, for a 23 nucleotide RNAi agent, the strand which is complementary to a region of an ABLIM3 gene, generally does not contain any mismatch within the central 13 nucleotides. The methods described herein or methods known in the art can be used to determine whether an RNAi agent containing a mismatch to a target sequence is effective in inhibiting the expression of an ABLIM3 gene. For example, Jackson et al. (*Nat. Biotechnol.* 2003; 21: 635-637) described an expression profile study where the expression of a small set of genes with sequence identity to the MAPK14 siRNA only at 12-18 nt of the sense strand, was down-regulated with similar kinetics to MAPK14. Similarly, Lin et al., (*Nucleic Acids Res.* 2005; 33(14): 4527-4535) using qPCR and reporter assays, showed that a 7 nt complementation between a siRNA and a target is sufficient to cause mRNA degradation of the target. Consideration of the efficacy of RNAi agents with mismatches in inhibiting expression of an ABLIM3 gene is important, especially if the particular region of complementarity in an ABLIM3 gene is known to have polymorphic sequence variation within the population.

[0151] An RNA target may have regions, or spans of the target RNA's nucleotide sequence, which are relatively more susceptible or amenable than other regions of the RNA target to mediating cleavage of the RNA target via RNA interference induced by the binding of an RNAi agent to that region. The increased susceptibility to RNA interference within such "hotspot regions" (or simply "hotspots") means that iRNA agents targeting the region will likely have higher efficacy in inducing iRNA interference than iRNA agents which target other regions of the target RNA. For example, without being bound by theory, the accessibility of a target region of a target RNA may influence the efficacy of iRNA agents which target that region, with some hotspot regions having increased accessibility. Secondary structures, for instance, that form in the RNA target (e.g., within or proximate to hotspot regions) may affect the ability of the iRNA agent to bind the target region and induce RNA interference.

[0152] According to certain aspects of the invention, an iRNA agent may be designed to target a hotspot region of any of the target RNAs described herein, including any identified portions of a target RNA (e.g., a particular exon). As used herein, a hotspot region may refer to an approximately 19-200, 19-150, 19-100, 19-75, 19-50, 21-200, 21-150, 21-100, 21-75, 21-50, 50-200, 50-150, 50-100, 50-75, 75-200, 75-150, 75-100, 100-200, or 100-150 nucleotide region of a target RNA sequence for which targeting using RNAi agents provides an observably higher probability of efficacious silencing relative to targeting other regions of the same target RNA. According to certain aspects of the invention, a hotspot region may comprise a limited region of the target RNA, and in some cases, a substantially limited region of the target, including for example, less than half of the length of the target RNA, such as about 5%, 10%, 15%, 20%, 25%, or 30% of the length of the target RNA. Conversely, the other regions against which a hotspot is compared may cumulatively comprise at least a majority of the length of the target RNA. For example, the other regions may cumulatively comprise at least about 60%, or at least about 70%, or at least about 80%, or at least about 90%, or at least about 95% of the length of the target RNA.

[0153] Compared regions of the target RNA may be empirically evaluated for identification of hotspots using efficacy data obtained from in vitro or in vivo screening assays. For example, RNAi agents targeting various regions that span a target RNA may be compared for frequency of efficacious iRNA agents (e.g., the amount by which target gene expression is inhibited, such as measured by mRNA expression or protein expression) that bind each region. In general, a hotspot can be recognized by observing clustering of multiple efficacious RNAi agents that bind to a limited region of the RNA target. A hotspot may be sufficiently characterized as such by observing efficacy of iRNA agents which cumulatively span at least about 60% of the target region identified as a hotspot, such as about 70%, about 80%, about 90%, or about 95% or more of the length of the region, including both ends of the region (i.e. at least about 60%, 70%, 80%, 90%, or 95% or more of the nucleotides within the region, including the nucleotides at each end of the region, were targeted by an iRNA agent).

According to some aspects of the invention, an iRNA agent which demonstrates at least about 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, or 95% inhibition over the region (e.g., no more than about 50%, 45%, 40%, 35%, 30%, 25%, 20%, 15%, 10%, or 5% mRNA remaining) may be identified as efficacious.

[0154] Amenability to targeting of RNA regions may also be assessed using quantitative comparison of inhibition measurements across different regions of a defined size (e.g., 25, 30, 40, 50, 60, 70, 80, 90, or 100, 110, 120, 130, 140, 150, 160, 170, 180, 190, or 200 nts). For example, an average level of inhibition may be determined for each region and the averages of each region may be compared. The average level of inhibition within a hotspot region may be substantially higher than the average of averages for all evaluated regions. According to some aspects, the average level of inhibition in a hotspot region may be at least about 10%, 20%, 30%, 40%, or 50% higher than the average of averages. According to some aspects, the average level of inhibition in a hotspot region may be at least about 1.0, 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, or 2.0 standard deviations above the average of averages. The average level of inhibition may be higher by a statistically significant (e.g., $p < 0.05$) amount. According to some aspects, each inhibition measurement within a hotspot region may be above a threshold amount (e.g., at or below a threshold amount of mRNA remaining). According to some aspects, each inhibition measurement within the region may be substantially higher than an average of all inhibition measurements across all the measured regions. For example, each inhibition measurement in a hotspot region may be at least about 10%, 20%, 30%, 40%, or 50% higher than the average of all inhibition measurements. According to some aspects, each inhibition measurement may be at least about 1.0, 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, or 2.0 standard deviations above the average of all inhibition measurements. Each inhibition measurement may be higher by a statistically significant (e.g., $p < 0.05$) amount than the average of all inhibition measurements. A standard for evaluating a hotspot may comprise various combinations of the above standards where compatible (e.g., an average level of inhibition of at least about a first amount and having no inhibition measurements below a threshold level of a second amount, lesser than the

first amount).

[0155] It is therefore expressly contemplated that any iRNA agent, including the specific exemplary iRNA agents described herein, which targets a hotspot region of a target RNA, may be preferably selected for inducing RNA interference of the target mRNA as targeting such a hotspot region is likely to exhibit a robust inhibitory response relative to targeting a region which is not a hotspot region. RNAi agents targeting target sequences that substantially overlap (e.g., by at least about 70%, 75%, 80%, 85%, 90%, 95% of the target sequence length) or, preferably, that reside fully within the hotspot region may be considered to target the hotspot region. Hotspot regions of the RNA target(s) of the instant invention may include any region for which the data disclosed herein demonstrates higher frequency of targeting by efficacious RNAi agents, including by any of the standards described elsewhere herein, whether or not the range(s) of such hotspot region(s) are explicitly specified.

[0156] In various embodiments, a dsRNA agent of the present invention targets a hotspot region of an mRNA encoding ABLIM3. In one embodiment, the hotspot region comprises nucleotides 3042-3081, 3420-3457, 4223-4261, 726-763, 1080-1137, 1342-1379, 1910-1955, 2167-2204, 480-520, 1080-1122, 1622-1665, 1744-1787, 2323-2363, 540-578, 1175-1212, 274-313, 349-401, or 2518-2555 of SEQ ID NO: 1. The dsRNA agent may be selected from the group consisting of AD-1809702, AD-1809719, AD-1810038, AD-1810053, AD-1810676, AD-1810692, AD-1807904, AD-1807919, AD-1808238, AD-1808258, AD-1808273, AD-1808418, AD-1808433, AD-1808842, AD-1808865, AD-1809051, AD-1809066, AD-1807707, AD-1807725, AD-1808628, AD-1808649, AD-1808727, AD-1808748, AD-1809187, AD-1809205, AD-1807764, AD-1807780, AD-1808312, AD-1808327, AD-1807561, AD-1807578, AD-1807616, AD-1807631, AD-1807646, AD-1809354, and AD-1809369.

[0157] As used herein, “substantially all of the nucleotides are modified” are largely but not wholly modified and can include not more than 5, 4, 3, 2, or 1 unmodified nucleotides.

[0158] The term “sense strand” or “passenger strand” as used herein, refers to the strand of an RNAi agent that includes a region that is substantially complementary to a region of the antisense strand as that term is defined herein.

[0159] As used herein, the term “cleavage region” refers to a region that is located immediately adjacent to the cleavage site. The cleavage site is the site on the target at which cleavage occurs. In some embodiments, the cleavage region comprises three bases on either end of, and immediately adjacent to, the cleavage site. In some embodiments, the cleavage region comprises two bases on either end of, and immediately adjacent to, the cleavage site. In some embodiments, the cleavage site specifically occurs at the site bound by nucleotides 10 and 11 of the antisense strand, and the cleavage region comprises nucleotides 11, 12 and 13.

[0160] As used herein, and unless otherwise indicated, the term “complementary,” when used to describe a first nucleotide sequence in relation to a second nucleotide sequence, refers to the ability of an oligonucleotide or polynucleotide comprising the first nucleotide sequence to hybridize and form a duplex structure under certain conditions with an oligonucleotide or polynucleotide comprising the second nucleotide sequence, as will be understood by the skilled person. Such conditions can be, for example, “stringent conditions”, including but not limited to, 400 mM NaCl, 40 mM PIPES pH 6.4, 1 mM EDTA, 50° C. or 70° C. for 12-16 hours followed by washing (see, e.g., “Molecular Cloning: A Laboratory Manual, Sambrook, et al. (1989) Cold Spring Harbor Laboratory Press). As used herein, “stringent conditions” or “stringent hybridization conditions” refers to conditions under which an antisense compound will hybridize to its target sequence, but to a minimal number of other sequences. Stringent conditions are sequence-dependent and will be different in different circumstances, and “stringent conditions” under which antisense compounds hybridize to a target sequence are determined by the nature and composition of the antisense compounds and the assays in which they are being investigated. Other conditions, such as physiologically relevant conditions as can be encountered inside an organism, can apply. The skilled person will be able to determine the set of conditions most appropriate for a test of complementarity of two sequences in accordance with the ultimate application of the hybridized nucleotides.

[0161] Complementary sequences within an RNAi agent, e.g., within a dsRNA as described herein, include base-pairing of the oligonucleotide or polynucleotide comprising a first nucleotide sequence to an oligonucleotide or polynucleotide comprising a second nucleotide sequence over the entire length of one or both nucleotide sequences. Such sequences can be referred to as “fully complementary” with respect to each other herein. However, where a first sequence is referred to as “substantially complementary” with respect to a second sequence herein, the two sequences can be fully complementary, or they can form one or more, but generally not more than 5, 4, 3 or 2 mismatched base pairs upon hybridization for a duplex up to 30 base pairs. In some embodiments, the “substantially complementary” sequences disclosed herein comprise a contiguous nucleotide sequence which is at least about 80% complementary over its entire length to the equivalent region of the target ABLIM3 sequence, such as about 85%, about 90%, about 91%, about 92%, about 93%, about 94%, about 95%, about 96%, about 97%, about 98%, or about 99% complementary. However, where two oligonucleotides are designed to form, upon hybridization, one or more single stranded overhangs, such overhangs shall not be regarded as mismatches with regard to the determination of complementarity. For example, a dsRNA comprising one oligonucleotide 21 nucleotides in length and another oligonucleotide 23 nucleotides in length, wherein the longer oligonucleotide comprises a sequence of 21 nucleotides that is fully complementary to the shorter oligonucleotide, can yet be referred to as “fully complementary” for the purposes described herein.

[0162] “Complementary” sequences, as used herein, can also include, or be formed entirely from, non-Watson-Crick base pairs or base pairs formed from non-natural and modified nucleotides, in so far as the above requirements with respect to their ability to hybridize are fulfilled. Such non-Watson-Crick base pairs include, but are not limited to, G:U Wobble or Hoogsteen base pairing.

[0163] The terms “complementary,” “fully complementary” and “substantially complementary” herein can be used with respect to the base matching between two oligonucleotides or polynucleotides, such as the sense strand and the antisense strand of a dsRNA, or between the antisense strand of an RNAi agent and a target sequence, as will be understood from the context of their use.

[0164] As used herein, a polynucleotide that is “substantially complementary to at least part of” a messenger RNA (mRNA) refers to a polynucleotide that is substantially complementary to a contiguous portion of the mRNA of interest (e.g., an mRNA encoding ABLIM3). For example, a polynucleotide is complementary to at least a part of an ABLIM3 mRNA if the sequence is substantially complementary to a non-interrupted portion of an mRNA encoding ABLIM3.

[0165] Accordingly, in some embodiments, the antisense polynucleotides disclosed herein are fully complementary to the target ABLIM3 sequence. In other embodiments, the antisense polynucleotides disclosed herein are substantially complementary to the target ABLIM3 sequence and comprise a contiguous nucleotide sequence which is at least 80% complementary over its entire length to the equivalent region of the nucleotide sequence of SEQ ID NO:1, or a fragment of any one of SEQ ID NOs: 1, 3, 5 and 7, such as about 85%, about 90%, about 91%, about 92%, about 93%, about 94%, about 95%, about 96%, about 97%, about 98%, or about 99% complementary.

[0166] In some embodiments, the antisense polynucleotides disclosed herein are substantially complementary to a fragment of a target ABLIM3 sequence and comprise a contiguous nucleotide sequence which is at least 80% complementary over its entire length to a fragment of SEQ ID NO: 1 selected from the group of nucleotides 80-102, 132-154, 230-252, 274-296, 291-313, 324-346, 349-371, 364-386, 379-401, 412-434, 480-502, 498-520, 540-562, 556-578, 580-602, 620-642, 672-694, 690-712, 711-733, 726-748, 741-763, 777-799, 885-907, 940-962, 981-1003, 999-1021, 1080-1102, 1100-1122, 1115-1137, 1175-1197, 1190-1212, 1240-1262, 1267-1289, 1311-1333, 1342-1364, 1357-1379, 1372-1394, 1406-1428, 1435-1457, 1548-1570, 1592-1614, 1622-1644, 1643-1665, 1672-1694, 1713-1735, 1744-1766, 1765-1787, 1810-1832, 1857-1879, 1882-1904, 1910-1932, 1933-1955, 1969-1991, 2037-2059, 2094-2116, 2125-2147, 2167-2189, 2182-2204, 2216-2238, 2243-2265, 2271-2293, 2323-2345, 2341-2363, 2383-2405, 2420-2442, 2457-2479, 2481-2503, 2518-2540, 2533-2555, 2559-2581, 2583-2605, 2665-2687, 2701-2723, 2727-2749, 2762-2784, 2786-2808, 2828-2850, 2859-2881, 2883-2905, 2935-2957, 2956-2978, 2972-2994, 3009-3031, 3042-3064, 3059-3081, 3077-3099, 3108-3130, 3144-3166, 3201-3223, 3218-3240, 3277-3299, 3301-3323, 3319-3341, 3336-3358, 3386-3408, 3405-3427, 3420-3442, 3435-3457, 3450-3472, 3530-3552, 3546-3568, 3577-3599, 3592-3614, 3624-3646, 3642-3664, 3680-3702, 3704-3726, 3719-3741, 3744-3766, 3761-3783, 3794-3816, 3812-3834, 3847-3869, 3864-3886, 3895-3917, 3917-3939, 3984-4006, 4016-4038, 4035-4057, 4056-4078, 4082-4104, 4097-4119, 4129-4151, 4157-4179, 4223-4245, 4239-4261, 4274-4296, 4289-4311, 4304-4326, 4334-4356, 4352-4374, 4367-4389, 4385-4407, 4402-4424, and 4419-4441 of SEQ ID NO: 1, such as about 85%, about 90%, about 91%, about 92%, about 93%, about 94%, about 95%, about 96%, about 97%, about 98%, or about 99% complementary. Ranges intermediate to the above recited ranges are also contemplated to be part of the disclosure.

[0167] In other embodiments, the antisense polynucleotides disclosed herein are substantially complementary to the target ABLIM3 sequence and comprise a contiguous nucleotide sequence which is at least about 80% complementary over its entire length to any one of the sense strand nucleotide sequences in Tables 3 or 4, or a fragment of any one of the sense strand nucleotide sequences in Tables 3 or 4, such as about 85%, about 90%, about 91%, about 92%, about 93%, about 94%, about 95%, about 96%, about 97%, about 98%, about 99%, or 100% complementary.

[0168] In one embodiment, an RNAi agent of the disclosure includes a sense strand that is substantially complementary to an antisense polynucleotide which, in turn, is the same as a target ABLIM3 sequence, and wherein the sense strand polynucleotide comprises a contiguous nucleotide sequence which is at least about 80% complementary over its entire length to the equivalent region of the nucleotide sequence of SEQ ID NO:1, or a fragment of any one of SEQ ID NO:1, such as about 85%, about 90%, about 91%, about 92%, about 93%, about 94%, about 95%, about 96%, about 97%, about 98%, about 99%, or 100% complementary.

[0169] In some embodiments, an iRNA of the invention includes a sense strand that is substantially complementary to an antisense polynucleotide which, in turn, is complementary to a target ABLIM3 sequence, and wherein the sense strand polynucleotide comprises a contiguous nucleotide sequence which is at least about 80% complementary over its entire length to any one of the antisense strand nucleotide sequences in Tables 3 or 4, or a fragment of any one of the antisense strand nucleotide sequences in Tables 3 or 4, such as about 85%, about 90%, about 91%, about 92%, about 93%, about 94%, about 95%, about 96%, about 97%, about 98%, about 99%, or 100% complementary.

[0170] In certain embodiments, the sense and antisense strands are selected from any one of duplexes AD-1807455, AD-1807481, AD-1807539, AD-1807561, AD-1807578, AD-1807591, AD-1807616, AD-1807631, AD-1807646, AD-1807659, AD-1807707, AD-1807725, AD-1807764, AD-1807780, AD-1807804, AD-1807844, AD-1807870, AD-1807888, AD-1807889, AD-1807904, AD-1807919, AD-1807955, AD-1808063, AD-1808098, AD-1808139, AD-1808157, AD-1808238, AD-1808258, AD-1808273, AD-1808312, AD-1808327, AD-1808356, AD-1808383, AD-1808407, AD-1808418, AD-1808433, AD-1808448, AD-1808482, AD-1808511, AD-1808561, AD-1808598, AD-1808628, AD-1808649, AD-1808678, AD-1808698, AD-1808727, AD-1808748, AD-1808773, AD-1808789, AD-1808814, AD-1808842, AD-1808865, AD-1808901, AD-1808941, AD-1808978, AD-1809009, AD-1809051, AD-1809066, AD-1809080, AD-1809107, AD-1809135, AD-1809187, AD-1809205, AD-1809246, AD-1809283, AD-1809293, AD-1809317, AD-1809354, AD-1809369, AD-1809395, AD-1809414, AD-1809456, AD-1809472, AD-1809498, AD-1809511, AD-1809535, AD-1809550, AD-1809561, AD-1809585, AD-1809596, AD-1809617, AD-1809633, AD-1809670, AD-1809702, AD-1809719, AD-1809737, AD-1809757, AD-1809773, AD-1809822, AD-

1809839, AD-1809895, AD-1809919, AD-1809937, AD-1809954, AD-1810004, AD-1810023, AD-1810038, AD-1810053, AD-1810068, AD-1810127, AD-1810143, AD-1810174, AD-1810189, AD-1810221, AD-1810239, AD-1810255, AD-1810277, AD-1810292, AD-1810296, AD-1810313, AD-1810345, AD-1810363, AD-1810398, AD-1810415, AD-1810446, AD-1810468, AD-1810494, AD-1810526, AD-1810545, AD-1810566, AD-1810591, AD-1810606, AD-1810638, AD-1810666, AD-1810676, AD-1810692, AD-1810727, AD-1810742, AD-1810757, AD-1810767, AD-1810785, AD-1810800, AD-1810818, AD-1810822, and AD-1810839

[0171] In one embodiment, at least partial suppression of the expression of an ABLIM3 gene, is assessed by a reduction of the amount of ABLIM3 mRNA, e.g., sense mRNA, antisense mRNA, total ABLIM3 mRNA, which can be isolated from or detected in a first cell or group of cells in which an ABLIM3 gene is transcribed and which has or have been treated such that the expression of an ABLIM3 gene is inhibited, as compared to a second cell or group of cells substantially identical to the first cell or group of cells but which has or have not been so treated (control cells). The degree of inhibition may be expressed in terms of:

$$[00001] \frac{(\text{mRNA}_{\text{in control cells}}) - (\text{mRNA}_{\text{in treated cells}})}{(\text{mRNA}_{\text{in control cells}})} \cdot \text{Math. } 100\%$$

[0172] The phrase “contacting a cell with an RNAi agent,” such as a dsRNA, as used herein, includes contacting a cell by any possible means. Contacting a cell with an RNAi agent includes contacting a cell in vitro with the RNAi agent or contacting a cell in vivo with the RNAi agent. The contacting may be done directly or indirectly. Thus, for example, the RNAi agent may be put into physical contact with the cell by the individual performing the method, or alternatively, the RNAi agent may be put into a situation that will permit or cause it to subsequently come into contact with the cell.

[0173] Contacting a cell in vitro may be done, for example, by incubating the cell with the RNAi agent. Contacting a cell in vivo may be done, for example, by injecting the RNAi agent into or near the tissue where the cell is located, or by injecting the RNAi agent into another area, e.g., the central nervous system (CNS), optionally via intrathecal, intravitreal or other injection, or to the bloodstream (i.e., intravenous) or the subcutaneous space, such that the agent will subsequently reach the tissue where the cell to be contacted is located. For example, the RNAi agent may contain or be coupled to a ligand, e.g., a lipophilic moiety or moieties as described below and further detailed, e.g., in PCT/US2019/031170, which is incorporated herein by reference, that directs or otherwise stabilizes the RNAi agent at a site of interest, e.g., the CNS. Combinations of in vitro and in vivo methods of contacting are also possible. For example, a cell may also be contacted in vitro with an RNAi agent and subsequently transplanted into a subject.

[0174] In one embodiment, contacting a cell with an RNAi agent includes “introducing” or “delivering the RNAi agent into the cell” by facilitating or effecting uptake or absorption into the cell. Absorption or uptake of an RNAi agent can occur through unaided diffusive or active cellular processes, or by auxiliary agents or devices. Introducing an RNAi agent into a cell may be in vitro or in vivo. For example, for in vivo introduction, an RNAi agent can be injected into a tissue site or administered systemically. In vitro introduction into a cell includes methods known in the art such as electroporation and lipofection. Further approaches are described herein below or are known in the art.

[0175] The term “lipophile” or “lipophilic moiety” broadly refers to any compound or chemical moiety having an affinity for lipids. One way to characterize the lipophilicity of the lipophilic moiety is by the octanol-water partition coefficient, $\log K_{\text{sub.ow}}$, where $K_{\text{sub.ow}}$ is the ratio of a chemical's concentration in the octanol-phase to its concentration in the aqueous phase of a two-phase system at equilibrium. The octanol-water partition coefficient is a laboratory-measured property of a substance. However, it may also be predicted by using coefficients attributed to the structural components of a chemical which are calculated using first-principle or empirical methods (see, for example, Tetko, et. al., *J. Chem. Inf. Comput. Sci.* 41:1407-21 (2001), which is incorporated herein by reference in its entirety). It provides a thermodynamic measure of the tendency of the substance to prefer a non-aqueous or oily milieu rather than water (i.e. its hydrophilic/lipophilic balance). In principle, a chemical substance is lipophilic in character when its $\log K_{\text{sub.ow}}$ exceeds 0. Typically, the lipophilic moiety possesses a $\log K_{\text{sub.ow}}$ exceeding 1, exceeding 1.5, exceeding 2, exceeding 3, exceeding 4, exceeding 5, or exceeding 10. For instance, the $\log K_{\text{sub.ow}}$ of 6-amino hexanol, for instance, is predicted to be approximately 0.7. Using the same method, the $\log K_{\text{sub.ow}}$ of cholesteryl N-(hexan-6-ol) carbamate is predicted to be 10.7.

[0176] The lipophilicity of a molecule can change with respect to the functional group it carries. For instance, adding a hydroxyl group or amine group to the end of a lipophilic moiety can increase or decrease the partition coefficient (e.g., $\log K_{\text{sub.ow}}$) value of the lipophilic moiety.

[0177] Alternatively, the hydrophobicity of the double-stranded RNAi agent, conjugated to one or more lipophilic moieties, can be measured by its protein binding characteristics. For instance, in certain embodiments, the unbound fraction in the plasma protein binding assay of the double-stranded RNAi agent could be determined to positively correlate to the relative hydrophobicity of the double-stranded RNAi agent, which could then positively correlate to the silencing activity of the double-stranded RNAi agent.

[0178] In one embodiment, the plasma protein binding assay determined is an electrophoretic mobility shift assay (EMSA) using human serum albumin protein. An exemplary protocol of this binding assay is illustrated in detail in, e.g., PCT/US2019/031170. The hydrophobicity of the double-stranded RNAi agent, measured by fraction of unbound siRNA in the binding assay, exceeds 0.15, exceeds 0.2, exceeds 0.25, exceeds 0.3, exceeds 0.35, exceeds 0.4, exceeds 0.45, or exceeds 0.5 for an enhanced in vivo delivery of siRNA.

[0179] Accordingly, conjugating the lipophilic moieties to the internal position(s) of the double-stranded RNAi agent provides optimal hydrophobicity for the enhanced in vivo delivery of siRNA.

[0180] The term “lipid nanoparticle” or “LNP” is a vesicle comprising a lipid layer encapsulating a pharmaceutically active molecule, such as a nucleic acid molecule, e.g., a RNAi agent or a plasmid from which an RNAi agent is transcribed. LNPs are described in, for example, U.S. Pat. Nos. 6,858,225, 6,815,432, 8,158,601, and 8,058,069, the entire contents of which are hereby incorporated herein by reference.

[0181] As used herein, a “subject” is an animal, such as a mammal, including a primate (such as a human, a non-human primate, e.g., a monkey, and a chimpanzee), or a non-primate (such as a rat, or a mouse). In one embodiment, the subject is a human, such as a human being treated or assessed for a disease, disorder, or condition that would benefit from reduction in ABLIM3 expression; a human at risk for a disease, disorder, or condition that would benefit from reduction in ABLIM3 expression; a human having a disease, disorder, or condition that would benefit from reduction in ABLIM3 expression; or human being treated for a disease, disorder, or condition that would benefit from reduction in ABLIM3 expression as described herein. In some embodiments, the subject is a female human. In other embodiments, the subject is a male human. In one embodiment, the subject is an adult subject. In one embodiment, the subject is a pediatric subject. In another embodiment, the subject is a juvenile subject, i.e., a subject below 20 years of age.

[0182] As used herein, the terms “treating” or “treatment” refer to a beneficial or desired result including, but not limited to, alleviation or amelioration of one or more signs or symptoms associated with ABLIM3 gene expression or ABLIM3 protein production, e.g., ABLIM3-associated disease, such as PTSD or age-related memory loss. “Treatment” can also mean prolonging survival as compared to expected survival in the absence of treatment.

[0183] The term “lower” in the context of the level of ABLIM3 in a subject or a disease marker or symptom refers to a statistically significant decrease in such level. The decrease can be, for example, at least 10%, 15%, 20%, 25%, 30%, %, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, or more. In certain embodiments, a decrease is at least about 20%. In certain embodiments, the decrease is at least about 30% in a disease marker, e.g., a decrease of 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99%, or more. In certain embodiments, the decrease is at least about 50% in a disease marker. “Lower” in the context of the level of ABLIM3 in a subject is preferably down to a level accepted as within the range of normal for an individual without such disorder. In certain embodiments, “lower” is the decrease in the difference between the level of a marker or symptom for a subject suffering from a disease and a level accepted within the range of normal for an individual, e.g., the level of memory loss as compared to the accepted normal level.

[0184] As used herein, “prevention” or “preventing,” when used in reference to a disease, disorder, or condition thereof, that would benefit from a reduction in expression of an ABLIM3 gene or production of an ABLIM3 protein, refers to a reduction in the likelihood that a subject will develop a symptom or a sign associated with such a disease, disorder, or condition, e.g., a symptom or a sign of an ABLIM3-associated disease, such as PTSD or age-related memory loss. The failure to develop a disease, disorder, or condition, or the reduction in the development of a symptom associated with such a disease, disorder, or condition (e.g., by at least about 10% on a clinically accepted scale for that disease or disorder), or the exhibition of delayed symptoms delayed (e.g., by days, weeks, months or years) is considered effective prevention.

[0185] As used herein, the term “ABLIM3-associated disease,” “ABLIM3-associated condition,” or “ABLIM3-associated disorder” includes any disease, condition, or disorder that would benefit from reduction in the expression and/or activity of ABLIM3. Exemplary ABLIM3-associated conditions include PTSD or age-related memory loss.

[0186] “Therapeutically effective amount,” as used herein, is intended to include the amount of an RNAi agent that, when administered to a subject having an ABLIM3-associated disease, is sufficient to effect treatment of the disease (e.g., by diminishing, ameliorating, or maintaining the existing disease or one or more symptoms of disease). The “therapeutically effective amount” may vary depending on the RNAi agent, how the agent is administered, the disease and its severity and the history, age, weight, family history, genetic makeup, the types of preceding or concomitant treatments, if any, and other individual characteristics of the subject to be treated.

[0187] “Prophylactically effective amount,” as used herein, is intended to include the amount of an RNAi agent that, when administered to a subject having an ABLIM3-associated disorder, is sufficient to prevent or ameliorate the disease or one or more symptoms of the disease. Ameliorating the disease includes slowing the course of the disease or reducing the severity of later-developing disease. The “prophylactically effective amount” may vary depending on the RNAi agent, how the agent is administered, the degree of risk of disease, and the history, age, weight, family history, genetic makeup, the types of preceding or concomitant treatments, if any, and other individual characteristics of the patient to be treated.

[0188] A “therapeutically-effective amount” or “prophylactically effective amount” also includes an amount of an RNAi agent that produces some desired local or systemic effect at a reasonable benefit/risk ratio applicable to any treatment. An RNAi agent employed in the methods of the present disclosure may be administered in a sufficient amount to produce a reasonable benefit/risk ratio applicable to such treatment.

[0189] The phrase “pharmaceutically acceptable” is employed herein to refer to those compounds, materials, compositions, or dosage forms which are, within the scope of sound medical judgment, suitable for use in contact with the tissues of human subjects and animal subjects without excessive toxicity, irritation, allergic response, or other problem or complication, commensurate with a reasonable benefit/risk ratio.

[0190] The phrase “pharmaceutically-acceptable carrier” as used herein means a pharmaceutically-acceptable material, composition or vehicle, such as a liquid or solid filler, diluent, excipient, manufacturing aid (e.g., lubricant, talc magnesium, calcium or zinc stearate, or steric acid), or solvent encapsulating material, involved in carrying or transporting the subject compound from one organ, or portion of the body, to another organ, or portion of the body. Each carrier must be “acceptable” in the sense of being compatible with the other ingredients of the formulation and not injurious to the subject

being treated. Some examples of materials which can serve as pharmaceutically-acceptable carriers include: (1) sugars, such as lactose, glucose and sucrose; (2) starches, such as corn starch and potato starch; (3) cellulose, and its derivatives, such as sodium carboxymethyl cellulose, ethyl cellulose and cellulose acetate; (4) powdered tragacanth; (5) malt; (6) gelatin; (7) lubricating agents, such as magnesium stearate, sodium lauryl sulfate and talc; (8) excipients, such as cocoa butter and suppository waxes; (9) oils, such as peanut oil, cottonseed oil, safflower oil, sesame oil, olive oil, corn oil and soybean oil; (10) glycols, such as propylene glycol; (11) polyols, such as glycerin, sorbitol, mannitol and polyethylene glycol; (12) esters, such as ethyl oleate and ethyl laurate; (13) agar; (14) buffering agents, such as magnesium hydroxide and aluminum hydroxide; (15) alginic acid; (16) pyrogen-free water; (17) isotonic saline; (18) Ringer's solution; (19) ethyl alcohol; (20) pH buffered solutions; (21) polyesters, polycarbonates or polyanhydrides; (22) bulking agents, such as polypeptides and amino acids (23) serum component, such as serum albumin, HDL and LDL; and (22) other non-toxic compatible substances employed in pharmaceutical formulations.

[0191] The term "sample," as used herein, includes a collection of similar fluids, cells, or tissues isolated from a subject, as well as fluids, cells, or tissues present within a subject. Examples of biological fluids include blood, serum and serosal fluids, plasma, cerebrospinal fluid, ocular fluids, lymph, urine, saliva, and the like. Tissue samples may include samples from tissues, organs or localized regions. For example, samples may be derived from particular organs, parts of organs, or fluids or cells within those organs. In certain embodiments, samples may be derived from the brain (e.g., whole brain or certain segments of brain, e.g., hippocampus, striatum, or certain types of cells in the brain, such as, e.g., dentate granule cells, neurons and glial cells (astrocytes, oligodendrocytes, microglial cells)). In some embodiments, a "sample derived from a subject" refers to blood drawn from the subject or plasma or serum derived therefrom. In further embodiments, a "sample derived from a subject" refers to brain tissue (or subcomponents thereof) or retinal tissue (or subcomponents thereof) derived from the subject.

[0192] The term "substituted" refers to the replacement of one or more hydrogen radicals in a given structure with the radical of a specified substituent including, but not limited to: alkyl, alkenyl, alkynyl, aryl, heterocyclyl, halo, thiol, alkylthio, arylthio, alkylthioalkyl, arylthioalkyl, alkylsulfonyl, alkylsulfonylalkyl, arylsulfonylalkyl, alkoxy, aryloxy, aralkoxy, aminocarbonyl, alkylaminocarbonyl, arylaminocarbonyl, alkoxycarbonyl, aryloxy carbonyl, haloalkyl, amino, trifluoromethyl, cyano, nitro, alkylamino, arylamino, alkylaminoalkyl, arylaminoalkyl, aminoalkylamino, hydroxy, alkoxyalkyl, carboxyalkyl, alkoxycarbonylalkyl, aminocarbonylalkyl, acyl, aralkoxycarbonyl, carboxylic acid, sulfonic acid, sulfonyl, phosphonic acid, aryl, heteroaryl, heterocyclic, and aliphatic. It is understood that the substituent can be further substituted.

[0193] The term "alkyl" refers to saturated and unsaturated non-aromatic hydrocarbon chains that may be a straight chain or branched chain, containing the indicated number of carbon atoms (these include without limitation propyl, allyl, or propargyl), which may be optionally inserted with N, O, or S. For example, "(C1-C6) alkyl" means a radical having from 1 to 6 carbon atoms in a linear or branched arrangement. "(C1-C6) alkyl" includes, for example, methyl, ethyl, propyl, isopropyl, n-butyl, tert-butyl, pentyl and hexyl. In certain embodiments, a lipophilic moiety of the instant disclosure can include a C6-C18 alkyl hydrocarbon chain.

[0194] The term "alkylene" refers to an optionally substituted saturated aliphatic branched or straight chain divalent hydrocarbon radical having the specified number of carbon atoms. For example, "(C1-C6) alkylene" means a divalent saturated aliphatic radical having from 1-6 carbon atoms in a linear arrangement, e.g., [(CH.sub.2).sub.n], where n is an integer from 1 to 6. "(C1-C6) alkylene" includes methylene, ethylene, propylene, butylene, pentylene and hexylene. Alternatively, "(C1-C6) alkylene" means a divalent saturated radical having from 1-6 carbon atoms in a branched arrangement, for example: [(CH.sub.2)CH.sub.2CH.sub.2CH.sub.2CH(CH.sub.3)], [(CH.sub.2)CH.sub.2CH.sub.2CH(CH.sub.3)CH.sub.2], [(CH.sub.2)CH(CH.sub.3)CH(CH.sub.3)CH(CH.sub.3)], and the like.

The term "alkylenedioxy" refers to a divalent species of the structure —O—R—O—, in which R represents an alkylene.

[0195] The term "mercapto" refers to an —SH radical. The term "thioalkoxy" refers to an —S-alkyl radical.

[0196] The term "halo" refers to any radical of fluorine, chlorine, bromine or iodine. "Halogen" and "halo" are used interchangeably herein.

[0197] As used herein, the term "cycloalkyl" means a saturated or unsaturated nonaromatic hydrocarbon ring group having from 3 to 14 carbon atoms, unless otherwise specified. For example, "(C3-C10) cycloalkyl" means a hydrocarbon radical of a (3-10)-membered saturated aliphatic cyclic hydrocarbon ring. Examples of cycloalkyl groups include, but are not limited to, cyclopropyl, methyl-cyclopropyl, 2,2-dimethyl-cyclobutyl, 2-ethyl-cyclopentyl, cyclohexyl, etc. Cycloalkyls may include multiple spiro- or fused rings. Cycloalkyl groups are optionally mono-, di-, tri-, tetra-, or penta-substituted on any position as permitted by normal valency.

[0198] As used herein, the term "alkenyl" refers to a non-aromatic hydrocarbon radical, straight or branched, containing at least one carbon-carbon double bond, and having from 2 to 10 carbon atoms unless otherwise specified. Up to five carbon-carbon double bonds may be present in such groups. For example, "C2-C6" alkenyl is defined as an alkenyl radical having from 2 to 6 carbon atoms. Examples of alkenyl groups include, but are not limited to, ethenyl, propenyl, butenyl, and cyclohexenyl. The straight, branched, or cyclic portion of the alkenyl group may contain double bonds and is optionally mono-, di-, tri-, tetra-, or penta-substituted on any position as permitted by normal valency. The term "cycloalkenyl" means a monocyclic hydrocarbon group having the specified number of carbon atoms and at least one carbon-carbon double bond.

[0199] As used herein, the term "alkynyl" refers to a hydrocarbon radical, straight or branched, containing from 2 to 10 carbon atoms, unless otherwise specified, and containing at least one carbon-carbon triple bond. Up to 5 carbon-carbon

triple bonds may be present. Thus, “C2-C6 alkynyl” means an alkynyl radical having from 2 to 6 carbon atoms. Examples of alkynyl groups include, but are not limited to, ethynyl, 2-propynyl, and 2-butynyl. The straight or branched portion of the alkynyl group may contain triple bonds as permitted by normal valency, and may be optionally mono-, di-, tri-, tetra-, or penta-substituted on any position as permitted by normal valency.

[0200] As used herein, “alkoxy” or “alkoxy” refers to an alkyl group as defined above with the indicated number of carbon atoms attached through an oxygen bridge. For example, “(C1-C3)alkoxy” includes methoxy, ethoxy and propoxy. For example, “(C1-C6)alkoxy”, is intended to include C1, C2, C3, C4, C5, and C6 alkoxy groups. For example, “(C1-C8)alkoxy”, is intended to include C1, C2, C3, C4, C5, C6, C7, and C8 alkoxy groups. Examples of alkoxy include, but are not limited to, methoxy, ethoxy, n-propoxy, i-propoxy, n-butoxy, s-butoxy, t-butoxy, n-pentoxy, s-pentoxy, n-heptoxy, and n-octoxy. “Alkylthio” means an alkyl radical attached through a sulfur linking atom. The terms “alkylamino” or “aminoalkyl”, means an alkyl radical attached through an NH linkage. “Dialkylamino” means two alkyl radical attached through a nitrogen linking atom. The amino groups may be unsubstituted, monosubstituted, or di-substituted. In some embodiments, the two alkyl radicals are the same (e.g., N,N-dimethylamino). In some embodiments, the two alkyl radicals are different (e.g., N-ethyl-N-methylamino).

[0201] As used herein, “aryl” or “aromatic” means any stable monocyclic or polycyclic carbon ring of up to 7 atoms in each ring, wherein at least one ring is aromatic. Examples of aryl groups include, but are not limited to, phenyl, naphthyl, anthracenyl, tetrahydronaphthyl, indanyl, and biphenyl. In cases where the aryl substituent is bicyclic and one ring is non-aromatic, it is understood that attachment is via the aromatic ring. Aryl groups are optionally mono-, di-, tri-, tetra-, or penta-substituted on any position as permitted by normal valency. The term “arylalkyl” or the term “aralkyl” refers to alkyl substituted with an aryl. The term “arylalkoxy” refers to an alkoxy substituted with aryl.

[0202] “Hetero” refers to the replacement of at least one carbon atom in a ring system with at least one heteroatom selected from N, S and O. “Hetero” also refers to the replacement of at least one carbon atom in an acyclic system. A hetero ring system or a hetero acyclic system may have, for example, 1, 2 or 3 carbon atoms replaced by a heteroatom.

[0203] As used herein, the term “heteroaryl” represents a stable monocyclic or polycyclic ring of up to 7 atoms in each ring, wherein at least one ring is aromatic and contains from 1 to 4 heteroatoms selected from the group consisting of O, N and S. Examples of heteroaryl groups include, but are not limited to, acridinyl, carbazolyl, cinnolinyl, quinoxalinyl, pyrazolyl, indolyl, benzotriazolyl, furanyl, thienyl, benzothienyl, benzofuranyl, benzimidazolonyl, benzoxazolonyl, quinolinyl, isoquinolinyl, dihydroisoindolonyl, imidazopyridinyl, isoindolonyl, indazolyl, oxazolyl, oxadiazolyl, isoxazolyl, indolyl, pyrazinyl, pyridazinyl, pyridinyl, pyrimidinyl, pyrrolyl, tetrahydroquinoline. “Heteroaryl” is also understood to include the N-oxide derivative of any nitrogen-containing heteroaryl. In cases where the heteroaryl substituent is bicyclic and one ring is non-aromatic or contains no heteroatoms, it is understood that attachment is via the aromatic ring or via the heteroatom containing ring. Heteroaryl groups are optionally mono-, di-, tri-, tetra-, or penta-substituted on any position as permitted by normal valency.

[0204] As used herein, the term “heterocycle,” “heterocyclic,” or “heterocyclyl” means a 3- to 14-membered aromatic or nonaromatic heterocycle containing from 1 to 4 heteroatoms selected from the group consisting of O, N and S, including polycyclic groups. As used herein, the term “heterocyclic” is also considered to be synonymous with the terms “heterocycle” and “heterocyclyl” and is understood as also having the same definitions set forth herein. “Heterocyclyl” includes the above mentioned heteroaryls, as well as dihydro and tetrahydro analogs thereof. Examples of heterocyclyl groups include, but are not limited to, azetidiny, benzoimidazolyl, benzofuranyl, benzofurazanyl, benzopyrazolyl, benzotriazolyl, benzothiophenyl, benzoxazolyl, carbazolyl, carbolinyl, cinnolinyl, furanyl, imidazolyl, indolinyl, indolyl, indolaziny, indazolyl, isobenzofuranyl, isoindolyl, isoquinolyl, isothiazolyl, isoxazolyl, naphthpyridinyl, oxadiazolyl, oxooxazolidinyl, oxazolyl, oxazoline, oxopiperazinyl, oxopyrrolidinyl, oxomorpholinyl, isoxazoline, oxetanyl, pyranyl, pyrazinyl, pyrazolyl, pyridazinyl, pyridopyridinyl, pyridazinyl, pyridyl, pyridinonyl, pyrimidyl, pyrimidinonyl, pyrrolyl, quinazolinyl, quinolyl, quinoxalinyl, tetrahydropyranyl, tetrahydrofuranyl, tetrahydrothiopyranyl, tetrahydroisoquinolinyl, tetrazolyl, tetrazolopyridyl, thiadiazolyl, thiazolyl, thienyl, triazolyl, 1,4-dioxanyl, hexahydroazepinyl, piperazinyl, piperidinyl, pyridin-2-onyl, pyrrolidinyl, morpholinyl, thiomorpholinyl, dihydrobenzoimidazolyl, dihydrobenzofuranyl, dihydrobenzothiophenyl, dihydrobenzoxazolyl, dihydrofuranyl, dihydroimidazolyl, dihydroindolyl, dihydroisooxazolyl, dihydroisothiazolyl, dihydrooxadiazolyl, dihydrooxazolyl, dihydropyrazinyl, dihydropyrazolyl, dihydropyridinyl, dihydropyrimidinyl, dihydropyrrolyl, dihydroquinolinyl, dihydrotetrazolyl, dihydrothiadiazolyl, dihydrothiazolyl, dihydrothienyl, dihydrotriazolyl, dihydroazetidiny, dioxidothiomorpholinyl, methylenedioxybenzoyl, tetrahydrofuranyl, and tetrahydrothienyl, and N-oxides thereof. Attachment of a heterocyclyl substituent can occur via a carbon atom or via a heteroatom. Heterocyclyl groups are optionally mono-, di-, tri-, tetra-, or penta-substituted on any position as permitted by normal valency.

[0205] “Heterocycloalkyl” refers to a cycloalkyl residue in which one to four of the carbons is replaced by a heteroatom such as oxygen, nitrogen or sulfur. Examples of heterocycles whose radicals are heterocyclyl groups include tetrahydropyran, morpholine, pyrrolidine, piperidine, thiazolidine, oxazole, oxazoline, isoxazole, dioxane, tetrahydrofuran and the like.

[0206] The term “heteroaryl” refers to an aromatic 5-8 membered monocyclic, 8-12 membered bicyclic, or 11-14 membered tricyclic ring system having 1-3 heteroatoms if monocyclic, 1-6 heteroatoms if bicyclic, or 1-9 heteroatoms if tricyclic, said heteroatoms selected from O, N, or S (e.g., carbon atoms and 1-3, 1-6, or 1-9 heteroatoms of N, O, or S if monocyclic, bicyclic, or tricyclic, respectively), wherein 0, 1, 2, 3, or 4 atoms of each ring may be substituted by a substituent. Examples of heteroaryl groups include pyridyl, furyl or furanyl, imidazolyl, benzimidazolyl, pyrimidinyl,

thiophenyl or thienyl, quinolinyl, indolyl, thiazolyl, and the like. The term “heteroarylalkyl” or the term “heteroalkyl” refers to an alkyl substituted with a heteroaryl. The term “heteroarylalkoxy” refers to an alkoxy substituted with heteroaryl. [0207] The term “cycloalkyl” as employed herein includes saturated and partially unsaturated cyclic hydrocarbon groups having 3 to 12 carbons, for example, 3 to 8 carbons, and, for example, 3 to 6 carbons, wherein the cycloalkyl group additionally may be optionally substituted. Cycloalkyl groups include, without limitation, cyclopropyl, cyclobutyl, cyclopentyl, cyclopentenyl, cyclohexyl, cyclohexenyl, cycloheptyl, and cyclooctyl. [0208] The term “acyl” refers to an alkylcarbonyl, cycloalkylcarbonyl, arylcarbonyl, heterocyclylcarbonyl, or heteroarylcarbonyl substituent, any of which may be further substituted by substituents. [0209] As used herein, “keto” refers to any alkyl, alkenyl, alkynyl, cycloalkyl, cycloalkenyl, heterocyclyl, heteroaryl, or aryl group as defined herein attached through a carbonyl bridge. [0210] Examples of keto groups include, but are not limited to, alkanoyl (e.g., acetyl, propionyl, butanoyl, pentanoyl, hexanoyl), alkenoyl (e.g., acryloyl) alkynoyl (e.g., ethynoyl, propynoyl, butynoyl, pentynoyl, hexynoyl), aryloyl (e.g., benzoyl), heteroaryloyl (e.g., pyrroloyl, imidazoloyl, quinolinoyl, pyridinoyl). [0211] As used herein, “alkoxycarbonyl” refers to any alkoxy group as defined above attached through a carbonyl bridge (i.e., —C(O)O-alkyl). Examples of alkoxycarbonyl groups include, but are not limited to, methoxycarbonyl, ethoxycarbonyl, iso-propoxycarbonyl, n-propoxycarbonyl, t-butoxycarbonyl, benzyloxycarbonyl or n-pentoxycarbonyl. [0212] As used herein, “aryloxycarbonyl” refers to any aryl group as defined herein attached through an oxycarbonyl bridge (i.e., —C(O)O-aryl). Examples of aryloxycarbonyl groups include, but are not limited to, phenoxycarbonyl and naphthylloxycarbonyl.

[0213] As used herein, “heteroaryloxycarbonyl” refers to any heteroaryl group as defined herein attached through an oxycarbonyl bridge (i.e., —C(O)O-heteroaryl). Examples of heteroaryloxycarbonyl groups include, but are not limited to, 2-pyridylloxycarbonyl, 2-oxazolylloxycarbonyl, 4-thiazolylloxycarbonyl, or pyrimidinylloxycarbonyl.

[0214] The term “oxo” refers to an oxygen atom, which forms a carbonyl when attached to carbon, an N-oxide when attached to nitrogen, and a sulfoxide or sulfone when attached to sulfur.

[0215] The person of ordinary skill in the art would readily understand and appreciate that the compounds and compositions disclosed herein may have certain atoms (e.g., N, O, or S atoms) in a protonated or deprotonated state, depending upon the environment in which the compound or composition is placed. Accordingly, as used herein, the structures disclosed herein envisage that certain functional groups, such as, for example, OH, SH, or NH, may be protonated or deprotonated. The disclosure herein is intended to cover the disclosed compounds and compositions regardless of their state of protonation based on the pH of the environment, as would be readily understood by the person of ordinary skill in the art.

II. RNAi Agents of the Disclosure

[0216] Described herein are RNAi agents that inhibit the expression of an ABLIM3 gene. In one embodiment, the RNAi agent includes double stranded ribonucleic acid (dsRNA) molecules for inhibiting the expression of an ABLIM3 gene in a cell, such as a cell within a subject, e.g., a mammal, such as a human having an ABLIM3-associated disease, e.g., PTSD or age-related memory loss. The dsRNA includes an antisense strand having a region of complementarity which is complementary to at least a part of an mRNA formed in the expression of an ABLIM3 gene. The region of complementarity is about 15-30 nucleotides or less in length. Upon contact with a cell expressing the ABLIM3 gene, the RNAi agent inhibits the expression of the ABLIM3 gene (e.g., a human gene, a primate gene, a non-primate gene) by at least 30% as compared to a similar cell not contacted with the RNAi agent or an RNAi agent not complementary to the ABLIM3 gene. Expression of the gene may be assayed by, for example, a PCR or branched DNA (bDNA)-based method, or by a protein-based method, such as by immunofluorescence analysis, using, for example, western blotting or flowcytometric techniques. In one embodiment, the level of knockdown is assayed in monkey Cos-7 cells using an assay method provided in Example 2 below. In another embodiment, the level of knockdown is assayed in human BE(2)-C cells. In some embodiments, the level of knockdown is assayed in DU145 cells.

[0217] A dsRNA includes two RNA strands that are complementary and hybridize to form a duplex structure under conditions in which the dsRNA will be used. One strand of a dsRNA (the antisense strand) includes a region of complementarity that is substantially complementary, or fully complementary, to a target sequence. The target sequence can be derived from the sequence of an mRNA formed during the expression of an ABLIM3 gene. The other strand (the sense strand) includes a region that is complementary to the antisense strand, such that the two strands hybridize and form a duplex structure when combined under suitable conditions. As described elsewhere herein and as known in the art, the complementary sequences of a dsRNA can also be contained as self-complementary regions of a single nucleic acid molecule, as opposed to being on separate oligonucleotides.

[0218] Generally, the duplex structure is 15 to 30 base pairs in length, e.g., 15-29, 15-28, 15-27, 15-26, 15-25, 15-24, 15-23, 15-22, 15-21, 15-20, 15-19, 15-18, 15-17, 18-30, 18-29, 18-28, 18-27, 18-26, 18-25, 18-24, 18-23, 18-22, 18-21, 18-20, 19-30, 19-29, 19-28, 19-27, 19-26, 19-25, 19-24, 19-23, 19-22, 19-21, 19-20, 20-30, 20-29, 20-28, 20-27, 20-26, 20-25, 20-24, 20-23, 20-22, 20-21, 21-30, 21-29, 21-28, 21-27, 21-26, 21-25, 21-24, 21-23, or 21-22 base pairs in length. In certain embodiments, the duplex structure is 18 to 25 base pairs in length, e.g., 18-25, 18-24, 18-23, 18-22, 18-21, 18-20, 19-25, 19-24, 19-23, 19-22, 19-21, 19-20, 20-25, 20-24, 20-23, 20-22, 20-21, 21-25, 21-24, 21-23, 21-22, 22-25, 22-24, 22-23, 23-25, 23-24 or 24-25 base pairs in length, for example, 19-21 base pairs in length. Ranges and lengths intermediate to the above recited ranges and lengths are also contemplated to be part of the disclosure.

[0219] Similarly, the region of complementarity to the target sequence is 15 to 30 nucleotides in length, e.g., 15-29, 15-28,

15-27, 15-26, 15-25, 15-24, 15-23, 15-22, 15-21, 15-20, 15-19, 15-18, 15-17, 18-30, 18-29, 18-28, 18-27, 18-26, 18-25, 18-24, 18-23, 18-22, 18-21, 18-20, 19-30, 19-29, 19-28, 19-27, 19-26, 19-25, 19-24, 19-23, 19-22, 19-21, 19-20, 20-30, 20-29, 20-28, 20-27, 20-26, 20-25, 20-24, 20-23, 20-22, 20-21, 21-30, 21-29, 21-28, 21-27, 21-26, 21-25, 21-24, 21-23, or 21-22 nucleotides in length, for example 19-23 nucleotides in length or 21-23 nucleotides in length. Ranges and lengths intermediate to the above recited ranges and lengths are also contemplated to be part of the disclosure.

[0220] In some embodiments, the duplex structure is 19 to 30 base pairs in length. Similarly, the region of complementarity to the target sequence is 19 to 30 nucleotides in length.

[0221] In some embodiments, the dsRNA is 15 to 23 nucleotides in length, 19 to 23 nucleotides in length, or 25 to 30 nucleotides in length. In general, the dsRNA is long enough to serve as a substrate for the Dicer enzyme. For example, it is well known in the art that dsRNAs longer than about 21-23 nucleotides can serve as substrates for Dicer. As the ordinarily skilled person will also recognize, the region of an RNA targeted for cleavage will most often be part of a larger RNA molecule, often an mRNA molecule. Where relevant, a "part" of an mRNA target is a contiguous sequence of an mRNA target of sufficient length to allow it to be a substrate for RNAi-directed cleavage (i.e., cleavage through a RISC pathway).

[0222] One of skill in the art will also recognize that the duplex region is a primary functional portion of a dsRNA, e.g., a duplex region of about 15 to 36 base pairs, e.g., 15-36, 15-35, 15-34, 15-33, 15-32, 15-31, 15-30, 15-29, 15-28, 15-27, 15-26, 15-25, 15-24, 15-23, 15-22, 15-21, 15-20, 15-19, 15-18, 15-17, 18-30, 18-29, 18-28, 18-27, 18-26, 18-25, 18-24, 18-23, 18-22, 18-21, 18-20, 19-30, 19-29, 19-28, 19-27, 19-26, 19-25, 19-24, 19-23, 19-22, 19-21, 19-20, 20-30, 20-29, 20-28, 20-27, 20-26, 20-25, 20-24, 20-23, 20-22, 20-21, 21-30, 21-29, 21-28, 21-27, 21-26, 21-25, 21-24, 21-23, or 21-22 base pairs, for example, 19-21 base pairs. Thus, in one embodiment, to the extent that it becomes processed to a functional duplex, of e.g., 15-30 base pairs, that targets a desired RNA for cleavage, an RNA molecule or complex of RNA molecules having a duplex region greater than 30 base pairs is a dsRNA. Thus, an ordinarily skilled artisan will recognize that in one embodiment, a miRNA is a dsRNA. In another embodiment, a dsRNA is not a naturally occurring miRNA. In another embodiment, an RNAi agent useful to target ABLIM3 expression is not generated in the target cell by cleavage of a larger dsRNA.

[0223] A dsRNA as described herein can further include one or more single-stranded nucleotide overhangs e.g., 1, 2, 3, or 4 nucleotides. A nucleotide overhang can comprise or consist of a nucleotide/nucleoside analog, including a deoxynucleotide/nucleoside. The overhang(s) can be on the sense strand, the antisense strand or any combination thereof. Furthermore, the nucleotide(s) of an overhang can be present on the 5'-end, 3'-end or both ends of either an antisense or sense strand of a dsRNA.

[0224] A dsRNA can be synthesized by standard methods known in the art. Double stranded RNAi compounds of the invention may be prepared using a two-step procedure. First, the individual strands of the double stranded RNA molecule are prepared separately. Then, the component strands are annealed. The individual strands of the dsRNA compound can be prepared using solution-phase or solid-phase organic synthesis or both. Organic synthesis offers the advantage that the oligonucleotide strands comprising unnatural or modified nucleotides can be easily prepared. Similarly, single-stranded oligonucleotides of the invention can be prepared using solution-phase or solid-phase organic synthesis or both.

[0225] In one aspect, a dsRNA of the disclosure includes at least two nucleotide sequences, a sense sequence and an antisense sequence. The sense strand sequence for ABLIM3 may be selected from the group of sequences provided in Tables 3 or 4, and the corresponding nucleotide sequence of the antisense strand of the sense strand may be selected from the group of sequences of Tables 3 or 4. In this aspect, one of the two sequences is complementary to the other of the two sequences, with one of the sequences being substantially complementary to a sequence of an mRNA generated in the expression of an ABLIM3 gene. As such, in this aspect, a dsRNA will include two oligonucleotides, where one oligonucleotide is described as the sense strand (passenger strand) in Tables 3 or 4, and the second oligonucleotide is described as the corresponding antisense strand (guide strand) of the sense strand in Tables 3 or 4.

[0226] In one embodiment, the substantially complementary sequences of the dsRNA are contained on separate oligonucleotides. In another embodiment, the substantially complementary sequences of the dsRNA are contained on a single oligonucleotide.

[0227] It will be understood that, although the sequences in Tables 3 or 4 are described as modified or conjugated sequences, the RNA of the RNAi agent of the disclosure e.g., a dsRNA of the disclosure, may comprise any one of the sequences set forth in Tables 3 or 4 that is un-modified, un-conjugated, or modified or conjugated differently than described therein. For example, although the sense strands of the agents of the invention may be conjugated to a GalNAc ligand, these agents may be conjugated to a moiety that directs delivery to the CNS, e.g., a C16 ligand, as described herein. A lipophilic ligand can be included in any of the positions provided in the instant application.

[0228] The skilled person is well aware that dsRNAs having a duplex structure of about 20 to 23 base pairs, e.g., 21, base pairs have been hailed as particularly effective in inducing RNA interference (Elbashir et al., (2001) EMBO J., 20:6877-6888). However, others have found that shorter or longer RNA duplex structures can also be effective (Chu and Rana (2007) RNA 14:1714-1719; Kim et al. (2005) Nat Biotech 23:222-226). In the embodiments described above, by virtue of the nature of the oligonucleotide sequences provided herein, dsRNAs described herein can include at least one strand of a length of minimally 21 nucleotides. It can be reasonably expected that shorter duplexes minus only a few nucleotides on one or both ends can be similarly effective as compared to the dsRNAs described above. Hence, dsRNAs having a sequence of at least 15, 16, 17, 18, 19, 20, or more contiguous nucleotides derived from one of the sequences provided herein, and differing in their ability to inhibit the expression of an ABLIM3 gene by not more than 10, 15, 20, 25, 30, 35, 40, 45 or 50% inhibition from a dsRNA comprising the full sequence using the in vitro assay with, e.g., BE(2)-C cells or

DU145 cells and a 10 nM concentration of the RNA agent and the PCR assay as provided in the examples herein, are contemplated to be within the scope of the present disclosure. In some embodiments, inhibition from a dsRNA comprising the full sequence is measured using the in vitro assay with primary mouse hepatocytes.

[0229] In addition, the RNA agents described herein identify a site(s) in an ABLIM3 mRNA transcript that is susceptible to RISC-mediated cleavage. As such, the present disclosure further features RNAi agents that target within this site(s). As used herein, an RNAi agent is said to “target within” a particular site of an mRNA transcript if the RNAi agent promotes cleavage of the mRNA transcript anywhere within that particular site. Such an RNAi agent will generally include at least about 15 contiguous nucleotides, preferably at least 19 nucleotides, from one of the sequences provided herein coupled to additional nucleotide sequences taken from the region contiguous to the selected sequence in an ABLIM3 gene.

III. Modified RNAi Agents of the Disclosure

[0230] In one embodiment, the RNA of the RNAi agent of the disclosure e.g., a dsRNA, is un-modified, and does not comprise modified nucleotides, e.g., chemical modifications or conjugations known in the art and described herein. In one embodiment, the RNA of an RNAi agent of the disclosure, e.g., a dsRNA, is chemically modified to enhance stability or other beneficial characteristics. In certain embodiments of the disclosure, substantially all of the nucleotides of an RNAi agent of the disclosure are modified. In other embodiments of the disclosure, all of the nucleotides of an RNAi agent of the disclosure are modified. RNAi agents of the disclosure in which “substantially all of the nucleotides are modified” are largely but not wholly modified and can include not more than 5, 4, 3, 2, or unmodified nucleotides. In still other embodiments of the disclosure, RNAi agents of the disclosure can include not more than 5, 4, 3, 2 or 1 modified nucleotides.

[0231] The nucleic acids featured in the disclosure can be synthesized or modified by methods well established in the art, such as those described in “Current protocols in nucleic acid chemistry,” Beaucage, S. L. et al. (Edrs.), John Wiley & Sons, Inc., New York, NY, USA, which is hereby incorporated herein by reference. Modifications include, for example, end modifications, e.g., 5'-end modifications (phosphorylation, conjugation, inverted linkages) or 3'-end modifications (conjugation, DNA nucleotides, inverted linkages, etc.); base modifications, e.g., replacement with stabilizing bases, destabilizing bases, or bases that base pair with an expanded repertoire of partners, removal of bases (abasic nucleotides), or conjugated bases; sugar modifications (e.g., at the 2'-position or 4'-position) or replacement of the sugar; or backbone modifications, including modification or replacement of the phosphodiester linkages. Specific examples of RNAi agents useful in the embodiments described herein include, but are not limited to, RNAs containing modified backbones or no natural internucleoside linkages. RNAs having modified backbones include, among others, those that do not have a phosphorus atom in the backbone. For the purposes of this specification, and as sometimes referenced in the art, modified RNAs that do not have a phosphorus atom in their internucleoside backbone can also be considered to be oligonucleosides. In some embodiments, a modified RNAi agent will have a phosphorus atom in its internucleoside backbone.

[0232] Modified RNA backbones include, for example, phosphorothioates, chiral phosphorothioates, phosphorodithioates, phosphotriesters, aminoalkylphosphotriesters, methyl and other alkyl phosphonates including 3'-alkylene phosphonates and chiral phosphonates, phosphinates, phosphoramidates including 3'-amino phosphoramidate and aminoalkylphosphoramidates, thionophosphoramidates, thionoalkylphosphonates, thionoalkylphosphotriesters, and boranophosphates having normal 3'-5' linkages, 2'-5'-linked analogs of these, and those having inverted polarity wherein the adjacent pairs of nucleoside units are linked 3'-5' to 5'-3' or 2'-5' to 5'-2'. Various salts, mixed salts and free acid forms are also included. In some embodiments of the invention, the dsRNA agents of the invention are in a free acid form. In other embodiments of the invention, the dsRNA agents of the invention are in a salt form. In one embodiment, the dsRNA agents of the invention are in a sodium salt form. In certain embodiments, when the dsRNA agents of the invention are in the sodium salt form, sodium ions are present in the agent as counterions for substantially all of the phosphodiester and/or phosphorothioate groups present in the agent. Agents in which substantially all of the phosphodiester and/or phosphorothioate linkages have a sodium counterion include not more than 5, 4, 3, 2, or 1 phosphodiester and/or phosphorothioate linkages without a sodium counterion. In some embodiments, when the dsRNA agents of the invention are in the sodium salt form, sodium ions are present in the agent as counterions for all of the phosphodiester and/or phosphorothioate groups present in the agent.

[0233] Representative U.S. patents that teach the preparation of the above phosphorus-containing linkages include, but are not limited to, U.S. Pat. Nos. 3,687,808; 4,469,863; 4,476,301; 5,023,243; 5,177,195; 5,188,897; 5,264,423; 5,276,019; 5,278,302; 5,286,717; 5,321,131; 5,399,676; 5,405,939; 5,453,496; 5,455,233; 5,466,677; 5,476,925; 5,519,126; 5,536,821; 5,541,316; 5,550,111; 5,563,253; 5,571,799; 5,587,361; 5,625,050; 6,028,188; 6,124,445; 6,160,109; 6,169,170; 6,172,209; 6,239,265; 6,277,603; 6,326,199; 6,346,614; 6,444,423; 6,531,590; 6,534,639; 6,608,035; 6,683,167; 6,858,715; 6,867,294; 6,878,805; 7,015,315; 7,041,816; 7,273,933; 7,321,029; and RE39,464, the entire contents of each of which are hereby incorporated herein by reference.

[0234] Modified RNA backbones that do not include a phosphorus atom therein have backbones that are formed by short chain alkyl or cycloalkyl internucleoside linkages, mixed heteroatoms and alkyl or cycloalkyl internucleoside linkages, or one or more short chain heteroatomic or heterocyclic internucleoside linkages. These include those having morpholino linkages (formed in part from the sugar portion of a nucleoside); siloxane backbones; sulfide, sulfoxide and sulfone backbones; formacetyl and thioformacetyl backbones; methylene formacetyl and thioformacetyl backbones; alkene containing backbones; sulfamate backbones; methyleneimino and methylenehydrazino backbones; sulfonate and sulfonamide backbones; amide backbones; and others having mixed N, O, S and CH₂ component parts.

[0235] Representative U.S. patents that teach the preparation of the above oligonucleosides include, but are not limited to,

U.S. Pat. Nos. 5,034,315; 5,166,315; 5,185,444; 5,214,134; 5,216,141; 5,235,033; 5,64,562; 5,264,564; 5,405,938; 5,434,257; 5,466,677; 5,470,967; 5,489,677; 5,541,307; 5,561,225; 5,596,086; 5,602,240; 5,608,046; 5,610,289; 5,618,704; 5,623,070; 5,663,312; 5,633,360; 5,677,437; and, 5,677,439, the entire contents of each of which are hereby incorporated herein by reference.

[0236] In other embodiments, suitable RNA mimetics are contemplated for use in RNAi agents, in which both the sugar and the internucleoside linkage, i.e., the backbone, of the nucleotide units are replaced with alternate groups. The nucleobase units are maintained for hybridization with an appropriate nucleic acid target compound. One such oligomeric compound, a RNA mimetic that has been shown to have excellent hybridization properties, is referred to as a peptide nucleic acid (PNA). In PNA compounds, the sugar backbone of an RNA is replaced with an amide containing backbone, in particular an aminoethylglycine backbone. The nucleobases are retained and are bound directly or indirectly to aza nitrogen atoms of the amide portion of the backbone. Representative U.S. patents that teach the preparation of PNA compounds include, but are not limited to, U.S. Pat. Nos. 5,539,082; 5,714,331; and 5,719,262, the entire contents of each of which are hereby incorporated herein by reference. Additional PNA compounds suitable for use in the RNAi agents of the disclosure are described in, for example, in Nielsen et al., *Science*, 1991, 254, 1497-1500.

[0237] Some embodiments featured in the disclosure include RNAs with phosphorothioate backbones and oligonucleosides with heteroatom backbones, and in particular —CH.sub.2—NH—CH.sub.2—, —CH.sub.2—N(CH.sub.3)—O—CH.sub.2— [known as a methylene (methylimino) or MMI backbone], —CH.sub.2—O—N(CH.sub.3)—CH.sub.2—, —CH.sub.2—N(CH.sub.3)—N(CH.sub.3)—CH.sub.2— and —N(CH.sub.3)—CH.sub.2—CH.sub.2— of the above-referenced U.S. Pat. No. 5,489,677, and the amide backbones of the above-referenced U.S. Pat. No. 5,602,240. In some embodiments, the RNAs featured herein have morpholino backbone structures of the above-referenced U.S. Pat. No. 5,034,506. The native phosphodiester backbone can be represented as —O—P(O)(OH)—OCH.sub.2—.

[0238] Modified RNAs can also contain one or more substituted sugar moieties. The RNAi agents, e.g., dsRNAs, featured herein can include one of the following at the 2'-position: OH; F; O—, S—, or N-alkyl; O—, S—, or N-alkenyl; O-, S- or N-alkynyl; or O-alkyl-O-alkyl, wherein the alkyl, alkenyl and alkynyl can be substituted or unsubstituted C.sub.1 to C.sub.10 alkyl or C.sub.2 to C.sub.10 alkenyl and alkynyl. Exemplary suitable modifications include O[(CH.sub.2).sub.nO].sub.mCH.sub.3, O(CH.sub.2).sub.nOCH.sub.3, O(CH.sub.2).sub.nNH.sub.2, O(CH.sub.2).sub.nCH.sub.3, O(CH.sub.2).sub.nONH.sub.2, and O(CH.sub.2).sub.nON[(CH.sub.2).sub.nCH.sub.3].sub.2, where n and m are from 1 to about 10. In other embodiments, dsRNAs include one of the following at the 2' position: C.sub.1 to C.sub.10 alkyl, substituted alkyl, alkaryl, aralkyl, O-alkaryl or O-aralkyl, SH, SCH.sub.3, OCN, Cl, Br, CN, CF.sub.3, OCF.sub.3, SOCH.sub.3, SO.sub.2CH.sub.3, ONO.sub.2, NO.sub.2, N.sub.3, NH.sub.2, heterocycloalkyl, heterocycloalkaryl, aminoalkylamino, polyalkylamino, substituted silyl, an RNA cleaving group, a reporter group, an intercalator, a group for improving the pharmacokinetic properties of an RNAi agent, or a group for improving the pharmacodynamic properties of an RNAi agent, and other substituents having similar properties. In some embodiments, the modification includes a 2'-methoxyethoxy (2'-O—CH.sub.2CH.sub.2OCH.sub.3, also known as 2'-O-(2-methoxyethyl) or 2'-MOE) (Martin et al., *Helv. Chm. Acta*, 1995, 78:486-504) i.e., an alkoxy-alkoxy group. Another exemplary modification is 2'-dimethylaminooxyethoxy, i.e., a O(CH.sub.2).sub.2ON(CH.sub.3).sub.2 group, also known as 2'-DMAOE, as described in examples herein below, and 2'-dimethylaminoethoxyethoxy (also known in the art as 2'-O-dimethylaminoethoxyethyl or 2'-DMAEOE), i.e., 2'-O—CH.sub.2—O—CH.sub.2—N(CH.sub.3).sub.2. Further exemplary modifications include: 5'-Me-2'-F nucleotides, 5'-Me-2'-OMe nucleotides, 5'-Me-2'-deoxynucleotides, (both R and S isomers in these three families); 2'-alkoxyalkyl; and 2'-NMA (N-methylacetamide).

[0239] Other modifications include 2'-methoxy (2'-OCH.sub.3), 2'-aminopropoxy (2'-OCH.sub.2CH.sub.2CH.sub.2NH.sub.2), 2'-O-hexadecyl, and 2'-fluoro (2'-F). Similar modifications can also be made at other positions on the RNA of an RNAi agent, particularly the 3' position of the sugar on the 3' terminal nucleotide or in 2'-5' linked dsRNAs and the 5' position of 5' terminal nucleotide. RNAi agents can also have sugar mimetics such as cyclobutyl moieties in place of the pentofuranosyl sugar. Representative U.S. patents that teach the preparation of such modified sugar structures include, but are not limited to, U.S. Pat. Nos. 4,981,957; 5,118,800; 5,319,080; 5,359,044; 5,393,878; 5,446,137; 5,466,786; 5,514,785; 5,519,134; 5,567,811; 5,576,427; 5,591,722; 5,597,909; 5,610,300; 5,627,053; 5,639,873; 5,646,265; 5,658,873; 5,670,633; and 5,700,920, certain of which are commonly owned with the instant application. The entire contents of each of the foregoing are hereby incorporated herein by reference.

[0240] An RNAi agent of the disclosure can also include nucleobase (often referred to in the art simply as "base") modifications or substitutions. As used herein, "unmodified" or "natural" nucleobases include the purine bases adenine (A) and guanine (G), and the pyrimidine bases thymine (T), cytosine (C) and uracil (U). Modified nucleobases include other synthetic and natural nucleobases such as 5-methylcytosine (5-me-C), 5-hydroxymethyl cytosine, xanthine, hypoxanthine, 2-aminoadenine, 6-methyl and other alkyl derivatives of adenine and guanine, 2-propyl and other alkyl derivatives of adenine and guanine, 2-thiouracil, 2-thiothymine and 2-thiocytosine, 5-halouracil and cytosine, 5-propynyl uracil and cytosine, 6-azo uracil, cytosine and thymine, 5-uracil (pseudouracil), 4-thiouracil, 8-halo, 8-amino, 8-thiol, 8-thioalkyl, 8-hydroxyl and other 8-substituted adenines and guanines, 5-halo, particularly 5-bromo, 5-trifluoromethyl and other 5-substituted uracils and cytosines, 7-methylguanine and 7-methyladenine, 8-azaguanine and 8-azaadenine, 7-deazaguanine and 7-daazaadenine and 3-deazaguanine and 3-deazaadenine. Further modified nucleobases include those disclosed in U.S. Pat. No. 3,687,808, those disclosed in *Modified Nucleosides in Biochemistry, Biotechnology and Medicine*, Herdewijn, P. ed. Wiley-VCH, 2008; those disclosed in *The Concise Encyclopedia Of Polymer Science And Engineering*, pages 858-859,

Kroschwitz, J. L, ed. John Wiley & Sons, 1990, these disclosed by Englisch et al., (1991) *Angewandte Chemie, International Edition*, 30:613, and those disclosed by Sanghvi, Y S., Chapter 15, dsRNA Research and Applications, pages 289-302, Crooke, S. T. and Lebleu, B., Ed., CRC Press, 1993. Certain of these modified nucleobases are particularly useful for increasing the binding affinity of the oligomeric compounds featured in the disclosure. These include 5-substituted pyrimidines, 6-azapyrimidines and N-2, N-6 and O-6 substituted purines, including 2-aminopropyladenine, 5-propynyluracil and 5-propynylcytosine. 5-methylcytosine substitutions have been shown to increase nucleic acid duplex stability by 0.6-1.2° C. (Sanghvi, Y. S., Crooke, S. T. and Lebleu, B., Eds., dsRNA Research and Applications, CRC Press, Boca Raton, 1993, pp. 276-278) and are exemplary base substitutions, even more particularly when combined with 2'-O-methoxyethyl sugar modifications.

[0241] Representative U.S. patents that teach the preparation of certain of the above noted modified nucleobases as well as other modified nucleobases include, but are not limited to, the above noted U.S. Pat. Nos. 3,687,808, 4,845,205; 5,130,302; 5,134,066; 5,175,273; 5,367,066; 5,432,272; 5,457,187; 5,459,255; 5,484,908; 5,502,177; 5,525,711; 5,552,540; 5,587,469; 5,594,121, 5,596,091; 5,614,617; 5,681,941; 5,750,692; 6,015,886; 6,147,200; 6,166,197; 6,222,025; 6,235,887; 6,380,368; 6,528,640; 6,639,062; 6,617,438; 7,045,610; 7,427,672; and 7,495,088, the entire contents of each of which are hereby incorporated herein by reference.

[0242] An RNAi agent of the disclosure can also be modified to include one or more bicyclic sugar moieties. A “bicyclic sugar” is a furanosyl ring modified by the bridging of two atoms. A “bicyclic nucleoside” (“BNA”) is a nucleoside having a sugar moiety comprising a bridge connecting two carbon atoms of the sugar ring, thereby forming a bicyclic ring system. In certain embodiments, the bridge connects the 4'-carbon and the 2'-carbon of the sugar ring. Thus, in some embodiments an agent of the disclosure may include one or more locked nucleic acids (LNA). A locked nucleic acid is a nucleotide having a modified ribose moiety in which the ribose moiety comprises an extra bridge connecting the 2' and 4' carbons. In other words, an LNA is a nucleotide comprising a bicyclic sugar moiety comprising a 4'-CH₂-O-2' bridge. This structure effectively “locks” the ribose in the 3'-endo structural conformation. The addition of locked nucleic acids to siRNAs has been shown to increase siRNA stability in serum, and to reduce off-target effects (Elmen, J. et al., (2005) *Nucleic Acids Research* 33(1):439-447; Mook, OR. et al., (2007) *Mol Canc Ther* 6(3):833-843; Grunweller, A. et al., (2003) *Nucleic Acids Research* 31(12):3185-3193). Examples of bicyclic nucleosides for use in the polynucleotides of the disclosure include without limitation nucleosides comprising a bridge between the 4' and the 2' ribosyl ring atoms. In certain embodiments, the antisense polynucleotide agents of the disclosure include one or more bicyclic nucleosides comprising a 4' to 2' bridge. Examples of such 4' to 2' bridged bicyclic nucleosides, include but are not limited to 4'-(CH_{sub.2})—O-2' (LNA); 4'-(CH_{sub.2})—S-2'; 4'-(CH_{sub.2})_{sub.2}—O-2' (ENA); 4'-CH(CH_{sub.3})—O-2' (also referred to as “constrained ethyl” or “cEt”) and 4'-CH(CH_{sub.2}OCH_{sub.3})—O-2' (and analogs thereof; see, e.g., U.S. Pat. No. 7,399,845); 4'-C(CH_{sub.3})(CH_{sub.3})O-2' (and analogs thereof; see e.g., U.S. Pat. No. 8,278,283); 4'-CH_{sub.2}—N(OCH_{sub.3})-2' (and analogs thereof; see e.g., U.S. Pat. No. 8,278,425); 4'-CH_{sub.2}—O—N(CH_{sub.3})-2' (see, e.g., U.S. Patent Publication No. 2004/0171570); 4'-CH_{sub.2}—N(R)—O-2', wherein R is H, C_{sub.1}-C_{sub.12} alkyl, or a protecting group (see, e.g., U.S. Pat. No. 7,427,672); 4'-CH_{sub.2}—C(H)(CH_{sub.3})-2' (see, e.g., Chattopadhyaya et al, *J. Org. Chem.*, 2009, 74, 118-134); and 4'-CH_{sub.2}—C(=CH_{sub.2})-2' (and analogs thereof; see, e.g., U.S. Pat. No. 8,278,426). The entire contents of each of the foregoing are hereby incorporated herein by reference.

[0243] Additional representative US patents and US Patent Publications that teach the preparation of locked nucleic acid nucleotides include, but are not limited to, the following: U.S. Pat. Nos. 6,268,490; 6,525,191; 6,670,461; 6,770,748; 6,794,499; 6,998,484; 7,053,207; 7,034,133; 7,084,125; 7,399,845; 7,427,672; 7,569,686; 7,741,457; 8,022,193; 8,030,467; 8,278,425; 8,278,426; 8,278,283; US 2008/0039618; and US 2009/0012281, the entire contents of each of which are hereby incorporated herein by reference.

[0244] Any of the foregoing bicyclic nucleosides can be prepared having one or more stereochemical sugar configurations including for example α-L-ribofuranose and β-D-ribofuranose (see WO 99/14226).

[0245] An RNAi agent of the disclosure can also be modified to include one or more constrained ethyl nucleotides. As used herein, a “constrained ethyl nucleotide” or “cEt” is a locked nucleic acid comprising a bicyclic sugar moiety comprising a 4'-CH(CH₃)-O-2' bridge. In one embodiment, a constrained ethyl nucleotide is in the S conformation referred to herein as “S-cEt.”

[0246] An RNAi agent of the disclosure may also include one or more “conformationally restricted nucleotides” (“CRN”). CRN are nucleotide analogs with a linker connecting the C2' and C4' carbons of ribose or the C3 and —C5' carbons of ribose. CRN lock the ribose ring into a stable conformation and increase the hybridization affinity to mRNA. The linker is of sufficient length to place the oxygen in an optimal position for stability and affinity resulting in less ribose ring puckering.

[0247] Representative publications that teach the preparation of certain of the above noted CRN include, but are not limited to, US 2013/0190383; and WO 2013/036868, the entire contents of each of which are hereby incorporated herein by reference.

[0248] In some embodiments, an RNAi agent of the disclosure comprises one or more monomers that are UNA (unlocked nucleic acid) nucleotides. UNA is unlocked acyclic nucleic acid, wherein any of the bonds of the sugar has been removed, forming an unlocked “sugar” residue. In one example, UNA also encompasses monomer with bonds between C1'-C4' have been removed (i.e. the covalent carbon-oxygen-carbon bond between the C1' and C4' carbons). In another example, the C2'-C3' bond (i.e. the covalent carbon-carbon bond between the C2' and C3' carbons) of the sugar has been removed (see *Nuc. Acids Symp. Series*, 52, 133-134 (2008) and Fluiter et al., *Mol. Biosyst.*, 2009, 10, 1039 hereby incorporated by

reference).

[0249] Representative U.S. publications that teach the preparation of UNA include, but are not limited to, U.S. Pat. No. 8,314,227; and US Patent Publication Nos. 2013/0096289; 2013/0011922; and 2011/0313020, the entire contents of each of which are hereby incorporated herein by reference.

[0250] An RNAi agent of the disclosure may also include one or more “cyclohexene nucleic acids” or (“CeNA”). CeNA are nucleotide analogs with a replacement of the furanose moiety of DNA by a cyclohexene ring. Incorporation of cyclohexenyl nucleosides in a DNA chain increases the stability of a DNA/RNA hybrid. CeNA is stable against degradation in serum and a CeNA/RNA hybrid is able to activate *E. coli* RNase H, resulting in cleavage of the RNA strand. (see Wang et al., Am. Chem. Soc. 2000, 122, 36, 8595-8602, hereby incorporated by reference).

[0251] Potentially stabilizing modifications to the ends of RNA molecules can include N-(acetylaminocaproyl)-4-hydroxyprolinol (Hyp-C6-NHAc), N-(caproyl-4-hydroxyprolinol (Hyp-C6), N-(acetyl-4-hydroxyprolinol (Hyp-NHAc), thymidine-2'-O-deoxythymidine (ether), N-(aminocaproyl)-4-hydroxyprolinol (Hyp-C6-amino), 2-docosanoyl-uridine-3'-phosphate, inverted base dT(idT) and others. Disclosure of this modification can be found in WO 2011/005861.

[0252] Other modifications of an RNAi agent of the disclosure include a 5' phosphate or 5' phosphate mimic, e.g., a 5'-terminal phosphate or phosphate mimic on the antisense strand of an RNAi agent. Suitable phosphate mimics are disclosed in, for example US 2012/0157511, the entire contents of which are incorporated herein by reference. In one embodiment, the double stranded RNAi agent of the invention further comprises a 5'-phosphate or a 5'-phosphate mimic at the 5' nucleotide of the antisense strand. In another embodiment, the double stranded RNAi agent further comprises a 5'-phosphate mimic at the 5' nucleotide of the antisense strand. In a specific embodiment, the 5'-phosphate mimic is a 5'-vinyl phosphonate (5'-VP). In one embodiment, the phosphate mimic is a 5'-cyclopropyl phosphonate (VP). In some embodiments, the 5'-end of the antisense strand of the double-stranded iRNA agent does not contain a 5'-vinyl phosphonate (VP).

[0253] In one embodiment, at least one of the modified nucleotides is selected from the group consisting of a deoxy-nucleotide, a 2'-O-methyl modified nucleotide, a 2'-fluoro modified nucleotide, a 2'-deoxy-modified nucleotide, a glycol modified nucleotide (GNA), e.g., Ggn, Cgn, Tgn, or Agn, a nucleotide with a 2' phosphate, e.g., G2p, C2p, A2p or U2p, and, a vinyl-phosphonate nucleotide; and combinations thereof. In other embodiments, each of the duplexes of Tables 3 and 4 may be particularly modified to provide another double-stranded iRNA agent of the present disclosure. In one example, the 3'-terminus of each sense duplex may be modified by removing the 3'-terminal L96 ligand and exchanging the two phosphodiester internucleotide linkages between the three 3'-terminal nucleotides with phosphorothioate internucleotide linkages. That is, the three 3'-terminal nucleotides (N) of a sense sequence of the formula:

5'-N.sub.1— . . . —N.sub.n-2N.sub.n-1N.sub.nL96 3'

may be replaced with

5'-N.sub.1— . . . —N.sub.n-2sN.sub.n-1sN.sub.n3';

while the antisense sequence remains unchanged to provide another double-stranded iRNA agent of the present disclosure.

A. Modified RNAi Agents Comprising Motifs of the Disclosure

[0254] In certain aspects of the disclosure, the double-stranded RNAi agents of the disclosure include agents with chemical modifications as disclosed, for example, in WO 2013/075035, the entire contents of which are incorporated herein by reference. As shown herein and in WO 2013/075035, a superior result may be obtained by introducing one or more motifs of three identical modifications on three consecutive nucleotides into a sense strand or antisense strand of an RNAi agent, particularly at or near the cleavage site. In some embodiments, the sense strand and antisense strand of the RNAi agent may otherwise be completely modified. The introduction of these motifs interrupts the modification pattern, if present, of the sense or antisense strand. The RNAi agent may be optionally conjugated with a lipophilic ligand, e.g., a C16 ligand, for instance on the sense strand. The RNAi agent may be optionally modified with a (S)-glycol nucleic acid (GNA) modification, for instance on one or more residues of the antisense strand. The resulting RNAi agents present superior gene silencing activity.

[0255] Accordingly, the disclosure provides double stranded RNAi agents capable of inhibiting the expression of a target gene (i.e., an ABLIM3 gene) in vivo. The RNAi agent comprises a sense strand and an antisense strand. Each strand of the RNAi agent may be 15-30 nucleotides in length. For example, each strand may be 16-30 nucleotides in length, 17-30 nucleotides in length, 25-30 nucleotides in length, 27-30 nucleotides in length, 17-23 nucleotides in length, 17-21 nucleotides in length, 17-19 nucleotides in length, 19-25 nucleotides in length, 19-23 nucleotides in length, 19-21 nucleotides in length, 21-25 nucleotides in length, or 21-23 nucleotides in length. In certain embodiments, each strand is 19-23 nucleotides in length.

[0256] The sense strand and antisense strand typically form a duplex double stranded RNA (“dsRNA”), also referred to herein as an “RNAi agent.” The duplex region of an RNAi agent may be 15-30 nucleotide pairs in length. For example, the duplex region can be 16-30 nucleotide pairs in length, 17-30 nucleotide pairs in length, 27-30 nucleotide pairs in length, 17-23 nucleotide pairs in length, 17-21 nucleotide pairs in length, 17-19 nucleotide pairs in length, 19-25 nucleotide pairs in length, 19-23 nucleotide pairs in length, 19-21 nucleotide pairs in length, 21-25 nucleotide pairs in length, or 21-23 nucleotide pairs in length. In another example, the duplex region is selected from 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, and 27 nucleotides in length. In another embodiment, the duplex region is 19-21 nucleotide pairs in length.

[0257] In one embodiment, the RNAi agent may contain one or more overhang regions or capping groups at the 3'-end, 5'-end, or both ends of one or both strands. The overhang can be 1-6 nucleotides in length, for instance 2-6 nucleotides in length, 1-5 nucleotides in length, 2-5 nucleotides in length, 1-4 nucleotides in length, 2-4 nucleotides in length, 1-3 nucleotides in length, 2-3 nucleotides in length, or 1-2 nucleotides in length. In another embodiment, the nucleotide overhang region is 2 nucleotides in length. The overhangs can be the result of one strand being longer than the other, or the result of two strands of the same length being staggered. The overhang can form a mismatch with the target mRNA or it can be complementary to the gene sequences being targeted or can be another sequence. The first and second strands can also be joined, e.g., by additional bases to form a hairpin, or by other non-base linkers.

[0258] In one embodiment, the nucleotides in the overhang region of the RNAi agent can each independently be a modified or unmodified nucleotide including, but not limited to 2'-sugar modified, such as, 2'-F, 2'-O-methyl, thymidine (T), and any combinations thereof.

[0259] For example, TT can be an overhang sequence for either end on either strand. The overhang can form a mismatch with the target mRNA or it can be complementary to the gene sequences being targeted or can be another sequence.

[0260] The 5'- or 3'-overhangs at the sense strand, antisense strand or both strands of the RNAi agent may be phosphorylated. In some embodiments, the overhang region(s) contains two nucleotides having a phosphorothioate between the two nucleotides, where the two nucleotides can be the same or different. In one embodiment, the overhang is present at the 3'-end of the sense strand, antisense strand, or both strands. In one embodiment, this 3'-overhang is present in the antisense strand. In one embodiment, this 3'-overhang is present in the sense strand.

[0261] The dsRNAi agent may contain only a single overhang, which can strengthen the interference activity of the RNAi, without affecting its overall stability. For example, the single-stranded overhang may be located at the 3'-terminal end of the sense strand or, alternatively, at the 3'-terminal end of the antisense strand. The RNAi may also have a blunt end, located at the 5'-end of the antisense strand (i.e., the 3'-end of the sense strand) or vice versa. Generally, the antisense strand of the RNAi has a nucleotide overhang at the 3'-end, and the 5'-end is blunt. While not wishing to be bound by theory, the asymmetric blunt end at the 5'-end of the antisense strand and 3'-end overhang of the antisense strand favor the guide strand loading into RISC process.

[0262] In one embodiment, the RNAi agent is a double blunt-ended of 19 nucleotides in length, wherein the sense strand contains at least one motif of three 2'-F modifications on three consecutive nucleotides at positions 7, 8, and 9 from the 5' end. The antisense strand contains at least one motif of three 2'-O-methyl modifications on three consecutive nucleotides at positions 11, 12, and 13 from the 5' end.

[0263] In another embodiment, the RNAi agent is a double blunt-ended of 20 nucleotides in length, wherein the sense strand contains at least one motif of three 2'-F modifications on three consecutive nucleotides at positions 8, 9, and 10 from the 5' end. The antisense strand contains at least one motif of three 2'-O-methyl modifications on three consecutive nucleotides at positions 11, 12, and 13 from the 5' end.

[0264] In yet another embodiment, the RNAi agent is a double blunt-ended of 21 nucleotides in length, wherein the sense strand contains at least one motif of three 2'-F modifications on three consecutive nucleotides at positions 9, 10, and 11 from the 5' end. The antisense strand contains at least one motif of three 2'-O-methyl modifications on three consecutive nucleotides at positions 11, 12, and 13 from the 5' end.

[0265] In one embodiment, the RNAi agent comprises a 21 nucleotide sense strand and a 23 nucleotide antisense strand, wherein the sense strand contains at least one motif of three 2'-F modifications on three consecutive nucleotides at positions 9, 10, and 11 from the 5' end; the antisense strand contains at least one motif of three 2'-O-methyl modifications on three consecutive nucleotides at positions 11, 12, and 13 from the 5' end, wherein one end of the RNAi agent is blunt, while the other end comprises a 2 nucleotide overhang. Preferably, the 2 nucleotide overhang is at the 3'-end of the antisense strand.

[0266] When the 2 nucleotide overhang is at the 3'-end of the antisense strand, there may be two phosphorothioate internucleotide linkages between the terminal three 3'-nucleotides of the antisense strand, wherein two of the three nucleotides are the overhang nucleotides, and the third nucleotide is a paired nucleotide next to the overhang nucleotide. In one embodiment, the RNAi agent additionally has two phosphorothioate internucleotide linkages between the terminal three nucleotides at both the 5'-end of the sense strand and at the 5'-end of the antisense strand. In one embodiment, every nucleotide in the sense strand and the antisense strand of the RNAi agent, including the nucleotides that are part of the motifs are modified nucleotides. In one embodiment each residue is independently modified with a 2'-O-methyl or 2'-fluoro, e.g., in an alternating motif. Optionally, the RNAi agent further comprises a ligand (e.g., a lipophilic ligand, optionally a C16 ligand).

[0267] In one embodiment, the RNAi agent comprises a sense and an antisense strand, wherein the sense strand is 25-30 nucleotide residues in length, wherein starting from the 5' terminal nucleotide (position 1) positions 1 to 23 of the first strand comprise at least 8 ribonucleotides; the antisense strand is 36-66 nucleotide residues in length and, starting from the 3' terminal nucleotide, comprises at least 8 ribonucleotides in the positions paired with positions 1-23 of sense strand to form a duplex; wherein at least the 3' terminal nucleotide of antisense strand is unpaired with sense strand, and up to 6 consecutive 3' terminal nucleotides are unpaired with sense strand, thereby forming a 3' single stranded overhang of 1-6 nucleotides; wherein the 5' terminus of antisense strand comprises from 10-30 consecutive nucleotides which are unpaired with sense strand, thereby forming a 10-30 nucleotide single stranded 5' overhang; wherein at least the sense strand 5' terminal and 3' terminal nucleotides are base paired with nucleotides of antisense strand when sense and antisense strands are aligned for maximum complementarity, thereby forming a substantially duplexed region between sense and antisense

strands; and antisense strand is sufficiently complementary to a target RNA along at least 19 ribonucleotides of antisense strand length to reduce target gene expression when the double stranded nucleic acid is introduced into a mammalian cell; and wherein the sense strand contains at least one motif of three 2'-F modifications on three consecutive nucleotides, where at least one of the motifs occurs at or near the cleavage site. The antisense strand contains at least one motif of three 2'-O-methyl modifications on three consecutive nucleotides at or near the cleavage site.

[0268] In one embodiment, the dsRNAi agent comprises sense and antisense strands, wherein the dsRNAi agent comprises a first strand having a length which is at least 25 and at most 29 nucleotides and a second strand having a length which is at most 30 nucleotides with at least one motif of three 2'-O-methyl modifications on three consecutive nucleotides at position 11, 12, and 13 from the 5' end; wherein the 3' end of the first strand and the 5' end of the second strand form a blunt end and the second strand is 1-4 nucleotides longer at its 3' end than the first strand, wherein the duplex region which is at least 25 nucleotides in length, and the second strand is sufficiently complementary to a target mRNA along at least 19 nucleotide of the second strand length to reduce target gene expression when the RNAi agent is introduced into a mammalian cell, and wherein dicer cleavage of the RNAi agent preferentially results in an siRNA comprising the 3' end of the second strand, thereby reducing expression of the target gene in the mammal. Optionally, the RNAi agent further comprises a ligand.

[0269] In one embodiment, the sense strand of the RNAi agent contains at least one motif of three identical modifications on three consecutive nucleotides, where one of the motifs occurs at the cleavage site in the sense strand.

[0270] In one embodiment, the antisense strand of the RNAi agent can also contain at least one motif of three identical modifications on three consecutive nucleotides, where one of the motifs occurs at or near the cleavage site in the antisense strand.

[0271] For an RNAi agent having a duplex region of 17-23 nucleotide in length, the cleavage site of the antisense strand is typically around the 10, 11 and 12 positions from the 5'-end. Thus the motifs of three identical modifications may occur at the 9, 10, and 11 positions; 10, 11, and 12 positions; 11, 12, and 13 positions; 12, 13, and 14 positions; or 13, 14, and 15 positions of the antisense strand, the count starting from the 1.sup.st nucleotide from the 5'-end of the antisense strand, or, the count starting from the 1.sup.st paired nucleotide within the duplex region from the 5'-end of the antisense strand. The cleavage site in the antisense strand may also change according to the length of the duplex region of the RNAi from the 5'-end.

[0272] The sense strand of the RNAi agent may contain at least one motif of three identical modifications on three consecutive nucleotides at the cleavage site of the strand; and the antisense strand may have at least one motif of three identical modifications on three consecutive nucleotides at or near the cleavage site of the strand. When the sense strand and the antisense strand form a dsRNA duplex, the sense strand and the antisense strand can be so aligned that one motif of the three nucleotides on the sense strand and one motif of the three nucleotides on the antisense strand have at least one nucleotide overlap, i.e., at least one of the three nucleotides of the motif in the sense strand forms a base pair with at least one of the three nucleotides of the motif in the antisense strand. Alternatively, at least two nucleotides may overlap, or all three nucleotides may overlap.

[0273] In one embodiment, the sense strand of the RNAi agent may contain more than one motif of three identical modifications on three consecutive nucleotides. The first motif may occur at or near the cleavage site of the strand and the other motifs may be a wing modification. The term "wing modification" herein refers to a motif occurring at another portion of the strand that is separated from the motif at or near the cleavage site of the same strand. The wing modification is either adjacent to the first motif or is separated by at least one or more nucleotides. When the motifs are immediately adjacent to each other, then the chemistry of the motifs are distinct from each other and when the motifs are separated by one or more nucleotide than the chemistries can be the same or different. Two or more wing modifications may be present. For instance, when two wing modifications are present, each wing modification may occur at one end relative to the first motif which is at or near cleavage site or on either side of the lead motif.

[0274] Like the sense strand, the antisense strand of the RNAi agent may contain more than one motif of three identical modifications on three consecutive nucleotides, with at least one of the motifs occurring at or near the cleavage site of the strand. This antisense strand may also contain one or more wing modifications in an alignment similar to the wing modifications that may be present on the sense strand.

[0275] In one embodiment, the wing modification on the sense strand or antisense strand of the RNAi agent typically does not include the first one or two terminal nucleotides at the 3'-end, 5'-end or both ends of the strand.

[0276] In another embodiment, the wing modification on the sense strand or antisense strand of the RNAi agent typically does not include the first one or two paired nucleotides within the duplex region at the 3'-end, 5'-end or both ends of the strand.

[0277] When the sense strand and the antisense strand of the RNAi agent each contain at least one wing modification, the wing modifications may fall on the same end of the duplex region, and have an overlap of one, two or three nucleotides.

[0278] When the sense strand and the antisense strand of the RNAi agent each contain at least two wing modifications, the sense strand and the antisense strand can be so aligned that two modifications each from one strand fall on one end of the duplex region, having an overlap of one, two or three nucleotides; two modifications each from one strand fall on the other end of the duplex region, having an overlap of one, two or three nucleotides; two modifications one strand fall on each side of the lead motif, having an overlap of one, two, or three nucleotides in the duplex region.

[0279] In one embodiment, the RNAi agent comprises mismatch(es) with the target, within the duplex, or combinations thereof. The mismatch may occur in the overhang region or the duplex region. The base pair may be ranked on the basis of

their propensity to promote dissociation or melting (e.g., on the free energy of association or dissociation of a particular pairing, the simplest approach is to examine the pairs on an individual pair basis, though next neighbor or similar analysis can also be used). In terms of promoting dissociation: A:U is preferred over G:C; G:U is preferred over G:C; and I:C is preferred over G:C (J=inosine). Mismatches, e.g., non-canonical or other than canonical pairings (as described elsewhere herein) are preferred over canonical (A:T, A:U, G:C) pairings; and pairings which include a universal base are preferred over canonical pairings.

[0280] In one embodiment, the RNAi agent comprises at least one of the first 1, 2, 3, 4, or 5 base pairs within the duplex regions from the 5'-end of the antisense strand independently selected from the group of: A:U, G:U, I:C, and mismatched pairs, e.g., non-canonical or other than canonical pairings or pairings which include a universal base, to promote the dissociation of the antisense strand at the 5'-end of the duplex.

[0281] In one embodiment, the nucleotide at the 1 position within the duplex region from the 5'-end in the antisense strand is selected from the group consisting of A, dA, dU, U, and dT. Alternatively, at least one of the first 1, 2 or 3 base pair within the duplex region from the 5'-end of the antisense strand is an AU base pair. For example, the first base pair within the duplex region from the 5'-end of the antisense strand is an AU base pair.

[0282] In another embodiment, the nucleotide at the 3'-end of the sense strand is deoxythymidine (dT). In another embodiment, the nucleotide at the 3'-end of the antisense strand is deoxythymidine (dT). In one embodiment, there is a short sequence of deoxythymidine nucleotides, for example, two dT nucleotides on the 3'-end of the sense or antisense strand.

[0283] In one embodiment, the sense strand sequence may be represented by formula (I):

5'*n.sub.p*-N.sub.a—(XXX).sub.i—N.sub.b—YYY—N.sub.b—(ZZZ).sub.j—N.sub.a-*n.sub.q*3' (I) [0284] wherein:

[0285] i and j are each independently 0 or 1; [0286] p and q are each independently 0-6; [0287] each N.sub.a independently represents an oligonucleotide sequence comprising 0-25 modified nucleotides, each sequence comprising at least two differently modified nucleotides; [0288] each N.sub.b independently represents an oligonucleotide sequence comprising 0-10 modified nucleotides; [0289] each *n.sub.p* and *n.sub.q* independently represent an overhang nucleotide; [0290] wherein Nb and Y do not have the same modification; and [0291] XXX, YYY and ZZZ each independently represent one motif of three identical modifications on three consecutive nucleotides. Preferably YYY is all 2'-F modified nucleotides.

[0292] In one embodiment, the N.sub.a or N.sub.b comprise modifications of alternating pattern.

[0293] In one embodiment, the YYY motif occurs at or near the cleavage site of the sense strand. For example, when the RNAi agent has a duplex region of 17-23 nucleotides in length, the YYY motif can occur at or the vicinity of the cleavage site (e.g.: can occur at positions 6, 7, 8, 7, 8, 9, 8, 9, 10, 9, 10, 11, 10, 11, 12 or 11, 12, 13) of – the sense strand, the count starting from the 1.sup.st nucleotide, from the 5'-end; or optionally, the count starting at the 1.sup.st paired nucleotide within the duplex region, from the 5'-end.

[0294] In one embodiment, i is 1 and j is 0, or i is 0 and j is 1, or both i and j are 1. The sense strand can therefore be represented by the following formulas:

5'*n.sub.p*-N.sub.a—YYY—N.sub.b—ZZZ—N.sub.a-*n.sub.q*3' (Ib);

5'*n.sub.p*-N.sub.a—XXX—N.sub.b—YYY—N.sub.a-*n.sub.q*3' (Ic); or

5'*n.sub.p*-N.sub.a—XXX—N.sub.b—YYY—N.sub.b—ZZZ—N.sub.a-*n.sub.q*3' (Id).

[0295] When the sense strand is represented by formula (Ib), N.sub.b represents an oligonucleotide sequence comprising 0-10, 0-7, 0-5, 0-4, 0-2 or 0 modified nucleotides.

[0296] Each N.sub.a independently can represent an oligonucleotide sequence comprising 2-20, 2-15, or 2-10 modified nucleotides.

[0297] When the sense strand is represented as formula (Ic), N.sub.b represents an oligonucleotide sequence comprising 0-10, 0-7, 0-10, 0-7, 0-5, 0-4, 0-2 or 0 modified nucleotides. Each N.sub.a can independently represent an oligonucleotide sequence comprising 2-20, 2-15, or 2-10 modified nucleotides.

[0298] When the sense strand is represented as formula (Id), each N.sub.b independently represents an oligonucleotide sequence comprising 0-10, 0-7, 0-5, 0-4, 0-2 or 0 modified nucleotides. Preferably, N.sub.b is 0, 1, 2, 3, 4, 5 or 6. Each N.sub.a can independently represent an oligonucleotide sequence comprising 2-20, 2-15, or 2-10 modified nucleotides.

[0299] Each of X, Y and Z may be the same or different from each other.

[0300] In other embodiments, i is 0 and j is 0, and the sense strand may be represented by the formula:

5'*n.sub.p*-N.sub.a—YYY—N.sub.a-*n.sub.q*3' (Ia).

[0301] When the sense strand is represented by formula (Ia), each N.sub.a independently can represent an oligonucleotide sequence comprising 2-20, 2-15, or 2-10 modified nucleotides.

[0302] In one embodiment, the antisense strand sequence of the RNAi may be represented by formula (Ie):

5'*n.sub.q*'-N.sub.a'—(Z'Z'Z').sub.k—N.sub.b'—Y'Y'Y'—N.sub.b'—(X'X'X').sub.l—N'.sub.a-*n.sub.p*3' (Ie) [0303]

wherein: [0304] k and l are each independently 0 or 1; [0305] p' and q' are each independently 0-6; [0306] each N.sub.a'

independently represents an oligonucleotide sequence comprising 0-25 modified nucleotides, each sequence comprising at least two differently modified nucleotides; [0307] each N.sub.b' independently represents an oligonucleotide sequence comprising 0-10 modified nucleotides; [0308] each n.sub.p' and n.sub.q' independently represent an overhang nucleotide; [0309] wherein N.sub.b' and Y' do not have the same modification; [0310] and X'X'X', Y'Y'Y' and Z'Z'Z' each independently represent one motif of three identical modifications on three consecutive nucleotides.

[0311] In one embodiment, the N.sub.a' or N.sub.b' comprise modifications of alternating pattern.

[0312] The Y'Y'Y' motif occurs at or near the cleavage site of the antisense strand. For example, when the RNAi agent has a duplex region of 17-23 nucleotide in length, the Y'Y'Y' motif can occur at positions 9, 10, 11; 10, 11, 12; 11, 12, 13; 12, 13, 14; or 13, 14, 15 of the antisense strand, with the count starting from the 1.sup.st nucleotide, from the 5'-end; or optionally, the count starting at the 1.sup.st paired nucleotide within the duplex region, from the 5'-end. Preferably, the Y'Y'Y' motif occurs at positions 11, 12, 13.

[0313] In one embodiment, Y'Y'Y' motif is all 2'-OMe modified nucleotides.

[0314] In one embodiment, k is 1 and l is 0, or k is 0 and l is 1, or both k and l are 1.

[0315] The antisense strand can therefore be represented by the following formulas:

5'n.sub.q'-N.sub.a'—Z'Z'Z'—N.sub.b'—Y'Y'Y'—N.sub.a'-n.sub.p'3' (Ig);

5'n.sub.q'-N.sub.a'—Y'Y'Y'—N.sub.b'—X'X'X'-n.sub.p'3' (Ih); or

5'n.sub.q'-N.sub.a'—Z'Z'Z'—N.sub.b'—Y'Y'Y'—N.sub.b'—X'X'X'—N.sub.a'-n.sub.p'3' (Ii).

[0316] When the antisense strand is represented by formula (Ig), N.sub.b' represents an oligonucleotide sequence comprising 0-10, 0-7, 0-10, 0-7, 0-5, 0-4, 0-2 or 0 modified nucleotides. Each N.sub.a' independently represents an oligonucleotide sequence comprising 2-20, 2-15, or 2-10 modified nucleotides.

[0317] When the antisense strand is represented as formula (Ih), N.sub.b' represents an oligonucleotide sequence comprising 0-10, 0-7, 0-10, 0-7, 0-5, 0-4, 0-2 or 0 modified nucleotides. Each N.sub.a' independently represents an oligonucleotide sequence comprising 2-20, 2-15, or 2-10 modified nucleotides.

[0318] When the antisense strand is represented as formula (Ii), each N.sub.b' independently represents an oligonucleotide sequence comprising 0-10, 0-7, 0-10, 0-7, 0-5, 0-4, 0-2 or 0 modified nucleotides. Each N.sub.a' independently represents an oligonucleotide sequence comprising 2-20, 2-15, or 2-10 modified nucleotides. Preferably, N.sub.b is 0, 1, 2, 3, 4, 5 or 6.

[0319] In other embodiments, k is 0 and l is 0 and the antisense strand may be represented by the formula:

5'n.sub.p'-N.sub.a'—Y'Y'Y'—N.sub.a'-n.sub.q'3' (If).

[0320] When the antisense strand is represented as formula (If), each N.sub.a' independently represents an oligonucleotide sequence comprising 2-20, 2-15, or 2-10 modified nucleotides.

[0321] Each of X', Y' and Z' may be the same or different from each other.

[0322] Each nucleotide of the sense strand and antisense strand may be independently modified with LNA, HNA, CeNA, 2'-methoxyethyl, 2'-O-methyl, 2'-O-allyl, 2'-C-allyl, 2'-hydroxyl, or 2'-fluoro. For example, each nucleotide of the sense strand and antisense strand is independently modified with 2'-O-methyl or 2'-fluoro. Each X, Y, Z, X', Y' and Z', in particular, may represent a 2'-O-methyl modification or a 2'-fluoro modification.

[0323] In one embodiment, the sense strand of the RNAi agent may contain YYY motif occurring at 9, 10 and 11 positions of the strand when the duplex region is 21 nt, the count starting from the 1.sup.st nucleotide from the 5'-end, or optionally, the count starting at the 1.sup.st paired nucleotide within the duplex region, from the 5'-end; and Y represents 2'-F modification. The sense strand may additionally contain XXX motif or ZZZ motifs as wing modifications at the opposite end of the duplex region; and XXX and ZZZ each independently represents a 2'-OMe modification or 2'-F modification.

[0324] In one embodiment the antisense strand may contain Y'Y'Y' motif occurring at positions 11, 12, 13 of the strand, the count starting from the 1.sup.st nucleotide from the 5'-end, or optionally, the count starting at the 1.sup.st paired nucleotide within the duplex region, from the 5'-end; and Y' represents 2'-O-methyl modification. The antisense strand may additionally contain X'X'X' motif or Z'Z'Z' motifs as wing modifications at the opposite end of the duplex region; and X'X'X' and Z'Z'Z' each independently represents a 2'-OMc modification or 2'-F modification.

[0325] The sense strand represented by any one of the above formulas (Ia), (Ib), (Ic), and (Id) forms a duplex with an antisense strand being represented by any one of formulas (If), (Ig), (Ih), and (Ii), respectively.

[0326] Accordingly, the RNAi agents for use in the methods of the disclosure may comprise a sense strand and an antisense strand, each strand having 14 to 30 nucleotides, the RNAi duplex represented by formula (Ij):

sense: 5'n.sub.p-N.sub.a—(XXX).sub.i—N.sub.b—YYY—N.sub.b—(ZZZ).sub.j—N.sub.a-n.sub.q3'

antisense: 3'n.sub.p'—N.sub.a'—(X'X'X').sub.k—N.sub.b'—Y'Y'Y'—N.sub.b'—(Z'Z'Z').sub.l—N.sub.a'-n.sub.q'5' (Ij) [0327] wherein: [0328] i, j, k, and l are each independently 0 or 1; [0329] p, p', q, and q' are each independently 0-6; [0330] each N.sub.a and N.sub.a independently represents an oligonucleotide sequence comprising 0-25 modified nucleotides, each sequence comprising at least two differently modified nucleotides; [0331] each N.sub.b and N.sub.b independently represents an oligonucleotide sequence comprising 0-10 modified nucleotides; [0332] wherein [0333] each

n.sub.p', n.sub.p, n.sub.q', and n.sub.q, each of which may or may not be present, independently represents an overhang nucleotide; and [0334] XXX, YYY, ZZZ, X'X'X', Y'Y'Y', and Z'Z'Z' each independently represent one motif of three identical modifications on three consecutive nucleotides.

[0335] In one embodiment, i is 0 and j is 0; or i is 1 and j is 0; or i is 0 and j is 1; or both i and j are 0; or both i and j are 1. In another embodiment, k is 0 and l is 0; or k is 1 and l is 0; k is 0 and l is 1; or both k and l are 0; or both k and l are 1.

[0336] Exemplary combinations of the sense strand and antisense strand forming an RNAi duplex include the formulas below:

5'n.sub.p-N.sub.a—YYY—N.sub.a-n.sub.q3'

3'n.sub.p'—N.sub.a'—Y'Y'Y'—N.sub.a'-n.sub.q'5' (Ik)

5'n.sub.p-N.sub.a—YYY—N.sub.b—ZZZ—N.sub.a-n.sub.q3'

3'n.sub.p'—N.sub.a'—Y'Y'Y'—N.sub.b'—Z'Z'Z'—N.sub.a'-n.sub.q'5' (II)

5'n.sub.p-N.sub.a—XXX—N.sub.b—YYY—N.sub.a-n.sub.q3'

3'n.sub.p'—N.sub.a'—X'X'X'—N.sub.b'—Y'Y'Y'—N.sub.a'-n.sub.q'5' (Im)

5'n.sub.p-N.sub.a—XXX—N.sub.b—YYY—N.sub.b—ZZZ—N.sub.a-n.sub.q3'

3'n.sub.p'—N.sub.a'—X'X'X'—N.sub.b'—Y'Y'Y'—N.sub.b'—Z'Z'Z'—N.sub.a'-n.sub.q'5' (In)

[0337] When the RNAi agent is represented by formula (Ik), each N.sub.a independently represents an oligonucleotide sequence comprising 2-20, 2-15, or 2-10 modified nucleotides.

[0338] When the RNAi agent is represented by formula (II), each N.sub.b independently represents an oligonucleotide sequence comprising 1-10, 1-7, 1-5 or 1-4 modified nucleotides. Each N.sub.a independently represents an oligonucleotide sequence comprising 2-20, 2-15, or 2-10 modified nucleotides.

[0339] When the RNAi agent is represented as formula (Im), each N.sub.b, N.sub.b' independently represents an oligonucleotide sequence comprising 0-10, 0-7, 0-10, 0-7, 0-5, 0-4, 0-2 or 0 modified nucleotides. Each N.sub.a independently represents an oligonucleotide sequence comprising 2-20, 2-15, or 2-10 modified nucleotides.

[0340] When the RNAi agent is represented as formula (In), each N.sub.b, N.sub.b' independently represents an oligonucleotide sequence comprising 0-10, 0-7, 0-10, 0-7, 0-5, 0-4, 0-2 or 0 modified nucleotides. Each N.sub.a, N.sub.a' independently represents an oligonucleotide sequence comprising 2-20, 2-15, or 2-10 modified nucleotides. Each of N.sub.a, N.sub.a', N.sub.b and N.sub.b' independently comprises modifications of alternating pattern.

[0341] In one embodiment, when the RNAi agent is represented by formula (In), the N.sub.a modifications are 2'-O-methyl or 2'-fluoro modifications. In another embodiment, when the RNAi agent is represented by formula (In), the N.sub.a modifications are 2'-O-methyl or 2'-fluoro modifications and n.sub.p>0 and at least one n.sub.p' is linked to a neighboring nucleotide a via phosphorothioate linkage. In yet another embodiment, when the RNAi agent is represented by formula (In), the N.sub.a modifications are 2'-O-methyl or 2'-fluoro modifications, n.sub.p>0 and at least one n.sub.p' is linked to a neighboring nucleotide via phosphorothioate linkage, and the sense strand is conjugated to one or more C16 (or related) moieties attached through a bivalent or trivalent branched linker (described below). In another embodiment, when the RNAi agent is represented by formula (In), the N.sub.a modifications are 2'-O-methyl or 2'-fluoro modifications, n'>0 and at least one n.sub.p' is linked to a neighboring nucleotide via phosphorothioate linkage, the sense strand comprises at least one phosphorothioate linkage, and the sense strand is conjugated to one or more lipophilic, e.g., C16 (or related) moieties, optionally attached through a bivalent or trivalent branched linker.

[0342] In one embodiment, when the RNAi agent is represented by formula (Ik), the N.sub.a modifications are 2'-O-methyl or 2'-fluoro modifications, n.sub.p>0 and at least one n.sub.p' is linked to a neighboring nucleotide via phosphorothioate linkage, the sense strand comprises at least one phosphorothioate linkage, and the sense strand is conjugated to one or more lipophilic, e.g., C16 (or related) moieties attached through a bivalent or trivalent branched linker.

[0343] In one embodiment, the RNAi agent is a multimer containing at least two duplexes represented by formula (Ij), (Ik), (II), (Im), and (In), wherein the duplexes are connected by a linker. The linker can be cleavable or non-cleavable. Optionally, the multimer further comprises a ligand. Each of the duplexes can target the same gene or two different genes; or each of the duplexes can target same gene at two different target sites.

[0344] In one embodiment, the RNAi agent is a multimer containing three, four, five, six or more duplexes represented by formula (Ij), (Ik), (II), (Im), and (In), wherein the duplexes are connected by a linker. The linker can be cleavable or non-cleavable. Optionally, the multimer further comprises a ligand. Each of the duplexes can target the same gene or two different genes; or each of the duplexes can target same gene at two different target sites.

[0345] In one embodiment, two RNAi agents represented by formula (Ij), (Ik), (II), (Im), and (In) are linked to each other at the 5' end, and one or both of the 3' ends and are optionally conjugated to a ligand. Each of the agents can target the same gene or two different genes; or each of the agents can target same gene at two different target sites.

[0346] Various publications describe multimeric RNAi agents that can be used in the methods of the disclosure. Such publications include WO2007/091269, WO2010/141511, WO2007/117686, WO2009/014887, and WO2011/031520; and U.S. Pat. No. 7,858,769, the entire contents of each of which are hereby incorporated herein by reference.

[0347] In certain embodiments, the compositions and methods of the disclosure include a vinyl phosphonate (VP) modification of an RNAi agent as described herein. In exemplary embodiments, a 5'-vinyl phosphonate modified nucleotide of the disclosure has the structure:

##STR00002## [0348] wherein X is O or S; [0349] R is hydrogen, hydroxy, fluoro, or C.sub.1-20alkoxy (e.g., methoxy or n-hexadecyloxy); [0350] R.sup.5' is =C(H)—P(O)(OH).sub.2 and the double bond between the C5' carbon and R.sup.5' is in the E or Z orientation (e.g., E orientation); and [0351] B is a nucleobase or a modified nucleobase, optionally where B is adenine, guanine, cytosine, thymine, or uracil.

[0352] In one embodiment, R.sup.5' is =C(H)—P(O)(OH).sub.2 and the double bond between the C5' carbon and R5' is in the E orientation. In another embodiment, R is methoxy and R.sup.5' is =C(H)—P(O)(OH).sub.2 and the double bond between the C5' carbon and R.sup.5' is in the E orientation. In another embodiment, X is S, R is methoxy, and R.sup.5' is =C(H)—P(O)(OH).sub.2 and the double bond between the C5' carbon and R5' is in the E orientation.

[0353] A vinyl phosphonate of the instant disclosure may be attached to either the antisense or the sense strand of a dsRNA of the disclosure. In certain embodiments, a vinyl phosphonate of the instant disclosure is attached to the antisense strand of a dsRNA, optionally at the 5' end of the antisense strand of the dsRNA. The dsRNA agent can comprise a phosphorus-containing group at the 5'-end of the sense strand or antisense strand. The 5'-end phosphorus-containing group can be 5'-end phosphate (5'-P), 5'-end phosphorothioate (5'-PS), 5'-end phosphorodithioate (5'-PS₂), 5'-end vinylphosphonate (5'-VP), 5'-end methylphosphonate (MePhos), or 5'-deoxy-5'-C-malonyl. When the 5'-end phosphorus-containing group is 5'-end vinylphosphonate (5'-VP), the 5'-VP can be either 5'-E-VP isomer (i.e., trans-vinylphosphonate,

##STR00003##

5'-Z—VP isomer (i.e., cis-vinylphosphonate,

##STR00004##

or mixtures thereof.

[0354] Vinyl phosphate modifications are also contemplated for the compositions and methods of the instant disclosure.

An exemplary vinyl phosphate structure is:

##STR00005##

For example, when the phosphate mimic is a 5'-vinyl phosphate, the 5'-terminal nucleotide can have the immediately structure, where the phosphonate group is replaced by a phosphate.

i. Thermally Destabilizing Modifications

[0355] In certain embodiments, a dsRNA molecule can be optimized for RNA interference by incorporating thermally destabilizing modifications in the seed region of the antisense strand. As used herein “seed region” means at positions 2-9 of the 5'-end of the referenced strand. For example, thermally destabilizing modifications can be incorporated in the seed region of the antisense strand to reduce or inhibit off-target gene silencing.

[0356] The term “thermally destabilizing modification(s)” includes modification(s) that would result with a dsRNA with a lower overall melting temperature (T_{sub.m}) than the T_{sub.m} of the dsRNA without having such modification(s). For example, the thermally destabilizing modification(s) can decrease the T_{sub.m} of the dsRNA by 1-4° C., such as one, two, three or four degrees Celsius. And, the term “thermally destabilizing nucleotide” refers to a nucleotide containing one or more thermally destabilizing modifications.

[0357] It has been discovered that dsRNAs with an antisense strand comprising at least one thermally destabilizing modification of the duplex within the first 9 nucleotide positions, counting from the 5' end, of the antisense strand have reduced off-target gene silencing activity. Accordingly, in some embodiments, the antisense strand comprises at least one (e.g., one, two, three, four, five or more) thermally destabilizing modification of the duplex within the first 9 nucleotide positions of the 5' region of the antisense strand. In some embodiments, one or more thermally destabilizing modification(s) of the duplex is/are located in positions 2-9, or preferably positions 4-8, from the 5'-end of the antisense strand. In some further embodiments, the thermally destabilizing modification(s) of the duplex is/are located at position 6, 7 or 8 from the 5'-end of the antisense strand. In still some further embodiments, the thermally destabilizing modification of the duplex is located at position 7 from the 5'-end of the antisense strand. In some embodiments, the thermally destabilizing modification of the duplex is located at position 2, 3, 4, 5 or 9 from the 5'-end of the antisense strand.

[0358] The thermally destabilizing modifications can include, but are not limited to, abasic modification; mismatch with the opposing nucleotide in the opposing strand; and sugar modification such as 2'-deoxy modification or acyclic nucleotide, e.g., unlocked nucleic acids (UNA) or glycol nucleic acid (GNA).

[0359] Exemplified abasic modifications include, but are not limited to the following:

##STR00006##

Wherein R=H, Me, Et or OMe; R'=H, Me, Et or OMe; R''=H, Me, Et or OMe

##STR00007##

wherein B is a modified or unmodified nucleobase.

[0360] Exemplified sugar modifications include, but are not limited to the following:

##STR00008##

##STR00009##

wherein B is a modified or unmodified nucleobase.

[0361] In some embodiments the thermally destabilizing modification of the duplex is selected from the group consisting of:

##STR00010##

wherein B is a modified or unmodified nucleobase and the asterisk on each structure represents either R, S or racemic.

[0362] In some embodiments the thermally destabilizing modification of the duplex is selected from the group consisting of:

##STR00011##

wherein B is a modified or unmodified nucleobase and the asterisk represents either R, S or racemic (e.g. S).

[0363] The term “acyclic nucleotide” refers to any nucleotide having an acyclic ribose sugar, for example, where any of bonds between the ribose carbons (e.g., C1'-C2', C2'-C3', C3'-C4', C4'-O4', or C1'-O4') is absent or at least one of ribose carbons or oxygen (e.g., C1', C2', C3', C4' or O4') are independently or in combination absent from the nucleotide. In some embodiments, acyclic nucleotide is

##STR00012##

wherein B is a modified or unmodified nucleobase, R.sub.1 and R.sub.2 independently are H, halogen, OR.sub.3, or alkyl; and R.sub.3 is H, alkyl, cycloalkyl, aryl, aralkyl, heteroaryl or sugar). The term “UNA” refers to unlocked acyclic nucleic acid, wherein any of the bonds of the sugar has been removed, forming an unlocked “sugar” residue. In one example, UNA also encompasses monomers with bonds between C1'-C4' being removed (i.e. the covalent carbon-oxygen-carbon bond between the C1' and C4' carbons). In another example, the C2'-C3' bond (i.e. the covalent carbon-carbon bond between the C2 and C3' carbons) of the sugar is removed (see Mikhailov et. al., Tetrahedron Letters, 26 (17): 2059 (1985); and Fluiter et al., Mol. Biosyst., 10: 1039 (2009), which are hereby incorporated by reference in their entirety). The acyclic derivative provides greater backbone flexibility without affecting the Watson-Crick pairings. The acyclic nucleotide can be linked via 2'-5' or 3'-5' linkage.

[0364] The term ‘GNA’ refers to glycol nucleic acid which is a polymer similar to DNA or RNA but differing in the composition of its “backbone” in that is composed of repeating glycerol units linked by phosphodiester bonds:

##STR00013##

[0365] The thermally destabilizing modification of the duplex can be mismatches (i.e., noncomplementary base pairs) between the thermally destabilizing nucleotide and the opposing nucleotide in the opposite strand within the dsRNA duplex. Exemplary mismatch base pairs include G:G, G:A, G:U, G:T, A:A, A:C, C:C, C:U, C:T, U:U, T:T, U:T, or a combination thereof. Other mismatch base pairings known in the art are also amenable to the present invention. A mismatch can occur between nucleotides that are either naturally occurring nucleotides or modified nucleotides, i.e., the mismatch base pairing can occur between the nucleobases from respective nucleotides independent of the modifications on the ribose sugars of the nucleotides. In certain embodiments, the dsRNA molecule contains at least one nucleobase in the mismatch pairing that is a 2'-deoxy nucleobase; e.g., the 2'-deoxy nucleobase is in the sense strand.

[0366] In some embodiments, the thermally destabilizing modification of the duplex in the seed region of the antisense strand includes nucleotides with impaired W—C H-bonding to complementary base on the target mRNA such as

##STR00014##

[0367] More examples of abasic nucleotide, acyclic nucleotide modifications (including UNA and GNA), and mismatch modifications have been described in detail in WO 2011/133876, which is herein incorporated by reference in its entirety.

[0368] The thermally destabilizing modifications may also include universal base with reduced or abolished capability to form hydrogen bonds with the opposing bases, and phosphate modifications.

[0369] In some embodiments, the thermally destabilizing modification of the duplex includes nucleotides with non-canonical bases such as, but not limited to, nucleobase modifications with impaired or completely abolished capability to form hydrogen bonds with bases in the opposite strand. These nucleobase modifications have been evaluated for destabilization of the central region of the dsRNA duplex as described in WO 2010/0011895, which is herein incorporated by reference in its entirety. Exemplary nucleobase modifications are:

##STR00015##

[0370] In some embodiments, the thermally destabilizing modification of the duplex in the seed region of the antisense strand includes one or more α -nucleotide complementary to the base on the target mRNA, such as:

##STR00016##

wherein R is H, OH, OCH.sub.3, F, NH.sub.2, NHMe, NMe.sub.2 or O-alkyl.

[0371] Exemplary phosphate modifications known to decrease the thermal stability of dsRNA duplexes compared to natural phosphodiester linkages are:

##STR00017##

[0372] The alkyl for the R group can be a C.sub.1-C.sub.6alkyl. Specific alkyls for the R group include, but are not limited to methyl, ethyl, propyl, isopropyl, butyl, pentyl and hexyl.

[0373] As the skilled artisan will recognize, in view of the functional role of nucleobases is defining specificity of an RNAi agent of the disclosure, while nucleobase modifications can be performed in the various manners as described herein, e.g., to introduce destabilizing modifications into an RNAi agent of the disclosure, e.g., for purpose of enhancing on-target effect relative to off-target effect, the range of modifications available and, in general, present upon RNAi agents of the disclosure tends to be much greater for non-nucleobase modifications, e.g., modifications to sugar groups or phosphate backbones of polyribonucleotides. Such modifications are described in greater detail in other sections of the instant disclosure and are expressly contemplated for RNAi agents of the disclosure, either possessing native nucleobases or

modified nucleobases as described above or elsewhere herein.

[0374] In addition to the antisense strand comprising a thermally destabilizing modification, the dsRNA can also comprise one or more stabilizing modifications. For example, the dsRNA can comprise at least two (e.g., two, three, four, five, six, seven, eight, nine, ten or more) stabilizing modifications. Without limitations, the stabilizing modifications all can be present in one strand. In some embodiments, both the sense and the antisense strands comprise at least two stabilizing modifications. The stabilizing modification can occur on any nucleotide of the sense strand or antisense strand. For instance, the stabilizing modification can occur on every nucleotide on the sense strand or antisense strand; each stabilizing modification can occur in an alternating pattern on the sense strand or antisense strand; or the sense strand or antisense strand comprises both stabilizing modification in an alternating pattern. The alternating pattern of the stabilizing modifications on the sense strand may be the same or different from the antisense strand, and the alternating pattern of the stabilizing modifications on the sense strand can have a shift relative to the alternating pattern of the stabilizing modifications on the antisense strand.

[0375] In some embodiments, the antisense strand comprises at least two (e.g., two, three, four, five, six, seven, eight, nine, ten or more) stabilizing modifications. Without limitations, a stabilizing modification in the antisense strand can be present at any positions. In some embodiments, the antisense comprises stabilizing modifications at positions 2, 6, 8, 9, 14, and 16 from the 5'-end. In some other embodiments, the antisense comprises stabilizing modifications at positions 2, 6, 14, and 16 from the 5'-end. In still some other embodiments, the antisense comprises stabilizing modifications at positions 2, 14, and 16 from the 5'-end.

[0376] In some embodiments, the antisense strand comprises at least one stabilizing modification adjacent to the destabilizing modification. For example, the stabilizing modification can be the nucleotide at the 5'-end or the 3'-end of the destabilizing modification, i.e., at position -1 or +1 from the position of the destabilizing modification. In some embodiments, the antisense strand comprises a stabilizing modification at each of the 5'-end and the 3'-end of the destabilizing modification, i.e., positions -1 and +1 from the position of the destabilizing modification.

[0377] In some embodiments, the antisense strand comprises at least two stabilizing modifications at the 3'-end of the destabilizing modification, i.e., at positions +1 and +2 from the position of the destabilizing modification.

[0378] In some embodiments, the sense strand comprises at least two (e.g., two, three, four, five, six, seven, eight, nine, ten or more) stabilizing modifications. Without limitations, a stabilizing modification in the sense strand can be present at any positions. In some embodiments, the sense strand comprises stabilizing modifications at positions 7, 10, and 11 from the 5'-end. In some other embodiments, the sense strand comprises stabilizing modifications at positions 7, 9, 10, and 11 from the 5'-end. In some embodiments, the sense strand comprises stabilizing modifications at positions opposite or complementary to positions 11, 12, and 15 of the antisense strand, counting from the 5'-end of the antisense strand. In some other embodiments, the sense strand comprises stabilizing modifications at positions opposite or complementary to positions 11, 12, 13, and 15 of the antisense strand, counting from the 5'-end of the antisense strand. In some embodiments, the sense strand comprises a block of two, three, or four stabilizing modifications.

[0379] In some embodiments, the sense strand does not comprise a stabilizing modification in position opposite or complementary to the thermally destabilizing modification of the duplex in the antisense strand.

[0380] Exemplary thermally stabilizing modifications include, but are not limited to, 2'-fluoro modifications. Other thermally stabilizing modifications include, but are not limited to, LNA.

[0381] In some embodiments, the dsRNA of the disclosure comprises at least four (e.g., four, five, six, seven, eight, nine, ten, or more) 2'-fluoro nucleotides. Without limitations, the 2'-fluoro nucleotides all can be present in one strand. In some embodiments, both the sense and the antisense strands comprise at least two 2'-fluoro nucleotides. The 2'-fluoro modification can occur on any nucleotide of the sense strand or antisense strand. For instance, the 2'-fluoro modification can occur on every nucleotide on the sense strand or antisense strand; each 2'-fluoro modification can occur in an alternating pattern on the sense strand or antisense strand; or the sense strand or antisense strand comprises both 2'-fluoro modifications in an alternating pattern. The alternating pattern of the 2'-fluoro modifications on the sense strand may be the same or different from the antisense strand, and the alternating pattern of the 2'-fluoro modifications on the sense strand can have a shift relative to the alternating pattern of the 2'-fluoro modifications on the antisense strand.

[0382] In some embodiments, the antisense strand comprises at least two (e.g., two, three, four, five, six, seven, eight, nine, ten, or more) 2'-fluoro nucleotides. Without limitations, a 2'-fluoro modification in the antisense strand can be present at any positions. In some embodiments, the antisense comprises 2'-fluoro nucleotides at positions 2, 6, 8, 9, 14, and 16 from the 5'-end. In some other embodiments, the antisense comprises 2'-fluoro nucleotides at positions 2, 6, 14, and 16 from the 5'-end. In still some other embodiments, the antisense comprises 2'-fluoro nucleotides at positions 2, 14, and 16 from the 5'-end.

[0383] In some embodiments, the antisense strand comprises at least one 2'-fluoro nucleotide adjacent to the destabilizing modification. For example, the 2'-fluoro nucleotide can be the nucleotide at the 5'-end or the 3'-end of the destabilizing modification, i.e., at position -1 or +1 from the position of the destabilizing modification. In some embodiments, the antisense strand comprises a 2'-fluoro nucleotide at each of the 5'-end and the 3'-end of the destabilizing modification, i.e., positions -1 and +1 from the position of the destabilizing modification.

[0384] In some embodiments, the antisense strand comprises at least two 2'-fluoro nucleotides at the 3'-end of the destabilizing modification, i.e., at positions +1 and +2 from the position of the destabilizing modification.

[0385] In some embodiments, the sense strand comprises at least two (e.g., two, three, four, five, six, seven, eight, nine, ten or more) 2'-fluoro nucleotides. Without limitations, a 2'-fluoro modification in the sense strand can be present at any

positions. In some embodiments, the sense strand comprises 2'-fluoro nucleotides at positions 7, 10, and 11 from the 5'-end. In some other embodiments, the sense strand comprises 2'-fluoro nucleotides at positions 7, 9, 10, and 11 from the 5'-end. In some embodiments, the sense strand comprises 2'-fluoro nucleotides at positions opposite or complementary to positions 11, 12, and 15 of the antisense strand, counting from the 5'-end of the antisense strand. In some other embodiments, the sense strand comprises 2'-fluoro nucleotides at positions opposite or complementary to positions 11, 12, 13, and 15 of the antisense strand, counting from the 5'-end of the antisense strand. In some embodiments, the sense strand comprises a block of two, three or four 2'-fluoro nucleotides.

[0386] In some embodiments, the sense strand does not comprise a 2'-fluoro nucleotide in position opposite or complementary to the thermally destabilizing modification of the duplex in the antisense strand.

[0387] In some embodiments, the dsRNA molecule of the disclosure comprises a 21 nucleotides (nt) sense strand and a 23 nucleotides (nt) antisense, wherein the antisense strand contains at least one thermally destabilizing nucleotide, where the at least one thermally destabilizing nucleotide occurs in the seed region of the antisense strand (i.e., at position 2-9 of the 5'-end of the antisense strand), wherein one end of the dsRNA is blunt, while the other end is comprises a 2 nt overhang, and wherein the dsRNA optionally further has at least one (e.g., one, two, three, four, five, six or all seven) of the following characteristics: (i) the antisense comprises 2, 3, 4, 5 or 6 2'-fluoro modifications; (ii) the antisense comprises 1, 2, 3, 4 or 5 phosphorothioate internucleotide linkages; (iii) the sense strand is conjugated with a ligand; (iv) the sense strand comprises 2, 3, 4 or 5 2'-fluoro modifications; (v) the sense strand comprises 1, 2, 3, 4 or 5 phosphorothioate internucleotide linkages; (vi) the dsRNA comprises at least four 2'-fluoro modifications; and (vii) the dsRNA comprises a blunt end at 5'-end of the antisense strand. Preferably, the 2 nt overhang is at the 3'-end of the antisense.

[0388] In some embodiments, the dsRNA molecule of the disclosure comprising a sense and antisense strands, wherein: the sense strand is 25-30 nucleotide residues in length, wherein starting from the 5' terminal nucleotide (position 1), positions 1 to 23 of said sense strand comprise at least 8 ribonucleotides; antisense strand is 36-66 nucleotide residues in length and, starting from the 3' terminal nucleotide, at least 8 ribonucleotides in the positions paired with positions 1-23 of sense strand to form a duplex; wherein at least the 3' terminal nucleotide of antisense strand is unpaired with sense strand, and up to 6 consecutive 3' terminal nucleotides are unpaired with sense strand, thereby forming a 3' single stranded overhang of 1-6 nucleotides; wherein the 5' terminus of antisense strand comprises from 10-30 consecutive nucleotides which are unpaired with sense strand, thereby forming a 10-30 nucleotide single stranded 5' overhang; wherein at least the sense strand 5' terminal and 3' terminal nucleotides are base paired with nucleotides of antisense strand when sense and antisense strands are aligned for maximum complementarity, thereby forming a substantially duplexed region between sense and antisense strands; and antisense strand is sufficiently complementary to a target RNA along at least 19 ribonucleotides of antisense strand length to reduce target gene expression when said double stranded nucleic acid is introduced into a mammalian cell; and wherein the antisense strand contains at least one thermally destabilizing nucleotide, where at least one thermally destabilizing nucleotide is in the seed region of the antisense strand (i.e. at position 2-9 of the 5'-end of the antisense strand). For example, the thermally destabilizing nucleotide occurs between positions opposite or complementary to positions 14-17 of the 5'-end of the sense strand, and wherein the dsRNA optionally further has at least one (e.g., one, two, three, four, five, six or all seven) of the following characteristics: (i) the antisense comprises 2, 3, 4, 5, or 6 2'-fluoro modifications; (ii) the antisense comprises 1, 2, 3, 4, or 5 phosphorothioate internucleotide linkages; (iii) the sense strand is conjugated with a ligand; (iv) the sense strand comprises 2, 3, 4, or 5 2'-fluoro modifications; (v) the sense strand comprises 1, 2, 3, 4, or 5 phosphorothioate internucleotide linkages; and (vi) the dsRNA comprises at least four 2'-fluoro modifications; and (vii) the dsRNA comprises a duplex region of 12-30 nucleotide pairs in length.

[0389] In some embodiments, the dsRNA molecule of the disclosure comprises a sense and antisense strands, wherein said dsRNA molecule comprises a sense strand having a length which is at least 25 and at most 29 nucleotides and an antisense strand having a length which is at most 30 nucleotides with the sense strand comprises a modified nucleotide that is susceptible to enzymatic degradation at position 11 from the 5' end, wherein the 3' end of said sense strand and the 5' end of said antisense strand form a blunt end and said antisense strand is 1-4 nucleotides longer at its 3' end than the sense strand, wherein the duplex region which is at least 25 nucleotides in length, and said antisense strand is sufficiently complementary to a target mRNA along at least 19 nt of said antisense strand length to reduce target gene expression when said dsRNA molecule is introduced into a mammalian cell, and wherein dicer cleavage of said dsRNA preferentially results in an siRNA comprising said 3' end of said antisense strand, thereby reducing expression of the target gene in the mammal, wherein the antisense strand contains at least one thermally destabilizing nucleotide, where the at least one thermally destabilizing nucleotide is in the seed region of the antisense strand (i.e. at position 2-9 of the 5'-end of the antisense strand), and wherein the dsRNA optionally further has at least one (e.g., one, two, three, four, five, six or all seven) of the following characteristics: (i) the antisense comprises 2, 3, 4, 5, or 6 2'-fluoro modifications; (ii) the antisense comprises 1, 2, 3, 4, or 5 phosphorothioate internucleotide linkages; (iii) the sense strand is conjugated with a ligand; (iv) the sense strand comprises 2, 3, 4, or 5 2'-fluoro modifications; (v) the sense strand comprises 1, 2, 3, 4, or 5 phosphorothioate internucleotide linkages; and (vi) the dsRNA comprises at least four 2'-fluoro modifications; and (vii) the dsRNA has a duplex region of 12-29 nucleotide pairs in length.

[0390] In some embodiments, every nucleotide in the sense strand and antisense strand of the dsRNA molecule may be modified. Each nucleotide may be modified with the same or different modification which can include one or more alteration of one or both of the non-linking phosphate oxygens or of one or more of the linking phosphate oxygens; alteration of a constituent of the ribose sugar, e.g., of the 2' hydroxyl on the ribose sugar; wholesale replacement of the

phosphate moiety with “dephospho” linkers; modification or replacement of a naturally occurring base; and replacement or modification of the ribose-phosphate backbone.

[0391] As nucleic acids are polymers of subunits, many of the modifications occur at a position which is repeated within a nucleic acid, e.g., a modification of a base, or a phosphate moiety, or a non-linking O of a phosphate moiety. In some cases, the modification will occur at all of the subject positions in the nucleic acid but in many cases it will not. By way of example, a modification may only occur at a 3' or 5' terminal position, may only occur in a terminal region, e.g., at a position on a terminal nucleotide or in the last 2, 3, 4, 5, or 10 nucleotides of a strand. A modification may occur in a double strand region, a single strand region, or in both. A modification may occur only in the double strand region of an RNA or may only occur in a single strand region of an RNA. E.g., a phosphorothioate modification at a non-linking O position may only occur at one or both termini, may only occur in a terminal region, e.g., at a position on a terminal nucleotide or in the last 2, 3, 4, 5, or 10 nucleotides of a strand, or may occur in double strand and single strand regions, particularly at termini. The 5' end or ends can be phosphorylated.

[0392] It may be possible, e.g., to enhance stability, to include particular bases in overhangs, or to include modified nucleotides or nucleotide surrogates, in single strand overhangs, e.g., in a 5' or 3' overhang, or in both. E.g., it can be desirable to include purine nucleotides in overhangs. In some embodiments all or some of the bases in a 3' or 5' overhang may be modified, e.g., with a modification described herein. Modifications can include, e.g., the use of modifications at the 2' position of the ribose sugar with modifications that are known in the art, e.g., the use of deoxyribonucleotides, 2'-deoxy-2'-fluoro (2'-F) or 2'-O-methyl modified instead of the ribosugar of the nucleobase, and modifications in the phosphate group, e.g., phosphorothioate modifications. Overhangs need not be homologous with the target sequence.

[0393] In some embodiments, each residue of the sense strand and antisense strand is independently modified with LNA, HNA, CeNA, 2'-methoxyethyl, 2'-O-methyl, 2'-O-allyl, 2'-C-allyl, 2'-deoxy, or 2'-fluoro. The strands can contain more than one modification. In some embodiments, each residue of the sense strand and antisense strand is independently modified with 2'-O-methyl or 2'-fluoro. It is to be understood that these modifications are in addition to the at least one thermally destabilizing modification of the duplex present in the antisense strand.

[0394] At least two different modifications are typically present on the sense strand and antisense strand. Those two modifications may be the 2'-deoxy, 2'-O-methyl or 2'-fluoro modifications, acyclic nucleotides or others. In some embodiments, the sense strand and antisense strand each comprises two differently modified nucleotides selected from 2'-O-methyl or 2'-deoxy. In some embodiments, each residue of the sense strand and antisense strand is independently modified with 2'-O-methyl nucleotide, 2'-deoxy nucleotide, 2'-deoxy-2'-fluoro nucleotide, 2'-O—N-methylacetamido (2'-O-NMA) nucleotide, a 2'-O-dimethylaminoethoxyethyl (2'-O-DMAEOE) nucleotide, 2'-O-aminopropyl (2'-O-AP) nucleotide, or 2'-ara-F nucleotide. Again, it is to be understood that these modifications are in addition to the at least one thermally destabilizing modification of the duplex present in the antisense strand.

[0395] In some embodiments, the dsRNA molecule of the disclosure comprises modifications of an alternating pattern, particular in the B1, B2, B3, B1', B2', B3, B4' regions. The term “alternating motif” or “alternative pattern” as used herein refers to a motif having one or more modifications, each modification occurring on alternating nucleotides of one strand. The alternating nucleotide may refer to one per every other nucleotide or one per every three nucleotides, or a similar pattern. For example, if A, B and C each represent one type of modification to the nucleotide, the alternating motif can be TABLE-US-00001 “ABABABABAB...,” “AABBAABBAABB...,” “AABAABAABAAB...,” “AAABAAABAAB...,” “AAABBBAAABBB...,” or “ABCABCABCABC...,” etc.

The type of modifications contained in the alternating motif may be the same or different. For example, if A, B, C, D each represent one type of modification on the nucleotide, the alternating pattern, i.e., modifications on every other nucleotide, may be the same, but each of the sense strand or antisense strand can be selected from several possibilities of modifications within the alternating motif such as “ABABAB . . .,” “ACACAC . . .,” “BDBDBD . . .” or “CDCDCD . . .,” etc.

[0396] In some embodiments, the dsRNA molecule of the disclosure comprises the modification pattern for the alternating motif on the sense strand relative to the modification pattern for the alternating motif on the antisense strand is shifted. The shift may be such that the modified group of nucleotides of the sense strand corresponds to a differently modified group of nucleotides of the antisense strand and vice versa. For example, the sense strand when paired with the antisense strand in the dsRNA duplex, the alternating motif in the sense strand may start with “ABABAB” from 5'-3' of the strand and the alternating motif in the antisense strand may start with “BABABA” from 3'-5' of the strand within the duplex region. As another example, the alternating motif in the sense strand may start with “AABBAABB” from 5'-3' of the strand and the alternating motif in the antisense strand may start with “BBAABBAA” from 3'-5' of the strand within the duplex region, so that there is a complete or partial shift of the modification patterns between the sense strand and the antisense strand.

[0397] The dsRNA molecule of the disclosure may further comprise at least one phosphorothioate or methylphosphonate internucleotide linkage. The phosphorothioate or methylphosphonate internucleotide linkage modification may occur on any nucleotide of the sense strand or antisense strand or both in any position of the strand. For instance, the internucleotide linkage modification may occur on every nucleotide on the sense strand or antisense strand; each internucleotide linkage modification may occur in an alternating pattern on the sense strand or antisense strand; or the sense strand or antisense strand comprises both internucleotide linkage modifications in an alternating pattern. The alternating pattern of the internucleotide linkage modification on the sense strand may be the same or different from the antisense strand, and the alternating pattern of the internucleotide linkage modification on the sense strand may have a shift relative to the alternating pattern of the internucleotide linkage modification on the antisense strand.

[0398] In some embodiments, the dsRNA molecule comprises the phosphorothioate or methylphosphonate internucleotide

linkage modification in the overhang region. For example, the overhang region comprises two nucleotides having a phosphorothioate or methylphosphonate internucleotide linkage between the two nucleotides. Internucleotide linkage modifications also may be made to link the overhang nucleotides with the terminal paired nucleotides within duplex region. For example, at least 2, 3, 4, or all the overhang nucleotides may be linked through phosphorothioate or methylphosphonate internucleotide linkage, and optionally, there may be additional phosphorothioate or methylphosphonate internucleotide linkages linking the overhang nucleotide with a paired nucleotide that is next to the overhang nucleotide. For instance, there may be at least two phosphorothioate internucleotide linkages between the terminal three nucleotides, in which two of the three nucleotides are overhang nucleotides, and the third is a paired nucleotide next to the overhang nucleotide. Preferably, these terminal three nucleotides may be at the 3'-end of the antisense strand.

[0399] In some embodiments, the sense strand of the dsRNA molecule comprises 1-10 blocks of two to ten phosphorothioate or methylphosphonate internucleotide linkages separated by 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, or 16 phosphate internucleotide linkages, wherein one of the phosphorothioate or methylphosphonate internucleotide linkages is placed at any position in the oligonucleotide sequence and the said sense strand is paired with an antisense strand comprising any combination of phosphorothioate, methylphosphonate and phosphate internucleotide linkages or an antisense strand comprising either phosphorothioate or methylphosphonate or phosphate linkage.

[0400] In some embodiments, the antisense strand of the dsRNA molecule comprises two blocks of two phosphorothioate or methylphosphonate internucleotide linkages separated by 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, or 18 phosphate internucleotide linkages, wherein one of the phosphorothioate or methylphosphonate internucleotide linkages is placed at any position in the oligonucleotide sequence and the said antisense strand is paired with a sense strand comprising any combination of phosphorothioate, methylphosphonate and phosphate internucleotide linkages or an antisense strand comprising either phosphorothioate or methylphosphonate or phosphate linkage.

[0401] In some embodiments, the antisense strand of the dsRNA molecule comprises two blocks of three phosphorothioate or methylphosphonate internucleotide linkages separated by 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, or 16 phosphate internucleotide linkages, wherein one of the phosphorothioate or methylphosphonate internucleotide linkages is placed at any position in the oligonucleotide sequence and the said antisense strand is paired with a sense strand comprising any combination of phosphorothioate, methylphosphonate and phosphate internucleotide linkages or an antisense strand comprising either phosphorothioate or methylphosphonate or phosphate linkage.

[0402] In some embodiments, the antisense strand of the dsRNA molecule comprises two blocks of four phosphorothioate or methylphosphonate internucleotide linkages separated by 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, or 14 phosphate internucleotide linkages, wherein one of the phosphorothioate or methylphosphonate internucleotide linkages is placed at any position in the oligonucleotide sequence and the said antisense strand is paired with a sense strand comprising any combination of phosphorothioate, methylphosphonate and phosphate internucleotide linkages or an antisense strand comprising either phosphorothioate or methylphosphonate or phosphate linkage.

[0403] In some embodiments, the antisense strand of the dsRNA molecule comprises two blocks of five phosphorothioate or methylphosphonate internucleotide linkages separated by 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, or 12 phosphate internucleotide linkages, wherein one of the phosphorothioate or methylphosphonate internucleotide linkages is placed at any position in the oligonucleotide sequence and the said antisense strand is paired with a sense strand comprising any combination of phosphorothioate, methylphosphonate and phosphate internucleotide linkages or an antisense strand comprising either phosphorothioate or methylphosphonate or phosphate linkage.

[0404] In some embodiments, the antisense strand of the dsRNA molecule comprises two blocks of six phosphorothioate or methylphosphonate internucleotide linkages separated by 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 phosphate internucleotide linkages, wherein one of the phosphorothioate or methylphosphonate internucleotide linkages is placed at any position in the oligonucleotide sequence and the said antisense strand is paired with a sense strand comprising any combination of phosphorothioate, methylphosphonate and phosphate internucleotide linkages or an antisense strand comprising either phosphorothioate or methylphosphonate or phosphate linkage.

[0405] In some embodiments, the antisense strand of the dsRNA molecule comprises two blocks of seven phosphorothioate or methylphosphonate internucleotide linkages separated by 1, 2, 3, 4, 5, 6, 7, or 8 phosphate internucleotide linkages, wherein one of the phosphorothioate or methylphosphonate internucleotide linkages is placed at any position in the oligonucleotide sequence and the said antisense strand is paired with a sense strand comprising any combination of phosphorothioate, methylphosphonate and phosphate internucleotide linkages or an antisense strand comprising either phosphorothioate or methylphosphonate or phosphate linkage.

[0406] In some embodiments, the antisense strand of the dsRNA molecule comprises two blocks of eight phosphorothioate or methylphosphonate internucleotide linkages separated by 1, 2, 3, 4, 5, or 6 phosphate internucleotide linkages, wherein one of the phosphorothioate or methylphosphonate internucleotide linkages is placed at any position in the oligonucleotide sequence and the said antisense strand is paired with a sense strand comprising any combination of phosphorothioate, methylphosphonate and phosphate internucleotide linkages or an antisense strand comprising either phosphorothioate or methylphosphonate or phosphate linkage.

[0407] In some embodiments, the antisense strand of the dsRNA molecule comprises two blocks of nine phosphorothioate or methylphosphonate internucleotide linkages separated by 1, 2, 3, or 4 phosphate internucleotide linkages, wherein one of the phosphorothioate or methylphosphonate internucleotide linkages is placed at any position in the oligonucleotide sequence and the said antisense strand is paired with a sense strand comprising any combination of phosphorothioate,

the Rp configuration. In some embodiments, a common pattern of backbone chiral centers comprises no more than 1 internucleotidic linkages in the Rp configuration. In some embodiments, a common pattern of backbone chiral centers comprises no more than 8 internucleotidic linkages which are not chiral (as a non-limiting example, a phosphodiester). In some embodiments, a common pattern of backbone chiral centers comprises no more than 7 internucleotidic linkages which are not chiral. In some embodiments, a common pattern of backbone chiral centers comprises no more than 6 internucleotidic linkages which are not chiral. In some embodiments, a common pattern of backbone chiral centers comprises no more than 5 internucleotidic linkages which are not chiral. In some embodiments, a common pattern of backbone chiral centers comprises no more than 4 internucleotidic linkages which are not chiral. In some embodiments, a common pattern of backbone chiral centers comprises no more than 3 internucleotidic linkages which are not chiral. In some embodiments, a common pattern of backbone chiral centers comprises no more than 2 internucleotidic linkages which are not chiral. In some embodiments, a common pattern of backbone chiral centers comprises no more than 1 internucleotidic linkages which are not chiral. In some embodiments, a common pattern of backbone chiral centers comprises at least 10 internucleotidic linkages in the Sp configuration, and no more than 8 internucleotidic linkages which are not chiral. In some embodiments, a common pattern of backbone chiral centers comprises at least 11 internucleotidic linkages in the Sp configuration, and no more than 7 internucleotidic linkages which are not chiral. In some embodiments, a common pattern of backbone chiral centers comprises at least 12 internucleotidic linkages in the Sp configuration, and no more than 6 internucleotidic linkages which are not chiral. In some embodiments, a common pattern of backbone chiral centers comprises at least 13 internucleotidic linkages in the Sp configuration, and no more than 6 internucleotidic linkages which are not chiral. In some embodiments, a common pattern of backbone chiral centers comprises at least 14 internucleotidic linkages in the Sp configuration, and no more than 5 internucleotidic linkages which are not chiral. In some embodiments, a common pattern of backbone chiral centers comprises at least 15 internucleotidic linkages in the Sp configuration, and no more than 4 internucleotidic linkages which are not chiral. In some embodiments, the internucleotidic linkages in the Sp configuration are optionally contiguous or not contiguous. In some embodiments, the internucleotidic linkages in the Rp configuration are optionally contiguous or not contiguous. In some embodiments, the internucleotidic linkages which are not chiral are optionally contiguous or not contiguous.

[0430] In some embodiments, compound of the disclosure comprises a block is a stereochemistry block. In some embodiments, a block is an Rp block in that each internucleotidic linkage of the block is Rp. In some embodiments, a 5'-block is an Rp block. In some embodiments, a 3'-block is an Rp block. In some embodiments, a block is an Sp block in that each internucleotidic linkage of the block is Sp. In some embodiments, a 5'-block is an Sp block. In some embodiments, a 3'-block is an Sp block. In some embodiments, provided oligonucleotides comprise both Rp and Sp blocks. In some embodiments, provided oligonucleotides comprise one or more Rp but no Sp blocks. In some embodiments, provided oligonucleotides comprise one or more Sp but no Rp blocks. In some embodiments, provided oligonucleotides comprise one or more PO blocks wherein each internucleotidic linkage in a natural phosphate linkage.

[0431] In some embodiments, compound of the disclosure comprises a 5'-block is an Sp block wherein each sugar moiety comprises a 2'-F modification. In some embodiments, a 5'-block is an Sp block wherein each of internucleotidic linkage is a modified internucleotidic linkage and each sugar moiety comprises a 2'-F modification. In some embodiments, a 5'-block is an Sp block wherein each of internucleotidic linkage is a phosphorothioate linkage and each sugar moiety comprises a 2'-F modification. In some embodiments, a 5'-block comprises 4 or more nucleoside units. In some embodiments, a 5'-block comprises 5 or more nucleoside units. In some embodiments, a 5'-block comprises 6 or more nucleoside units. In some embodiments, a 5'-block comprises 7 or more nucleoside units. In some embodiments, a 3'-block is an Sp block wherein each sugar moiety comprises a 2'-F modification. In some embodiments, a 3'-block is an Sp block wherein each of internucleotidic linkage is a modified internucleotidic linkage and each sugar moiety comprises a 2'-F modification. In some embodiments, a 3'-block is an Sp block wherein each of internucleotidic linkage is a phosphorothioate linkage and each sugar moiety comprises a 2'-F modification. In some embodiments, a 3'-block comprises 4 or more nucleoside units. In some embodiments, a 3'-block comprises 5 or more nucleoside units. In some embodiments, a 3'-block comprises 6 or more nucleoside units. In some embodiments, a 3'-block comprises 7 or more nucleoside units.

[0432] In some embodiments, compound of the disclosure comprises a type of nucleoside in a region or an oligonucleotide is followed by a specific type of internucleotidic linkage, e.g., natural phosphate linkage, modified internucleotidic linkage, Rp chiral internucleotidic linkage, Sp chiral internucleotidic linkage, etc. In some embodiments, A is followed by Sp. In some embodiments, A is followed by Rp. In some embodiments, A is followed by natural phosphate linkage (PO). In some embodiments, U is followed by Sp. In some embodiments, U is followed by Rp. In some embodiments, U is followed by natural phosphate linkage (PO). In some embodiments, C is followed by Sp. In some embodiments, C is followed by Rp. In some embodiments, C is followed by natural phosphate linkage (PO). In some embodiments, G is followed by Sp. In some embodiments, G is followed by Rp. In some embodiments, G is followed by natural phosphate linkage (PO). In some embodiments, C and U are followed by Sp. In some embodiments, C and U are followed by Rp. In some embodiments, C and U are followed by natural phosphate linkage (PO). In some embodiments, A and G are followed by Sp. In some embodiments, A and G are followed by Rp.

[0433] In some embodiments, the antisense strand comprises phosphorothioate internucleotide linkages between nucleotide positions 21 and 22, and between nucleotide positions 22 and 23, wherein the antisense strand contains at least one thermally destabilizing modification of the duplex located in the seed region of the antisense strand (i.e., at position 2-9 of the 5'-end of the antisense strand), and wherein the dsRNA optionally further has at least one (e.g., one, two, three, four, five, six, seven or all eight) of the following characteristics: (i) the antisense comprises 2, 3, 4, 5 or 6 2'-fluoro

modifications; (ii) the antisense comprises 3, 4 or 5 phosphorothioate internucleotide linkages; (iii) the sense strand is conjugated with a ligand; (iv) the sense strand comprises 2, 3, 4 or 5 2'-fluoro modifications; (v) the sense strand comprises 1, 2, 3, 4 or 5 phosphorothioate internucleotide linkages; (vi) the dsRNA comprises at least four 2'-fluoro modifications; (vii) the dsRNA comprises a duplex region of 12-40 nucleotide pairs in length; and (viii) the dsRNA has a blunt end at 5'-end of the antisense strand.

[0434] In some embodiments, the antisense strand comprises phosphorothioate internucleotide linkages between nucleotide positions 1 and 2, between nucleotide positions 2 and 3, between nucleotide positions 21 and 22, and between nucleotide positions 22 and 23, wherein the antisense strand contains at least one thermally destabilizing modification of the duplex located in the seed region of the antisense strand (i.e., at position 2-9 of the 5'-end of the antisense strand), and wherein the dsRNA optionally further has at least one (e.g., one, two, three, four, five, six, seven or all eight) of the following characteristics: (i) the antisense comprises 2, 3, 4, 5 or 6 2'-fluoro modifications; (ii) the sense strand is conjugated with a ligand; (iii) the sense strand comprises 2, 3, 4 or 5 2'-fluoro modifications; (iv) the sense strand comprises 1, 2, 3, 4 or 5 phosphorothioate internucleotide linkages; (v) the dsRNA comprises at least four 2'-fluoro modifications; (vi) the dsRNA comprises a duplex region of 12-40 nucleotide pairs in length; (vii) the dsRNA comprises a duplex region of 12-40 nucleotide pairs in length; and (viii) the dsRNA has a blunt end at 5'-end of the antisense strand.

[0435] In some embodiments, the sense strand comprises phosphorothioate internucleotide linkages between nucleotide positions 1 and 2, and between nucleotide positions 2 and 3, wherein the antisense strand contains at least one thermally destabilizing modification of the duplex located in the seed region of the antisense strand (i.e., at position 2-9 of the 5'-end of the antisense strand), and wherein the dsRNA optionally further has at least one (e.g., one, two, three, four, five, six, seven or all eight) of the following characteristics: (i) the antisense comprises 2, 3, 4, 5 or 6 2'-fluoro modifications; (ii) the antisense comprises 1, 2, 3, 4 or 5 phosphorothioate internucleotide linkages; (iii) the sense strand is conjugated with a ligand; (iv) the sense strand comprises 2, 3, 4 or 5 2'-fluoro modifications; (v) the sense strand comprises 3, 4 or 5 phosphorothioate internucleotide linkages; (vi) the dsRNA comprises at least four 2'-fluoro modifications; (vii) the dsRNA comprises a duplex region of 12-40 nucleotide pairs in length; and (viii) the dsRNA has a blunt end at 5'-end of the antisense strand.

[0436] In some embodiments, the sense strand comprises phosphorothioate internucleotide linkages between nucleotide positions 1 and 2, and between nucleotide positions 2 and 3, the antisense strand comprises phosphorothioate internucleotide linkages between nucleotide positions 1 and 2, between nucleotide positions 2 and 3, between nucleotide positions 21 and 22, and between nucleotide positions 22 and 23, wherein the antisense strand contains at least one thermally destabilizing modification of the duplex located in the seed region of the antisense strand (i.e., at position 2-9 of the 5'-end of the antisense strand), and wherein the dsRNA optionally further has at least one (e.g., one, two, three, four, five, six or all seven) of the following characteristics: (i) the antisense comprises 2, 3, 4, 5 or 6 2'-fluoro modifications; (ii) the sense strand is conjugated with a ligand; (iii) the sense strand comprises 2, 3, 4 or 5 2'-fluoro modifications; (iv) the sense strand comprises 3, 4 or 5 phosphorothioate internucleotide linkages; (v) the dsRNA comprises at least four 2'-fluoro modifications; (vi) the dsRNA comprises a duplex region of 12-40 nucleotide pairs in length; and (vii) the dsRNA has a blunt end at 5'-end of the antisense strand.

[0437] In some embodiments, the dsRNA molecule of the disclosure comprises mismatch(es) with the target, within the duplex, or combinations thereof. The mismatch can occur in the overhang region or the duplex region. The base pair can be ranked on the basis of their propensity to promote dissociation or melting (e.g., on the free energy of association or dissociation of a particular pairing, the simplest approach is to examine the pairs on an individual pair basis, though next neighbor or similar analysis can also be used). In terms of promoting dissociation: A:U is preferred over G:C; G:U is preferred over G:C; and I:C is preferred over G:C (I=inosine). Mismatches, e.g., non-canonical or other than canonical pairings (as described elsewhere herein) are preferred over canonical (A:T, A:U, G:C) pairings; and pairings which include a universal base are preferred over canonical pairings.

[0438] In some embodiments, the dsRNA molecule of the disclosure comprises at least one of the first 1, 2, 3, 4, or 5 base pairs within the duplex regions from the 5'-end of the antisense strand can be chosen independently from the group of: A:U, G:U, I:C, and mismatched pairs, e.g., non-canonical or other than canonical pairings or pairings which include a universal base, to promote the dissociation of the antisense strand at the 5'-end of the duplex.

[0439] In some embodiments, the nucleotide at the 1 position within the duplex region from the 5'-end in the antisense strand is selected from the group consisting of A, dA, dU, U, and dT. Alternatively, at least one of the first 1, 2 or 3 base pair within the duplex region from the 5'-end of the antisense strand is an AU base pair. For example, the first base pair within the duplex region from the 5'-end of the antisense strand is an AU base pair.

[0440] It was found that introducing 4'-modified or 5'-modified nucleotide to the 3'-end of a phosphodiester (PO), phosphorothioate (PS), or phosphorodithioate (PS2) linkage of a dinucleotide at any position of single stranded or double stranded oligonucleotide can exert steric effect to the internucleotide linkage and, hence, protecting or stabilizing it against nucleases.

[0441] In some embodiments, 5'-modified nucleoside is introduced at the 3'-end of a dinucleotide at any position of single stranded or double stranded siRNA. For instance, a 5'-alkylated nucleoside may be introduced at the 3'-end of a dinucleotide at any position of single stranded or double stranded siRNA. The alkyl group at the 5' position of the ribose sugar can be racemic or chirally pure R or 5' isomer. An exemplary 5'-alkylated nucleoside is 5'-methyl nucleoside. The 5'-methyl can be either racemic or chirally pure R or S isomer.

[0442] In some embodiments, 4'-modified nucleoside is introduced at the 3'-end of a dinucleotide at any position of single

stranded or double stranded siRNA. For instance, a 4'-alkylated nucleoside may be introduced at the 3'-end of a dinucleotide at any position of single stranded or double stranded siRNA. The alkyl group at the 4' position of the ribose sugar can be racemic or chirally pure R or S isomer. An exemplary 4'-alkylated nucleoside is 4'-methyl nucleoside. The 4'-methyl can be either racemic or chirally pure R or S isomer. Alternatively, a 4'-O-alkylated nucleoside may be introduced at the 3'-end of a dinucleotide at any position of single stranded or double stranded siRNA. The 4'-O-alkyl of the ribose sugar can be racemic or chirally pure R or S isomer. An exemplary 4'-O-alkylated nucleoside is 4'-O-methyl nucleoside. The 4'-O-methyl can be either racemic or chirally pure R or S isomer.

[0443] In some embodiments, 5'-alkylated nucleoside is introduced at any position on the sense strand or antisense strand of a dsRNA, and such modification maintains or improves potency of the dsRNA. The 5'-alkyl can be either racemic or chirally pure R or S isomer. An exemplary 5'-alkylated nucleoside is 5'-methyl nucleoside. The 5'-methyl can be either racemic or chirally pure R or S isomer.

[0444] In some embodiments, 4'-alkylated nucleoside is introduced at any position on the sense strand or antisense strand of a dsRNA, and such modification maintains or improves potency of the dsRNA. The 4'-alkyl can be either racemic or chirally pure R or S isomer. An exemplary 4'-alkylated nucleoside is 4'-methyl nucleoside. The 4'-methyl can be either racemic or chirally pure R or S isomer.

[0445] In some embodiments, 4'-O-alkylated nucleoside is introduced at any position on the sense strand or antisense strand of a dsRNA, and such modification maintains or improves potency of the dsRNA. The 5'-alkyl can be either racemic or chirally pure R or S isomer. An exemplary 4'-O-alkylated nucleoside is 4'-O-methyl nucleoside. The 4'-O-methyl can be either racemic or chirally pure R or S isomer.

[0446] In some embodiments, the dsRNA molecule of the disclosure can comprise 2'-5' linkages (with 2'-H, 2'-OH and 2'-OMe and with P=O or P=S). For example, the 2'-5' linkages modifications can be used to promote nuclease resistance or to inhibit binding of the sense to the antisense strand, or can be used at the 5' end of the sense strand to avoid sense strand activation by RISC.

[0447] In another embodiment, the dsRNA molecule of the disclosure can comprise L sugars (e.g., L ribose, L-arabinose with 2'-H, 2'-OH and 2'-OMe). For example, these L sugars modifications can be used to promote nuclease resistance or to inhibit binding of the sense to the antisense strand, or can be used at the 5' end of the sense strand to avoid sense strand activation by RISC.

[0448] Various publications describe multimeric siRNA which can all be used with the dsRNA of the disclosure. Such publications include WO2007/091269, U.S. Pat. No. 7,858,769, WO2010/141511, WO2007/117686, WO2009/014887, and WO2011/031520 which are hereby incorporated by their entirety.

[0449] As described in more detail below, the RNAi agent that contains conjugations of one or more carbohydrate moieties to an RNAi agent may improve one or more properties of the RNAi agent. In many cases, the carbohydrate moiety will be attached to a modified subunit of the RNAi agent. For example, the ribose sugar of one or more ribonucleotide subunits of a dsRNA agent can be replaced with another moiety, e.g., a non-carbohydrate (preferably cyclic) carrier to which is attached a carbohydrate ligand. A ribonucleotide subunit in which the ribose sugar of the subunit has been so replaced is referred to herein as a ribose replacement modification subunit (RRMS). A cyclic carrier may be a carbocyclic ring system, i.e., all ring atoms are carbon atoms, or a heterocyclic ring system, i.e., one or more ring atoms may be a heteroatom, e.g., nitrogen, oxygen, sulfur. The cyclic carrier may be a monocyclic ring system, or may contain two or more rings, e.g. fused rings. The cyclic carrier may be a fully saturated ring system, or it may contain one or more double bonds.

[0450] The ligand may be attached to the polynucleotide via a carrier. The carriers include (i) at least one "backbone attachment point," preferably two "backbone attachment points" and (ii) at least one "tethering attachment point." A "backbone attachment point" as used herein refers to a functional group, e.g. a hydroxyl group, or generally, a bond available for, and that is suitable for incorporation of the carrier into the backbone, e.g., the phosphate, or modified phosphate, e.g., sulfur containing, backbone, of a ribonucleic acid. A "tethering attachment point" (TAP) in some embodiments refers to a constituent ring atom of the cyclic carrier, e.g., a carbon atom or a heteroatom (distinct from an atom which provides a backbone attachment point), that connects a selected moiety. The moiety can be, e.g., a carbohydrate, e.g. monosaccharide, disaccharide, trisaccharide, tetrasaccharide, oligosaccharide and polysaccharide. Optionally, the selected moiety is connected by an intervening tether to the cyclic carrier. Thus, the cyclic carrier will often include a functional group, e.g., an amino group, or generally, provide a bond, that is suitable for incorporation or tethering of another chemical entity, e.g., a ligand to the constituent ring.

[0451] The RNAi agents may be conjugated to a ligand via a carrier, wherein the carrier can be cyclic group or acyclic group. The cyclic group can be selected from pyrrolidinyl, pyrazolinyl, pyrazolidinyl, imidazolinyl, imidazolidinyl, piperidinyl, piperazinyl, [1,3]dioxolane, oxazolidinyl, isoxazolidinyl, morpholinyl, thiazolidinyl, isothiazolidinyl, quinoxalinyl, pyridazinonyl, tetrahydrofuryl and decalinyl. The acyclic group can be a serinol backbone or diethanolamine backbone.

[0452] In certain specific embodiments, the RNAi agent for use in the methods of the disclosure is an agent selected from the group of agents listed in Tables 3 or 4. These agents may further comprise a ligand.

IV. iRNAs Conjugated to Ligands

[0453] Another modification of the RNA of an iRNA of the invention involves chemically linking to the iRNA one or more ligands, moieties or conjugates that enhance the activity, cellular distribution or cellular uptake of the iRNA, e.g., into a cell. Such moieties include but are not limited to lipid moieties such as a cholesterol moiety (Letsinger et al., *Proc. Natl. Acad. Sci. USA*, 1989, 86: 6553-6556), cholic acid (Manoharan et al., *Biorg. Med. Chem. Let.*, 1994, 4:1053-1060), a

thioether, e.g., beryl-S-tritylthiol (Manoharan et al., *Ann. N.Y. Acad. Sci.*, 1992, 660:306-309; Manoharan et al., *Biorg. Med. Chem. Lett.*, 1993, 3:2765-2770), a thiocholesterol (Oberhauser et al., *Nucl. Acids Res.*, 1992, 20:533-538), an aliphatic chain, e.g., dodecandiol or undecyl residues (Saison-Behmoaras et al., *EMBO J*, 1991, 10:1111-1118; Kabanov et al., *FEBS Lett.*, 1990, 259:327-330; Svinarchuk et al., *Biochimie*, 1993, 75:49-54), a phospholipid, e.g., di-hexadecyl-rac-glycerol or triethyl-ammonium 1,2-di-G-hexadecyl-rac-glycero-3-phosphonate (Manoharan et al., *Tetrahedron Lett.*, 1995, 36:3651-3654; Shea et al., *Nucl. Acids Res.*, 1990, 18:3777-3783), a polyamine or a polyethylene glycol chain (Manoharan et al., *Nucleosides & Nucleotides*, 1995, 14:969-973), or adamantane acetic acid (Manoharan et al., *Tetrahedron Lett.*, 1995, 36:3651-3654), a palmityl moiety (Mishra et al., *Biochim. Biophys. Acta*, 1995, 1264:229-237), or an octadecylamine or hexylamino-carbonyloxycholesterol moiety (Crooke et al., *J. Pharmacol. Exp. Ther.*, 1996, 277:923-937).

[0454] In certain embodiments, a ligand alters the distribution, targeting or lifetime of an iRNA agent into which it is incorporated. In some embodiments, a ligand provides an enhanced affinity for a selected target, e.g., molecule, cell or cell type, compartment, e.g., a cellular or organ compartment, tissue, organ or region of the body, as, e.g., compared to a species absent such a ligand. Typical ligands will not take part in duplex pairing in a duplexed nucleic acid.

[0455] Ligands can include a naturally occurring substance, such as a protein (e.g., human serum albumin (HSA), low-density lipoprotein (LDL), or globulin); carbohydrate (e.g., a dextran, pullulan, chitin, chitosan, inulin, cyclodextrin or hyaluronic acid); or a lipid. The ligand may also be a recombinant or synthetic molecule, such as a synthetic polymer, e.g., a synthetic polyamino acid. Examples of polyamino acids include polyamino acid is a polylysine (PLL), poly L-aspartic acid, poly L-glutamic acid, styrene-maleic acid anhydride copolymer, poly(L-lactide-co-glycolid) copolymer, divinyl ether-maleic anhydride copolymer, N-(2-hydroxypropyl)methacrylamide copolymer (HMPA), polyethylene glycol (PEG), poly vinyl alcohol (PVA), polyurethane, poly(2-ethylacrylic acid), N-isopropylacrylamide polymers, or polyphosphazene. Example of polyamines include: polyethylenimine, polylysine (PLL), spermine, spermidine, polyamine, pseudopeptide-polyamine, peptidomimetic polyamine, dendrimer polyamine, arginine, amidine, protamine, cationic lipid, cationic porphyrin, quaternary salt of a polyamine, or an α helical peptide.

[0456] Ligands can also include targeting groups, e.g., a cell or tissue targeting agent, e.g., a lectin, glycoprotein, lipid or protein, e.g., an antibody, that binds to a specified cell type such as a kidney cell. A targeting group can be a thyrotropin, melanotropin, lectin, glycoprotein, surfactant protein A, Mucin carbohydrate, multivalent lactose, multivalent galactose, N-acetyl-galactosamine, N-acetyl-glucosamine multivalent mannose, multivalent fucose, glycosylated polyaminoacids, multivalent galactose, transferrin, bisphosphonate, polyglutamate, polyaspartate, a lipid, cholesterol, a steroid, bile acid, folate, vitamin B12, biotin, or an RGD peptide or RGD peptide mimetic. In certain embodiments, the ligand is a multivalent galactose, e.g., an N-acetyl-galactosamine.

[0457] Other examples of ligands include dyes, intercalating agents (e.g. acridines), cross-linkers (e.g. psoralene, mitomycin C), porphyrins (TPPC4, texaphyrin, Sapphyrin), polycyclic aromatic hydrocarbons (e.g., phenazine, dihydrophenazine), artificial endonucleases (e.g. EDTA), lipophilic molecules, e.g., cholesterol, cholic acid, adamantane acetic acid, 1-pyrene butyric acid, dihydrotestosterone, 1,3-Bis-O(hexadecyl)glycerol, geranyloxyhexyl group, hexadecylglycerol, borneol, menthol, 1,3-propanediol, heptadecyl group, palmitic acid, myristic acid, O3-(oleoyl)lithocholic acid, O3-(oleoyl)cholenic acid, dimethoxytrityl, or phenoxazine) and peptide conjugates (e.g., antennapedia peptide, Tat peptide), alkylating agents, phosphate, amino, mercapto, PEG (e.g., PEG-40K), mPEG, [mPEG].sub.2, polyamino, alkyl, substituted alkyl, radiolabeled markers, enzymes, haptens (e.g. biotin), transport/absorption facilitators (e.g., aspirin, vitamin E, folic acid), synthetic ribonucleases (e.g., imidazole, bisimidazole, histamine, imidazole clusters, acridine-imidazole conjugates, Eu3+ complexes of tetraazamacrocycles), dinitrophenyl, HRP, or AP.

[0458] Ligands can be proteins, e.g., glycoproteins, or peptides, e.g., molecules having a specific affinity for a co-ligand, or antibodies e.g., an antibody, that binds to a specified cell type such as a cancer cell, endothelial cell, or bone cell. Ligands may also include hormones and hormone receptors. They can also include non-peptidic species, such as lipids, lectins, carbohydrates, vitamins, cofactors, multivalent lactose, multivalent galactose, N-acetyl-galactosamine, N-acetyl-glucosamine multivalent mannose, or multivalent fucose. The ligand can be, for example, a lipopolysaccharide, an activator of p38 MAP kinase, or an activator of NF- κ B.

[0459] The ligand can be a substance, e.g., a drug, which can increase the uptake of the iRNA agent into the cell, for example, by disrupting the cell's cytoskeleton, e.g., by disrupting the cell's microtubules, microfilaments, or intermediate filaments. The drug can be, for example, taxon, vincristine, vinblastine, cytochalasin, nocodazole, japlakinolide, latrunculin A, phalloidin, swinholide A, indanocine, or myoservin.

[0460] In some embodiments, a ligand attached to an iRNA as described herein acts as a pharmacokinetic modulator (PK modulator). PK modulators include lipophiles, bile acids, steroids, phospholipid analogues, peptides, protein binding agents, polyethylene glycol (PEG), vitamins etc. Exemplary PK modulators include, but are not limited to, cholesterol, fatty acids, cholic acid, lithocholic acid, dialkylglycerides, diacylglyceride, phospholipids, sphingolipids, naproxen, ibuprofen, vitamin E, biotin etc. Oligonucleotides that comprise a number of phosphorothioate linkages are also known to bind to serum protein, thus short oligonucleotides, e.g., oligonucleotides of about 5 bases, 10 bases, 15 bases or 20 bases, comprising multiple of phosphorothioate linkages in the backbone are also amenable to the present invention as ligands (e.g. as PK modulating ligands). In addition, aptamers that bind serum components (e.g. serum proteins) are also suitable for use as PK modulating ligands in the embodiments described herein.

[0461] Ligand-conjugated iRNAs of the invention may be synthesized by the use of an oligonucleotide that bears a

pendant reactive functionality, such as that derived from the attachment of a linking molecule onto the oligonucleotide (described below). This reactive oligonucleotide may be reacted directly with commercially-available ligands, ligands that are synthesized bearing any of a variety of protecting groups, or ligands that have a linking moiety attached thereto. [0462] The oligonucleotides used in the conjugates of the present invention may be conveniently and routinely made through the well-known technique of solid-phase synthesis. Equipment for such synthesis is sold by several vendors including, for example, Applied Biosystems® (Foster City, Calif.). Any other means for such synthesis known in the art may additionally or alternatively be employed. It is also known to use similar techniques to prepare other oligonucleotides, such as the phosphorothioates and alkylated derivatives.

[0463] In the ligand-conjugated oligonucleotides and ligand-molecule bearing sequence-specific linked nucleosides of the present invention, the oligonucleotides and oligonucleosides may be assembled on a suitable DNA synthesizer utilizing standard nucleotide or nucleoside precursors, or nucleotide or nucleoside conjugate precursors that already bear the linking moiety, ligand-nucleotide or nucleoside-conjugate precursors that already bear the ligand molecule, or non-nucleoside ligand-bearing building blocks.

[0464] When using nucleotide-conjugate precursors that already bear a linking moiety, the synthesis of the sequence-specific linked nucleosides is typically completed, and the ligand molecule is then reacted with the linking moiety to form the ligand-conjugated oligonucleotide. In some embodiments, the oligonucleotides or linked nucleosides of the present invention are synthesized by an automated synthesizer using phosphoramidites derived from ligand-nucleoside conjugates in addition to the standard phosphoramidites and non-standard phosphoramidites that are commercially available and routinely used in oligonucleotide synthesis.

A. Lipid Conjugates

[0465] In certain embodiments, the ligand or conjugate is a lipid or lipid-based molecule. Such a lipid or lipid-based molecule can typically bind a serum protein, such as human serum albumin (HSA). An HSA binding ligand allows for distribution of the conjugate to a target tissue, e.g., a non-kidney target tissue of the body. For example, the target tissue can be the brain or the liver. Other molecules that can bind HSA can also be used as ligands. For example, naproxen or aspirin can be used. A lipid or lipid-based ligand can (a) increase resistance to degradation of the conjugate, (b) increase targeting or transport into a target cell or cell membrane, or (c) can be used to adjust binding to a serum protein, e.g., HSA. [0466] A lipid-based ligand can be used to modulate, e.g., control (e.g., inhibit) the binding of the conjugate to a target tissue. For example, a lipid or lipid-based ligand that binds to HSA more strongly will be less likely to be targeted to the kidney and therefore less likely to be cleared from the body. A lipid or lipid-based ligand that binds to HSA less strongly can be used to target the conjugate to the kidney.

[0467] In certain embodiments, the lipid-based ligand binds HSA. For example, the ligand can bind HSA with a sufficient affinity such that distribution of the conjugate to a non-kidney tissue is enhanced. However, the affinity is typically not so strong that the HSA-ligand binding cannot be reversed.

[0468] In certain embodiments, the lipid-based ligand binds HSA weakly or not at all, such that distribution of the conjugate to the kidney is enhanced. Other moieties that target to kidney cells can also be used in place of or in addition to the lipid-based ligand.

[0469] In another aspect, the ligand is a moiety, e.g., a vitamin, which is taken up by a target cell, e.g., a proliferating cell. These are particularly useful for treating disorders characterized by unwanted cell proliferation, e.g., of the malignant or non-malignant type, e.g., cancer cells. Exemplary vitamins include vitamin A, E, and K. Other exemplary vitamins include B vitamin, e.g., folic acid, B12, riboflavin, biotin, pyridoxal or other vitamins or nutrients taken up by cancer cells. Also included are HSA and low density lipoprotein (LDL).

B. Cell Permeation Agents

[0470] In another aspect, the ligand is a cell-permeation agent, such as a helical cell-permeation agent. In certain embodiments, the agent is amphipathic. An exemplary agent is a peptide such as tat or antennopodia. If the agent is a peptide, it can be modified, including a peptidylmimetic, invertomers, non-peptide or pseudo-peptide linkages, and use of D-amino acids. The helical agent is typically an α -helical agent and can have a lipophilic and a lipophobic phase.

[0471] The ligand can be a peptide or peptidomimetic. A peptidomimetic (also referred to herein as an oligopeptidomimetic) is a molecule capable of folding into a defined three-dimensional structure similar to a natural peptide. The attachment of peptide and peptidomimetics to iRNA agents can affect pharmacokinetic distribution of the iRNA, such as by enhancing cellular recognition and absorption. The peptide or peptidomimetic moiety can be about 5-50 amino acids long, e.g., about 5, 10, 15, 20, 25, 30, 35, 40, 45, or 50 amino acids long.

[0472] A peptide or peptidomimetic can be, for example, a cell permeation peptide, cationic peptide, amphipathic peptide, or hydrophobic peptide (e.g., consisting primarily of Tyr, Trp, or Phe). The peptide moiety can be a dendrimer peptide, constrained peptide or crosslinked peptide. In another alternative, the peptide moiety can include a hydrophobic membrane translocation sequence (MTS). An exemplary hydrophobic MTS-containing peptide is RFGF having the amino acid sequence AAVALLPAVLLALLAP (SEQ ID NO: 3). An RFGF analogue (e.g., amino acid sequence AALLPVLLAAP (SEQ ID NO: 4)) containing a hydrophobic MTS can also be a targeting moiety. The peptide moiety can be a "delivery" peptide, which can carry large polar molecules including peptides, oligonucleotides, and protein across cell membranes. For example, sequences from the HIV Tat protein (GRKKRRQRRPPQ (SEQ ID NO: 5)) and the *Drosophila* Antennapedia protein (RQIKIWFQNRRMKWKK (SEQ ID NO: 6)) have been found to be capable of functioning as delivery peptides. A peptide or peptidomimetic can be encoded by a random sequence of DNA, such as a peptide identified from a phage-display library, or one-bead-one-compound (OBOC) combinatorial library (Lain et al., *Nature*, 354:82-84,

1991). Typically, the peptide or peptidomimetic tethered to a dsRNA agent via an incorporated monomer unit is a cell targeting peptide such as an arginine-glycine-aspartic acid (RGD)-peptide, or RGD mimic. A peptide moiety can range in length from about 5 amino acids to about 40 amino acids. The peptide moieties can have a structural modification, such as to increase stability or direct conformational properties. Any of the structural modifications described below can be utilized.

[0473] An RGD peptide for use in the compositions and methods of the invention may be linear or cyclic, and may be modified, e.g., glycosylated or methylated, to facilitate targeting to a specific tissue(s). RGD-containing peptides and peptidomimetics may include D-amino acids, as well as synthetic RGD mimics. In addition to RGD, one can use other moieties that target the integrin ligand. In certain embodiments, conjugates of this ligand target PECAM-1 or VEGF.

[0474] An RGD peptide moiety can be used to target a particular cell type, e.g., a tumor cell, such as an endothelial tumor cell or a breast cancer tumor cell (Zitzmann et al., *Cancer Res.*, 62:5139-43, 2002). An RGD peptide can facilitate targeting of an dsRNA agent to tumors of a variety of other tissues, including the lung, kidney, spleen, or liver (Aoki et al., *Cancer Gene Therapy* 8:783-787, 2001). Typically, the RGD peptide will facilitate targeting of an iRNA agent to the kidney. The RGD peptide can be linear or cyclic, and can be modified, e.g., glycosylated or methylated to facilitate targeting to specific tissues. For example, a glycosylated RGD peptide can deliver an iRNA agent to a tumor cell expressing α .sub.v β .sub.3 (Haubner et al., *Jour. Nucl. Med.*, 42:326-336, 2001).

[0475] A "cell permeation peptide" is capable of permeating a cell, e.g., a microbial cell, such as a bacterial or fungal cell, or a mammalian cell, such as a human cell. A microbial cell-permeating peptide can be, for example, an α -helical linear peptide (e.g., LL-37 or Ceropin P1), a disulfide bond-containing peptide (e.g., α -defensin, D-defensin or bactenecin), or a peptide containing only one or two dominating amino acids (e.g., PR-39 or indolicidin). A cell permeation peptide can also include a nuclear localization signal (NLS). For example, a cell permeation peptide can be a bipartite amphipathic peptide, such as MPG, which is derived from the fusion peptide domain of HIV-1 gp41 and the NLS of SV40 large T antigen (Simeoni et al., *Nucl. Acids Res.* 31:2717-2724, 2003).

C. Carbohydrate Conjugates

[0476] In some embodiments of the compositions and methods of the invention, an iRNA further comprises a carbohydrate. The carbohydrate conjugated iRNA are advantageous for the in vivo delivery of nucleic acids, as well as compositions suitable for in vivo therapeutic use, as described herein. As used herein, "carbohydrate" refers to a compound which is either a carbohydrate per se made up of one or more monosaccharide units having at least 6 carbon atoms (which can be linear, branched or cyclic) with an oxygen, nitrogen or sulfur atom bonded to each carbon atom; or a compound having as a part thereof a carbohydrate moiety made up of one or more monosaccharide units each having at least six carbon atoms (which can be linear, branched or cyclic), with an oxygen, nitrogen or sulfur atom bonded to each carbon atom. Representative carbohydrates include the sugars (mono-, di-, tri- and oligosaccharides containing from about 4, 5, 6, 7, 8, or 9 monosaccharide units), and polysaccharides such as starches, glycogen, cellulose and polysaccharide gums. Specific monosaccharides include C5 and above (e.g., C5, C6, C7, or C8) sugars; di- and tri-saccharides include sugars having two or three monosaccharide units (e.g., C5, C6, C7, or C8).

[0477] In certain embodiments, a carbohydrate conjugate comprises a monosaccharide.

[0478] In certain embodiments, the monosaccharide is an N-acetylgalactosamine (GalNAc). GalNAc conjugates, which comprise one or more N-acetylgalactosamine (GalNAc) derivatives, are described, for example, in U.S. Pat. No. 8,106,022, the entire content of which is hereby incorporated herein by reference. In some embodiments, the GalNAc conjugate serves as a ligand that targets the iRNA to particular cells. In some embodiments, the GalNAc conjugate targets the iRNA to liver cells, e.g., by serving as a ligand for the asialoglycoprotein receptor of liver cells (e.g., hepatocytes).

[0479] In some embodiments, the carbohydrate conjugate comprises one or more GalNAc derivatives. The GalNAc derivatives may be attached via a linker, e.g., a bivalent or trivalent branched linker. In some embodiments the GalNAc conjugate is conjugated to the 3' end of the sense strand. In some embodiments, the GalNAc conjugate is conjugated to the iRNA agent (e.g., to the 3' end of the sense strand) via a linker, e.g., a linker as described herein. In some embodiments the GalNAc conjugate is conjugated to the 5' end of the sense strand. In some embodiments, the GalNAc conjugate is conjugated to the iRNA agent (e.g., to the 5' end of the sense strand) via a linker, e.g., a linker as described herein.

[0480] In certain embodiments of the invention, the GalNAc or GalNAc derivative is attached to an iRNA agent of the invention via a monovalent linker. In some embodiments, the GalNAc or GalNAc derivative is attached to an iRNA agent of the invention via a bivalent linker. In yet other embodiments of the invention, the GalNAc or GalNAc derivative is attached to an iRNA agent of the invention via a trivalent linker. In other embodiments of the invention, the GalNAc or GalNAc derivative is attached to an iRNA agent of the invention via a tetravalent linker.

[0481] In certain embodiments, the double stranded RNAi agents of the invention comprise one GalNAc or GalNAc derivative attached to the iRNA agent. In certain embodiments, the double stranded RNAi agents of the invention comprise a plurality (e.g., 2, 3, 4, 5, or 6) GalNAc or GalNAc derivatives, each independently attached to a plurality of nucleotides of the double stranded RNAi agent through a plurality of monovalent linkers.

[0482] In some embodiments, for example, when the two strands of an iRNA agent of the invention are part of one larger molecule connected by an uninterrupted chain of nucleotides between the 3'-end of one strand and the 5'-end of the respective other strand forming a hairpin loop comprising, a plurality of unpaired nucleotides, each unpaired nucleotide within the hairpin loop may independently comprise a GalNAc or GalNAc derivative attached via a monovalent linker. The hairpin loop may also be formed by an extended overhang in one strand of the duplex.

[0483] In some embodiments, for example, when the two strands of an iRNA agent of the invention are part of one larger

molecule connected by an uninterrupted chain of nucleotides between the 3'-end of one strand and the 5'-end of the respective other strand forming a hairpin loop comprising, a plurality of unpaired nucleotides, each unpaired nucleotide within the hairpin loop may independently comprise a GalNAc or GalNAc derivative attached via a monovalent linker. The hairpin loop may also be formed by an extended overhang in one strand of the duplex.

[0484] In some embodiments, the GalNAc conjugate is

##STR00018##

[0485] In some embodiments, the RNAi agent is attached to the carbohydrate conjugate via a linker as shown in the following schematic, wherein X is O or S

##STR00019##

[0486] In some embodiments, the RNAi agent is conjugated to L96 as defined in Table 2 and shown below:

##STR00020##

[0487] In certain embodiments, a carbohydrate conjugate for use in the compositions and methods of the invention is selected from the group consisting of:

##STR00021## ##STR00022## ##STR00023## ##STR00024## ##STR00025##

wherein Y is O or S and n is 3-6 (Formula XXIV);

##STR00026##

wherein Y is O or S and n is 3-6 (Formula XXV);

##STR00027##

wherein X is O or S (Formula XXVII);

##STR00028##

Formula XXXIV.

[0488] In certain embodiments, a carbohydrate conjugate for use in the compositions and methods of the invention is a monosaccharide. In certain embodiments, the monosaccharide is an N-acetylgalactosamine, such as

##STR00029##

[0489] Another representative carbohydrate conjugate for use in the embodiments described herein includes, but is not limited to,

##STR00030##

[0490] when one of X or Y is an oligonucleotide, the other is a hydrogen.

[0491] In some embodiments, a suitable ligand is a ligand disclosed in WO 2019/055633, the entire contents of which are incorporated herein by reference. In one embodiment the ligand comprises the structure below:

##STR00031##

[0492] In certain embodiments, the RNAi agents of the disclosure may include GalNAc ligands, even if such GalNAc ligands are currently projected to be of limited value for intrathecal/CNS delivery route(s) of the instant disclosure.

[0493] In certain embodiments of the invention, the GalNAc or GalNAc derivative is attached to an iRNA agent of the invention via a monovalent linker. In some embodiments, the GalNAc or GalNAc derivative is attached to an iRNA agent of the invention via a bivalent linker. In yet other embodiments of the invention, the GalNAc or GalNAc derivative is attached to an iRNA agent of the invention via a trivalent linker.

[0494] In one embodiment, the double stranded RNAi agents of the invention comprise one or more GalNAc or GalNAc derivative attached to the iRNA agent. The GalNAc may be attached to any nucleotide via a linker on the sense strand or antisense strand. The GalNAc may be attached to the 5'-end of the sense strand, the 3' end of the sense strand, the 5'-end of the antisense strand, or the 3'-end of the antisense strand. In one embodiment, the GalNAc is attached to the 3' end of the sense strand, e.g., via a trivalent linker.

[0495] In other embodiments, the double stranded RNAi agents of the invention comprise a plurality (e.g., 2, 3, 4, 5, or 6) GalNAc or GalNAc derivatives, each independently attached to a plurality of nucleotides of the double stranded RNAi agent through a plurality of linkers, e.g., monovalent linkers.

[0496] In some embodiments, for example, when the two strands of an iRNA agent of the invention is part of one larger molecule connected by an uninterrupted chain of nucleotides between the 3'-end of one strand and the 5'-end of the respective other strand forming a hairpin loop comprising, a plurality of unpaired nucleotides, each unpaired nucleotide within the hairpin loop may independently comprise a GalNAc or GalNAc derivative attached via a monovalent linker.

[0497] In some embodiments, the carbohydrate conjugate further comprises one or more additional ligands as described above, such as, but not limited to, a PK modulator or a cell permeation peptide.

[0498] Additional carbohydrate conjugates and linkers suitable for use in the present invention include those described in WO 2014/179620 and WO 2014/179627, the entire contents of each of which are incorporated herein by reference.

D. Linkers

[0499] In some embodiments, the conjugate or ligand described herein can be attached to an iRNA oligonucleotide with various linkers that can be cleavable or non-cleavable.

[0500] The term "linker" or "linking group" means an organic moiety that connects two parts of a compound, e.g., covalently attaches two parts of a compound. Linkers typically comprise a direct bond or an atom such as oxygen or sulfur, a unit such as NR8, C(O), C(O)NH, SO, SO.sub.2, SO.sub.2NH or a chain of atoms, such as, but not limited to, substituted or unsubstituted alkyl, substituted or unsubstituted alkenyl, substituted or unsubstituted alkynyl, arylalkyl, arylalkenyl, arylalkynyl, heteroarylalkyl, heteroarylalkenyl, heteroarylalkynyl, heterocyclalkyl, heterocyclalkenyl, heterocyclalkynyl, aryl, heteroaryl, heterocycl, cycloalkyl, cycloalkenyl, alkylarylalkyl, alkylarylalkenyl,

alkynylalkynyl, alkenylarylalkynyl, alkenylarylalkenyl, alkynylarylalkynyl, alkynylarylalkenyl, alkynylarylalkynyl, alkylheteroarylalkynyl, alkylheteroarylalkenyl, alkylheteroarylalkynyl, alkenylheteroarylalkynyl, alkenylheteroarylalkenyl, alkenylheteroarylalkynyl, alkynylheteroarylalkynyl, alkynylheteroarylalkenyl, alkynylheteroarylalkynyl, alkylheterocyclalkynyl, alkylheterocyclalkenyl, alkylheterocyclalkynyl, alkenylheterocyclalkynyl, alkenylheterocyclalkenyl, alkenylheterocyclalkynyl, alkynylheterocyclalkynyl, alkynylheterocyclalkenyl, alkynylheterocyclalkynyl, alkylaryl, alkenylaryl, alkynylaryl, alkylheteroaryl, alkenylheteroaryl, alkynylheteroaryl, which one or more methylenes can be interrupted or terminated by O, S, S(O), SO₂, N(R₈), C(O), substituted or unsubstituted aryl, substituted or unsubstituted heteroaryl, substituted or unsubstituted heterocyclic; where R₈ is hydrogen, acyl, aliphatic or substituted aliphatic. In certain embodiments, the linker is between about 1-24 atoms, 2-24, 3-24, 4-24, 5-24, 6-24, 6-18, 7-18, 8-18 atoms, 7-17, 8-17, 6-16, 7-16, or 8-16 atoms.

[0501] A cleavable linking group is one which is sufficiently stable outside the cell, but which upon entry into a target cell is cleaved to release the two parts the linker is holding together. In another embodiment, the cleavable linking group is cleaved at least about 10 times, 20, times, 30 times, 40 times, 50 times, 60 times, 70 times, 80 times, 90 times or more, or at least about 100 times faster in a target cell or under a first reference condition (which can, e.g., be selected to mimic or represent intracellular conditions) than in the blood of a subject, or under a second reference condition (which can, e.g., be selected to mimic or represent conditions found in the blood or serum).

[0502] Cleavable linking groups are susceptible to cleavage agents, e.g., pH, redox potential or the presence of degradative molecules. Generally, cleavage agents are more prevalent or found at higher levels or activities inside cells than in serum or blood. Examples of such degradative agents include: redox agents which are selected for particular substrates or which have no substrate specificity, including, e.g., oxidative or reductive enzymes or reductive agents such as mercaptans, present in cells, that can degrade a redox cleavable linking group by reduction; esterases; endosomes or agents that can create an acidic environment, e.g., those that result in a pH of five or lower; enzymes that can hydrolyze or degrade an acid cleavable linking group by acting as a general acid, peptidases (which can be substrate specific), and phosphatases.

[0503] A cleavable linkage group, such as a disulfide bond can be susceptible to pH. The pH of human serum is 7.4, while the average intracellular pH is slightly lower, ranging from about 7.1-7.3. Endosomes have a more acidic pH, in the range of 5.5-6.0, and lysosomes have an even more acidic pH at around 5.0. Some linkers will have a cleavable linking group that is cleaved at a selected pH, thereby releasing a cationic lipid from the ligand inside the cell, or into the desired compartment of the cell.

[0504] A linker can include a cleavable linking group that is cleavable by a particular enzyme. The type of cleavable linking group incorporated into a linker can depend on the cell to be targeted. For example, a liver-targeting ligand can be linked to a cationic lipid through a linker that includes an ester group. Liver cells are rich in esterases, and therefore the linker will be cleaved more efficiently in liver cells than in cell types that are not esterase-rich. Other cell-types rich in esterases include cells of the lung, renal cortex, and testis.

[0505] Linkers that contain peptide bonds can be used when targeting cell types rich in peptidases, such as liver cells and synoviocytes.

[0506] In general, the suitability of a candidate cleavable linking group can be evaluated by testing the ability of a degradative agent (or condition) to cleave the candidate linking group. It will also be desirable to also test the candidate cleavable linking group for the ability to resist cleavage in the blood or when in contact with other non-target tissue. Thus, one can determine the relative susceptibility to cleavage between a first and a second condition, where the first is selected to be indicative of cleavage in a target cell and the second is selected to be indicative of cleavage in other tissues or biological fluids, e.g., blood or serum. The evaluations can be carried out in cell free systems, in cells, in cell culture, in organ or tissue culture, or in whole animals. It can be useful to make initial evaluations in cell-free or culture conditions and to confirm by further evaluations in whole animals. In certain embodiments, useful candidate compounds are cleaved at least about 2, 4, 10, 20, 30, 40, 50, 60, 70, 80, 90, or about 100 times faster in the cell (or under in vitro conditions selected to mimic intracellular conditions) as compared to blood or serum (or under in vitro conditions selected to mimic extracellular conditions).

i. Redox Cleavable Linking Groups

[0507] In certain embodiments, a cleavable linking group is a redox cleavable linking group that is cleaved upon reduction or oxidation. An example of reductively cleavable linking group is a disulphide linking group (—S—S—). To determine if a candidate cleavable linking group is a suitable “reductively cleavable linking group,” or for example is suitable for use with a particular iRNA moiety and particular targeting agent one can look to methods described herein. For example, a candidate can be evaluated by incubation with dithiothreitol (DTT), or other reducing agent using reagents known in the art, which mimic the rate of cleavage which would be observed in a cell, e.g., a target cell. The candidates can also be evaluated under conditions which are selected to mimic blood or serum conditions. In one, candidate compounds are cleaved by at most about 10% in the blood. In other embodiments, useful candidate compounds are degraded at least about 2, 4, 10, 20, 30, 40, 50, 60, 70, 80, 90, or about 100 times faster in the cell (or under in vitro conditions selected to mimic intracellular conditions) as compared to blood (or under in vitro conditions selected to mimic extracellular conditions). The rate of cleavage of candidate compounds can be determined using standard enzyme kinetics assays under conditions chosen to mimic intracellular media and compared to conditions chosen to mimic extracellular media.

ii. Phosphate-Based Cleavable Linking Groups

[0508] In certain embodiments, a cleavable linker comprises a phosphate-based cleavable linking group. A phosphate-

based cleavable linking group is cleaved by agents that degrade or hydrolyze the phosphate group. An example of an agent that cleaves phosphate groups in cells are enzymes such as phosphatases in cells. Examples of phosphate-based linking groups are —O—P(O)(ORk)—O— , —O—P(S)(ORk)—O— , —O—P(S)(SRk)—O— , —S—P(O)(ORk)—O— , —O—P(O)(ORk)—S— , —S—P(O)(ORk)—S— , —O—P(S)(ORk)—S— , —S—P(S)(ORk)—O— , —O—P(O)(Rk)—O— , —O—P(S)(Rk)—O— , —S—P(O)(Rk)—O— , —S—P(S)(Rk)—O— , —S—P(O)(Rk)—S— , —O—P(S)(Rk)—S— , wherein Rk at each occurrence can be, independently, C1-C20 alkyl, C1-C20 haloalkyl, C6-C10 aryl, or C7-C12 aralkyl. Additional embodiments include —O—P(O)(OH)—O— , —O—P(S)(OH)—O— , —O—P(S)(SH)—O— , —S—P(O)(OH)—O— , —O—P(O)(OH)—S— , —S—P(O)(OH)—S— , —O—P(S)(OH)—S— , —S—P(S)(OH)—O— , —O—P(O)(H)—O— , —O—P(S)(H)—O— , —S—P(O)(H)—O— , —S—P(S)(H)—O— , —S—P(O)(H)—S— , —O—P(S)(H)—S— , wherein Rk at each occurrence can be, independently, C1-C20 alkyl, C1-C20 haloalkyl, C6-C10 aryl, or C7-C12 aralkyl. In certain embodiments, a phosphate-based linking group is —O—P(O)(OH)—O— . These candidates can be evaluated using methods analogous to those described above.

iii. Acid Cleavable Linking Groups

[0509] In certain embodiments, a cleavable linker comprises an acid cleavable linking group. An acid cleavable linking group is a linking group that is cleaved under acidic conditions. In one embodiment, acid cleavable linking groups are cleaved in an acidic environment with a pH of about 6.5 or lower (e.g., about 6.0, 5.75, 5.5, 5.25, 5.0, or lower), or by agents such as enzymes that can act as a general acid. In a cell, specific low pH organelles, such as endosomes and lysosomes can provide a cleaving environment for acid cleavable linking groups. Examples of acid cleavable linking groups include but are not limited to hydrazones, esters, and esters of amino acids. Acid cleavable groups can have the general formula —C=NN— , C(O)O , or —OC(O) . Another embodiment is when the carbon attached to the oxygen of the ester (the alkoxy group) is an aryl group, substituted alkyl group, or tertiary alkyl group such as dimethyl pentyl or t-butyl. These candidates can be evaluated using methods analogous to those described above.

iv. Ester-Based Cleavable Linking Groups

[0510] In certain embodiments, a cleavable linker comprises an ester-based cleavable linking group. An ester-based cleavable linking group is cleaved by enzymes such as esterases and amidases in cells. Examples of ester-based cleavable linking groups include but are not limited to esters of alkylene, alkenylene and alkynylene groups. Ester cleavable linking groups have the general formula —C(O)O— , or —OC(O)— . These candidates can be evaluated using methods analogous to those described above.

v. Peptide-Based Cleavable Linking Groups

[0511] In yet another embodiment, a cleavable linker comprises a peptide-based cleavable linking group. A peptide-based cleavable linking group is cleaved by enzymes such as peptidases and proteases in cells. Peptide-based cleavable linking groups are peptide bonds formed between amino acids to yield oligopeptides (e.g., dipeptides, tripeptides etc.) and polypeptides. Peptide-based cleavable groups do not include the amide group (—C(O)NH—). The amide group can be formed between any alkylene, alkenylene or alkynylene. A peptide bond is a special type of amide bond formed between amino acids to yield peptides and proteins. The peptide based cleavage group is generally limited to the peptide bond (i.e., the amide bond) formed between amino acids yielding peptides and proteins and does not include the entire amide functional group. Peptide-based cleavable linking groups have the general formula $\text{—NHCHR.sup.AC(O)NHCHR.sup.BC(O)—}$, where R.sup.A and R.sup.B are the R groups of the two adjacent amino acids. These candidates can be evaluated using methods analogous to those described above.

[0512] In some embodiments, an iRNA of the invention is conjugated to a carbohydrate through a linker. Non-limiting examples of iRNA carbohydrate conjugates with linkers of the compositions and methods of the invention include, but are not limited to,

##STR00032## ##STR00033## ##STR00034##

when one of X or Y is an oligonucleotide, the other is a hydrogen.

[0513] In certain embodiments of the compositions and methods of the invention, a ligand is one or more “GalNAc” (N-acetylgalactosamine) derivatives attached through a bivalent or trivalent branched linker.

[0514] In certain embodiments, a dsRNA of the invention is conjugated to a bivalent or trivalent branched linker selected from the group of structures shown in any of formula (XLV)-(XLVI):

##STR00035## [0515] wherein: [0516] q2A, q2B, q3A, q3B, q4A, q4B, q5A, q5B and q5C represent independently for each occurrence 0-20 and wherein the repeating unit can be the same or different; [0517] P.sup.2A, P.sup.2B, P.sup.3A, P.sup.3B, P.sup.4A, P.sup.4B, P.sup.5A, P.sup.5B, P.sup.5C, T.sup.2A, T.sup.2B, T.sup.3A, T.sup.3B, T.sup.4A, T.sup.4B, T.sup.4A, T.sup.5B, T.sup.5C are each independently for each occurrence absent, CO, NH, O, S, OC(O), NHC(O), CH.sub.2, CH.sub.2NH or CH.sub.2O; [0518] Q.sup.2A, Q.sup.2B, Q.sup.3A, Q.sup.3B, Q.sup.4A, Q.sup.4B, Q.sup.5A, Q.sup.5B, Q.sup.5C are independently for each occurrence absent, alkylene, substituted alkylene wherein one or more methylenes can be interrupted or terminated by one or more of O, S, S(O), SO.sub.2, N(R.sup.N), C(R')=C(R''), C≡C or C(O); [0519] R.sup.2A, R.sup.2B, R.sup.3A, R.sup.3B, R.sup.4A, R.sup.4B, R.sup.5A, R.sup.5B, R.sup.5C are each independently for each occurrence absent, NH, O, S, CH.sub.2, C(O)O, C(O)NH, NHCH(R.sup.a)C(O), $\text{—C(O)—CH(R.sup.a)—NH—}$, CO, CH=N—O,

##STR00036##

or heterocyclyl; [0520] L.sup.2A, L.sup.2B, L.sup.3A, L.sup.3B, L.sup.4A, L.sup.4B, L.sup.5A, L.sup.5B and L.sup.5C represent the ligand; i.e. each independently for each occurrence a monosaccharide (such as GalNAc), disaccharide, trisaccharide, tetrasaccharide, oligosaccharide, or polysaccharide; and R.sup.a is H or amino acid side chain. Trivalent

conjugating GalNAc derivatives are particularly useful for use with RNAi agents for inhibiting the expression of a target gene, such as those of formula (XLIX):

##STR00037## [0521] wherein L.sup.5A, L.sup.5B and L.sup.5C represent a monosaccharide, such as GalNAc derivative.

[0522] Examples of suitable bivalent and trivalent branched linker groups conjugating GalNAc derivatives include, but are not limited to, the structures recited above as formulas II, VII, XI, X, and XIII.

[0523] Representative U.S. patents that teach the preparation of RNA conjugates include, but are not limited to, U.S. Pat. Nos. 4,828,979; 4,948,882; 5,218,105; 5,525,465; 5,541,313; 5,545,730; 5,552,538; 5,578,717; 5,580,731; 5,591,584; 5,109,124; 5,118,802; 5,138,045; 5,414,077; 5,486,603; 5,512,439; 5,578,718; 5,608,046; 4,587,044; 4,605,735; 4,667,025; 4,762,779; 4,789,737; 4,824,941; 4,835,263; 4,876,335; 4,904,582; 4,958,013; 5,082,830; 5,112,963; 5,214,136; 5,082,830; 5,112,963; 5,214,136; 5,245,022; 5,254,469; 5,258,506; 5,262,536; 5,272,250; 5,292,873; 5,317,098; 5,371,241; 5,391,723; 5,416,203; 5,451,463; 5,510,475; 5,512,667; 5,514,785; 5,565,552; 5,567,810; 5,574,142; 5,585,481; 5,587,371; 5,595,726; 5,597,696; 5,599,923; 5,599,928; 5,688,941; 6,294,664; 6,320,017; 6,576,752; 6,783,931; 6,900,297; 7,037,646; and 8,106,022, the entire contents of each of which are hereby incorporated herein by reference.

[0524] It is not necessary for all positions in a given compound to be uniformly modified, and in fact more than one of the aforementioned modifications can be incorporated in a single compound or even at a single nucleoside within an iRNA. The present invention also includes iRNA compounds that are chimeric compounds.

[0525] "Chimeric" iRNA compounds or "chimeras," in the context of this invention, are iRNA compounds, preferably dsRNA agents, that contain two or more chemically distinct regions, each made up of at least one monomer unit, i.e., a nucleotide in the case of a dsRNA compound. These iRNAs typically contain at least one region wherein the RNA is modified so as to confer upon the iRNA increased resistance to nuclease degradation, increased cellular uptake, or increased binding affinity for the target nucleic acid. An additional region of the iRNA can serve as a substrate for enzymes capable of cleaving RNA:DNA or RNA:RNA hybrids. By way of example, RNase H is a cellular endonuclease which cleaves the RNA strand of an RNA:DNA duplex. Activation of RNase H, therefore, results in cleavage of the RNA target, thereby greatly enhancing the efficiency of iRNA inhibition of gene expression. Consequently, comparable results can often be obtained with shorter iRNAs when chimeric dsRNAs are used, compared to phosphorothioate deoxy dsRNAs hybridizing to the same target region. Cleavage of the RNA target can be routinely detected by gel electrophoresis and, if necessary, associated nucleic acid hybridization techniques known in the art.

[0526] In certain instances, the RNA of an iRNA can be modified by a non-ligand group. A number of non-ligand molecules have been conjugated to iRNAs in order to enhance the activity, cellular distribution or cellular uptake of the iRNA, and procedures for performing such conjugations are available in the scientific literature. Such non-ligand moieties have included lipid moieties, such as cholesterol (Kubo, T. et al., *Biochem. Biophys. Res. Comm.*, 2007, 365(1):54-61; Letsinger et al., *Proc. Natl. Acad. Sci. USA*, 1989, 86:6553), cholic acid (Manoharan et al., *Bioorg. Med. Chem. Lett.*, 1994, 4:1053), a thioether, e.g., hexyl-S-tritylthiol (Manoharan et al., *Ann. N Y Acad. Sci.*, 1992, 660:306; Manoharan et al., *Bioorg. Med. Chem. Lett.*, 1993, 3:2765), a thiocholesterol (Oberhauser et al., *Nucl. Acids Res.*, 1992, 20:533), an aliphatic chain, e.g., dodecandiol or undecyl residues (Saison-Behmoaras et al., *EMBO J.*, 1991, 10:111; Kabanov et al., *FEBS Lett.*, 1990, 259:327; Svinarchuk et al., *Biochimie*, 1993, 75:49), a phospholipid, e.g., di-hexadecyl-rac-glycerol or triethylammonium 1,2-di-O-hexadecyl-rac-glycero-3-H-phosphonate (Manoharan et al., *Tetrahedron Lett.*, 1995, 36:3651; Shea et al., *Nucl. Acids Res.*, 1990, 18:3777), a polyamine or a polyethylene glycol chain (Manoharan et al., *Nucleosides & Nucleotides*, 1995, 14:969), or adamantane acetic acid (Manoharan et al., *Tetrahedron Lett.*, 1995, 36:3651), a palmityl moiety (Mishra et al., *Biochim. Biophys. Acta*, 1995, 1264:229), or an octadecylamine or hexylamino-carbonyl-oxycholesterol moiety (Crooke et al., *J. Pharmacol. Exp. Ther.*, 1996, 277:923). Representative United States patents that teach the preparation of such RNA conjugates have been listed above. Typical conjugation protocols involve the synthesis of RNAs bearing an aminolinker at one or more positions of the sequence. The amino group is then reacted with the molecule being conjugated using appropriate coupling or activating reagents. The conjugation reaction can be performed either with the RNA still bound to the solid support or following cleavage of the RNA, in solution phase. Purification of the RNA conjugate by HPLC typically affords the pure conjugate.

V. Delivery of an RNAi Agent of the Disclosure

[0527] The delivery of an RNAi agent of the disclosure to a cell e.g., a cell within a subject, such as a human subject (e.g., a subject in need thereof, such as a subject having an ABLIM3-associated disorder, e.g., PTSD or age-related memory loss, can be achieved in a number of different ways. For example, delivery may be performed by contacting a cell with an RNAi agent of the disclosure either in vitro or in vivo. In vivo delivery may also be performed directly by administering a composition comprising an RNAi agent, e.g., a dsRNA, to a subject. Alternatively, in vivo delivery may be performed indirectly by administering one or more vectors that encode and direct the expression of the RNAi agent. These alternatives are discussed further below.

[0528] In general, any method of delivering a nucleic acid molecule (in vitro or in vivo) can be adapted for use with an RNAi agent of the disclosure (see e.g., Akhtar S. and Julian R L., (1992) *Trends Cell. Biol.* 2(5):139-144 and WO94/02595, which are incorporated herein by reference in their entireties). For in vivo delivery, factors to consider in order to deliver an RNAi agent include, for example, biological stability of the delivered agent, prevention of non-specific effects, and accumulation of the delivered agent in the target tissue. The non-specific effects of an RNAi agent can be minimized by local administration, for example, by direct injection or implantation into a tissue or topically administering the preparation. Local administration to a treatment site maximizes local concentration of the agent, limits the exposure of

the agent to systemic tissues or otherwise harmed by the agent or that can degrade the agent, and permits a lower total dose of the RNAi agent to be administered. Several studies have shown successful knockdown of gene products when an RNAi agent is administered locally. For example, intraocular delivery of a VEGF dsRNA by intravitreal injection in cynomolgus monkeys (Tolentino, M J. et al., (2004) *Retina* 24:132-138) and subretinal injections in mice (Reich, S J. et al. (2003) *Mol. Vis.* 9:210-216) were both shown to prevent neovascularization in an experimental model of age-related macular degeneration. In addition, direct intratumoral injection of a dsRNA in mice reduces tumor volume (Pille, J. et al. (2005) *Mol. Ther.* 11:267-274) and can prolong survival of tumor-bearing mice (Kim, W J. et al., (2006) *Mol. Ther.* 14:343-350; Li, S. et al., (2007) *Mol. Ther.* 15:515-523). RNA interference has also shown success with local delivery to the CNS by direct injection (Don, G. et al., (2004) *Nucleic Acids* 32:e49; Tan, P H. et al. (2005) *Gene Ther.* 12:59-66; Makimura, H. et al. (2002) *BMC Neurosci.* 3:18; Shishkina, G T., et al. (2004) *Neuroscience* 129:521-528; Thakker, E R., et al. (2004) *Proc. Natl. Acad. Sci. U.S.A.* 101:17270-17275; Akaneya, Y., et al. (2005) *J Neurophysiol.* 93:594-602) and to the lungs by intranasal administration (Howard, K A. et al., (2006) *Mol. Ther.* 14:476-484; Zhang, X. et al., (2004) *J. Biol. Chem.* 279:10677-10684; Bitko, V. et al., (2005) *Nat. Med.* 11:50-55). For administering an RNAi agent systemically for the treatment of a disease, the RNA can be modified or alternatively delivered using a drug delivery system; both methods act to prevent the rapid degradation of the dsRNA by endo- and exo-nucleases in vivo. Modification of the RNA or the pharmaceutical carrier can also permit targeting of the RNAi agent to the target tissue and avoid undesirable off-target effects (e.g., without wishing to be bound by theory, use of GNAs as described herein has been identified to destabilize the seed region of a dsRNA, resulting in enhanced preference of such dsRNAs for on-target effectiveness, relative to off-target effects, as such off-target effects are significantly weakened by such seed region destabilization). RNAi agents can be modified by chemical conjugation to lipophilic groups such as cholesterol to enhance cellular uptake and prevent degradation. For example, an RNAi agent directed against ApoB conjugated to a lipophilic cholesterol moiety was injected systemically into mice and resulted in knockdown of apoB mRNA in both the liver and jejunum (Soutschek, J. et al., (2004) *Nature* 432:173-178). Conjugation of an RNAi agent to an aptamer has been shown to inhibit tumor growth and mediate tumor regression in a mouse model of prostate cancer (McNamara, J O. et al., (2006) *Nat. Biotechnol.* 24:1005-1015). In an alternative embodiment, the RNAi agent can be delivered using drug delivery systems such as a nanoparticle, a dendrimer, a polymer, liposomes, or a cationic delivery system. Positively charged cationic delivery systems facilitate binding of molecule RNAi agent (negatively charged) and also enhance interactions at the negatively charged cell membrane to permit efficient uptake of an RNAi agent by the cell. Cationic lipids, dendrimers, or polymers can either be bound to an RNAi agent, or induced to form a vesicle or micelle (see e.g., Kim S H. et al., (2008) *Journal of Controlled Release* 129(2):107-116) that encases an RNAi agent. The formation of vesicles or micelles further prevents degradation of the RNAi agent when administered systemically. Methods for making and administering cationic-RNAi agent complexes are well within the abilities of one skilled in the art (see e.g., Sorensen, D R., et al. (2003) *J. Mol. Biol.* 327:761-766; Verma, U N. et al., (2003) *Clin. Cancer Res.* 9:1291-1300; Arnold, A S et al. (2007) *J. Hypertens.* 25:197-205, which are incorporated herein by reference in their entirety). Some non-limiting examples of drug delivery systems useful for systemic delivery of RNAi agents include DOTAP (Sorensen, D R., et al (2003), supra; Verma, U N. et al., (2003), supra), Oligofectamine, "solid nucleic acid lipid particles" (Zimmermann, T S. et al., (2006) *Nature* 441:111-114), cardiolipin (Chien, P Y. et al., (2005) *Cancer Gene Ther.* 12:321-328; Pal, A. et al., (2005) *IntJ. Oncol.* 26:1087-1091), polyethyleneimine (Bonnet M E. et al., (2008) *Pharm. Res.* August 16 Epub ahead of print; Aigner, A. (2006) *J. Biomed Biotechnol.* 71659), Arg-Gly-Asp (RGD) peptides (Liu. S. (2006) *Mol. Pharm.* 3:472-487), and polyamidoamines (Tomalia, D A. et al., (2007) *Biochem. Soc. Trans.* 35:61-67; Yoo, H. et al., (1999) *Pharm. Res.* 16:1799-1804). In some embodiments, an RNAi agent forms a complex with cyclodextrin for systemic administration. Methods for administration and pharmaceutical compositions of RNAi agents and cyclodextrins can be found in U.S. Pat. No. 7,427,605, which is herein incorporated by reference in its entirety.

[0529] Certain aspects of the instant disclosure relate to a method of reducing the expression of an ABLIM3 target gene in a cell, comprising contacting said cell with the double-stranded RNAi agent of the disclosure. In one embodiment, the cell is a CNS cell, such as a brain cell. In other embodiment, the cell is a dentate granule cell, neuron, an astrocyte, or an oligodendrocyte.

[0530] Another aspect of the disclosure relates to a method of reducing the expression of an ABLIM3 target gene in a subject, comprising administering to the subject the double-stranded RNAi agent of the disclosure.

[0531] Another aspect of the disclosure relates to a method of treating a subject having PTSD or age-related memory loss, comprising administering to the subject a therapeutically effective amount of the double-stranded ABLIM3-targeting RNAi agent of the disclosure, thereby treating the subject.

[0532] Another aspect of the disclosure relates to a method of preventing a sign or a symptom in a subject having PTSD or age-related memory loss, comprising administering to the subject a therapeutically effective amount of the double-stranded ABLIM3-targeting RNAi agent of the disclosure, thereby preventing a sign or symptom in the subject that would benefit from reduction in ABLIM3 expression. Exemplary PTSD signs or symptoms include anxiety, fear, agitation, depression, distress, hostility, vivid flashbacks, insomnia, nightmares, social isolation, and physical sensations such as pain, sweating, nausea, and trembling. Exemplary age-related memory loss signs and symptoms include memory loss, frequently losing or misplacing things, frequently forgetting conversations, appointments, and events, difficultly remembering names, difficulty following a conversation, and difficulty coming up with the desired words.

[0533] Another aspect of the disclosure relates to a method of preventing a subject from having PTSD or age-related memory loss, comprising administering to the subject a therapeutically effective amount of the double-stranded ABLIM3-

targeting RNAi agent of the disclosure, thereby preventing the subject from having PTSD or age-related memory loss. [0534] In one embodiment, the double-stranded RNAi agent is administered intrathecally. By intrathecal administration of the double-stranded RNAi agent, the method can reduce the expression of an ABLIM3 target gene in a brain (e.g., hippocampus and cerebrum).

[0535] For ease of exposition the formulations, compositions and methods in this section are discussed largely with regard to modified siRNA compounds. It may be understood, however, that these formulations, compositions and methods can be practiced with other siRNA compounds, e.g., unmodified siRNA compounds, and such practice is within the disclosure. A composition that includes an RNAi agent can be delivered to a subject by a variety of routes. Exemplary routes include: intrathecal, intravenous, topical, rectal, anal, vaginal, nasal, pulmonary, and ocular.

[0536] The RNAi agents of the disclosure can be incorporated into pharmaceutical compositions suitable for administration. Such compositions typically include one or more species of RNAi agent and a pharmaceutically acceptable carrier. As used herein the language “pharmaceutically acceptable carrier” is intended to include any and all solvents, dispersion media, coatings, antibacterial and antifungal agents, isotonic and absorption delaying agents, and the like, compatible with pharmaceutical administration. The use of such media and agents for pharmaceutically active substances is well known in the art. Except insofar as any conventional media or agent is incompatible with the active compound, use thereof in the compositions is contemplated. Supplementary active compounds can also be incorporated into the compositions.

[0537] The pharmaceutical compositions of the present disclosure may be administered in a number of ways depending upon whether local or systemic treatment is desired and upon the area to be treated. Administration may be topical (including ophthalmic, vaginal, rectal, intranasal, transdermal), intrathecal, oral, or parenteral. Parenteral administration includes intravenous drip, subcutaneous, intraperitoneal or intramuscular injection, or intrathecal or intraventricular administration.

[0538] The route and site of administration may be chosen to enhance targeting. For example, to target neural or spinal tissue, intrathecal injection would be a logical choice. Lung cells might be targeted by administering the RNAi agent in aerosol form. The vascular endothelial cells could be targeted by coating a balloon catheter with the RNAi agent and mechanically introducing the RNA.

[0539] Formulations for topical administration may include transdermal patches, ointments, lotions, creams, gels, drops, suppositories, sprays, liquids, and powders. Conventional pharmaceutical carriers, aqueous, powder or oily bases, thickeners and the like may be necessary or desirable. Coated condoms, gloves and the like may also be useful.

[0540] Compositions for oral administration include powders or granules, suspensions or solutions in water, syrups, elixirs or non-aqueous media, tablets, capsules, lozenges, or troches. In the case of tablets, carriers that can be used include lactose, sodium citrate and salts of phosphoric acid. Various disintegrants such as starch, and lubricating agents such as magnesium stearate, sodium lauryl sulfate and talc, are commonly used in tablets. For oral administration in capsule form, useful diluents are lactose and high molecular weight polyethylene glycols. When aqueous suspensions are required for oral use, the nucleic acid compositions can be combined with emulsifying and suspending agents. If desired, certain sweetening or flavoring agents can be added.

[0541] Compositions for intrathecal or intraventricular administration may include sterile aqueous solutions which may also contain buffers, diluents, and other suitable additives.

[0542] Formulations for parenteral administration may include sterile aqueous solutions which may also contain buffers, diluents, and other suitable additives. Intraventricular injection may be facilitated by an intraventricular catheter, for example, attached to a reservoir. For intravenous use, the total concentration of solutes may be controlled to render the preparation isotonic.

[0543] In one embodiment, the administration of the siRNA compound, e.g., a double-stranded siRNA compound, or siRNA compound, composition is parenteral, e.g., intravenous (e.g., as a bolus or as a diffusible infusion), intradermal, intraperitoneal, intramuscular, intrathecal, intraventricular, intracranial, subcutaneous, transmucosal, buccal, sublingual, endoscopic, rectal, oral, vaginal, topical, pulmonary, intranasal, urethral, or ocular. Administration can be provided by the subject or by another person, e.g., a health care provider. The medication can be provided in measured doses or in a dispenser which delivers a metered dose. Selected modes of delivery are discussed in more detail below.

A. Intrathecal Administration.

[0544] In one embodiment, the double-stranded RNAi agent is delivered by intrathecal injection (i.e., injection into the spinal fluid which bathes the brain and spinal cord tissue). Intrathecal injection of RNAi agents into the spinal fluid can be performed as a bolus injection or via minipumps which can be implanted beneath the skin, providing a regular and constant delivery of siRNA into the spinal fluid. The circulation of the spinal fluid from the choroid plexus, where it is produced, down around the spinal cord and dorsal root ganglia and subsequently up past the cerebellum and over the cortex to the arachnoid granulations, where the fluid can exit the CNS, that, depending upon size, stability, and solubility of the compounds injected, molecules delivered intrathecally could hit targets throughout the entire CNS.

[0545] In some embodiments, the intrathecal administration is via a pump. The pump may be a surgically implanted osmotic pump. In one embodiment, the osmotic pump is implanted into the subarachnoid space of the spinal canal to facilitate intrathecal administration.

[0546] In some embodiments, the intrathecal administration is via an intrathecal delivery system for a pharmaceutical including a reservoir containing a volume of the pharmaceutical agent, and a pump configured to deliver a portion of the pharmaceutical agent contained in the reservoir. More details about this intrathecal delivery system may be found in WO

2015/116658, which is incorporated by reference in its entirety.

[0547] The amount of intrathecally injected RNAi agents may vary from one target gene to another target gene and the appropriate amount that has to be applied may have to be determined individually for each target gene. Typically, this amount ranges from 10 µg to 2 mg, preferably 50 µg to 1500 µg, more preferably 100 µg to 1000 µg.

B. Vector Encoded RNAi Agents of the Disclosure

[0548] RNAi agents targeting the ABLIM3 gene can be expressed from transcription units inserted into DNA or RNA vectors (see, e.g., Couture, A, et al., *TIG.* (1996), 12:5-10; WO 00/22113, WO 00/22114, and U.S. Pat. No. 6,054,299). Expression is preferably sustained (months or longer), depending upon the specific construct used and the target tissue or cell type. These transgenes can be introduced as a linear construct, a circular plasmid, or a viral vector, which can be an integrating or non-integrating vector. The transgene can also be constructed to permit it to be inherited as an extrachromosomal plasmid (Gassmann, et al., (1995) *Proc. Natl. Acad. Sci. USA* 92:1292).

[0549] The individual strand or strands of an RNAi agent can be transcribed from a promoter on an expression vector. Where two separate strands are to be expressed to generate, for example, a dsRNA, two separate expression vectors can be co-introduced (e.g., by transfection or infection) into a target cell. Alternatively, each individual strand of a dsRNA can be transcribed by promoters both of which are located on the same expression plasmid. In one embodiment, a dsRNA is expressed as inverted repeat polynucleotides joined by a linker polynucleotide sequence such that the dsRNA has a stem and loop structure.

[0550] RNAi agent expression vectors are generally DNA plasmids or viral vectors. Expression vectors compatible with eukaryotic cells, preferably those compatible with vertebrate cells, can be used to produce recombinant constructs for the expression of an RNAi agent as described herein. Delivery of RNAi agent expressing vectors can be systemic, such as by intravenous or intramuscular administration, by administration to target cells ex-planted from the patient followed by reintroduction into the patient, or by any other means that allows for introduction into a desired target cell.

[0551] Viral vector systems which can be utilized with the methods and compositions described herein include, but are not limited to, (a) adenovirus vectors; (b) retrovirus vectors, including but not limited to lentiviral vectors, moloney murine leukemia virus, etc.; (c) adeno-associated virus vectors; (d) herpes simplex virus vectors; (e) SV 40 vectors; (f) polyoma virus vectors; (g) papilloma virus vectors; (h) picornavirus vectors; (i) pox virus vectors such as an orthopox, e.g., vaccinia virus vectors or avipox, e.g. canary pox or fowl pox; and (j) a helper-dependent or gutless adenovirus. Replication-defective viruses can also be advantageous. Different vectors will or will not become incorporated into the cells' genome. The constructs can include viral sequences for transfection, if desired. Alternatively, the construct can be incorporated into vectors capable of episomal replication, e.g. EPV and EBV vectors. Constructs for the recombinant expression of an RNAi agent will generally require regulatory elements, e.g., promoters, enhancers, etc., to ensure the expression of the RNAi agent in target cells. Other aspects to consider for vectors and constructs are known in the art.

VI. Pharmaceutical Compositions of the Invention

[0552] The present disclosure also includes pharmaceutical compositions and formulations which include the RNAi agents of the disclosure. In one embodiment, provided herein are pharmaceutical compositions containing an RNAi agent, as described herein, and a pharmaceutically acceptable carrier. The pharmaceutical compositions containing the RNAi agent are useful for treating a disease or condition treatable by reduction or inhibition of the expression or activity of ABLIM3, e.g., ABLIM3-associated disease, such PTSD or age-related memory loss.

[0553] In some embodiments, the pharmaceutical compositions of the invention are sterile. In another embodiment, the pharmaceutical compositions of the invention are pyrogen free.

[0554] Such pharmaceutical compositions are formulated based on the mode of delivery. One example is compositions that are formulated for systemic administration via parenteral delivery, e.g., by intravenous (IV), intramuscular (IM), or for subcutaneous (subQ) delivery. Another example is compositions that are formulated for direct delivery into the CNS, e.g., by intrathecal or intravitreal routes of injection, optionally by infusion into the brain (e.g., striatum), such as by continuous pump infusion.

[0555] The pharmaceutical compositions of the disclosure may be administered in dosages sufficient to inhibit expression of an ABLIM3 gene. In general, a suitable dose of an RNAi agent of the disclosure will be in the range of about 0.001 to about 200.0 milligrams per kilogram body weight of the recipient per day, generally in the range of about 1 to 50 mg per kilogram body weight per day.

[0556] A repeat-dose regimen may include administration of a therapeutic amount of an RNAi agent on a regular basis, such as monthly to once every six months. In certain embodiments, the RNAi agent is administered about once per quarter (i.e., about once every three months) to about twice per year.

[0557] After an initial treatment regimen (e.g., loading dose), the treatments can be administered on a less frequent basis.

[0558] In other embodiments, a single dose of the pharmaceutical compositions can be long lasting, such that subsequent doses are administered at not more than 1, 2, 3, or 4 or more month intervals. In some embodiments of the disclosure, a single dose of the pharmaceutical compositions of the disclosure is administered once per month. In other embodiments of the disclosure, a single dose of the pharmaceutical compositions of the disclosure is administered once per quarter to twice per year.

[0559] The skilled artisan will appreciate that certain factors can influence the dosage and timing required to effectively treat a subject, including but not limited to the severity of the disease or disorder, previous treatments, the general health or age of the subject, and other diseases present. Moreover, treatment of a subject with a therapeutically effective amount of a composition can include a single treatment or a series of treatments.

[0560] Advances in genetics have generated a number of rodent models for the study of various human diseases, including PTSD or age-related memory loss, that would benefit from reduction in the expression of ABLIM3. Such models can be used for in vivo testing of RNAi agents, as well as for determining a therapeutically effective dose. Suitable rodent models are known in the art and include, for example, those described in, for example, Borhgans and Homberg (2015) *World J. Psychiatry* 5(4):387-96. The pharmaceutical compositions of the present disclosure can be administered in a number of ways depending upon whether local or systemic treatment is desired and upon the area to be treated. Administration can be topical (e.g., by a transdermal patch), pulmonary, e.g., by inhalation or insufflation of powders or aerosols, including by nebulizer; intratracheal, intranasal, epidermal and transdermal, oral or parenteral. Parenteral administration includes intravenous, intraarterial, subcutaneous, intraperitoneal or intramuscular injection or infusion; subdermal, e.g., via an implanted device; or intracranial, e.g., by intraparenchymal, intrathecal or intraventricular, administration.

[0561] The RNAi agents can be delivered in a manner to target a particular tissue, such as the CNS (e.g., neuronal, glial or vascular tissue of the brain).

[0562] Pharmaceutical compositions and formulations for topical administration can include transdermal patches, ointments, lotions, creams, gels, drops, suppositories, sprays, liquids and powders. Conventional pharmaceutical carriers, aqueous, powder or oily bases, thickeners and the like can be necessary or desirable. Coated condoms, gloves and the like can also be useful. Suitable topical formulations include those in which the RNAi agents featured in the disclosure are in admixture with a topical delivery agent such as lipids, liposomes, fatty acids, fatty acid esters, steroids, chelating agents and surfactants. Suitable lipids and liposomes include neutral (e.g., dioleoylphosphatidyl DOPE ethanolamine, dimyristoylphosphatidyl choline DMPC, distearoylphosphatidyl choline) negative (e.g., dimyristoylphosphatidyl glycerol DMPG) and cationic (e.g., dioleoyltetramethylaminopropyl DOTAP and dioleoylphosphatidyl ethanolamine DOTMA). RNAi agents featured in the disclosure can be encapsulated within liposomes or can form complexes thereto, in particular to cationic liposomes. Alternatively, RNAi agents can be complexed to lipids, in particular to cationic lipids. Suitable fatty acids and esters include but are not limited to arachidonic acid, oleic acid, eicosanoic acid, lauric acid, caprylic acid, capric acid, myristic acid, palmitic acid, stearic acid, linoleic acid, linolenic acid, dicaprate, tricaprate, monoolein, dilaurin, glyceryl 1-monocaprate, 1-dodecylazacycloheptan-2-one, an acylcarnitine, an acylcholine, or a C.sub.1-20 alkyl ester (e.g., isopropylmyristate TPM), monoglyceride, diglyceride or pharmaceutically acceptable salt thereof. Topical formulations are described in detail in U.S. Pat. No. 6,747,014, which is incorporated herein by reference.

A. RNAi Agent Formulations Comprising Membranous Molecular Assemblies

[0563] An RNAi agent for use in the compositions and methods of the disclosure can be formulated for delivery in a membranous molecular assembly, e.g., a liposome or a micelle. As used herein, the term “liposome” refers to a vesicle composed of amphiphilic lipids arranged in at least one bilayer, e.g., one bilayer or a plurality of bilayers. Liposomes include unilamellar and multilamellar vesicles that have a membrane formed from a lipophilic material and an aqueous interior. The aqueous portion contains the RNAi agent composition. The lipophilic material isolates the aqueous interior from an aqueous exterior, which typically does not include the RNAi agent composition, although in some examples, it may. Liposomes are useful for the transfer and delivery of active ingredients to the site of action. Because the liposomal membrane is structurally similar to biological membranes, when liposomes are applied to a tissue, the liposomal bilayer fuses with bilayer of the cellular membranes. As the merging of the liposome and cell progresses, the internal aqueous contents that include the RNAi agent are delivered into the cell where the RNAi agent can specifically bind to a target RNA and can mediate RNAi. In some cases, the liposomes are also specifically targeted, e.g., to direct the RNAi agent to particular cell types.

[0564] A liposome containing an RNAi agent can be prepared by a variety of methods. In one example, the lipid component of a liposome is dissolved in a detergent so that micelles are formed with the lipid component. For example, the lipid component can be an amphipathic cationic lipid or lipid conjugate. The detergent can have a high critical micelle concentration and may be nonionic. Exemplary detergents include cholate, CHAPS, octylglucoside, deoxycholate, and lauroyl sarcosine. The RNAi agent preparation is then added to the micelles that include the lipid component. The cationic groups on the lipid interact with the RNAi agent and condense around the RNAi agent to form a liposome. After condensation, the detergent is removed, e.g., by dialysis, to yield a liposomal preparation of RNAi agent.

[0565] If necessary a carrier compound that assists in condensation can be added during the condensation reaction, e.g., by controlled addition. For example, the carrier compound can be a polymer other than a nucleic acid (e.g., spermine or spermidine). pH can also be adjusted to favor condensation.

[0566] Methods for producing stable polynucleotide delivery vehicles, which incorporate a polynucleotide/cationic lipid complex as structural components of the delivery vehicle, are further described in, e.g., WO 96/37194, the entire contents of which are incorporated herein by reference. Liposome formation can also include one or more aspects of exemplary methods described in Felgner, P. L. et al., (1987) *Proc. Natl. Acad. Sci. USA* 8:7413-7417; U.S. Pat. Nos. 4,897,355; 5,171,678; Bangham et al., (1965) *M. Mol. Biol.* 23:238; Olson et al., (1979) *Biochim. Biophys. Acta* 557:9; Szoka et al., (1978) *Proc. Natl. Acad. Sci.* 75: 4194; Mayhew et al., (1984) *Biochim. Biophys. Acta* 775:169; Kim et al., (1983) *Biochim. Biophys. Acta* 728:339; and Fukunaga et al., (1984) *Endocrinol.* 115:757. Commonly used techniques for preparing lipid aggregates of appropriate size for use as delivery vehicles include sonication and freeze-thaw plus extrusion (see, e.g., Mayer et al., (1986) *Biochim. Biophys. Acta* 858:161. Microfluidization can be used when consistently small (50 to 200 nm) and relatively uniform aggregates are desired (Mayhew et al., (1984) *Biochim. Biophys. Acta* 775:169. These methods are readily adapted to packaging RNAi agent preparations into liposomes.

[0567] Liposomes fall into two broad classes. Cationic liposomes are positively charged liposomes which interact with the

negatively charged nucleic acid molecules to form a stable complex. The positively charged nucleic acid/liposome complex binds to the negatively charged cell surface and is internalized in an endosome. Due to the acidic pH within the endosome, the liposomes are ruptured, releasing their contents into the cell cytoplasm (Wang et al. (1987) *Biochem. Biophys. Res. Commun.*, 147:980-985).

[0568] Liposomes, which are pH-sensitive or negatively charged, entrap nucleic acids rather than complex with them. Since both the nucleic acid and the lipid are similarly charged, repulsion rather than complex formation occurs. Nevertheless, some nucleic acid is entrapped within the aqueous interior of these liposomes. pH sensitive liposomes have been used to deliver nucleic acids encoding the thymidine kinase gene to cell monolayers in culture. Expression of the exogenous gene was detected in the target cells (Zhou et al. (1992) *Journal of Controlled Release*, 19:269-274).

[0569] One major type of liposomal composition includes phospholipids other than naturally-derived phosphatidylcholine. Neutral liposome compositions, for example, can be formed from dimyristoyl phosphatidylcholine (DMPC) or dipalmitoyl phosphatidylcholine (DPPC). Anionic liposome compositions generally are formed from dimyristoyl phosphatidylglycerol, while anionic fusogenic liposomes are formed primarily from dioleoyl phosphatidylethanolamine (DOPE). Another type of liposomal composition is formed from phosphatidylcholine (PC) such as, for example, soybean PC, and egg PC. Another type is formed from mixtures of phospholipid or phosphatidylcholine or cholesterol.

[0570] Examples of other methods to introduce liposomes into cells in vitro and in vivo include U.S. Pat. Nos. 5,283,185; 5,171,678; WO 94/00569; WO 93/24640; WO 91/16024; Felgner, (1994) *J. Biol. Chem.* 269:2550; Nabel, (1993) *Proc. Natl. Acad. Sci.* 90:11307; Nabel, (1992) *Human Gene Ther.* 3:649; Gershon, (1993) *Biochem.* 32:7143; and Strauss, (1992) *EMBO J.* 11:417.

[0571] Non-ionic liposomal systems have also been examined to determine their utility in the delivery of drugs to the skin, in particular systems comprising non-ionic surfactant and cholesterol. Non-ionic liposomal formulations comprising Novasome™ I (glyceryl dilaurate/cholesterol/polyoxyethylene-10-stearyl ether) and Novasome™ II (glyceryl distearate/cholesterol/polyoxyethylene-10-stearyl ether) were used to deliver cyclosporin-A into the dermis of mouse skin. Results indicated that such non-ionic liposomal systems were effective in facilitating the deposition of cyclosporine A into different layers of the skin (Hu et al., (1994) *S.T.P. Pharma. Sci.*, 4(6):466).

[0572] Liposomes also include “sterically stabilized” liposomes, a term which, as used herein, refers to liposomes comprising one or more specialized lipids that, when incorporated into liposomes, result in enhanced circulation lifetimes relative to liposomes lacking such specialized lipids. Examples of sterically stabilized liposomes are those in which part of the vesicle-forming lipid portion of the liposome (A) comprises one or more glycolipids, such as monosialoganglioside GM1, or (B) is derivatized with one or more hydrophilic polymers, such as a polyethylene glycol (PEG) moiety. While not wishing to be bound by any particular theory, it is thought in the art that, at least for sterically stabilized liposomes containing gangliosides, sphingomyelin, or PEG-derivatized lipids, the enhanced circulation half-life of these sterically stabilized liposomes derives from a reduced uptake into cells of the reticuloendothelial system (RES) (Allen et al., (1987) *FEBS Letters*, 223:42; Wu et al., (1993) *Cancer Research*, 53:3765).

[0573] Various liposomes comprising one or more glycolipids are known in the art. Papahadjopoulos et al. (*Ann. N.Y. Acad. Sci.*, (1987), 507:64) reported the ability of monosialoganglioside GM1, galactocerebroside sulfate and phosphatidylinositol to improve blood half-lives of liposomes. These findings were expounded upon by Gabizon et al. (*Proc. Natl. Acad. Sci. U.S.A.*, (1988), 85:6949). U.S. Pat. No. 4,837,028 and WO 88/04924, both to Allen et al., disclose liposomes comprising (1) sphingomyelin and (2) the ganglioside GM1 or a galactocerebroside sulfate ester. U.S. Pat. No. 5,543,152 (Webb et al.) discloses liposomes comprising sphingomyelin. Liposomes comprising 1,2-sn-dimyristoylphosphatidylcholine are disclosed in WO 97/13499 (Lim et al.).

[0574] In one embodiment, cationic liposomes are used. Cationic liposomes possess the advantage of being able to fuse to the cell membrane. Non-cationic liposomes, although not able to fuse as efficiently with the plasma membrane, are taken up by macrophages in vivo and can be used to deliver RNAi agents to macrophages.

[0575] Further advantages of liposomes include: liposomes obtained from natural phospholipids are biocompatible and biodegradable; liposomes can incorporate a wide range of water and lipid soluble drugs; liposomes can protect encapsulated RNAi agents in their internal compartments from metabolism and degradation (Rosoff, in “Pharmaceutical Dosage Forms,” Lieberman, Rieger and Banker (Eds.), 1988, volume 1, p. 245). Important considerations in the preparation of liposome formulations are the lipid surface charge, vesicle size and the aqueous volume of the liposomes.

[0576] A positively charged synthetic cationic lipid, N-[1-(2,3-dioleoyloxy)propyl]-N,N,N-trimethylammonium chloride (DOTMA) can be used to form small liposomes that interact spontaneously with nucleic acid to form lipid-nucleic acid complexes which are capable of fusing with the negatively charged lipids of the cell membranes of tissue culture cells, resulting in delivery of RNAi agent (see, e.g., Felgner, P. L. et al., (1987) *Proc. Natl. Acad. Sci. USA* 8:7413-7417, and U.S. Pat. No. 4,897,355 for a description of DOTMA and its use with DNA).

[0577] A DOTMA analogue, 1,2-bis(oleoyloxy)-3-(trimethylammonia)propane (DOTAP) can be used in combination with a phospholipid to form DNA-complexing vesicles. Lipofectin™ Bethesda Research Laboratories, Gaithersburg, Md.) is an effective agent for the delivery of highly anionic nucleic acids into living tissue culture cells that comprise positively charged DOTMA liposomes which interact spontaneously with negatively charged polynucleotides to form complexes. When enough positively charged liposomes are used, the net charge on the resulting complexes is also positive. Positively charged complexes prepared in this way spontaneously attach to negatively charged cell surfaces, fuse with the plasma membrane, and efficiently deliver functional nucleic acids into, for example, tissue culture cells. Another commercially available cationic lipid, 1,2-bis(oleoyloxy)-3,3-(trimethylammonia)propane (“DOTAP”) (Boehringer Mannheim,

Indianapolis) differs from DOTMA in that the oleoyl moieties are linked by ester, rather than ether linkages.

[0578] Other reported cationic lipid compounds include those that have been conjugated to a variety of moieties including, for example, carboxyspermine which has been conjugated to one of two types of lipids and includes compounds such as 5-carboxyspermylglycine dioctaoyleamide (“DOGS”) (Transfectam™, Promega, Madison, Wisconsin) and dipalmitoylphosphatidylethanolamine 5-carboxyspermyl-amide (“DPPE”) (see, e.g., U.S. Pat. No. 5,171,678).

[0579] Another cationic lipid conjugate includes derivatization of the lipid with cholesterol (“DC-Chol”) which has been formulated into liposomes in combination with DOPE (See, Gao, X. and Huang, L., (1991) *Biochim. Biophys. Res. Commun.* 179:280). Lipopolylysine, made by conjugating polylysine to DOPE, has been reported to be effective for transfection in the presence of serum (Zhou, X. et al., (1991) *Biochim. Biophys. Acta* 1065:8). For certain cell lines, these liposomes containing conjugated cationic lipids, are said to exhibit lower toxicity and provide more efficient transfection than the DOTMA-containing compositions. Other commercially available cationic lipid products include DMRIE and DMRIE-HP (Vical, La Jolla, California) and Lipofectamine (DOSPA) (Life Technology, Inc., Gaithersburg, Maryland). Other cationic lipids suitable for the delivery of oligonucleotides are described in WO 98/39359 and WO 96/37194.

[0580] Liposomal formulations are particularly suited for topical administration; liposomes present several advantages over other formulations. Such advantages include reduced side effects related to high systemic absorption of the administered drug, increased accumulation of the administered drug at the desired target, and the ability to administer RNAi agent into the skin. In some implementations, liposomes are used for delivering RNAi agent to epidermal cells and also to enhance the penetration of RNAi agent into dermal tissues, e.g., into skin. For example, the liposomes can be applied topically. Topical delivery of drugs formulated as liposomes to the skin has been documented (see, e.g., Weiner et al., (1992) *Journal of Drug Targeting*, vol. 2, 405-410 and du Plessis et al., (1992) *Antiviral Research*, 18:259-265; Mannino, R. J. and Fould-Fogerite, S., (1998) *Biotechniques* 6:682-690; Itani, T. et al., (1987) *Gene* 56:267-276; Nicolau, C. et al. (1987) *Meth. Enzymol.* 149:157-176; Straubinger, R. M. and Papahadjopoulos, D. (1983) *Meth. Enzymol.* 101:512-527; Wang, C. Y. and Huang, L., (1987) *Proc. Natl. Acad. Sci. USA* 84:7851-7855).

[0581] Non-ionic liposomal systems have also been examined to determine their utility in the delivery of drugs to the skin, in particular systems comprising non-ionic surfactant and cholesterol. Non-ionic liposomal formulations comprising Novasome I (glyceryl dilaurate/cholesterol/polyoxyethylene-10-stearyl ether) and Novasome II (glyceryl distearate/cholesterol/polyoxyethylene-10-stearyl ether) were used to deliver a drug into the dermis of mouse skin. Such formulations with RNAi agent are useful for treating a dermatological disorder.

[0582] Liposomes that include RNAi agents can be made highly deformable. Such deformability can enable the liposomes to penetrate through pore that are smaller than the average radius of the liposome. For example, transfersomes are a type of deformable liposomes. Transfersomes can be made by adding surface edge activators, usually surfactants, to a standard liposomal composition. Transfersomes that include RNAi agent can be delivered, for example, subcutaneously by infection in order to deliver RNAi agent to keratinocytes in the skin. In order to cross intact mammalian skin, lipid vesicles must pass through a series of fine pores, each with a diameter less than 50 nm, under the influence of a suitable transdermal gradient. In addition, due to the lipid properties, these transfersomes can be self-optimizing (adaptive to the shape of pores, e.g., in the skin), self-repairing, and can frequently reach their targets without fragmenting, and often self-loading.

[0583] Other formulations amenable to the present disclosure are described in PCT application number PCT/US2008/088588, filed Dec. 13, 2008; PCT/US2008/088587, filed Dec. 13, 2008; PCT/US2009/041442, filed Apr. 22, 2009; and PCT/US2007/080331, filed Oct. 3, 2007.

[0584] Transfersomes, yet another type of liposomes, are highly deformable lipid aggregates which are attractive candidates for drug delivery vehicles. Transfersomes can be described as lipid droplets which are so highly deformable that they are easily able to penetrate through pores which are smaller than the droplet. Transfersomes are adaptable to the environment in which they are used, e.g., they are self-optimizing (adaptive to the shape of pores in the skin), self-repairing, frequently reach their targets without fragmenting, and often self-loading. To make transfersomes it is possible to add surface edge-activators, usually surfactants, to a standard liposomal composition. Transfersomes have been used to deliver serum albumin to the skin. The transfersome-mediated delivery of serum albumin has been shown to be as effective as subcutaneous injection of a solution containing serum albumin.

[0585] Surfactants find wide application in formulations such as those described herein, particularly in emulsions (including microemulsions) and liposomes. The most common way of classifying and ranking the properties of the many different types of surfactants, both natural and synthetic, is by the use of the hydrophile/lipophile balance (HLB). The nature of the hydrophilic group (also known as the “head”) provides the most useful means for categorizing the different surfactants used in formulations (Rieger, in *Pharmaceutical Dosage Forms*, Marcel Dekker, Inc., New York, N.Y., 1988, p. 285).

[0586] If the surfactant molecule is not ionized, it is classified as a nonionic surfactant. Nonionic surfactants find wide application in pharmaceutical and cosmetic products and are usable over a wide range of pH values. In general, their HLB values range from 2 to about 18 depending on their structure. Nonionic surfactants include nonionic esters such as ethylene glycol esters, propylene glycol esters, glyceryl esters, polyglyceryl esters, sorbitan esters, sucrose esters, and ethoxylated esters. Nonionic alkanolamides and ethers such as fatty alcohol ethoxylates, propoxylated alcohols, and ethoxylated/propoxylated block polymers are also included in this class. The polyoxyethylene surfactants are the most popular members of the nonionic surfactant class.

[0587] If the surfactant molecule carries a negative charge when it is dissolved or dispersed in water, the surfactant is classified as anionic. Anionic surfactants include carboxylates such as soaps, acyl lactylates, acyl amides of amino acids,

esters of sulfuric acid such as alkyl sulfates and ethoxylated alkyl sulfates, sulfonates such as alkyl benzene sulfonates, acyl isethionates, acyl taurates and sulfosuccinates, and phosphates. The most important members of the anionic surfactant class are the alkyl sulfates and the soaps.

[0588] If the surfactant molecule carries a positive charge when it is dissolved or dispersed in water, the surfactant is classified as cationic. Cationic surfactants include quaternary ammonium salts and ethoxylated amines. The quaternary ammonium salts are the most used members of this class.

[0589] If the surfactant molecule has the ability to carry either a positive or negative charge, the surfactant is classified as amphoteric. Amphoteric surfactants include acrylic acid derivatives, substituted alkylamides, N-alkylbetaines and phosphatides. The use of surfactants in drug products, formulations and in emulsions has been reviewed (Rieger, in *Pharmaceutical Dosage Forms*, Marcel Dekker, Inc., New York, N.Y., 1988, p. 285).

[0590] The RNAi agent for use in the methods of the disclosure can also be provided as micellar formulations. "Micelles" are defined herein as a particular type of molecular assembly in which amphipathic molecules are arranged in a spherical structure such that all the hydrophobic portions of the molecules are directed inward, leaving the hydrophilic portions in contact with the surrounding aqueous phase. The converse arrangement exists if the environment is hydrophobic.

[0591] A mixed micellar formulation suitable for delivery through transdermal membranes may be prepared by mixing an aqueous solution of the siRNA composition, an alkali metal C.sub.8 to C.sub.22 alkyl sulphate, and a micelle forming compounds. Exemplary micelle forming compounds include lecithin, hyaluronic acid, pharmaceutically acceptable salts of hyaluronic acid, glycolic acid, lactic acid, chamomile extract, cucumber extract, oleic acid, linoleic acid, linolenic acid, monoolein, monooleates, monolaurates, borage oil, evening of primrose oil, menthol, trihydroxy oxo cholanyl glycine and pharmaceutically acceptable salts thereof, glycerin, polyglycerin, lysine, polylysine, triolein, polyoxyethylene ethers and analogues thereof, polidocanol alkyl ethers and analogues thereof, chenodeoxycholate, deoxycholate, and mixtures thereof. The micelle forming compounds may be added at the same time or after addition of the alkali metal alkyl sulphate. Mixed micelles will form with substantially any kind of mixing of the ingredients but vigorous mixing in order to provide smaller size micelles.

[0592] In one method a first micellar composition is prepared which contains the siRNA composition and at least the alkali metal alkyl sulphate. The first micellar composition is then mixed with at least three micelle forming compounds to form a mixed micellar composition. In another method, the micellar composition is prepared by mixing the siRNA composition, the alkali metal alkyl sulphate and at least one of the micelle forming compounds, followed by addition of the remaining micelle forming compounds, with vigorous mixing.

[0593] Phenol or m-cresol may be added to the mixed micellar composition to stabilize the formulation and protect against bacterial growth. Alternatively, phenol or m-cresol may be added with the micelle forming ingredients. An isotonic agent such as glycerin may also be added after formation of the mixed micellar composition.

[0594] For delivery of the micellar formulation as a spray, the formulation can be put into an aerosol dispenser and the dispenser is charged with a propellant. The propellant, which is under pressure, is in liquid form in the dispenser. The ratios of the ingredients are adjusted so that the aqueous and propellant phases become one, i.e., there is one phase. If there are two phases, it is necessary to shake the dispenser prior to dispensing a portion of the contents, e.g., through a metered valve. The dispensed dose of pharmaceutical agent is propelled from the metered valve in a fine spray.

[0595] Propellants may include hydrogen-containing chlorofluorocarbons, hydrogen-containing fluorocarbons, dimethyl ether and diethyl ether. In certain embodiments, HFA 134a (1,1,1,2 tetrafluoroethane) may be used.

[0596] The specific concentrations of the essential ingredients can be determined by relatively straightforward experimentation. For absorption through the oral cavities, it is often desirable to increase, e.g., at least double or triple, the dosage for through injection or administration through the gastrointestinal tract.

B. Lipid Particles

[0597] RNAi agents, e.g., dsRNAs of in the disclosure may be fully encapsulated in a lipid formulation, e.g., a LNP, or other nucleic acid-lipid particle.

[0598] As used herein, the term "LNP" refers to a stable nucleic acid-lipid particle. LNPs typically contain a cationic lipid, a non-cationic lipid, and a lipid that prevents aggregation of the particle (e.g., a PEG-lipid conjugate). LNPs are extremely useful for systemic applications, as they exhibit extended circulation lifetimes following intravenous (i.v.) injection and accumulate at distal sites (e.g., sites physically separated from the administration site). LNPs include "pSPLP," which include an encapsulated condensing agent-nucleic acid complex as set forth in WO 00/03683. The particles of the present disclosure typically have a mean diameter of about 50 nm to about 150 nm, more typically about 60 nm to about 130 nm, more typically about 70 nm to about 110 nm, most typically about 70 nm to about 90 nm, and are substantially nontoxic. In addition, the nucleic acids when present in the nucleic acid-lipid particles of the present disclosure are resistant in aqueous solution to degradation with a nuclease. Nucleic acid-lipid particles and their method of preparation are disclosed in, e.g., U.S. Pat. Nos. 5,976,567; 5,981,501; 6,534,484; 6,586,410; 6,815,432; United States Patent Publication No. 2010/0324120 and WO 96/40964.

[0599] In one embodiment, the lipid to drug ratio (mass/mass ratio) (e.g., lipid to dsRNA ratio) will be in the range of from about 1:1 to about 50:1, from about 1:1 to about 25:1, from about 3:1 to about 15:1, from about 4:1 to about 10:1, from about 5:1 to about 9:1, or about 6:1 to about 9:1. Ranges intermediate to the above recited ranges are also contemplated to be part of the disclosure.

[0600] Certain specific LNP formulations for delivery of RNAi agents have been described in the art, including, e.g., "LNP01" formulations as described in, e.g., WO 2008/042973, which is hereby incorporated by reference.

[0601] Additional exemplary lipid-dsRNA formulations are identified in Table 1 below.

TABLE-US-00002 TABLE 1 cationic lipid/non-cationic lipid/cholesterol/PEG-lipid conjugate Ionizable/Cationic Lipid Lipid:siRNA ratio SNALP-1 1,2-Dilinolenyloxy-N,N-DLinDMA/DPPC/Cholesterol/PEG- dimethylaminopropane (DLinDMA) cDMA (57.1/7.1/34.4/1.4) lipid:siRNA ~7:1 2-XTC 2,2-Dilinoleyl-4-dimethylaminoethyl-[1,3]- XTC/DPPC/Cholesterol/PEG-cDMA dioxolane (XTC) 57.1/7.1/34.4/1.4 lipid:siRNA ~7:1 LNP05 2,2-Dilinoleyl-4-dimethylaminoethyl-[1,3]- XTC/DSPC/Cholesterol/PEG-DMG dioxolane (XTC) 57.5/7.5/31.5/3.5 lipid:siRNA ~6:1 LNP06 2,2-Dilinoleyl-4-dimethylaminoethyl-[1,3]- XTC/DSPC/Cholesterol/PEG-DMG dioxolane (XTC) 57.5/7.5/31.5/3.5 lipid:siRNA ~11:1 LNP07 2,2-Dilinoleyl-4-dimethylaminoethyl-[1,3]- XTC/DSPC/Cholesterol/PEG-DMG dioxolane (XTC) 60/7.5/31/1.5, lipid:siRNA ~6:1 LNP08 2,2-Dilinoleyl-4-dimethylaminoethyl-[1,3]- XTC/DSPC/Cholesterol/PEG-DMG dioxolane (XTC) 60/7.5/31/1.5, lipid:siRNA ~11:1 LNP09 2,2-Dilinoleyl-4-dimethylaminoethyl-[1,3]- XTC/DSPC/Cholesterol/PEG-DMG dioxolane (XTC) 50/10/38.5/1.5 Lipid:siRNA 10:1 LNP10 (3aR,5s,6aS)-N,N-dimethyl-2,2- ALN100/DSPC/Cholesterol/PEG- di((9Z,12Z)-octadeca-9,12- DMG dienyl)tetrahydro-3aH- 50/10/38.5/1.5 cyclopenta[d][1,3]dioxol-5-amine Lipid:siRNA 10:1 (ALN100) LNP11 (6Z,9Z,28Z,31Z)-heptatriaconta-6,9,28,31- MC-3/DSPC/Cholesterol/PEG-DMG tetraen-19-yl 4-(dimethylamino)butanoate 50/10/38.5/1.5 (MC3) Lipid:siRNA 10:1 LNP12 1,1'-(2-(4-(2-((2-(bis(2- Tech G1/DSPC/Cholesterol/PEG- hydroxydodecyl)amino)ethyl) (2- DMG hydroxydodecyl)amino)ethyl)piperazin-1- 50/10/38.5/1.5 yl)ethylazanediy)didodecan-2-ol (Tech Lipid:siRNA 10:1 G1) LNP13 XTC XTC/DSPC/Chol/PEG-DMG 50/10/38.5/1.5 Lipid:siRNA: 33:1 LNP14 MC3 MC3/DSPC/Chol/PEG-DMG 40/15/40/5 Lipid:siRNA: 11:1 LNP15 MC3 MC3/DSPC/Chol/PEG-DSG/GalNAc- PEG-DSG 50/10/35/4.5/0.5 Lipid:siRNA: 11:1 LNP16 MC3 MC3/DSPC/Chol/PEG-DMG 50/10/38.5/1.5 Lipid:siRNA: 7:1 LNP17 MC3 MC3/DSPC/Chol/PEG-DSG 50/10/38.5/1.5 Lipid:siRNA: 10:1 LNP18 MC3 MC3/DSPC/Chol/PEG-DMG 50/10/38.5/1.5 Lipid:siRNA: 12:1 LNP19 MC3 MC3/DSPC/Chol/PEG-DMG 50/10/35/5 Lipid:siRNA: 8:1 LNP20 MC3 MC3/DSPC/Chol/PEG-DPG 50/10/38.5/1.5 Lipid:siRNA: 10:1 LNP21 C12-200 C12-200/DSPC/Chol/PEG-DSG 50/10/38.5/1.5 Lipid:siRNA: 7:1 LNP22 XTC XTC/DSPC/Chol/PEG-DSG 50/10/38.5/1.5 Lipid:siRNA: 10:1 DSPC: distearoylphosphatidylcholine DPPC: dipalmitoylphosphatidylcholine PEG-DMG: PEG-didimyristoyl glycerol (C14-PEG, or PEG-C14) (PEG with avg mol wt of 2000) PEG-DSG: PEG-distyryl glycerol (C18-PEG, or PEG-C18) (PEG with avg mol wt of 2000) PEG-cDMA: PEG-carbamoyl-1,2-dimyristyloxypropylamine (PEG with avg mol wt of 2000) SNALP (1,2-Dilinolenyloxy-N,N-dimethylaminopropane (DLinDMA)) comprising formulations are described in WO 2009/127060, which is hereby incorporated by reference. XTC comprising formulations are described in WO 2010/088537, the entire contents of which are hereby incorporated herein by reference. MC3 comprising formulations are described, e.g., in United States Patent Publication No. 2010/0324120, the entire contents of which are hereby incorporated by reference. ALNY-100 comprising formulations are described in WO 2010/054406, the entire contents of which are hereby incorporated herein by reference. C12-200 comprising formulations are described in WO 2010/129709, the entire contents of which are hereby incorporated herein by reference.

[0602] Compositions and formulations for oral administration include powders or granules, microparticulates, nanoparticulates, suspensions or solutions in water or non-aqueous media, capsules, gel capsules, sachets, tablets or minitabets. Thickeners, flavoring agents, diluents, emulsifiers, dispersing aids or binders can be desirable. In some embodiments, oral formulations are those in which dsRNAs featured in the disclosure are administered in conjunction with one or more penetration enhancer surfactants and chelators. Suitable surfactants include fatty acids or esters or salts thereof, bile acids or salts thereof. Suitable bile acids/salts include chenodeoxycholic acid (CDCA) and ursodeoxychenodeoxycholic acid (UDCA), cholic acid, dehydrocholic acid, deoxycholic acid, glucolic acid, glycholic acid, glycodeoxycholic acid, taurocholic acid, taurodeoxycholic acid, sodium tauro-24,25-dihydro-fusidate and sodium glycodihydrofusidate. Suitable fatty acids include arachidonic acid, undecanoic acid, oleic acid, lauric acid, caprylic acid, capric acid, myristic acid, palmitic acid, stearic acid, linoleic acid, linolenic acid, dicaprinate, tricaprinate, monoolein, dilaurin, glyceryl 1-monocaprinate. 1-dodecylazacycloheptan-2-one, an acylcarnitine, an acylcholine, or a monoglyceride, a diglyceride or a pharmaceutically acceptable salt thereof (e.g., sodium). In some embodiments, combinations of penetration enhancers are used, for example, fatty acids/salts in combination with bile acids/salts. One exemplary combination is the sodium salt of lauric acid, capric acid and UDCA. Further penetration enhancers include polyoxyethylene-9-lauryl ether, polyoxyethylene-20-cetyl ether. DsRNAs featured in the disclosure can be delivered orally, in granular form including sprayed dried particles, or complexed to form micro or nanoparticles. DsRNA complexing agents include poly-amino acids; polyimines; polyacrylates; polyalkylacrylates, polyoxethanes, polyalkylcyanoacrylates; cationized gelatins, albumins, starches, acrylates, polyethyleneglycols (PEG) and starches; polyalkylcyanoacrylates; DEAE-derivatized polyimines, pollulans, celluloses and starches. Suitable complexing agents include chitosan, N-trimethylchitosan, poly-L-lysine, polyhistidine, polyornithine, polyspermines, protamine, polyvinylpyridine, polythiodiethylaminomethylethylene P(TDAE), polyaminostyrene (e.g., p-amino), poly(methylcyanoacrylate), poly(ethylcyanoacrylate), poly(butylcyanoacrylate), poly(isobutylcyanoacrylate), poly(isohexylcyanoacrylate), DEAE-methacrylate, DEAE-hexylacrylate, DEAE-acrylamide, DEAE-albumin and DEAE-dextran, polymethylacrylate, polyhexylacrylate, poly(D,L-lactic acid), poly(DL-lactic-co-glycolic acid (PLGA), alginate, and polyethyleneglycol (PEG). Oral formulations for dsRNAs and their preparation are described in detail in U.S. Pat. No. 6,887,906, U.S. 2003/0027780, and U.S. Pat. No. 6,747,014, each of which is incorporated herein by reference.

[0603] Compositions and formulations for parenteral, intraparenchymal (into the brain), intrathecal, intraventricular or intrahepatic administration can include sterile aqueous solutions which can also contain buffers, diluents and other suitable additives such as, but not limited to, penetration enhancers, carrier compounds and other pharmaceutically acceptable

carriers or excipients.

[0604] Pharmaceutical compositions of the present disclosure include, but are not limited to, solutions, emulsions, and liposome-containing formulations. These compositions can be generated from a variety of components that include, but are not limited to, preformed liquids, self-emulsifying solids and self-emulsifying semisolids. Exemplary formulations include those that target the brain when treating or preventing ABLIM3-associated diseases or disorders.

[0605] The pharmaceutical formulations of the present disclosure, which can conveniently be presented in unit dosage form, can be prepared according to conventional techniques well known in the pharmaceutical industry. Such techniques include the step of bringing into association the active ingredients with the pharmaceutical carrier(s) or excipient(s). In general, the formulations are prepared by uniformly and intimately bringing into association the active ingredients with liquid carriers or finely divided solid carriers or both, and then, if necessary, shaping the product.

[0606] The compositions of the present disclosure can be formulated into any of many possible dosage forms such as, but not limited to, tablets, capsules, gel capsules, liquid syrups, soft gels, suppositories, and enemas. The compositions of the present disclosure can also be formulated as suspensions in aqueous, non-aqueous or mixed media. Aqueous suspensions can further contain substances which increase the viscosity of the suspension including, for example, sodium carboxymethylcellulose, sorbitol or dextran. The suspension can also contain stabilizers.

C. Additional Formulations

i. Emulsions

[0607] The compositions of the present disclosure can be prepared and formulated as emulsions. Emulsions are typically heterogeneous systems of one liquid dispersed in another in the form of droplets usually exceeding 0.1 μm in diameter (see e.g., Ansel's Pharmaceutical Dosage Forms and Drug Delivery Systems, Allen, L V., Popovich N G., and Ansel H C., 2004, Lippincott Williams & Wilkins (8th ed.), New York, NY; Idson, in Pharmaceutical Dosage Forms, Lieberman, Rieger and Banker (Eds.), 1988, Marcel Dekker, Inc., New York, N.Y., volume 1, p. 199; Rosoff, in Pharmaceutical Dosage Forms, Lieberman, Rieger and Banker (Eds.), 1988, Marcel Dekker, Inc., New York, N.Y., Volume 1, p. 245; Block in Pharmaceutical Dosage Forms, Lieberman, Rieger and Banker (Eds.), 1988, Marcel Dekker, Inc., New York, N.Y., volume 2, p. 335; Higuchi et al., in Remington's Pharmaceutical Sciences, Mack Publishing Co., Easton, Pa., 1985, p. 301).

Emulsions are often biphasic systems comprising two immiscible liquid phases intimately mixed and dispersed with each other. In general, emulsions can be of either the water-in-oil (w/o) or the oil-in-water (o/w) variety. When an aqueous phase is finely divided into and dispersed as minute droplets into a bulk oily phase, the resulting composition is called a water-in-oil (w/o) emulsion. Alternatively, when an oily phase is finely divided into and dispersed as minute droplets into a bulk aqueous phase, the resulting composition is called an oil-in-water (o/w) emulsion. Emulsions can contain additional components in addition to the dispersed phases, and the active drug which can be present as a solution in either aqueous phase, oily phase or itself as a separate phase. Pharmaceutical excipients such as emulsifiers, stabilizers, dyes, and anti-oxidants can also be present in emulsions as needed. Pharmaceutical emulsions can also be multiple emulsions that are comprised of more than two phases such as, for example, in the case of oil-in-water-in-oil (o/w/o) and water-in-oil-in-water (w/o/w) emulsions. Such complex formulations often provide certain advantages that simple binary emulsions do not. Multiple emulsions in which individual oil droplets of an o/w emulsion enclose small water droplets constitute a w/o/w emulsion. Likewise, a system of oil droplets enclosed in globules of water stabilized in an oily continuous phase provides an o/w/o emulsion.

[0608] Emulsions are characterized by little or no thermodynamic stability. Often, the dispersed or discontinuous phase of the emulsion is well dispersed into the external or continuous phase and maintained in this form through the means of emulsifiers or the viscosity of the formulation. Either of the phases of the emulsion can be a semisolid or a solid, as is the case of emulsion-style ointment bases and creams. Other means of stabilizing emulsions entail the use of emulsifiers that can be incorporated into either phase of the emulsion. Emulsifiers can broadly be classified into four categories: synthetic surfactants, naturally occurring emulsifiers, absorption bases, and finely dispersed solids (see e.g., Ansel's Pharmaceutical Dosage Forms and Drug Delivery Systems, Allen, L V., Popovich N G., and Ansel H C., 2004, Lippincott Williams & Wilkins (8th ed.), New York, NY; Idson, in Pharmaceutical Dosage Forms, Lieberman, Rieger and Banker (Eds.), 1988, Marcel Dekker, Inc., New York, N.Y., volume 1, p. 199).

[0609] Synthetic surfactants, also known as surface active agents, have found wide applicability in the formulation of emulsions and have been reviewed in the literature (see e.g., Ansel's Pharmaceutical Dosage Forms and Drug Delivery Systems, Allen, L V., Popovich N G., and Ansel H C., 2004, Lippincott Williams & Wilkins (8th ed.), New York, NY; Rieger, in Pharmaceutical Dosage Forms, Lieberman, Rieger and Banker (Eds.), 1988, Marcel Dekker, Inc., New York, N.Y., volume 1, p. 285; Idson, in Pharmaceutical Dosage Forms, Lieberman, Rieger and Banker (Eds.), Marcel Dekker, Inc., New York, N.Y., 1988, volume 1, p. 199). Surfactants are typically amphiphilic and comprise a hydrophilic and a hydrophobic portion. The ratio of the hydrophilic to the hydrophobic nature of the surfactant has been termed the hydrophile/lipophile balance (HLB) and is a valuable tool in categorizing and selecting surfactants in the preparation of formulations. Surfactants can be classified into different classes based on the nature of the hydrophilic group: nonionic, anionic, cationic and amphoteric (see e.g., Ansel's Pharmaceutical Dosage Forms and Drug Delivery Systems, Allen, LV., Popovich N G., and Ansel H C., 2004, Lippincott Williams & Wilkins (8th ed.), New York, NY Rieger, in Pharmaceutical Dosage Forms, Lieberman, Rieger and Banker (Eds.), 1988, Marcel Dekker, Inc., New York, N.Y., volume 1, p. 285).

[0610] Naturally occurring emulsifiers used in emulsion formulations include lanolin, beeswax, phosphatides, lecithin and acacia. Absorption bases possess hydrophilic properties such that they can soak up water to form w/o emulsions yet retain their semisolid consistencies, such as anhydrous lanolin and hydrophilic petrolatum. Finely divided solids have also been

used as good emulsifiers especially in combination with surfactants and in viscous preparations. These include polar inorganic solids, such as heavy metal hydroxides, nonswelling clays such as bentonite, attapulgite, hectorite, kaolin, montmorillonite, colloidal aluminum silicate and colloidal magnesium aluminum silicate, pigments and nonpolar solids such as carbon or glyceryl tristearate.

[0611] A large variety of non-emulsifying materials are also included in emulsion formulations and contribute to the properties of emulsions. These include fats, oils, waxes, fatty acids, fatty alcohols, fatty esters, humectants, hydrophilic colloids, preservatives and antioxidants (Block, in *Pharmaceutical Dosage Forms*, Lieberman, Rieger and Banker (Eds.), 1988, Marcel Dekker, Inc., New York, N.Y., volume 1, p. 335; Idson, in *Pharmaceutical Dosage Forms*, Lieberman, Rieger and Banker (Eds.), 1988, Marcel Dekker, Inc., New York, N.Y., volume 1, p. 199).

[0612] Hydrophilic colloids or hydrocolloids include naturally occurring gums and synthetic polymers such as polysaccharides (for example, acacia, agar, alginic acid, carrageenan, guar gum, karaya gum, and tragacanth), cellulose derivatives (for example, carboxymethylcellulose and carboxypropylcellulose), and synthetic polymers (for example, carbomers, cellulose ethers, and carboxyvinyl polymers). These disperse or swell in water to form colloidal solutions that stabilize emulsions by forming strong interfacial films around the dispersed-phase droplets and by increasing the viscosity of the external phase.

[0613] Since emulsions often contain a number of ingredients such as carbohydrates, proteins, sterols and phosphatides that can readily support the growth of microbes, these formulations often incorporate preservatives. Commonly used preservatives included in emulsion formulations include methyl paraben, propyl paraben, quaternary ammonium salts, benzalkonium chloride, esters of p-hydroxybenzoic acid, and boric acid. Antioxidants are also commonly added to emulsion formulations to prevent deterioration of the formulation. Antioxidants used can be free radical scavengers such as tocopherols, alkyl gallates, butylated hydroxyanisole, butylated hydroxytoluene, or reducing agents such as ascorbic acid and sodium metabisulfite, and antioxidant synergists such as citric acid, tartaric acid, and lecithin.

[0614] The application of emulsion formulations via dermatological, oral and parenteral routes and methods for their manufacture have been reviewed in the literature (see e.g., Ansel's *Pharmaceutical Dosage Forms and Drug Delivery Systems*, Allen, LV., Popovich N G., and Ansel H C., 2004, Lippincott Williams & Wilkins (8th ed.), New York, NY; Idson, in *Pharmaceutical Dosage Forms*, Lieberman, Rieger and Banker (Eds.), 1988, Marcel Dekker, Inc., New York, N.Y., volume 1, p. 199). Emulsion formulations for oral delivery have been very widely used because of ease of formulation, as well as efficacy from an absorption and bioavailability standpoint (see e.g., Ansel's *Pharmaceutical Dosage Forms and Drug Delivery Systems*, Allen, LV., Popovich N G., and Ansel H C., 2004, Lippincott Williams & Wilkins (8th ed.), New York, NY; Rosoff, in *Pharmaceutical Dosage Forms*, Lieberman, Rieger and Banker (Eds.), 1988, Marcel Dekker, Inc., New York, N.Y., volume 1, p. 245; Idson, in *Pharmaceutical Dosage Forms*, Lieberman, Rieger and Banker (Eds.), 1988, Marcel Dekker, Inc., New York, N.Y., volume 1, p. 199). Mineral-oil base laxatives, oil-soluble vitamins and high fat nutritive preparations are among the materials that have commonly been administered orally as o/w emulsions.

ii. Microemulsions

[0615] In one embodiment of the present disclosure, the compositions of RNAi agents and nucleic acids are formulated as microemulsions. A microemulsion can be defined as a system of water, oil and amphiphile which is a single optically isotropic and thermodynamically stable liquid solution (see e.g., Ansel's *Pharmaceutical Dosage Forms and Drug Delivery Systems*, Allen, LV., Popovich N G., and Ansel H C., 2004, Lippincott Williams & Wilkins (8th ed.), New York, NY; Rosoff, in *Pharmaceutical Dosage Forms*, Lieberman, Rieger and Banker (Eds.), 1988, Marcel Dekker, Inc., New York, N.Y., volume 1, p. 245). Typically, microemulsions are systems that are prepared by first dispersing an oil in an aqueous surfactant solution and then adding a sufficient amount of a fourth component, generally an intermediate chain-length alcohol to form a transparent system. Therefore, microemulsions have also been described as thermodynamically stable, isotropically clear dispersions of two immiscible liquids that are stabilized by interfacial films of surface-active molecules (Leung and Shah, in: *Controlled Release of Drugs: Polymers and Aggregate Systems*, Rosoff, M., Ed., 1989, VCH Publishers, New York, pages 185-215). Microemulsions commonly are prepared via a combination of three to five components that include oil, water, surfactant, cosurfactant and electrolyte. Whether the microemulsion is of the water-in-oil (w/o) or an oil-in-water (o/w) type is dependent on the properties of the oil and surfactant used, and on the structure and geometric packing of the polar heads and hydrocarbon tails of the surfactant molecules (Schott, in *Remington's Pharmaceutical Sciences*, Mack Publishing Co., Easton, Pa., 1985, p. 271).

[0616] The phenomenological approach utilizing phase diagrams has been extensively studied and has yielded a comprehensive knowledge, to one skilled in the art, of how to formulate microemulsions (see e.g., Ansel's *Pharmaceutical Dosage Forms and Drug Delivery Systems*, Allen, LV., Popovich N G., and Ansel H C., 2004, Lippincott Williams & Wilkins (8th ed.), New York, NY; Rosoff, in *Pharmaceutical Dosage Forms*, Lieberman, Rieger and Banker (Eds.), 1988, Marcel Dekker, Inc., New York, N.Y., volume 1, p. 245; Block, in *Pharmaceutical Dosage Forms*, Lieberman, Rieger and Banker (Eds.), 1988, Marcel Dekker, Inc., New York, N.Y., volume 1, p. 335). Compared to conventional emulsions, microemulsions offer the advantage of solubilizing water-insoluble drugs in a formulation of thermodynamically stable droplets that are formed spontaneously.

[0617] Surfactants used in the preparation of microemulsions include, but are not limited to, ionic surfactants, non-ionic surfactants, Brij 96, polyoxyethylene oleyl ethers, polyglycerol fatty acid esters, tetraglycerol monolaurate (ML310), tetraglycerol monooleate (MO310), hexaglycerol monooleate (PO310), hexaglycerol pentaoleate (P0500), decaglycerol monocaprate (MCA750), decaglycerol monooleate (M0750), decaglycerol sequioleate (S0750), decaglycerol decaoleate (DA0750), alone or in combination with cosurfactants. The cosurfactant, usually a short-chain alcohol such as ethanol, 1-

propanol, and 1-butanol, serves to increase the interfacial fluidity by penetrating into the surfactant film and consequently creating a disordered film because of the void space generated among surfactant molecules. Microemulsions can, however, be prepared without the use of cosurfactants and alcohol-free self-emulsifying microemulsion systems are known in the art. The aqueous phase can typically be, but is not limited to, water, an aqueous solution of the drug, glycerol, PEG300, PEG400, polyglycerols, propylene glycols, and derivatives of ethylene glycol. The oil phase can include, but is not limited to, materials such as Captex 300, Captex 355, Capmul MCM, fatty acid esters, medium chain (C8-C12) mono, di, and triglycerides, polyoxyethylated glyceryl fatty acid esters, fatty alcohols, polyglycolized glycerides, saturated polyglycolized C8-C10 glycerides, vegetable oils and silicone oil.

[0618] Microemulsions are particularly of interest from the standpoint of drug solubilization and the enhanced absorption of drugs. Lipid based microemulsions (both o/w and w/o) have been proposed to enhance the oral bioavailability of drugs, including peptides (see e.g., U.S. Pat. Nos. 6,191,105; 7,063,860; 7,070,802; 7,157,099; Constantinides et al., *Pharmaceutical Research*, 1994, 11, 1385-1390; Ritschel, *Meth. Find. Exp. Clin. Pharmacol.*, 1993, 13, 205).

Microemulsions afford advantages of improved drug solubilization, protection of drug from enzymatic hydrolysis, possible enhancement of drug absorption due to surfactant-induced alterations in membrane fluidity and permeability, ease of preparation, ease of oral administration over solid dosage forms, improved clinical potency, and decreased toxicity (see e.g., U.S. Pat. Nos. 6,191,105; 7,063,860; 7,070,802; 7,157,099; Constantinides et al., *Pharmaceutical Research*, 1994, 11, 1385; Ho et al., *J. Pharm. Sci.*, 1996, 85, 138-143). Often microemulsions can form spontaneously when their components are brought together at ambient temperature. This can be particularly advantageous when formulating thermolabile drugs, peptides or RNAi agents. Microemulsions have also been effective in the transdermal delivery of active components in both cosmetic and pharmaceutical applications. It is expected that the microemulsion compositions and formulations of the present disclosure will facilitate the increased systemic absorption of RNAi agents and nucleic acids from the gastrointestinal tract, as well as improve the local cellular uptake of RNAi agents and nucleic acids.

[0619] Microemulsions of the present disclosure can also contain additional components and additives such as sorbitan monostearate (Grill 3), Labrasol, and penetration enhancers to improve the properties of the formulation and to enhance the absorption of the RNAi agents and nucleic acids of the present disclosure. Penetration enhancers used in the microemulsions of the present disclosure can be classified as belonging to one of five broad categories—surfactants, fatty acids, bile salts, chelating agents, and non-chelating non-surfactants (Lee et al., *Critical Reviews in Therapeutic Drug Carrier Systems*, 1991, p. 92). Each of these classes has been discussed above.

iii. Microparticles

[0620] An RNAi agent of the disclosure may be incorporated into a particle, e.g., a microparticle. Microparticles can be produced by spray-drying, but may also be produced by other methods including lyophilization, evaporation, fluid bed drying, vacuum drying, or a combination of these techniques.

iv. Penetration Enhancers

[0621] In one embodiment, the present disclosure employs various penetration enhancers to effect the efficient delivery of nucleic acids, particularly RNAi agents, to the skin of animals. Most drugs are present in solution in both ionized and nonionized forms. However, usually only lipid soluble or lipophilic drugs readily cross cell membranes. It has been discovered that even non-lipophilic drugs can cross cell membranes if the membrane to be crossed is treated with a penetration enhancer. In addition to aiding the diffusion of non-lipophilic drugs across cell membranes, penetration enhancers also enhance the permeability of lipophilic drugs.

[0622] Penetration enhancers can be classified as belonging to one of five broad categories, i.e., surfactants, fatty acids, bile salts, chelating agents, and non-chelating non-surfactants (see e.g., Malmsten, M. *Surfactants and polymers in drug delivery*, Informa Health Care, New York, NY, 2002; Lee et al., *Critical Reviews in Therapeutic Drug Carrier Systems*, 1991, p. 92). Each of the above mentioned classes of penetration enhancers are described below in greater detail.

[0623] Surfactants (or “surface-active agents”) are chemical entities which, when dissolved in an aqueous solution, reduce the surface tension of the solution or the interfacial tension between the aqueous solution and another liquid, with the result that absorption of RNAi agents through the mucosa is enhanced. In addition to bile salts and fatty acids, these penetration enhancers include, for example, sodium lauryl sulfate, polvoxyethylene-9-lauryl ether and polvoxyethylene-20-cetyl ether) (see e.g., Malmsten, M. *Surfactants and polymers in drug delivery*, Informa Health Care, New York, NY, 2002; Lee et al., *Critical Reviews in Therapeutic Drug Carrier Systems*, 1991, p. 92); and perfluorochemical emulsions, such as FC-43. Takahashi et al., *J. Pharm. Pharmacol.*, 1988, 40, 252).

[0624] Various fatty acids and their derivatives which act as penetration enhancers include, for example, oleic acid, lauric acid, capric acid (n-decanoic acid), myristic acid, palmitic acid, stearic acid, linoleic acid, linolenic acid, dicaprate, tricaprate, monoolein (1-monooleoyl-rac-glycerol), dilaurin, caprylic acid, arachidonic acid, glycerol 1-monocaprate, 1-dodecylazacycloheptan-2-one, acylcarnitines, acylcholines, C.sub.1-20 alkyl esters thereof (e.g., methyl, isopropyl and t-butyl), and mono- and di-glycerides thereof (i.e., oleate, laurate, caprate, myristate, palmitate, stearate, linoleate, etc.) (see e.g., Touitou, E., et al. *Enhancement in Drug Delivery*, CRC Press, Danvers, MA, 2006; Lee et al., *Critical Reviews in Therapeutic Drug Carrier Systems*, 1991, p. 92; Muranishi, *Critical Reviews in Therapeutic Drug Carrier Systems*, 1990, 7, 1-33; El Hariri et al., *J. Pharm. Pharmacol.*, 1992, 44, 651-654).

[0625] The physiological role of bile includes the facilitation of dispersion and absorption of lipids and fat-soluble vitamins (see e.g., Malmsten, M. *Surfactants and polymers in drug delivery*, Informa Health Care, New York, NY, 2002; Brunton, Chapter 38 in: Goodman & Gilman's *The Pharmacological Basis of Therapeutics*, 9th Ed., Hardman et al. Eds., McGraw-Hill, New York, 1996, pp. 934-935). Various natural bile salts, and their synthetic derivatives, act as penetration

enhancers. Thus the term “bile salts” includes any of the naturally occurring components of bile as well as any of their synthetic derivatives. Suitable bile salts include, for example, cholic acid (or its pharmaceutically acceptable sodium salt, sodium cholate), dehydrocholic acid (sodium dehydrocholate), deoxycholic acid (sodium deoxycholate), glucolic acid (sodium glucolate), glycholic acid (sodium glycocholate), glycodeoxycholic acid (sodium glycodeoxycholate), taurocholic acid (sodium taurocholate), taurodeoxycholic acid (sodium taurodeoxycholate), chenodeoxycholic acid (sodium chenodeoxycholate), ursodeoxycholic acid (UDCA), sodium tauro-24,25-dihydro-fusidate (STDHF), sodium glycodihydrofusidate and polyoxyethylene-9-lauryl ether (POE) (see e.g., Malmsten, M. *Surfactants and polymers in drug delivery*, Informa Health Care, New York, NY, 2002; Lee et al., *Critical Reviews in Therapeutic Drug Carrier Systems*, 1991, page 92; Swinyard. Chapter 39 In: *Remington's Pharmaceutical Sciences*, 18th Ed., Gennaro, ed., Mack Publishing Co., Easton, Pa., 1990, pages 782-783; Muranishi, *Critical Reviews in Therapeutic Drug Carrier Systems*, 1990, 7, 1-33; Yamamoto et al., *J. Pharm. Exp. Ther.*, 1992, 263, 25; Yamashita et al., *J. Pharm. Sci.*, 1990, 79, 579-583).

[0626] Chelating agents, as used in connection with the present disclosure, can be defined as compounds that remove metallic ions from solution by forming complexes therewith, with the result that absorption of RNAi agents through the mucosa is enhanced. With regards to their use as penetration enhancers in the present disclosure, chelating agents have the added advantage of also serving as DNase inhibitors, as most characterized DNA nucleases require a divalent metal ion for catalysis and are thus inhibited by chelating agents (Jarrett, *J. Chromatogr.*, 1993, 618, 315-339). Suitable chelating agents include but are not limited to disodium ethylenediaminetetraacetate (EDTA), citric acid, salicylates (e.g., sodium salicylate, 5-methoxysalicylate and homovanilate), N-acyl derivatives of collagen, laureth-9 and N-amino acyl derivatives of beta-diketones (enamines)(see e.g., Katdare, A. et al., *Excipient development for pharmaceutical, biotechnology, and drug delivery*, CRC Press, Danvers, MA, 2006; Lee et al., *Critical Reviews in Therapeutic Drug Carrier Systems*, 1991, page 92; Muranishi, *Critical Reviews in Therapeutic Drug Carrier Systems*, 1990, 7, 1-33; Buur et al., *J. Control Rel.*, 1990, 14, 43-51).

[0627] As used herein, non-chelating non-surfactant penetration enhancing compounds can be defined as compounds that demonstrate insignificant activity as chelating agents or as surfactants but that nonetheless enhance absorption of RNAi agents through the alimentary mucosa (see e.g., Muranishi, *Critical Reviews in Therapeutic Drug Carrier Systems*, 1990, 7, 1-33). This class of penetration enhancers includes, for example, unsaturated cyclic ureas, 1-alkyl- and 1-alkenylazacyclo-alkanone derivatives (Lee et al., *Critical Reviews in Therapeutic Drug Carrier Systems*, 1991, page 92); and non-steroidal anti-inflammatory agents such as diclofenac sodium, indomethacin and phenylbutazone (Yamashita et al., *J. Pharm. Pharmacol.*, 1987, 39, 621-626).

[0628] Agents that enhance uptake of RNAi agents at the cellular level can also be added to the pharmaceutical and other compositions of the present disclosure. For example, cationic lipids, such as lipofectin (Junichi et al, U.S. Pat. No. 5,705,188), cationic glycerol derivatives, and polycationic molecules, such as polylysine (WO 97/30731), are also known to enhance the cellular uptake of dsRNAs.

[0629] Other agents can be utilized to enhance the penetration of the administered nucleic acids, including glycols such as ethylene glycol and propylene glycol, pyrrols such as 2-pyrrol, azones, and terpenes such as limonene and menthone.

vi. Excipients

[0630] In contrast to a carrier compound, a “pharmaceutical carrier” or “excipient” is a pharmaceutically acceptable solvent, suspending agent or any other pharmacologically inert vehicle for delivering one or more nucleic acids to an animal. The excipient can be liquid or solid and is selected, with the planned manner of administration in mind, so as to provide for the desired bulk, consistency, etc., when combined with a nucleic acid and the other components of a given pharmaceutical composition. Typical pharmaceutical carriers include, but are not limited to, binding agents (e.g., pregelatinized maize starch, polyvinylpyrrolidone or hydroxypropyl methylcellulose, etc.); fillers (e.g., lactose and other sugars, microcrystalline cellulose, pectin, gelatin, calcium sulfate, ethyl cellulose, polyacrylates or calcium hydrogen phosphate, etc.); lubricants (e.g., magnesium stearate, talc, silica, colloidal silicon dioxide, stearic acid, metallic stearates, hydrogenated vegetable oils, corn starch, polyethylene glycols, sodium benzoate, sodium acetate, etc.); disintegrants (e.g., starch, sodium starch glycolate, etc.); and wetting agents (e.g., sodium lauryl sulphate, etc.).

[0631] Pharmaceutically acceptable organic or inorganic excipients suitable for non-parenteral administration which do not deleteriously react with nucleic acids can also be used to formulate the compositions of the present disclosure. Suitable pharmaceutically acceptable carriers include, but are not limited to, water, salt solutions, alcohols, polyethylene glycols, gelatin, lactose, amylose, magnesium stearate, talc, silicic acid, viscous paraffin, hydroxymethylcellulose, polyvinylpyrrolidone and the like.

[0632] Formulations for topical administration of nucleic acids can include sterile and non-sterile aqueous solutions, non-aqueous solutions in common solvents such as alcohols, or solutions of the nucleic acids in liquid or solid oil bases. The solutions can also contain buffers, diluents and other suitable additives. Pharmaceutically acceptable organic or inorganic excipients suitable for non-parenteral administration which do not deleteriously react with nucleic acids can be used.

[0633] Suitable pharmaceutically acceptable excipients include, but are not limited to, water, salt solutions, alcohol, polyethylene glycols, gelatin, lactose, amylose, magnesium stearate, talc, silicic acid, viscous paraffin, hydroxymethylcellulose, polyvinylpyrrolidone and the like.

vii. Other Components

[0634] The compositions of the present disclosure can additionally contain other adjunct components conventionally found in pharmaceutical compositions, at their art-established usage levels. Thus, for example, the compositions can contain additional, compatible, pharmaceutically-active materials such as, for example, antipruritics, astringents, local anesthetics

or anti-inflammatory agents, or can contain additional materials useful in physically formulating various dosage forms of the compositions of the present disclosure, such as dyes, flavoring agents, preservatives, antioxidants, opacifiers, thickening agents and stabilizers. However, such materials, when added, should not unduly interfere with the biological activities of the components of the compositions of the present disclosure. The formulations can be sterilized and, if desired, mixed with auxiliary agents, e.g., lubricants, preservatives, stabilizers, wetting agents, emulsifiers, salts for influencing osmotic pressure, buffers, colorings, flavorings or aromatic substances and the like which do not deleteriously interact with the nucleic acid(s) of the formulation.

[0635] Aqueous suspensions can contain substances which increase the viscosity of the suspension including, for example, sodium carboxymethylcellulose, sorbitol or dextran. The suspension can also contain stabilizers.

[0636] In some embodiments, pharmaceutical compositions featured in the disclosure include (a) one or more RNAi agents and (b) one or more agents which function by a non-RNAi mechanism and which are useful in treating an ABLIM3-associated disorder, e.g., PTSD or age-related memory loss. Examples of such agents include, are provided elsewhere herein.

[0637] Toxicity and therapeutic efficacy of such compounds can be determined by standard pharmaceutical procedures in cell cultures or experimental animals, e.g., for determining the LD₅₀ (the dose lethal to 50% of the population) and the ED₅₀ (the dose therapeutically effective in 50% of the population). The dose ratio between toxic and therapeutic effects is the therapeutic index and it can be expressed as the ratio LD₅₀/ED₅₀. Compounds that exhibit high therapeutic indices are preferred.

[0638] The data obtained from cell culture assays and animal studies can be used in formulating a range of dosage for use in humans. The dosage of compositions featured herein in the disclosure lies generally within a range of circulating concentrations that include the ED₅₀ with little or no toxicity. The dosage can vary within this range depending upon the dosage form employed and the route of administration utilized. For any compound used in the methods featured in the disclosure, the therapeutically effective dose can be estimated initially from cell culture assays. A dose can be formulated in animal models to achieve a circulating plasma concentration range of the compound or, when appropriate, of the polypeptide product of a target sequence (e.g., achieving a decreased concentration of the polypeptide) that includes the IC₅₀ (i.e., the concentration of the test compound which achieves a half-maximal inhibition of symptoms) as determined in cell culture. Such information can be used to more accurately determine useful doses in humans. Levels in plasma can be measured, for example, by high performance liquid chromatography.

[0639] In addition to their administration, as discussed above, the RNAi agents featured in the disclosure can be administered in combination with other known agents effective in treatment of pathological processes mediated by nucleotide repeat expression. In any event, the administering physician can adjust the amount and timing of RNAi agent administration on the basis of results observed using standard measures of efficacy known in the art or described herein.

VII. Kits

[0640] In certain aspects, the instant disclosure provides kits that include a suitable container containing a pharmaceutical formulation of a siRNA compound, e.g., a double-stranded siRNA compound, or ssiRNA compound, (e.g., a precursor, e.g., a larger siRNA compound which can be processed into a ssiRNA compound, or a DNA which encodes an siRNA compound, e.g., a double-stranded siRNA compound, or ssiRNA compound, or precursor thereof).

[0641] Such kits include one or more dsRNA agent(s) and instructions for use, e.g., instructions for administering a prophylactically or therapeutically effective amount of a dsRNA agent(s). The dsRNA agent may be in a vial or a pre-filled syringe. The kits may optionally further comprise means for administering the dsRNA agent (e.g., an injection device, such as a pre-filled syringe or an intrathecal pump), or means for measuring the inhibition of C3 (e.g., means for measuring the inhibition of ABLIM3 mRNA, ABLIM3 protein, and/or ABLIM3 activity). Such means for measuring the inhibition of ABLIM3 may comprise a means for obtaining a sample from a subject, such as, e.g., a CSF and/or plasma sample. The kits of the invention may optionally further comprise means for determining the therapeutically effective or prophylactically effective amount.

[0642] In certain embodiments the individual components of the pharmaceutical formulation may be provided in one container. Alternatively, it may be desirable to provide the components of the pharmaceutical formulation separately in two or more containers, e.g., one container for a siRNA compound preparation, and at least another for a carrier compound. The kit may be packaged in a number of different configurations such as one or more containers in a single box. The different components can be combined, e.g., according to instructions provided with the kit. The components can be combined according to a method described herein, e.g., to prepare and administer a pharmaceutical composition. The kit can also include a delivery device.

VIII. Methods for Inhibiting ABLIM3 Expression

[0643] The present disclosure also provides methods of inhibiting expression of an ABLIM3 gene in a cell. The methods include contacting a cell with an RNAi agent, e.g., double stranded RNAi agent, in an amount effective to inhibit expression and/or activity of ABLIM3 in the cell, thereby inhibiting expression and/or activity of ABLIM3 in the cell. In certain embodiments of the disclosure, ABLIM3 expression and/or activity is inhibited by at least 30% preferentially in CNS (e.g., brain) cells. In specific embodiments, ABLIM3 expression and/or activity is inhibited by at least 30%.

[0644] Contacting of a cell with an RNAi agent, e.g., a double stranded RNAi agent, may be done in vitro or in vivo.

Contacting a cell in vivo with the RNAi agent includes contacting a cell or group of cells within a subject, e.g., a human subject, with the RNAi agent. Combinations of in vitro and in vivo methods of contacting a cell are also possible.

[0645] Contacting a cell may be direct or indirect, as discussed above. Furthermore, contacting a cell may be accomplished

via a targeting ligand, including any ligand described herein or known in the art. In some embodiments, the targeting ligand is a carbohydrate moiety, e.g., a GalNAc ligand, or any other ligand that directs the RNAi agent to a site of interest. [0646] The term “inhibiting,” as used herein, is used interchangeably with “reducing,” “silencing,” “downregulating,” “suppressing” and other similar terms, and includes any level of inhibition. In certain embodiments, a level of inhibition, e.g., for an RNAi agent of the instant disclosure, can be assessed in cell culture conditions, e.g., wherein cells in cell culture are transfected via Lipofectamine™-mediated transfection at a concentration in the vicinity of a cell of 10 nM or less, 1 nM or less, etc. Knockdown of a given RNAi agent can be determined via comparison of pre-treated levels in cell culture versus post-treated levels in cell culture, optionally also comparing against cells treated in parallel with a scrambled or other form of control RNAi agent. Knockdown in cell culture of, e.g., at least about 30%, can thereby be identified as indicative of “inhibiting” or “reducing,” “downregulating” or “suppressing”, etc. having occurred. It is expressly contemplated that assessment of targeted mRNA or encoded protein levels (and therefore an extent of “inhibiting”, etc. caused by an RNAi agent of the disclosure) can also be assessed in in vivo systems for the RNAi agents of the instant disclosure, under properly controlled conditions as described in the art.

[0647] The phrase “inhibiting ABLIM3,” “inhibiting expression of an ABLIM3 gene” or “inhibiting expression of ABLIM3,” as used herein, includes inhibition of expression of any ABLIM3 gene (such as, e.g., a mouse ABLIM3 gene, a rat ABLIM3 gene, a monkey ABLIM3 gene, or a human ABLIM3 gene) as well as variants or mutants of an ABLIM3 gene that encode an ABLIM3 protein. Thus, the ABLIM3 gene may be a wild-type ABLIM3 gene, a mutant ABLIM3 gene, or a transgenic ABLIM3 gene in the context of a genetically manipulated cell, group of cells, or organism.

[0648] “Inhibiting expression of an ABLIM3 gene” includes any level of inhibition of an ABLIM3 gene, e.g., at least partial suppression of the expression of an ABLIM3 gene, such as an inhibition by at least 30%. In certain embodiments, inhibition is by at least 35%, at least 40%, at least 45%, by at least 50%, at least 55%, by at least 60%, by at least 65%, by at least 70%, by at least 75%, by at least 80%, by at least 85%, by at least 90%, by at least 95%, or by at least 99%.

ABLIM3 inhibition can be measured using the in vitro assay with, e.g., BE(2)-C cells and a 10 nM concentration of the RNA agent and the PCR assay as provided in the examples herein, are contemplated to be within the scope of the present disclosure. In some embodiments, ABLIM3 inhibition can be measured using the in vitro assay with primary mouse hepatocytes. In another embodiment, ABLIM3 inhibition can be measured using the in vitro assay with Cos-7 (Dual-Luciferase psiCHECK2 vector). In yet another embodiment, ABLIM3 inhibition can be measured using the in vitro assay with BE(2)-C cells. In some embodiments, ABLIM3 inhibition can be measured using the in vitro assay with Neuro-2a cells.

[0649] The expression of an ABLIM3 gene may be assessed based on the level of any variable associated with ABLIM3 gene expression, e.g., ABLIM3 mRNA level (e.g., sense mRNA, antisense mRNA, and/or total ABLIM3 mRNA) or ABLIM3 protein level (e.g., total ABLIM3 protein and/or wild-type ABLIM3 protein).

[0650] Inhibition may be assessed by a decrease in an absolute or relative level of one or more of these variables compared with a control level. The control level may be any type of control level that is utilized in the art, e.g., a pre-dose baseline level, or a level determined from a similar subject, cell, or sample that is untreated or treated with a control (such as, e.g., buffer only control or inactive agent control).

[0651] For example, in some embodiments of the methods of the disclosure, expression of an ABLIM3 gene is inhibited by at least 30%, 40%, 50%, 60%, 70%, 80%, 85%, 90%, or 95%, or to below the level of detection of the assay. In certain embodiments, the methods include a clinically relevant inhibition of expression of ABLIM3, e.g. as demonstrated by a clinically relevant outcome after treatment of a subject with an agent to reduce the expression of ABLIM3.

[0652] Inhibition of the expression of an ABLIM3 gene may be manifested by a reduction of the amount of mRNA expressed by a first cell or group of cells (such cells may be present, for example, in a sample derived from a subject) in which an ABLIM3 gene is transcribed and which has or have been treated (e.g., by contacting the cell or cells with an RNAi agent of the disclosure, or by administering an RNAi agent of the disclosure to a subject in which the cells are or were present) such that the expression of an ABLIM3 gene is inhibited, as compared to a second cell or group of cells substantially identical to the first cell or group of cells but which has not or have not been so treated (control cell(s) not treated with an RNAi agent or not treated with an RNAi agent targeted to the gene of interest). The degree of inhibition may be expressed in terms of:

[00002]
$$\frac{(\text{mRNA}_{\text{in control cells}}) - (\text{mRNA}_{\text{in treated cells}})}{(\text{mRNA}_{\text{in control cells}})} \cdot \text{Math. } 100\%$$

[0653] In other embodiments, inhibition of the expression of an ABLIM3 gene may be assessed in terms of a reduction of a parameter that is functionally linked to an ABLIM3 gene expression, e.g., ABLIM3 protein expression. ABLIM3 gene silencing may be determined in any cell expressing ABLIM3, either endogenous or heterologous from an expression construct, and by any assay known in the art.

[0654] Inhibition of the expression of an ABLIM3 protein may be manifested by a reduction in the level of the ABLIM3 protein (or functional parameter, e.g., activity) that is expressed by a cell or group of cells (e.g., the level of protein expressed in a sample derived from a subject). As explained above, for the assessment of mRNA suppression, the inhibition of protein expression levels in a treated cell or group of cells may similarly be expressed as a percentage of the level of protein in a control cell or group of cells. In some embodiments, the phrase “inhibiting ABLIM3”, can also refer to the inhibition of the activity of ABLIM3, e.g., at least partial suppression of the ABLIM3 activity, such as an inhibition by at least 30%. In certain embodiments, inhibition of the ABLIM3 activity is by at least 35%, at least 40%, at least 45%, by at least 50%, at least 55%, by at least 60%, by at least 65%, by at least 70%, by at least 75%, by at least 80%, by at least

85%, by at least 90%, or by at least 95%, or by at least 99%. ABLIM3 activity can be measured using an in vitro assay, such as an actin binding assay, e.g., the assay described in (Srivastava et al., 2008, *J Vis Exp*, 13:690).

[0655] A control cell or group of cells that may be used to assess the inhibition of the expression of an ABLIM3 gene includes a cell or group of cells that has not yet been contacted with an RNAi agent of the disclosure. For example, the control cell or group of cells may be derived from an individual subject (e.g., a human or animal subject) prior to treatment of the subject with an RNAi agent.

[0656] The level of ABLIM3 mRNA that is expressed by a cell or group of cells may be determined using any method known in the art for assessing mRNA expression. In one embodiment, the level of expression of ABLIM3 in a sample is determined by detecting a transcribed polynucleotide, or portion thereof, e.g., mRNA of the ABLIM3 gene. RNA may be extracted from cells using RNA extraction techniques including, for example, using acid phenol/guanidine isothiocyanate extraction (RNAzol B; Biogenesis), RNeasy™ RNA preparation kits (Qiagen®) or PAXgene (PreAnalytix, Switzerland). Typical assay formats utilizing ribonucleic acid hybridization include nuclear run-on assays, RT-PCR, RNase protection assays, northern blotting, in situ hybridization, and microarray analysis. Strand specific ABLIM3 mRNAs may be detected using the quantitative RT-PCR and or droplet digital PCR methods described in, for example, Jiang, et al. supra, Lagier-Tourenne, et al., supra and Jiang, et al., supra. Circulating ABLIM3 mRNA may be detected using methods the described in WO2012/177906, the entire contents of which are hereby incorporated herein by reference.

[0657] In some embodiments, the level of expression of ABLIM3 is determined using a nucleic acid probe. The term “probe”, as used herein, refers to any molecule that is capable of selectively binding to a specific ABLIM3 nucleic acid or protein, or fragment thereof. Probes can be synthesized by one of skill in the art, or derived from appropriate biological preparations. Probes may be specifically designed to be labeled. Examples of molecules that can be utilized as probes include, but are not limited to, RNA, DNA, proteins, antibodies, and organic molecules.

[0658] Isolated mRNA can be used in hybridization or amplification assays that include, but are not limited to, Southern or northern analyses, polymerase chain reaction (PCR) analyses and probe arrays. One method for the determination of mRNA levels involves contacting the isolated mRNA with a nucleic acid molecule (probe) that can hybridize to ABLIM3 mRNA. In one embodiment, the mRNA is immobilized on a solid surface and contacted with a probe, for example by running the isolated mRNA on an agarose gel and transferring the mRNA from the gel to a membrane, such as nitrocellulose. In an alternative embodiment, the probe(s) are immobilized on a solid surface and the mRNA is contacted with the probe(s), for example, in an Affymetrix® gene chip array. A skilled artisan can readily adapt known mRNA detection methods for use in determining the level of ABLIM3 mRNA.

[0659] An alternative method for determining the level of expression of ABLIM3 in a sample involves the process of nucleic acid amplification or reverse transcriptase (to prepare cDNA) of for example mRNA in the sample, e.g., by RT-PCR (the experimental embodiment set forth in Mullis, 1987, U.S. Pat. No. 4,683,202), ligase chain reaction (Barany (1991) *Proc. Natl. Acad. Sci. USA* 88:189-193), self-sustained sequence replication (Guatelli et al. (1990) *Proc. Natl. Acad. Sci. USA* 87:1874-1878), transcriptional amplification system (Kwoh et al. (1989) *Proc. Natl. Acad. Sci. USA* 86:1173-1177), Q-Beta Replicase (Lizardi et al. (1988) *Bio/Technology* 6:1197), rolling circle replication (Lizardi et al., U.S. Pat. No. 5,854,033) or any other nucleic acid amplification method, followed by the detection of the amplified molecules using techniques well known to those of skill in the art. These detection schemes are especially useful for the detection of nucleic acid molecules if such molecules are present in very low numbers. In particular aspects of the disclosure, the level of expression of ABLIM3 is determined by quantitative fluorogenic RT-PCR (i.e., the TaqMan™ System), by a Dual-Glo® Luciferase assay, or by other art-recognized method for measurement of ABLIM3 expression or mRNA level.

[0660] The expression level of ABLIM3 mRNA may be monitored using a membrane blot (such as used in hybridization analysis such as northern, Southern, dot, and the like), or microwells, sample tubes, gels, beads or fibers (or any solid support comprising bound nucleic acids). See U.S. Pat. Nos. 5,770,722, 5,874,219, 5,744,305, 5,677,195 and 5,445,934, which are incorporated herein by reference. The determination of ABLIM3 expression level may also comprise using nucleic acid probes in solution.

[0661] In some embodiments, the level of mRNA expression is assessed using branched DNA (bDNA) assays or real time PCR (qPCR). The use of this PCR method is described and exemplified in the Examples presented herein. Such methods can also be used for the detection of ABLIM3 nucleic acids.

[0662] The level of ABLIM3 protein expression may be determined using any method known in the art for the measurement of protein levels. Such methods include, for example, electrophoresis, capillary electrophoresis, high performance liquid chromatography (HPLC), thin layer chromatography (TLC), hyperdiffusion chromatography, fluid or gel precipitin reactions, absorption spectroscopy, a colorimetric assays, spectrophotometric assays, flow cytometry, immunodiffusion (single or double), immunoelectrophoresis, western blotting, radioimmunoassay (RIA), enzyme-linked immunosorbent assays (ELISAs), immunofluorescent assays, electrochemiluminescence assays, and the like. Such assays can also be used for the detection of proteins indicative of the presence or replication of ABLIM3 proteins. In some embodiments, the efficacy of the methods of the disclosure in the treatment of an ABLIM3-associated disease, e.g., PTSD or age-related memory loss, is assessed by a decrease in ABLIM3 mRNA level (e.g., by assessment of a CSF sample and/or plasma sample for ABLIM3 level, by brain biopsy, or otherwise).

[0663] In some embodiments of the methods of the disclosure, the RNAi agent is administered to a subject such that the RNAi agent is delivered to a specific site within the subject. The inhibition of expression of ABLIM3 may be assessed using measurements of the level or change in the level of ABLIM3 mRNA (e.g., sense mRNA, antisense mRNA, total ABLIM3 mRNA) and/or ABLIM3 protein (e.g., total ABLIM3 protein, wild-type ABLIM3 protein) in a sample derived

a specific site within the subject, e.g., CNS cells. In certain embodiments, the methods include a clinically relevant inhibition of expression of ABLIM3, e.g. as demonstrated by a clinically relevant outcome after treatment of a subject with an agent to reduce the expression of ABLIM3, such as, for example, in a subject. In certain embodiments, a clinically relevant inhibition of expression of ABLIM3 is demonstrated by improved memory; a decrease in anxiety, depression, fear, agitation, hostility, and/or distress; fewer nightmares, intrusive thoughts, and/or flashbacks; and/or a reduction in physical symptoms including pain, sweating, nausea, and trembling.

[0664] As used herein, the terms detecting or determining a level of an analyte are understood to mean performing the steps to determine if a material, e.g., protein, RNA, is present. As used herein, methods of detecting or determining include detection or determination of an analyte level that is below the level of detection for the method used.

IX. Methods of Treating or Preventing ABLIM3-Associated Diseases

[0665] The present disclosure also provides methods of using an RNAi agent of the disclosure or a composition containing an RNAi agent of the disclosure to reduce or inhibit ABLIM3 expression in a cell. The methods include contacting the cell with a dsRNA of the disclosure and maintaining the cell for a time sufficient to obtain degradation of the mRNA transcript of an ABLIM3 gene, thereby inhibiting expression of the ABLIM3 gene in the cell.

[0666] Reduction in gene expression can be assessed by any methods known in the art. For example, a reduction in the expression of ABLIM3 may be determined by determining the mRNA expression level of ABLIM3 using methods routine to one of ordinary skill in the art. e.g., northern blotting, qRT-PCR; by determining the protein level of ABLIM3 using methods routine to one of ordinary skill in the art, such as western blotting, immunological techniques.

[0667] In the methods of the disclosure the cell may be contacted in vitro or in vivo, i.e., the cell may be within a subject.

[0668] A cell suitable for treatment using the methods of the disclosure may be any cell that expresses an ABLIM3 gene. A cell suitable for use in the methods of the disclosure may be a mammalian cell, e.g., a primate cell (such as a human cell or a non-human primate cell, e.g., a monkey cell or a chimpanzee cell), a non-primate cell (such as a rat cell, or a mouse cell). In one embodiment, the cell is a human cell, e.g., a human CNS cell.

[0669] ABLIM3 expression (e.g., as assessed by sense mRNA, antisense mRNA, total ABLIM3 mRNA, total ABLIM3 protein) is inhibited in the cell by at least 30%, 40%, preferably at least 50%, 60%, 70%, 80%, 85%, 90%, or 95%, 99%, or to below the level of detection of the assay.

[0670] The in vivo methods of the disclosure may include administering to a subject a composition containing an RNAi agent, where the RNAi agent includes a nucleotide sequence that is complementary to at least a part of an RNA transcript of the ABLIM3 gene of the mammal to be treated. When the organism to be treated is a mammal such as a human, the composition can be administered by any means known in the art including, but not limited to oral, intraperitoneal, or parenteral routes, including intracranial (e.g., intraventricular, intraparenchymal, and intrathecal), intravenous, intramuscular, intravitreal, subcutaneous, transdermal, airway (aerosol), nasal, rectal, and topical (including buccal and sublingual) administration. In certain embodiments, the compositions are administered by intravenous infusion or injection. In certain embodiments, the compositions are administered by subcutaneous injection. In certain embodiments, the compositions are administered by intrathecal injection.

[0671] In some embodiments, the administration is via a depot injection. A depot injection may release the RNAi agent in a consistent way over a prolonged time period. Thus, a depot injection may reduce the frequency of dosing needed to obtain a desired effect, e.g., a desired inhibition of ABLIM3, or a therapeutic or prophylactic effect. A depot injection may also provide more consistent serum concentrations. Depot injections may include subcutaneous injections or intramuscular injections. For example, the depot injection may be a subcutaneous injection.

[0672] In some embodiments, the administration is via a pump. The pump may be an external pump or a surgically implanted pump. In certain embodiments, the pump is a subcutaneously implanted osmotic pump. In other embodiments, the pump is an infusion pump. An infusion pump may be used for intracranial, intravenous, subcutaneous, arterial, or epidural infusions. In one embodiment, the infusion pump is a subcutaneous infusion pump. In other embodiments, the pump is a surgically implanted pump that delivers the RNAi agent to the CNS.

[0673] The mode of administration may be chosen based upon whether local or systemic treatment is desired and based upon the area to be treated. The route and site of administration may be chosen to enhance targeting.

[0674] In one aspect, the present disclosure also provides methods for inhibiting the expression of an ABLIM3 gene in a mammal. The methods include administering to the mammal a composition comprising a dsRNA that targets an ABLIM3 gene in a cell of the mammal, thereby inhibiting expression of the ABLIM3 gene in the cell. Reduction in gene expression can be assessed by any methods known in the art and by methods, e.g. qRT-PCR, described herein. Reduction in protein production can be assessed by any methods known in the art and by methods, e.g. ELISA, described herein. In one embodiment, a CNS biopsy sample or a cerebrospinal fluid (CSF) sample serves as the tissue material for monitoring the reduction in ABLIM3 gene or protein expression (or of a proxy therefore).

[0675] The present disclosure further provides methods of treatment of a subject in need thereof. The treatment methods of the disclosure include administering an RNAi agent of the disclosure to a subject, e.g., a subject that would benefit from inhibition of ABLIM3 expression, such as a subject having PTSD or age-related memory loss, in a therapeutically effective amount of an RNAi agent targeting an ABLIM3 gene or a pharmaceutical composition comprising an RNAi agent targeting an ABLIM3 gene.

[0676] In addition, the present disclosure provides methods of preventing, treating or inhibiting the progression of an ABLIM3-associated disease or condition in a subject. The methods include administering to the subject a therapeutically effective amount of any of the RNAi agent, e.g., dsRNA agents, or the pharmaceutical composition provided herein,

thereby preventing or inhibiting the progression of an ABLIM3-associated disease or condition in the subject. [0677] An RNAi agent of the disclosure may be administered as a “free RNAi agent.” A free RNAi agent is administered in the absence of a pharmaceutical composition. The naked RNAi agent may be in a suitable buffer solution. The buffer solution may comprise acetate, citrate, prolamine, carbonate, or phosphate, or any combination thereof. In one embodiment, the buffer solution is phosphate buffered saline (PBS). The pH and osmolarity of the buffer solution containing the RNAi agent can be adjusted such that it is suitable for administering to a subject.

[0678] Alternatively, an RNAi agent of the disclosure may be administered as a pharmaceutical composition, such as a dsRNA liposomal formulation.

[0679] In one embodiment, the subject is a human. Subjects that would benefit from a reduction or inhibition of ABLIM3 gene expression include those having an ABLIM3-associated disease, such as PTSD or age-related memory loss.

[0680] In one embodiment, an ABLIM3-associated disease is post-traumatic stress disorder (PTSD). PTSD is a disorder that is triggered by experiencing or witnessing a traumatic, terrifying, or dangerous event. People with PTSD may experience severe anxiety, nightmares, uncontrollable thoughts, and flashbacks and these symptoms can last for months to years and interfere with daily life. Little is known about the molecular mechanisms of PTSD, although studies suggest dysfunction of contextual processing, which is a function of the hippocampal-prefrontal-thalamic circuitry, may contribute to PTSD pathophysiology (Liberzon and Abelson, 2017, *Neuron*, 92(1): 14-30).

[0681] Treatment for PTSD includes therapies, such as cognitive, behavioral, and exposure therapy, or psychopharmacology treatments, such as Fluoxetine (Prozac), Paroxetine (Paxil), Sertraline (Zoloft), Venlafaxine (Effexor), antidepressants, monoamine oxidase inhibitors (MAOIs), antipsychotics or second generation antipsychotics (SGAs), beta-blockers, and benzodiazepines. However, the nonresponse to psychological therapies can be as high as 50% and psychopharmacology treatments have a low efficacy for treating PTSD.

[0682] In one embodiment, an ABLIM3-associated disease is age-related memory loss. Changes in memory occur naturally during aging and age-related cognitive changes (e.g., memory loss) can present as frequently losing or misplacing things, frequently forgetting conversations, appointments, and events, difficulty remembering names, difficulty following a conversation, and difficulty coming up with the desired words. Over time, memories are generalized and the details of the memories are lost. Little is known about the mechanism of memory precision and generalization and the loss of details over time. There is evidence that the hippocampus, through hippocampal-cortical interactions, plays an important role in maintenance of memory specificity. Memories are thought to be encoded in the hippocampus in dentate granule cells called engram-bearing dentate granule cells.

[0683] A recent study by Guo et al., found that the levels of ABLIM3 decreased upon learning (Guo, et al., 2018, *Nat Med*. 24(4):438-449). ABLIM3 was highly expressed in dentate granule cells and localized to the mossy fiber terminals, at sites of structural stabilization, that form excitatory synapses with stratum lucidum interneurons. The study found that downregulation of ABLIM3 in dentate granule cells decreased remote memory generalization and increased remote memory precision, which could help alleviate memory impairments and PTSD.

[0684] The disclosure further provides methods for the use of an RNAi agent or a pharmaceutical composition thereof, e.g., for treating a subject that would benefit from reduction or inhibition of ABLIM3 expression, e.g., a subject having an ABLIM3-associated disorder, in combination with other pharmaceuticals or other therapeutic methods, e.g., with known pharmaceuticals or known therapeutic methods, such as, for example, those which are currently employed for treating these disorders. For example, in certain embodiments, an RNAi agent targeting ABLIM3 is administered in combination with, e.g., an agent useful in treating an ABLIM3-associated disorder as described elsewhere herein or as otherwise known in the art. For example, additional agents suitable for treating a subject that would benefit from reduction in ABLIM3 expression, e.g., a subject having an ABLIM3-associated disorder, such as PTSD or age-related memory loss, may include agents currently used to treat or prevent these ABLIM3-associated disorders or treat or prevent symptoms of these ABLIM3-associated disorders. The RNAi agent and additional therapeutic agents or methods may be administered at the same time or in the same combination, e.g., intrathecally. Alternatively, the additional therapeutic agent or method can be administered as part of a separate composition or at separate times or by another method known in the art or described herein, in conjunction with or separately from administration of the RNAi agent of the present invention.

[0685] For example, additional therapeutic agents and therapeutic methods suitable for preventing or treating PTSD include Fluoxetine (Prozac), Paroxetine (Paxil), Sertraline (Zoloft), Venlafaxine (Effexor), antidepressants, monoamine oxidase inhibitors (MAOIs), antipsychotics or second generation antipsychotics (SGAs), beta-blockers, benzodiazepines, trauma-focused psychotherapies, prolonged exposure (PE) therapy, cognitive processing therapy (CPT), and eye movement desensitization and reprocessing (EMDR).

[0686] For example, additional therapeutic agents and therapeutic methods suitable for preventing or treating age-related memory loss include cholinesterase inhibitors (Aricept, Exelon, Razadyne), memantine (Namenda), Adderall, and Ritalin.

[0687] In addition, agents that decrease or otherwise affect the ABLIM3 activity may be used in combination with the agent or method of the present invention. Exemplary agents that decrease or otherwise affect the ABLIM3 activity include, but are not limited to, small molecule ABLIM3 inhibitors and anti-ABLIM3 antibodies.

[0688] In one embodiment, the method includes administering a composition featured herein such that expression of the target ABLIM3 gene is decreased, for at least one month. In certain embodiments, expression is decreased for at least 2 months, 3 months, or 6 months.

[0689] In certain embodiments, the RNAi agents useful for the methods and compositions featured herein specifically target RNAs (primary or processed) of the target ABLIM3 gene. Compositions and methods for inhibiting the expression

of these genes using RNAi agents can be prepared and performed as described herein.

[0690] Administration of the dsRNA according to the methods of the disclosure may result in a reduction of the severity, signs, symptoms, or markers of such diseases or disorders in a patient with an ABLIM3-associated disorder. By “reduction” in this context is meant a statistically significant or clinically significant decrease in such level. The reduction can be, for example, at least 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, or about 100%.

[0691] Efficacy of treatment or prevention of disease can be assessed, for example by measuring disease progression, disease remission, symptom severity, reduction in pain, quality of life, dose of a medication required to sustain a treatment effect, level of a disease marker or any other measurable parameter appropriate for a given disease being treated or targeted for prevention. It is well within the ability of one skilled in the art to monitor efficacy of treatment or prevention by measuring any one of such parameters, or any combination of parameters. For example, efficacy of treatment of an ABLIM3-associated disorder may be assessed, for example, by periodic monitoring of a subject's marker's and/or symptoms. Comparisons of the later readings with the initial readings provide a physician an indication of whether the treatment is effective. It is well within the ability of one skilled in the art to monitor efficacy of treatment or prevention by measuring any one of such parameters, or any combination of parameters. In connection with the administration of an RNAi agent targeting ABLIM3 or pharmaceutical composition thereof, “effective against” an ABLIM3-associated disorder indicates that administration in a clinically appropriate manner results in a beneficial effect for at least a statistically significant fraction of patients, such as an improvement of symptoms, a cure, a reduction in disease, extension of life, improvement in quality of life, or other effect generally recognized as positive by medical doctors familiar with treating ABLIM3-associated disorders and the related causes.

[0692] A treatment or preventive effect is evident when there is a statistically significant improvement in one or more parameters of disease status, or by a failure to worsen or to develop symptoms where they would otherwise be anticipated. As an example, a favorable change of at least 10% in a measurable parameter of disease, and preferably at least 20%, 30%, 40%, 50% or more can be indicative of effective treatment. Efficacy for a given RNAi agent drug or formulation of that drug can also be judged using an experimental animal model for the given disease as known in the art. When using an experimental animal model, efficacy of treatment is evidenced when a statistically significant reduction in a marker or symptom is observed.

[0693] Alternatively, the efficacy can be measured by a reduction in the severity of disease as determined by one skilled in the art of diagnosis based on a clinically accepted disease severity grading scale. Any positive change resulting in e.g., lessening of severity of disease measured using the appropriate scale, represents adequate treatment using an RNAi agent or RNAi agent formulation as described herein.

[0694] In certain embodiments, subjects can be administered a therapeutic amount of dsRNA, such as about 0.01 mg/kg to about 200 mg/kg. In other embodiments, subjects can be administered a therapeutic amount of dsRNA, such as about 0.01 mg/kg to about 500 mg/kg. In yet other embodiments, subjects can be administered a therapeutic amount of dsRNA of about 500 mg/kg or more.

[0695] The RNAi agent can be administered intrathecally, via intravitreal injection, or by intravenous infusion over a period of time, on a regular basis. In certain embodiments, after an initial treatment regimen, the treatments can be administered on a less frequent basis. Administration of the RNAi agent can reduce ABLIM3 levels, e.g., in a cell, tissue, blood, CSF sample or other compartment of the patient.

[0696] Before administration of a full dose of the RNAi agent, patients can be administered a smaller dose, such as a 5% infusion reaction, and monitored for adverse effects, such as an allergic reaction. In another example, the patient can be monitored for unwanted immunostimulatory effects, such as increased cytokine (e.g., TNF-alpha or INF-alpha) levels.

[0697] Alternatively, the RNAi agent can be administered subcutaneously, i.e., by subcutaneous injection. One or more injections may be used to deliver the desired, e.g., monthly dose of RNAi agent to a subject. The injections may be repeated over a period of time. The administration may be repeated on a regular basis. In certain embodiments, after an initial treatment regimen, the treatments can be administered on a less frequent basis. A repeat-dose regimen may include administration of a therapeutic amount of RNAi agent on a regular basis, such as monthly or extending to once a quarter, twice per year, once per year. In certain embodiments, the RNAi agent is administered about once per month to about once per quarter (i.e., about once every three months).

[0698] Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. Although methods and materials similar or equivalent to those described herein can be used in the practice or testing of the RNAi agents and methods featured in the invention, suitable methods and materials are described below. All publications, patent applications, patents, and other references mentioned herein are incorporated by reference in their entirety. In case of conflict, the present specification, including definitions, will control. In addition, the materials, methods, and examples are illustrative only and not intended to be limiting.

[0699] An informal Sequence Listing is filed herewith and forms part of the specification as filed.

EXAMPLES

Example 1. iRNA Synthesis

Source of Reagents

[0700] Where the source of a reagent is not specifically given herein, such reagent can be obtained from any supplier of reagents for molecular biology at a quality/purity standard for application in molecular biology.

siRNA Design

[0701] siRNAs targeting the human Ablim3 gene (human: NCBI refseqID NM_001301015.3; NCBI GeneID: 22885) were designed using custom R and Python scripts. The human NM_001301015.3 REFSEQ mRNA has a length of 4452 bases.

[0702] A detailed list of the unmodified Ablim3 sense and antisense strand nucleotide sequences is shown in Table 3. A detailed list of the modified Ablim3 sense and antisense strand nucleotide sequences is shown in Table 4.

[0703] It is to be understood that, throughout the application, a duplex name without a decimal is equivalent to a duplex name with a decimal which merely references the batch number of the duplex. For example, AD-1810068 is equivalent to AD-1810068.1.

siRNA Synthesis

[0704] siRNAs were synthesized and annealed using routine methods known in the art.

[0705] Briefly, siRNA sequences were synthesized at 1 μ mol scale on a Mermade 192 synthesizer (BioAutomation) using the solid support mediated phosphoramidite chemistry. The solid support was controlled pore glass (500 Å) loaded with custom GalNAc ligand or universal solid support (AM biochemical). Ancillary synthesis reagents, 2'-F and 2'-O-Methyl RNA and deoxy phosphoramidites were obtained from Thermo-Fisher (Milwaukee, WI) and Hongene (China). 2'F 2'-O-Methyl, GNA (glycol nucleic acids), 5' phosphate and other modifications were introduced using the corresponding phosphoramidites. Synthesis of 3' GalNAc conjugated single strands was performed on a GalNAc modified CPG support. Custom CPG universal solid support was used for the synthesis of antisense single strands. Coupling time for all phosphoramidites (100 mM in acetonitrile) was 5 minutes employing 5-Ethylthio-1H-tetrazole (ETT) as activator (0.6 M in acetonitrile). Phosphorothioate linkages were generated using a 50 mM solution of 3-((Dimethylamino-methylidene)amino)-3H-1,2,4-dithiazole-3-thione (DDTT, obtained from Chemgenes (Wilmington, MA, USA)) in anhydrous acetonitrile/pyridine (1:1 v/v). Oxidation time was 3 minutes. All sequences were synthesized with final removal of the DMT group ("DMT off").

[0706] Upon completion of the solid phase synthesis, oligoribonucleotides were cleaved from the solid support and deprotected in sealed 96 deep well plates using 200 μ L Aqueous Methylamine reagents at 60° C. for 20 minutes. For sequences containing 2' ribo residues (2'-OH) that are protected with a tert-butyl dimethyl silyl (TBDMS) group, a second step deprotection was performed using TEA.Math.3HF (triethylaniline trihydro fluoride) reagent. To the methylamine deprotection solution, 200 μ L of dimethyl sulfoxide (DMSO) and 300 μ L TEA.Math.3HF reagent was added and the solution was incubated for additional 20 minutes at 60° C. At the end of cleavage and deprotection step, the synthesis plate was allowed to come to room temperature and was precipitated by addition of 1 mL of acetonitrile:ethanol mixture (9:1). The plates were cooled at -80° C. for 2 hours, supernatant decanted carefully with the aid of a multi-channel pipette. The oligonucleotide pellet was re-suspended in 20 mM NaOAc buffer and were desalted using a 5 mL HiTrap size exclusion column (GE Healthcare) on an AKTA Purifier System equipped with an A905 autosampler and a Frac 950 fraction collector. Desalted samples were collected in 96-well plates. Samples from each sequence were analyzed by LC-MS to confirm the identity, UV (260 nm) for quantification and a selected set of samples by IEX chromatography to determine purity.

[0707] Annealing of single strands was performed on a Tecan liquid handling robot. Equimolar mixture of sense and antisense single strands were combined and annealed in 96 well plates. After combining the complementary single strands, the 96-well plate was sealed tightly and heated in an oven at 100° C. for 10 minutes and allowed to come slowly to room temperature over a period 2-3 hours. The concentration of each duplex was normalized to 10 M in 1×PBS and then submitted for in vitro screening assays.

Example 2. In Vitro Screening Methods

Cell Culture and 384-Well Transfections

[0708] Cos-7 (ATCC) are transfected by adding 5 μ L of 1 ng/KL, diluted in Opti-MEM, C9orf72 intron 1 psiCHECK2 vector (Blue Heron Biotechnology), 4.9 μ L of Opti-MEM plus 0.1 μ L of Lipofectamine 2000 per well (Invitrogen, Carlsbad CA. cat #11668-019) to 5 μ L of siRNA duplexes per well, with 4 replicates of each siRNA duplex, into a 384-well plate, and incubated at room temperature for 15 minutes. Thirty-five L of Dulbecco's Modified Eagle Medium (ThermoFisher) containing $\sim 5 \times 10^5$ cells are then added to the siRNA mixture. Cells are incubated for 48 hours followed by Firefly (transfection control) and *Renilla* (fused to target sequence) luciferase measurements. Three dose experiments are performed at 10 nM, 1 nM, and 0.1 nM.

[0709] BE(2)-C (ATCC) are transfected by adding 4.9 μ L of Opti-MEM plus 0.1 μ L of RNAiMAX per well (Invitrogen, Carlsbad CA. cat #13778-150) to 5 μ L of siRNA duplexes per well, with 4 replicates of each siRNA duplex, into a 384-well plate, and incubated at room temperature for 15 minutes. Forty L of 1:1 mixture of Minimum Essential Medium and F12 Medium (ThermoFisher) containing $\sim 5 \times 10^5$ cells are then added to the siRNA mixture. Cells are incubated for 48 hours prior to RNA purification. Three dose experiments are performed at 10 nM, 1 nM, and 0.1 nM.

Total RNA Isolation Using DYNABEADS mRNA Isolation Kit (Invitrogen™, Part #: 610-12)

[0710] Cells are lysed in 75 μ L of Lysis/Binding Buffer containing 3 μ L of beads per well and are mixed for 10 minutes on an electrostatic shaker. The washing steps are automated on a Biotek EL406, using a magnetic plate support. Beads are washed (in 90 μ L) once in Buffer A, once in Buffer B, and twice in Buffer E, with aspiration steps in between. Following a final aspiration, complete 10 μ L RT mixture is added to each well, as described below.

cDNA Synthesis Using ABI High Capacity cDNA Reverse Transcription Kit (Applied Biosystems, Foster City, CA, Cat #4368813)

[0711] A master mix of 1 μ L 10× Buffer, 0.4 μ L 25×dNTPs, 1 μ L Random primers, 0.5 μ L Reverse Transcriptase, 0.5 μ L

RNase inhibitor and 6.6 μ L of H.sub.2O per reaction is added per well. Plates are sealed, are agitated for 10 minutes on an electrostatic shaker, and then are incubated at 37° C. for 2 hours. Following this, the plates are agitated at 80° C. for 8 minutes.

Real Time PCR

[0712] Two microliter (μ L) of cDNA is added to a master mix containing 0.5 μ L of human GAPDH TaqMan Probe and 0.5 μ L human Ablim3 probe, 2 μ L nuclease-free water and 5 μ L Lightcycler 480 probe master mix (Roche Cat #04887301001) per well in a 384 well plates (Roche cat #04887301001). Real time PCR is done in a LightCycler480 Real Time PCR system (Roche). Each duplex is tested at least two times and data are normalized to cells transfected with a non-targeting control siRNA. To calculate relative fold change, real time data are analyzed using the $\Delta\Delta C_t$ method and are normalized to assays performed with cells transfected with a non-targeting control siRNA.

Example 3. Single Dose In Vitro Screening of ABLIM3 siRNAs

[0713] Hepa1-6 mouse hepatoma cells were transfected by adding 5 μ L of 1 ng/ μ L of each siRNA duplex diluted in Opti-MEM, 100 ng V190 or V191 human ABLIM3 reporter construct, plus 0.5 μ L of Lipofectamine 2000 to an individual well in a 96-well plate, with 4 replicates of each siRNA duplex. The mixture was incubated at room temperature for 15 minutes. One-hundred μ L of complete growth media without antibiotic containing $\sim 2 \times 10^5$ Hepa1-6 cells was then added to the siRNA mixture. Cells were incubated for 24 hours followed by Firefly (transfection control) and *Renilla* (fused to target sequence) luciferase measurements. A single dose experiment was performed at 1 nM final duplex concentration. The results are shown in Table 5 and are presented as the average percent ABLIM3 mRNA remaining as compared to a negative control. A positive control of Rluc directed siRNA was also performed.

[0714] It is expressly contemplated that nucleotides 3042-3081, 3420-3457, 4223-4261, 726-763, 1080-1137, 1342-1379, 1910-1955, 2167-2204, 480-520, 1080-1122, 1622-1665, 1744-1787, 2323-2363, 540-578, 1175-1212, 274-313, 349-401, or 2518-2555 of NM_001301015.3 comprise hotspot regions. These regions are targeted by AD-1809702, AD-1809719, AD-1810038, AD-1810053, AD-1810676, AD-1810692, AD-1807904, AD-1807919, AD-1808238, AD-1808258, AD-1808273, AD-1808418, AD-1808433, AD-1808842, AD-1808865, AD-1809051, AD-1809066, AD-1807707, AD-1807725, AD-1808628, AD-1808649, AD-1808727, AD-1808748, AD-1809187, AD-1809205, AD-1807764, AD-1807780, AD-1808312, AD-1808327, AD-1807561, AD-1807578, AD-1807616, AD-1807631, AD-1807646, AD-1809354, and AD-1809369. Table 6 provides a summary of the hotspot regions in NM_001301015.3 and the duplexes that target the hotspot regions.

TABLE-US-00003 TABLE 2 Abbreviations of nucleotide monomers used in nucleic acid sequence representation. It will be understood that these monomers, when present in an oligonucleotide, are mutually linked by 5'-3'- phosphodiester bonds; and it is understood that when the nucleotide contains a 2'-fluoro modification, then the fluoro replaces the hydroxy at that position in the parent nucleotide (i.e., it is a 2'-deoxy-2'- fluoronucleotide). Abbreviation Nucleotide(s) A Adenosine-3'-phosphate Ab beta-L-adenosine-3'-phosphate Abs beta-L-adenosine-3'-phosphorothioate Af 2'-fluoroadenosine-3'-phosphate Afs 2'-fluoroadenosine-3'-phosphorothioate As adenosine-3'-phosphorothioate (A2p) adenosine-2'-phosphate (A2ps) adenosine-2'-phosphorothioate C cytidine-3'-phosphate Cb beta-L-cytidine-3'-phosphate Cbs beta-L-cytidine-3'-phosphorothioate Cf 2'-fluorocytidine-3'-phosphate Cfs 2'-fluorocytidine-3'-phosphorothioate Cs cytidine-3'-phosphorothioate (C2p) cytidine-2'-phosphate (C2ps) cytidine-2'-phosphorothioate G guanosine-3'-phosphate Gb beta-L-guanosine-3'-phosphate Gbs beta-L-guanosine-3'-phosphorothioate Gf 2'-fluoroguanosine-3'-phosphate Gfs 2'-fluoroguanosine-3'-phosphorothioate Gs guanosine-3'-phosphorothioate (G2p) guanosine-2'-phosphate (G2ps) guanosine-2'-phosphorothioate T 5'-methyluridine-3'-phosphate Tf 2'-fluoro-5-methyluridine-3'-phosphate Tfs 2'-fluoro-5-methyluridine-3'-phosphorothioate Ts 5-methyluridine-3'-phosphorothioate U Uridine-3'-phosphate Uf 2'-fluorouridine-3'-phosphate Ufs 2'-fluorouridine-3'-phosphorothioate Us uridine-3'-phosphorothioate (U2p) uridine-2'-phosphate (U2ps) uridine-2'-phosphorothioate N any nucleotide, modified or unmodified a 2'-O-methyladenosine-3'-phosphate as 2'-O-methyladenosine-3'-phosphorothioate c 2'-O-methylcytidine-3'-phosphate cs 2'-O-methylcytidine-3'-phosphorothioate g 2'-O-methylguanosine-3'-phosphate gs 2'-O-methylguanosine-3'-phosphorothioate t 2'-O-methyl-5-methyluridine-3'-phosphate ts 2'-O-methyl-5-methyluridine-3'-phosphorothioate u 2'-O-methyluridine-3'-phosphate us 2'-O-methyluridine-3'-phosphorothioate s phosphorothioate linkage L96 N-[tris(GalNAc-alkyl)-amido-dodecanoyl]-4-hydroxyprolinol [Hyp-(GalNAc-alkyl)3] [00038]  embedded image (Agn) Adenosine-glycol nucleic acid (GNA) S-Isomer (Cgn) Cytidine-glycol nucleic acid (GNA) S-Isomer (Ggn) Guanosine-glycol nucleic acid (GNA) S-Isomer (Tgn) Thymidine-glycol nucleic acid (GNA) S-Isomer P Phosphate VP Vinyl-phosphonate dA 2'-deoxyadenosine-3'-phosphate dAs 2'-deoxyadenosine-3'-phosphorothioate dC 2'-deoxycytidine-3'-phosphate dCs 2'-deoxycytidine-3'-phosphorothioate dG 2'-deoxyguanosine-3'-phosphate dGs 2'-deoxyguanosine-3'-phosphorothioate dT 2'-deoxythymidine-3'-phosphate dTs 2'-deoxythymidine-3'-phosphorothioate dU 2'-deoxyuridine dUs 2'-deoxyuridine-3'-phosphorothioate

TABLE-US-00004 TABLE 3 Unmodified Sense and Antisense Strand Sequences of Human ABLIM3 dsRNA Agents Range in Range in Duplex Sense Sequence SEQ ID NM_ Antisense Sequence SEQ ID NM_ Name 5' to 3' NO: 001301015.3 5' to 3' NO: 001301015.3 AD- CCGGGAUUCCAAUCUG 8 82-102 UUCCUCAGAUUGGA 143 80-102 1807455 AGGAA AUCCCGGGC AD- CUCCUUCUCCUUGCUC 9 134-154 UGUGAGAGCAAGGA 144 132-154 1807481 UCACA GAAGGAGGC AD- CACCUGAGGAUACAGA 10 232-252 UCAGCUCUGUAUCCU 145 230-252 1807539 GCUGA CAGGUGCU AD- CAGGUGAUGAGUGAG 11 276-296 UCGAACCUCACUCAU 146 274-296 1807561 GUUCGA CACCUGGG AD- UCGAAGAACGGAAGA 12 293-313 UUUAAAUCUCCGU 147 291-313 1807578 UUUAAA UCUUCGAAC AD- CCUCCGUUAUUGAAUGA 13 326-346 UUCUUUCAUCAAU 148 324-346 1807591 AAGAA ACGGAGGCC AD-

UGCAACAACAUACCA 14 351-371 UUGGUGGUGGUGA 149 349-371 1807616 UGAAA CUUUGCACU
AD- AUGAACACUAGCAUUC 15 366-386 UUAAGGAAUGCUAG 150 364-386 1807631 CUUAA
UGUUCAUGG AD- CCUUAUCAGCAGAAUC 16 381-401 UUAAGGAUUCUGCU 151 379-401 1807646
CUUAA GAUAAGGAA AD- AGCUCCAAUGUCAUCC 17 414-434 UCACUGGAUGACAU 152 412-434
1807659 AGUGA UGGAGCUGC AD- CAACCACUCCACAUC 18 482-502 UAUCUGAUGUGGAA 153 480-
502 1807707 AGAUA GUGGUUGUU AD- AUGCUUACCUGUCA 19 500-520 UAUACUUGACAGGU 154
498-520 1807725 GUAUA GAAGCAUCU AD- AGGCUUCUUCUUCAAG 20 542-562 UGGUUCUUGAAGAA 155
540-562 1807764 AACCA GAAGCCUGA AD- AACCAGGAGUACAUCU 21 558-578 UGUGCAGAUGUACU
156 556-578 1807780 GCACA CCUGGUUCU AD- GACUACCAGCAACUCU 22 582-602
UCCAUAAGAGUUGCU 157 580-602 1807804 AUGGA GGUAGUCCU AD- GGGACUUCAUCACAGG 23 622-
642 UUUCGCCUGUGAUG 158 620-642 1807844 CGAAA AAGUCCCGG AD- CAAGUGCUUCGUGUGC 24
674-694 UAGCUGCACACGAA 159 672-694 1807870 AGCUA GCACUUGGG AD- CUUGUGCAGGAAGCCU 25
692-712 UGGAAAGGCCUUCU 160 690-712 1807888 UUCA GCACAAGCU AD- CAUUGGAGACAAGGU 26
713-733 UAGGUCACCUUGUC 161 711-733 1807889 GACCUA UCCAAUGGG AD- GACCUUCAGCGGUAAA
27 728-748 UAUUCUUUACCGCU 162 726-748 1807904 GAUA GAAGGUCAC AD- AGAAUGUGUGUGCCA
28 743-763 UACGUUUGGCACAC 163 741-763 1807919 AACGUA ACAUUCUUU AD-
CAGCAGUAAGCCCAUC 29 779-799 UUCUUGAUGGGCUU 164 777-799 1807955 AAGAA ACUGCUGGC
AD- CUGCUUCAAGUGCCAG 30 887-907 UAGGUCUGGCACUU 165 885-907 1808063 ACCUA
GAAGCAGCU AD- GAUGGUGUCCAUAUCU 31 942-962 UUCACAGUAUGGAA 166 940-962 1808098
GUGAA CACCAUCCU AD- UGGCAUAAAUGUGA 32 983-1003 UAAGUCUCACAUUU 167 981-1003
1808139 GACUUA AAUGCCAAA AD- UUGUGACCGAUACAUC 33 1001-1021 UCACUGAUGUAUCG 168 999-
1021 1808157 AGUGA GUCACAAGU AD- CCACCAGAUGUUCACC 34 1082-1102 UCUUCGGUGAACAU 169
1080-1102 1808238 GAAGA CUGGUGGCA AD- GAGAGGAAUGUACC 35 1102-1122 UUGUGAGGUACAUC 170
1100-1122 1808258 UCACAA UCCUCUCCU AD- UCACAGGUUCCGAGGU 36 1117-1137 UCCAAACCUCGGAAC
171 1115-1137 1808273 UUGGA CUGUGAGG AD- UAAAGCAUAGACGGAC 37 1177-1197 UAGAUGUCCGUCUA
172 1175-1197 1808312 AUCUA UGCUUUAAC AD- CAUCUGAAACCUCCAU 38 1192-1212
UUGAGAUGGAGGUU 173 1190-1212 1808327 CUCAA UCAGAUGUC AD- CGAGUCAUCUGCGCUA 39 1242-
1262 UACUUUAGCGCAGA 174 1240-1262 1808356 AAGUA UGACUCGGU AD- GAGAUCUUAUUACA 40
1269-1289 UUCUUUGUAAUUA 175 1267-1289 1808383 AAGAA GGAUCUCAU AD- GUCUAUCUACGAGGUA
41 1313-1333 UGUUGUACCUCGUA 176 1311-1333 1808407 CAACA GAUAGACUU AD-
AUUUCUAUGAGCCUC 42 1344-1364 UGAAUGAGGCUCAU 177 1342-1364 1808418 AUUCA AGGAAAUGA
AD- CAUUCAGAUACAUGU 43 1359-1379 UUCGGACAUGUAUC 178 1357-1379 1808433 CCGAA
UGGAAUGAG AD- UCCGACGAGAUGCUGG 44 1374-1394 UCUCUCCAGCAUCUC 179 1372-1394 1808448
AGAGA GUCGGACA AD- AGUCGCUGGGAACAUC 45 1408-1428 UAGAUAAUGUUCUCC 180 1406-1428 1808482
AUCUA AGCGACUCU AD- CAGGACAUCUACGAGA 46 1437-1457 UAGGUUCUCGUAGA 181 1435-1457
1808511 ACCUA UGUCCUGGG AD- UCACCACUACUACCGC 47 1550-1570 UCAGAGCGGUAGUA 182 1548-
1570 1808561 UCUGA GUGGUGAGG AD- CAUACCAUAGCCAGUU 48 1594-1614 UAUCUAAUCUGGCUA 183
1592-1614 1808598 AGAUA UGGUAUGGA AD- CCACUCCAACCUCUUA 49 1624-1644 UCUGGUAAGAGGUU
184 1622-1644 1808628 CCAGA GGAGUGGAG AD- CUCCAAGCACUUUA 50 1645-1665
UGAUGUGAAAGUGC 185 1643-1665 1808649 CAUCA UUGGGAGCC AD- GACAGUAACAUCUACC 51 1674-
1694 UUUCCGGUAGAUGU 186 1672-1694 1808678 GGAAA UACUGUCUC AD- UGGUGAUUUGUCUAC 52
1715-1735 UUUGCUGUAGACAA 187 1713-1735 1808698 AGCAA AUCACCAUG AD- ACAAGUGAAGACAUA
53 1746-1766 UUGGCUGAUGUCUU 188 1744-1766 1808727 GCCAA CACUUGUUU AD-
ACCUCCAAGUACAGUC 54 1767-1787 UAUGGGACUGUACU 189 1765-1787 1808748 CCAUA UGGAGGUCU
AD- UCGGAGUCUGAGUACU 55 1812-1832 UGUCCAGUACUCAG 190 1810-1832 1808773 GGACA
ACUCCGAAG AD- AGCCAGAAGGUUCUCG 56 1859-1879 UCAGACGAGAACCU 191 1857-1879 1808789
UCUGA UCUGGCUCG AD- GAGGAGGAUGAUUUU 57 1884-1904 UCGGUCAAAAUCAU 192 1882-1904 1808814
GACCGA CCUCCUCUC AD- ACAAGCUCCAAAGUGG 58 1912-1932 UAAUCCACUUUGG 193 1910-1932
1808842 AAUUA AGCUUGUGC AD- CGGCUGAUUCUGAAGG 59 1935-1955 UUCUCCUUCAGAA 194 1933-
1955 1808865 AAGAA UCAGCCGGC AD- AGCUCCUAUGCAGAU 60 1971-1991 UCAGGGAUCUGCAU 195
1969-1991 1808901 CCUGA AGGAGCUCG AD- UGGCUAUGAGAUGUCC 61 2039-2059 UUGAGGGACAUCUC
196 2037-2059 1808941 CUCAA AUAGCCAGC AD- UCCUCUCAUCUCCAAA 62 2096-2116
UCAGAUUUGGAGAU 197 2094-2116 1808978 UCUGA GAGAGGAUC AD- GCCUACCGAAGAAAUG 63 2127-
2147 UAGCCCAUUUCUUCG 198 2125-2147 1809009 GGCUA GUAGGCAG AD- CUCUCCACUACGACA 64
2169-2189 UAUGCUGUCGUAGU 199 2167-2189 1809051 GCAUA GGAAGAGGU AD- AGCAUGAACGCAGUCA
65 2184-2204 UCAGUUGACUGCGU 200 2182-2204 1809066 ACUGA UCAUGCUGU AD-
ACAAGAUCUACCCUUA 66 2218-2238 UUUCAUAAGGGUAG 201 2216-2238 1809080 UGAAA AUCUUGUAC
AD- UGGUGACUACAAGAG 67 2245-2265 UUCUCCUCUUGUA 202 2243-2265 1809107 GAAGAA
GUCACCAGC AD- GCCCAAGGAUGUAGAC 68 2273-2293 UUCCUGUCUACAUC 203 2271-2293 1809135
AGGAA UUGGGCAG AD- UUCUACCAAGUCUUUG 69 2325-2345 UAUGCCAAAGACUU 204 2323-2345 1809187
GCAUA GGUAGAACU AD- AUGACCAUCUCUGAGU 70 2343-2363 UUCAACUCAGAGA 205 2341-2363
1809205 UUGAA UGGUCAUGC AD- AAUGAACUGAAGAAG 71 2385-2405 UGCUUGCUUCUUA 206 2383-2405

1809246 CAAGCA AUUUAUUUUAU AD- AGGCUUAUUAUUUUAU 207 2420-2442 1809283 AUAUGA AGAGCCUCU AD- AUGUAAAUCUCUCUA 73 2459-2479 UUCAGUAGAGAGAU 208 2457-2479 1809293 CUGAA UUUACAUAU AD- CGGUAAUAUCCUCUCU 74 2483-2503 UCACAAGAGAGGAU 209 2481-2503 1809317 UGUGA UAUACCGAG AD- UGCCAUGAGACUUGCU 75 2520-2540 UGAAAAGCAAGUCU 210 2518-2540 1809354 UUUCA CAUGGCAGG AD- UUUUCUGUACUGUCAG 76 2535-2555 UUUGCCUGACAGUA 211 2533-2555 1809369 GCAA CAGAAAAGC AD- GUCAUCGAGAUUUU 77 2561-2581 UCAUAAAAUAUCU 212 2559-2581 1809395 UUAUGA CGAUGACGU AD- UUACUUUCUCUUUUUCU 78 2585-2605 UACUUAGAAAAGAG 213 2583-2605 1809414 AAGUA AAAGUAAGG AD- CUGAAACAUCUGUCCU 79 2667-2687 UAGUUAGGACAGAU 214 2665-2687 1809456 AACUA GUUUCAGUA AD- CAACUCUCUUUCCUAA 80 2703-2723 UUUCUUUAGGAAAG 215 2701-2723 1809472 JAGAAA AGAGUUGGA AD- UGAAGAAGUCUCUCUU 81 2729-2749 UAGAGAAGAGAGAC 216 2727-2749 1809498 CUCUA UUCUUCAGG AD- UUUCUUAUUUUCU 82 2764-2784 UAGCCCUAGAAAU 217 2762-2784 1809511 GGGCUA UAAGAAAGG AD- CUGACCAUGUGGUUUC 83 2788-2808 UUGUGGAAACCACA 218 2786-2808 1809535 CACAA UGGUCAGCA AD- CUCCCAUGGGAAGAUC 84 2830-2850 UUGCAGAUUCUCCCA 219 2828-2850 1809550 UGCAA UGGGAGGG AD- AAUCAGUGAGCACCU 85 2861-2881 UUCAAGGUGUCUA 220 2859-2881 1809561 UUGAA CUGAUUUGG AD- CCACGAAGAACUUUCU 86 2885-2905 UGUUGAGAAAGUUC 221 2883-2905 1809585 CAACA UUCGUGGGC AD- GCAAUGCAGUAAAA 87 2937-2957 UCACCCUUUUAACUG 222 2935-2957 1809596 GGGUGA CAUUGCCC AD- GCCUCAAUCUAGUCA 88 2958-2978 UGUAAUGACUAGAU 223 2956-2978 1809617 UUACA UUGAGGCUC AD- UUACACCAGUCAACAG 89 2974-2994 UACUUCUGUUGACU 224 2972-2994 1809633 AAGUA GGUGUAAUG AD- CUCCUCAGCUCCUUA 90 3011-3031 UAGGGUUAAGGAGC 225 3009-3031 1809670 CCCUA UGAGGAGAG AD- UGGAUUGUAACCUCUC 91 3044-3064 UAAGGGAGAGGUUA 226 3042-3064 1809702 CCUUA CAUCCAGG AD- CUUGUCCAAUUCUAGG 92 3061-3081 UGAAUCCUAGAUUU 227 3059-3081 1809719 AUUCA GGACAAGGG AD- UCCUGGUAGGAAAAG 93 3079-3099 UUUUCCUUUUCU 228 3077-3099 1809737 GAAAAA ACCAGGAU AD- UCCUCCACCACUCC 94 3110-3130 UAGUUGGAAGUGGU 229 3108-3130 1809757 AACUA GGAGGGAAG AD- CUGGACUUGGAGAACC 95 3146-3166 UCUCUGGUUCUCCAA 230 3144-3166 1809773 AGAGA GUCCAGGU AD- CACCUGUUAGAAUCAG 96 3203-3223 UAGCUCUGAUUUCUA 231 3201-3223 1809822 AGCUA ACAGGUGCC AD- GCUGCAGGAUUUCUUG 97 3220-3240 UGUCCCAAGAAAUCC 232 3218-3240 1809839 GGACA UGCAGCUC AD- UUCCCUCUUUCAAGGA 98 3279-3299 UCUUCUCCUUGAAA 233 3277-3299 1809895 GAAGA GAGGGAAGG AD- UGAUUGCAGCUUGUA 99 3303-3323 UAAGAAUACAAGCU 234 3301-3323 1809919 UUCUUA GCAAUCAUG AD- UUUAGCCUUAUUACAA 100 3321-3341 UUAGAUUGUAAUUA 235 3319-3341 1809937 UCUAA GGCUAAAGA AD- CUAUGUGCCUGACAAC 101 3338-3358 UUUGAGUUGUCAGG 236 3336-3358 1809954 UCAAA CACAUAGAU AD- CUCCAACUGAACAACC 102 3388-3408 UGUCUGGUUGUUA 237 3386-3408 1810004 AGACA GUUGGAGCU AD- CAGACAACUCUCAUA 103 3407-3427 UAGGAUGAUGAGAG 238 3405-3427 1810023 UCCUA UUGUCUGUC AD- AUCCUCCAGAGAGAAA 104 3422-3442 UCUAUUUUCUCUCU 239 3420-3442 1810038 AUAGA GGAGGAUGA AD- AAUAGGCCGUGUCUA 105 3437-3457 UUCUUUGAGACACG 240 3435-3457 1810053 AAGAA GCCUAUUUU AD- AAAGAAAGGUUCUUG 106 3452-3472 UUAGACCAAGAACC 241 3450-3472 1810068 GUCUAA UUUCUUUGA AD- UGCUGCAAAGUUGGCC 107 3532-3552 UACAUGGCCAACUU 242 3530-3552 1810127 AUGUA UGCAGCAGG AD- AUGUUACACAGGGAA 108 3548-3568 UAAAGUUUCCUGU 243 3546-3568 1810143 ACUUUA GAAACAUGG AD- CUGCUUAUGAGAUUCC 109 3579-3599 UUUUUGGAAUCUA 244 3577-3599 1810174 AAAAA UAAGCAGCC AD- CAAAUGAAGUGUUG 110 3594-3614 UUUGGCCAACACUUC 245 3592-3614 1810189 GCCAAA AUUUUGGA AD- CCAUCCUGGAUUUUC 111 3626-3646 UACUGGGAAAAUCC 246 3624-3646 1810221 CAGUA AGGAUGGCC AD- GUGGCUUCCCUUCCUG 112 3644-3664 UCGAGCAGGAAGGG 247 3642-3664 1810239 CUCGA AAGCCACUG AD- AAAGCUUAACCUCUCU 113 3682-3702 UGAGAAGAGAGGUU 248 3680-3702 1810255 UCUCU AAGCUUUCU AD- CCAAACCUUUCACCUU 114 3706-3726 UAUUCAAGGUGAAA 249 3704-3726 1810277 GAAUA GGUUUGGAG AD- UGA AUGGGUAAUGUU 115 3721-3741 UCACCAAACAUUACC 250 3719-3741 1810292 UGGUGA CAUUCAAG AD- UGUUCCUUCUUGGAGA 116 3746-3766 UGGCUUCUCCAAGA 251 3744-3766 1810296 AGCCA AGGAACAGC AD- GCCUUGAGUCGGACCA 117 3763-3783 UAAAUGGUCCGAC 252 3761-3783 1810313 UUUUA UCAAGGCUU AD- AAGGAUGAAGAAGUG 118 3796-3816 UAUUUUCACUUCU 253 3794-3816 1810345 AAAAUA CAUCCUUC AD- AUGACAAUAAUGACUC 119 3814-3834 UUUGAGAGUCAUUA 254 3812-3834 1810363 UCAAA UUGUCAUUU AD- ACAUGGCAAUGUAG 120 3849-3869 UCAGUUCUACAUUU 255 3847-3869 1810398 AACUGA GCCAUGUCA AD- CUGACUAAAUAUGAAC 121 3866-3886 UGUUUGUCAAUUU 256 3864-3886 1810415 AAACA AAGUCAGUU AD- ACCUCUGAUGUUGAGC 122 3897-3917 UAGGUGCUCACAU 257 3895-3917 1810446 ACCUA CAGAGGUGC AD- UGAAUACUGAGCACUG 123 3919-3939 UCAUUCAGUGCUCU 258 3917-3939 1810468 AAUGA GUAUUCAGC AD- CAGGGAUAUGCCUACC 124 3986-4006 UUAUUGGUAGGCAU 259 3984-4006 1810494 AAUAA AUCCCUGAG AD- AGGCAUGUACCCAAAC 125 4018-4038 UUAUUGUUUGGGUA 260 4016-4038 1810526 AUAAA CAUGCCUUA AD- ACGGAUGUAAGGCAG 126 4037-4057 UACUUUCUGCCUUA 261 4035-4057 1810545 AAAGUA AUCCGUUA AD- AUCGGAGAAGGAAUG 127 4058-4078 UUUUCUCAUUCUUC 262 4056-4078 1810566 AGAAAA UCCGAUCA AD- GUGAUGUUAUGAAA 128 4084-4104 UUGACUUUUCAUUA 263 4082-4104 1810591

AGCAAACAACACGCG AD- AGUCAUUCUGCAGCUAG 129 4099-4119 UUGCUCUAGCUGCA 264 4097-4119
 1810606 AGCAA UAUGACUUU AD- CUUUCUGGAAGAGAU 130 4131-4151 UAUGCAAUCUCUCC 265 4129-
 4151 1810638 UGCAUA AGAAAGCU AD- AUUCAGGAAGGAUCU 131 4159-4179 UUACAAAGAUCUU 266
 4157-4179 1810666 UUGUAA CCUGAAUUU AD- AGAGGGAAGUCUGCU 132 4225-4245 UAACACAGCAGACU
 267 4223-4245 1810676 GUGUUA UCCCUCUGG AD- UGUUCUCAUGUAGGA 133 4241-4261
 UUGACAUCUACA 268 4239-4261 1810692 UGUCAA GAGAACACA AD- UCUCUUUUUGGCCAAU 134 4276-
 4296 UAGACAUUGGCCAA 269 4274-4296 1810727 GUCUA AAAGAGAAG AD- UGUCUUUUCACUUCC 135
 4291-4311 UGUCAGGAAAGUGA 270 4289-4311 1810742 UGACA AAAGACAUU AD- CUGACCCUUUAGAAUC
 136 4306-4326 UGGAUGAUUCUAAA 271 4304-4326 1810757 AUCCA GGGUCAGGA AD-
 GCAAUCAUGGAAGUU 137 4336-4356 UAAGGCAACUCCA 272 4334-4356 1810767 GCCUUA UGAUUGCGU
 AD- UUAUUGUCACUGGU 138 4354-4374 UUUCUUAACCAGUG 273 4352-4374 1810785 AAGAAA
 ACAUAAGG AD- AAGAACUUGGCGAGA 139 4369-4389 UUCAAUCUCGCCAA 274 4367-4389 1810800
 UUGAAA GUUCUUA AD- AAGGGCUUUUGUUU 140 4387-4407 UCAACAAUACAAA 275 4385-4407
 1810818 UGUUGA AGCCCUUCA AD- UUGUUGGAUUAUUUU 141 4404-4424 UGAAACAAAAUAU 276 4402-
 4424 1810822 GUUUA CCAACAACA AD- UUCCCAUAAAAGCACA 142 4421-4441 UAUGAUGUGCUUU 277
 4419-4441 1810839 UCAUA AUGGGAAAC
 TABLE-US-00005 TABLE 4 Modified Sense and Antisense Strand Sequences of Human ABLIM3
 dsRNA Agents SEQ SEQ mRNA Target SEQ Duplex Sense Sequence ID Antisense Sequence ID Sequence ID ID
 5' to 3' NO: 5' to 3' NO: 5' to 3' NO: AD-1807455 cscsggg(Ahd)UfuCf 278 VPusUfscuCfaGfAfu 413
 GCGCGGAUCCA 548 CfAfaucugaggsasa uggAfaUfccgsgsgsc AUCUGAGGAG AD-1807481 csusccu(Uhd)CfuCfC
 279 VPusGfsugaGfaGfCfaa 414 GCCUCCUUCUCCU 549 fUfugcucucascsa ggAfgAfaggagsgsc UGCUCUCACC AD-
 1807539 csasccu(Ghd)AfgGf 280 VPusCfsagcUfcUfGfua 415 AGCACCUGAGGAU 550 AfUfacagagcusgsa
 ucCfuCfaggugscsu ACAGAGCUGG AD-1807561 csasggg(Ghd)AfuGf 281 VPusCfsgaaCfcUfCfac 416
 CCCAGGUGAUGAG 551 AfGfugagguucsgsa ucAfuCfaccugsgsg UGAGGUUCA AD-1807578 uscsгаа(Ghd)AfaCfG
 282 VPusUfsuaaAfuCfUfu 417 GUUCGAAGAACGG 552 fGfaagauuuasasa ccgUfuCfuucgasasc AAGAUUUAAA AD-
 1807591 cscsucc(Ghd)UfaUfu 283 VPusUfscuuUfcAfUfu 418 GGCCUCCGUUUUG 553 fGfaugaagsasa
 caaUfaCfaggagsgsc AAUGAAAGAC AD-1807616 usgscaa(Ahd)GfaCfa 284 VPusUfsucaUfgGfUfg 419
 AGUGCAAAGACAU 554 fUfcaccaugasasa augUfcUfuucgascsu CACCAUGAAC AD-1807631 asusгаа(Chd)AfcUfA
 285 VPusUfsaagGfaAfUfg 420 CCAUGAACACUAG 555 fGfcauuccusasa cuaGfuGfucausgsg CAUUCUUAU AD-
 1807646 cscsuua(Uhd)CfaGfC 286 VPusUfsaagGfaUfUfc 421 UUCUUAUCAGCA 556 fAfgaauccusasa
 ugcUfgAfuaaggsasa GAAUCCUUAU AD-1807659 asgscuc(Chd)AfaUfG 287 VPusCfsacuGfgAfUfg 422
 GCAGCUCAAUGU 557 fUfcauccagusgsa acaUfuGfgagcugsgsc CAUCCAGUGC AD-1807707 csasacc(Ahd)CfuUfC
 288 VPusAfsucuGfaUfGfu 423 AACAACCACUUC 558 fCfacaucagasasa ggaAfgUfgguugsusu ACAUCAGAUG AD-
 1807725 asusgcu(Uhd)CfaCfC 289 VPusAfsuacUfuGfAfc 424 AGAUGCUUCACCU 559 fUfgucaaguasasa
 aggUfgAfagcauscsu GUCAAGUAUG AD-1807764 asgsgcu(Uhd)CfuUf 290 VPusGfsguuCfuUfGfa 425
 UCAGGCUUCUUCU 560 CfUfucaagaacscsa agaAfgAfagccusgsa UCAAGAACCA AD-1807780 asascca(Ghd)GfaGfU
 291 VPusGfsugcAfgAfUfg 426 AGAACCAGGAGUA 561 fAfcaucugcascsa uacUfcCfugguuscsu CAUCUGCACC AD-
 1807804 gsascua(Chd)CfaGfC 292 VPusCfscuuAfgAfGfu 427 AGGACUACCAGCA 562 fAfacucuaugsgsa
 ugcUfgGfuagucscsu ACUCUAUGGC AD-1807844 gsgsgac(Uhd)UfcAf 293 VPusUfsucGfcUfGfu 428
 CCGGACUUAUC 563 Ufcfagggcgasasa gauGfaAfguccsgsg ACAGGCGAAG AD-1807870 csasagu(Ghd)CfuUfC
 294 VPusAfsugcuGfcAfCfac 429 CCAAGUGCUUCG 564 fGfugugcagsusa gaAfgCfaguugsgsg UGUGCAGCUU AD-
 1807888 csusugu(Ghd)CfaGf 295 VPusGfsgaaAfgGfCfu 430 AGCUUGUGCAGGA 565 GfAfagccuuucscsa
 uccUfgCfacaagscsu AGCCUUUCCC AD-1807889 csasuug(Ghd)AfgAf 296 VPusAfsgguCfaCfCfu 431
 CCCAUUGGAGACA 566 CfAfaggugaccsusa uguCfuCfcaaugsgsg AGGUGACCUU AD-1807904 gsasccu(Uhd)CfaGfC
 297 VPusAfsuucUfuUfAfc 432 GUGACCUUCAGCG 567 fGfguaaagaasasa cgcUfgAfaggucsasc GUAAAGAAUG AD-
 1807919 asgsaa(Uhd)UfgUf 298 VPusAfsugcuUfuGfGfc 433 AAAGAAUGUGUGU 568 GfUfgccaaacgsusa
 acaCfaCfauucusus GCAAACGUG AD-1807955 csasgca(Ghd)UfaAfG 299 VPusUfscuuGfaUfGfg 434
 GCCAGCAGUAAGC 569 fCfcaucaagsasa gcuUfaCfugcugsgsc CCAUCAAGAU AD-1808063 csusgcu(Uhd)CfaAf 300
 VPusAfsgguCfuGfGfc 435 AGCUGCUUCAAGU 570 GfUfgccagaccsusa acuUfgAfagcagcsu GCCAGACCUG AD-
 1808098 gsasugg(Uhd)GfuUf 301 VPusUfscacAfgUfAfu 436 AGGAUGGUGUUC 571 CfCfauacugugsasa
 ggaAfcAfcaucscsu AUACUGUGAG AD-1808139 usgsgca(Uhd)UfaAf 302 VPusAfsaguCfuCfAfa 437
 UUUGGCAUUAU 572 AfUfgugagacususa uuUfaAfugccasasa GUGAGACUUG AD-1808157 ususgug(Ahd)CfcGf
 303 VPusCfsacuGfaUfGfua 438 ACUUGUGACCGAU 573 AfUfacaucagusgsa ucGfgUfcacaasgsu ACAUCAGUGG AD-
 1808238 cscsacc(Ahd)GfaUfG 304 VPusCfsuucGfgUfGfa 439 UGCCACCAGAUGU 574 fUfucaccgaasgsa
 acaUfcUfggugsgsc UACCGAAGG AD-1808258 gsasgag(Ghd)AfaAf 305 VPusUfsgugAfgGfUfa 440
 AGGAGAGGAAUG 575 UfGfuaccucacsasa cauUfuCfcucucscsu UACCUCACAG AD-1808273 uscsaca(Ghd)GfuUfC
 306 VPusCfscaaAfcCfUfcg 441 CCUCACAGGUUC \$76 fCfagguuugsgsa gaAfcCfugugagsgsg GAGGUUUGGC AD-
 1808312 usasaag(Chd)AfuAf 307 VPusAfsgauGfuCfCfG 442 GUUAAAGCAUAGA 577 GfAfcggacaucsusa
 ucuAfuGfuuuasasc CGGACAUCUG AD-1808327 csasucu(Ghd)AfaAfC 308 VPusUfsgagAfuGfGfa 443
 GACAUCUGAAACC 578 fCfuccaucucsasa gguUfuCfagaugsusc UCCAUCUCAC AD-1808356 csgsagu(Chd)AfuCf 309
 VPusAfsuuUfaGfCfG 444 ACCGAGUCAUCUG 579 UfGfcgcuaaagsusa cagAfuGfacucgsgsu CGCUAAAGUG AD-
 1808383 gsasgau(Chd)CfuUf 310 VPusUfscuuUfgUfAfa 445 AUGAGAUCUUA 580 AfAfuuacaagsasa

uaaAfgGfGaaugsasu UUAACAAAGAC AD-1808407 gsusca(Uhd)CfuAf 311 VPusGfsuugUfaCfAf 316
AAGUCUAUCUACG 581 CfGfagguaacaascsa cguAfgAfuagacsusu AGGUACAACG AD-1808418 asusuuc(Chd)UfaUf
312 VPusGfsaauGfaGfGfc 447 UCAUUUCCUAUGA 582 GfAfgccucauuscsa ucaUfaGfgaausgsa GCCUCAUUEC AD-
1808433 csasuuc(Chd)AfgAf 313 VPusUfscggAfcAfUfg 448 CUCAUUECAGAU 583 UfaAfauguccgsasa
uauCfuGfgaaugsasg CAUGUCCGAC AD-1808448 uscsca(Chd)GfaGfA 314 VPusCfsucuCfcAfGfca 449
UGUCCGACGAGAU 584 fUfgcuggagasgsa ucUfcGfucggascsa GCUGGAGAGA AD-1808482 asgsucg(Chd)UfgGf 315
VPusAfsgauAfaUfGfu 450 AGAGUCGCUGGGA 585 GfAfacauuaucsusa uccCfaGfcgacuscsu ACAUUAUCUC AD-
1808511 csasgga(Chd)AfuCfu 316 VPusAfsggUfcUfCfG 451 CCCAGGACAUCUA 586 fAfcgagaaccsusa
uagAfuGfuccugsgsg CGAGAACCUG AD-1808561 uscsacc(Ahd)CfuAfc 317 VPusCfsagaGfcGfGfua 452
CCUCACCACUACU 587 fUfaccgcucusgsa guAfgUfggugasgsg ACCGCUCUGG AD-1808598 csasuac(Chd)AfuAfG 318
VPusAfsucuAfaCfUfg 453 UCCAUAACCAUAGC 588 fCfcaguuagasusa gcuAfuGfguaugsgsa CAGUUAGAUG AD-
1808628 cscsacu(Chd)CfaAfc 319 VPusCfsuggUfaAfGfa 454 CUCCACUCCAACC 589 fCfucuuaccasgsa
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GGCUCCCAAGCAC 590 fCfuucacauscsa gugCfuUfgggagsgsc UUUCACAUC AD-1808678 gsascag(Uhd)AfaCfA
321 VPusUfsuccGfgUfAfg 456 GAGACAGUAACAU 591 fUfcuaccggasasa augUfuAfcugucsusc CUACCGGAAA AD-
1808698 usgsug(Ahd)UfuUf 322 VPusUfsugcUfgUfAfg 457 CAUGGUGAUUUGU 592 GfUfcuacagcasasa
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AAACAAGUGAAGA 593 fAfaucagccsasa ucuUfcAfcuugususu CAUCAGCCAG AD-1808748 ascsuc(Chd)AfaGfU
324 VPusAfsuggGfaCfUfg 459 AGACCUCCAAGUA 594 fAfcaguccasasa uacUfuGfgagguscsu CAGUCCCAUC AD-
1808773 uscsnga(Ghd)UfcUf 325 VPusGfsuccAfgUfAfc 460 CUUCGGAGUCUGA 595 GfAfguacuggascsa
ucaGfaCfuccgasasg GUACUGGACC AD-1808789 asgscca(Ghd)AfaGfG 326 VPusCfsagaCfGfGfAfa 461
CGAGCCAGAAGGU 596 fUfucucgucusgsa ccUfuCfuggcuscsg UCUCGUCUGG AD-1808814 gsasgga(Ghd)GfaUf 327
VPusCfsgguCfaAfAfa 462 GAGAGGAGGAUGA 597 GfAfuuuugaccsgsa ucaUfcCfuccuscusc UUUUGACCGC AD-
1808842 ascsaag(Chd)UfcCfA 328 VPusAfsauuCfcAfCfu 463 GCACAAGCUCCAA 598 fAfauggaasusa
uugGfaGfcuugsgsc AGUGGAAUUG AD-1808865 csgsgcu(Ghd)AfuUf 329 VPusUfscuuCfcUfUfca 464
GCCGGCUGAUUCU 599 CfUfgaaggaagsasa gaAfuCfagccgsgsg GAAGGAAGAA AD-1808901 asgsuc(Chd)UfaUfG
330 VPusCfsaggGfaUfCfu 465 CGAGCUCCUAUGC 600 fCfagaucuccsgsa gcaUfaGfgagcuscsg AGAUCCUGG AD-
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333 VPusAfsgccCfaUfUfuc 468 CUGCCUACCGAAG 603 fGfaaagggsusa uuCfGfuaggcsasg AAAUGGGCUG AD-
1809051 csuscuu(Chd)CfaCfu 334 VPusAfsugcUfgUfCfG 469 ACCUCUUECACUA 604 fAfcgacagcasasa
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ACAGCAUGAACGC 605 fCfagucaacugsgsa gogUfuCfaugcugsgu AGUCAACUGG AD-1809080 ascsaag(Ahd)UfcUfA
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1809107 usgsng(Ahd)CfuAf 337 VPusUfscuuCfcUfCfu 472 GCUGGUGACUACA 607 CfAfaaggaagsasa
uguAfgUfcaccasgsg AGAGGAAGAA AD-1809135 gscscca(Ahd)GfgAf 338 VPusUfscuuGfuCfUfca 473
CUGCCCAAGGAUG 608 UfGfuagacaggsasa auCfcUfugggcsasg UAGACAGGAC AD-1809187 usscua(Chd)CfaAfG
339 VPusAfsugcCfaAfAfg 474 AGUUCUACCAAGU 609 fUfcuuuggcasasa acuUfgGfuagaascsu CUUUGGCAUG AD-
1809205 asusgac(Chd)AfuCfu 340 VPusUfscaaAfcUfCfag 475 GCAUGACCAUCUC 610 fCfugaguugsasa
agAfuGfgucausgsg UGAGUUUGAC AD-1809246 asasuga(Ahd)CfuGf 341 VPusGfscuuGfcUfUfc 476
GGAAUGAACUGAA 611 AfAfgaagcaagscsa uucAfgUfucuuuscsc GAAGCAAGCC AD-1809283 asgsngu(Chd)UfaUf
342 VPusCfsauaUfaUfAfu 477 AGAGGCUCUAUAA 612 AfAfaauauausgsa uuaUfaGfagccuscsc AUUAUAUAGC AD-
1809293 asusgua(Ahd)AfaUf 343 VPusUfscagUfaGfAfg 478 AUUAUGUAAAUCU 613 CfUfcucuacugsasa
agaUfuUfuacausasu CUCUACUGAA AD-1809317 csgsgua(Uhd)AfaUf 344 VPusCfsacaAfgAfGfag 479
CUCGGUAUAAUCC 614 CfCfucucuugsgsa gaUfuAfuaccgsasg UCUCUUGUGU AD-1809354 usgscca(Uhd)GfaGf
345 VPusGfsaaaAfgCfAfa 480 CCUGCCAUGAGAC 615 AfCfuugcuuuuscsc ucUfcAfuggcasgsg UUGCUUUUCU AD-
1809369 ususuuc(Uhd)GfuAf 346 VPusUfsugcCfuGfAfc 481 GCUUUUCUGUACU 616 CfUfgucaggcasasa
aguAfcAfgaaaasgsg GUCAGGCAAG AD-1809395 gsuscau(Chd)GfaGf 347 VPusCfsauaAfaAfAfa 482
ACGUCAUCGAGAU 617 AfUfauuuuuauausgsa ucUfcGfaugacsgsu AUUUUUAUGC AD-1809414 ususacu(Uhd)UfcUf
348 VPusAfsuuAfgAfAfa 483 CCUUAUCUUCUCU 618 CfUfuuuuauausgsa agaGfaAfaguaasgsg UUUUAAGUG AD-
1809456 csusgaa(Ahd)CfaUfc 349 VPusAfsuuAfgGfAfc 484 UACUGAAACAUCU 619 fUfguccuacscasa
agaUfgUfuucagsasa GUCCUAAUCU AD-1809472 csasacu(Chd)UfcUfu 350 VPusUfsucuUfuAfGfg 485
UCCAACUCUCUUU 620 fUfccuaaagasasa aaaGfaGfaguugsgsa CCUAAAGAAG AD-1809498 usgsaag(Ahd)AfgUf 351
VPusAfsagAfaGfAfg 486 CCUGAAGAAGUCU 621 CfUfcucuucscasa agaCfuUfcuucasgsg CUCUUCUCUC AD-
1809511 ususcu(Uhd)AfaAf 352 VPusAfsgccCfuAfGfaa 487 CCUUCUUAUAAU 622 UfUfucuaaggcsusa
auUfuAfaagaasgsg UCUAGGGCUG AD-1809535 csusgac(Chd)AfuGf 353 VPusUfsgugGfaAfAfc 488
UGCUGACCAUGUG 623 UfGfguuuaccasasa cacAfuGfgucagcsa GUUCCACAC AD-1809550 csusccc(Ahd)UfgGf
354 VPusUfsgcaGfaUfCfu 489 CCCUCCCAUGGGA 624 GfAfaugucscasa uccCfaUfgggagsgsg AGAUCUGCAG AD-
1809561 asasauc(Ahd)GfuGf 355 VPusUfscaaAfgGfUfg 490 CCAAUUCAGUGAG \$25 AfGfcaccuuugsasa
cucAfcUfgauuugsgsg CACCUUUGAG AD-1809585 cscsacg(Ahd)AfgAf 356 VPusGfsuugAfgAfAfa 491
GCCACGAAGAAC 626 AfCfuucucaascsa guuCfuUfcgugsgsg UUCUCAACA AD-1809596 gscsaau(Ghd)CfaGfU

357 VPusCfscuUfaa492 GGCACCAAGGUSGSA acUfgCfauugsgsgs AAAAGGGUGA AD-1809617 gscscuc(Ahd)AfaUfc 358 VPusGfsuaaUfgAfCfu 493 GAGCCUCAAUCU 628 fUfagucauuascsa agaUfuUfgaggcsusc AGUCAUUACA AD-1809633 ususaca(Chd)CfaGfu 359 VPusAfscuuCfuGfUfu 494 CAUUACACCAGUC 629 fCfaacagaagsusa gacUfgGfuguaasusg AACAGAAGUG AD-1809670 csusccu(Chd)AfgCfu 360 VPusAfsgggUfuAfAfg 495 CUCUCCUCAGCUC 630 fCfcuaaaccsusa gagCfuGfaggagsasg CUUAACCCUC AD-1809702 usgsgau(Uhd)GfuAf 361 VPusAfsaggGfaGfAfg 496 CCUGGAUUGUAAC 631 AfCfcucuccususa guuAfcAfauccasgsg CUCUCCCUUG AD-1809719 csusugu(Chd)CfaAf 362 VPusGfsaauCfcUfAfga 497 CCCUUGUCCAAAU 632 AfUfcuaggauuscsa uuUfgGfacaagsgsg CUAGGAUUEC AD-1809737 uscscug(Ghd)UfaGf 363 VPusUfsuuuCfcUfUfu 498 AUUCCUGGUAGGA 633 GfAfaaaggaaasasa uccUfaCfcaggasasu AAAGGAAAAG AD-1809757 uscscu(Chd)CfaCfc 364 VPusAfsguuGfgAfAfg 499 CUUCCCUCCACCA 634 fAfcuuccaacsusa uggUfgGfagggasasg CUUCCAACUG AD-1809773 csusgga(Chd)UfuGf 365 VPusCfsucuGfgUfUfc 500 ACCUGGACUUGGA 635 GfAfgaaccagasgsa uccAfaGfuccagsgsu GAACCAGAGG AD-1809822 csasccu(Ghd)UfuAf 366 VPusAfsgcuCfuGfAfu 501 GGCACCUGUUAGA 636 GfAfaucagagcsusa ucuAfaCfaggugscsc AUCAGAGCUG AD-1809839 gscsugc(Ahd)GfgAf 367 VPusGfsuccCfaAfGfaa 502 GAGCUGCAGGAUU 637 UfUfucuugggascsa auCfcUfgcagcsusc UCUUGGGACC AD-1809895 ususccc(Uhd)CfuUf 368 VPusCfsuucUfcCfUfu 503 CCUUCCCUUUUC 638 UfCfaaggagaasgsa gaaAfgAfgggaasgsg AAGGAGAAGC AD-1809919 usgsauu(Ghd)CfaGf 369 VPusAfsagaAfuAfCfaa 504 CAUGAUUGCAGCU 639 CfUfuguaauuscsa gcUfgCfaaucasusg UGUAUUCUUU AD-1809937 ususuag(Chd)CfuUf 370 VPusUfsagaUfuGfUfa 505 UCUUUAGCCUUU 640 AfUfuacaucusasa auaAfgGfcuaaasgsa UACAAUCUAU AD-1809954 csusaug(Uhd)GfcCf 371 VPusUfsugaGfuUfGfu 506 AUCUAUGUGCCUG 641 UfGfacaacucasasa cagGfcAfcuagsasu ACAACUCAAC AD-1810004 csuscca(Ahd)CfuGfa 372 VPusGfsucuGfgUfUfg 507 AGCUCCAACUGAA 642 fAfcaaccagascsa uucAfgUfuggagcsu CAACCAGACA AD-1810023 csasgac(Ahd)AfcUfc 373 VPusAfsggaUfgAfUfg 508 GACAGACAACUCU 643 fUfcaucauccsusa agaGfuUfgucugsusc CAUCAUCCUC AD-1810038 asusccu(Chd)CfaGfa 374 VPusCfsuauUfuUfCfu 509 UCAUCCUCCAGAG 644 fGfagaaaauasgsa cucUfgGfaggausgsa AGAAAAUAGG AD-1810053 asasuag(Ghd)CfcGfu 375 VPusUfscuuUfgAfGfa 510 AAAAUAGGCCGUG 645 fGfucuaaasgsa cacGfgCfcuaauususu UCUCAAAGAA AD-1810068 asasaga(Ahd)AfgGf 376 VPusUfsagaCfcAfAfga 511 UCAAAGAAAGGUU 646 UfUfcuuggucusasa acCfuUfucuuusgsa CUUGGUCUAU AD-1810127 usgscug(Chd)AfaAf 377 VPusAfscauGfgCfCfaa 512 CCUGCUGCAAAGU 647 GfUfuggccaugsusa cuUfuGfcagcasgsg UGGCCAUGUU AD-1810143 asusguu(Uhd)CfaCf 378 VPusAfsaagUfuUfCfcc 513 CCAUGUUUCACAG 648 AfGfggaaacuususa ugUfgAfaacausgsg GGAAACUUUU AD-1810174 csusgcu(Uhd)AfuGf 379 VPusUfsuuuGfgAfAfu 514 GGCUGCUUAUGAG 649 AfGfauuccaasasa cucAfuAfagcagcsusc AUUCCAAAAU AD-1810189 csasaaa(Uhd)GfaAfG 380 VPusUfsuggCfcAfAfc 515 UCCAAAAUGAAGU 650 fUfguuggccasasa acuUfcAfuuuugsgsa GUUGGCCAAC AD-1810221 cscsauc(Chd)UfgGfa 381 VPusAfscugGfgAfAfa 516 GGCAUCCUGGAU 651 fUfuuucccagsusa aucCfaGfgauggscsc UUUCCAGUG AD-1810239 gsusggc(Uhd)UfcCf 382 VPusCfsgagCfaGfGfaa 517 CAGUGGCUUCCCU 652 CfUfuccugcucsgsa ggGfaAfgccacsusg UCCUGCUCGC AD-1810255 asasagc(Uhd)UfaAfC 383 VPusGfsagaAfgAfGfa 518 AGAAAGCUUAACC 653 fCfucucuucuscscs gguUfaAfgcuuuscscu UCUCUUCUCC AD-1810277 cscsaaa(Chd)CfuUfu 384 VPusAfsuucAfaGfGfu 519 CUCCAAACCUUUC 654 fCfaccuugaasusa gaaAfgGfuuuggsasg ACCUUGAAUG AD-1810292 usgsaau(Ghd)GfgUf 385 VPusCfsaccAfaAfCfau 520 CUUGAAUGGGUAA 655 AfAfuguuugsgsusa uaCfcCfauucasasg UGUUUGGUGG AD-1810296 usgsuuc(Chd)UfuCf 386 VPusGfsgcuUfcUfCfca 521 GCUGUCCUUCUU 656 UfUfggagaagcsusa agAfaGfgaacasgsg GGAGAAGCCU AD-1810313 gscscuu(Ghd)AfgUf 387 VPusAfsaaaUfgGfUfcc 522 AAGCCUUGAGUCG 657 CfGfgaccauuususa gaCfuCfaaggcsusu GACCAUUUUG AD-1810345 asasgga(Uhd)GfaAf 388 VPusAfsuuuUfcAfCfu 523 GGAAGGAUGAAGA 658 GfAfagugaaaasusa ucuUfcAfuccuuscsc AGUGAAAAUG AD-1810363 asusgac(Ahd)AfuAf 389 VPusUfsugaGfaGfUfc 524 AAAUGACAAUAAU 659 AfUfgacucucasasa auuAfuUfgucaususu GACUCUCAAG AD-1810398 ascsaug(Ghd)CfaAfA 390 VPusCfsaguUfcUfAfca 525 UGACAUGGCAAAU 660 fUfguagaacusgsa uuUfgCfcauguscsa GUAGAACUGA AD-1810415 csusgac(Uhd)UfaAf 391 VPusGfsuuuGfuUfCfa 526 AACUGACUUAUU 661 AfUfugaacaaascsa auuUfaAfgucagsusu UGAACAAACC AD-1810446 ascscuc(Uhd)GfaUfG 392 VPusAfsgguGfcUfCfa 527 GCACCUCUGAUGU 662 fUfugagcaccsusa acaUfcAfgaggusgsg UGAGCACCUG AD-1810468 usgsaau(Ahd)CfuGf 393 VPusCfsaauCfaGfUfgc 528 GCUGAAUACUGAG 663 AfGfcacugaasgsa ucAfgUfauucasgsg CACUGAAUGG AD-1810494 csasggg(Ahd)UfaUf 394 VPusUfsaauGfgUfAfg 529 CUCAGGGAUAUGC 664 GfCfcuaccausasa gcaUfaUfccugsasg CUACCAAUAG AD-1810526 asgsgca(Uhd)GfuAf 395 VPusUfsuauGfuUfUfg 530 UAAGGCAUGUACC 665 CfCfcaaacaasasa gguAfcAfugccususa CAAACAUAAAC AD-1810545 ascsgga(Uhd)GfuAf 396 VPusAfscuuUfcUfGfc 531 UAACGGAUGUAAG 666 AfGfgcagaaasgsa cuuAfcAfuccgususa GCAGAAAGUG AD-1810566 asuscgg(Ahd)GfaAf 397 VPusUfsuucUfcAfUfu 532 UGAUCGGAGAAGG 667 GfGfaaugagaasasa ccuUfcUfccgauscsa AAUGAGAAAG AD-1810591 gsusgau(Ghd)UfuAf 398 VPusUfsgacUfuUfUfc 533 GCGUGAUGUUAAU 668 AfUfgaaaagucsasa auuAfaCfaucacsgsg GAAAAGUCAU AD-1810606 asgsuca(Uhd)AfuGf 399 VPusUfsgcuCfuAfGfc 534 AAAGUCAUAUGCA 669 CfAfgcuagagcsasa ugcAfuAfugacususu GCUAGAGCAG AD-1810638 csusuuc(Uhd)GfgAf 400 VPusAfsugcAfaUfCfu 535 AGCUUUCUGGAAG 670 AfGfagauucasusa cuuCfcAfgaaagcsu AGAUUGCAUC AD-1810666 asusuca(Ghd)GfaAf 401 VPusUfsacaAfaGfAfuc 536 AAUUCAGGAAGG 671 GfGfaucuuugusasa cuUfcCfugaasusu AUCUUUGUAG AD-1810676 asgsagg(Ghd)AfaGf 402 VPusAfsacaCfaGfCfag 537 CCAGAGGGAAGUC 672 UfCfugcugugusasa acUfuCfccucsgsg UGCUGUGUUC AD-1810692 usgsuuc(Uhd)CfaUf 403 VPusUfsgacAfuCfCfua 538 UGUGUUCUCAUGU 673 GfUfaggaugcsasa

CAUfgAfgaacascas AGAUGUCAG AD-1810727 uscsu(Uhd)UfuUf 404 VPusAfgacAfuUfGfg 539
CUUCUCUUUUUGG 674 GfGfccaugucsusa ccaAfaAfagagasg CCAAUGUCUU AD-1810742 usgsucu(Uhd)UfuCf
405 VPusGfsucaGfgAfAfa 540 AAUGUCUUUUUCAC 675 AfCfuuccugascsa gugAfaAfagacasusu UUUCCUGACC AD-
1810757 csusgac(Chd)CfuUfU 406 VPusGfsgauGfaUfUfc 541 UCCUGACCCUUUA 676 fAfgaaucaucscsa
uaaAfgGfgucagsgsa GAAUCAUCCC AD-1810767 gscsaau(Chd)AfuGf 407 VPusAfsaggCfaAfCfu 542
ACGCAAUCAUGGA 677 GfAfaguugccususa uccAfuGfauugcsgsu AGUUGCCUUA AD-1810785 ususauu(Ghd)UfcAf
408 VPusUfsucuUfaAfCfca 543 CCUUAUUGUCACU 678 CfUfgguuaagasasa guGfaCfaauaasgsg GGUAAGAAC AD-
1810800 asasgaa(Chd)UfuGfG 409 VPusUfsucaAfuCfUfc 544 UUAAGAACUUGGC 679 fCfagauugasasa
gccAfaGfuucuuasasa GAGAUUGAAG AD-1810818 asasggg(Chd)UfuUf 410 VPusCfsaacAfaUfAfac 545
UGAAGGGCUUUUG 680 UfGfuuauguusgsa aaAfaGfcccuscsa UUAUUGUUGU AD-1810822 ususguu(Ghd)GfaUf
411 VPusGfsaaaCfaAfAfaa 546 UGUUGUUGGAUUAU 681 AfUfuuuuguuuscsa uaUfcCfaacaascsa UUUUGUUUCC AD-
1810839 ususccc(Ahd)UfaAf 412 VPusAfsugaUfgUfGfc 547 GUUUCCCAUAAAA 682 AfAfgcacaucasusa
uuuUfaUfgggaasasc GCACAUCAU

TABLE-US-00006 TABLE 5 ABLIM3 Single Dose Screen in Hepa1-6 Cells % ABLIM3 % ABLIM3 Message Message
Remaining Remaining Duplex Name Mean SD Duplex Name Mean SD AD-1810839.1 49.940 1.488 AD-1809293.1
18.881 1.411 AD-1810822.1 47.037 1.563 AD-1809283.1 2.900 0.279 AD-1810818.1 48.205 1.013 AD-1809246.1 8.773
0.999 AD-1810800.1 51.029 3.275 AD-1809205.1 8.458 0.483 AD-1810785.1 50.509 2.233 AD-1809187.1 4.142 0.183
AD-1810767.1 48.138 2.366 AD-1809135.1 18.581 0.778 AD-1810757.1 62.353 1.932 AD-1809107.1 13.513 1.581 AD-
1810742.1 54.296 3.748 AD-1809205.1 17.023 1.176 AD-1810727.1 59.165 1.980 AD-1809187.1 11.397 0.772 AD-
1810692.1 55.551 1.804 AD-1809135.1 34.202 3.523 AD-1810676.1 66.324 1.823 AD-1809107.1 30.671 1.055 AD-
1810666.1 51.279 1.499 AD-1809080.1 94.268 2.234 AD-1810638.1 50.405 1.370 AD-1809066.1 37.454 0.708 AD-
1810606.1 49.318 1.427 AD-1809051.1 37.525 2.644 AD-1810591.1 47.445 1.020 AD-1809009.1 50.593 2.955 AD-
1810566.1 46.991 1.020 AD-1808978.1 28.150 0.731 AD-1810545.1 77.584 2.483 AD-1808941.1 31.554 1.946 AD-
1810526.1 48.566 2.463 AD-1808901.1 38.254 0.646 AD-1810494.1 52.776 2.132 AD-1808865.1 30.500 1.028 AD-
1810468.1 46.448 1.662 AD-1808842.1 38.960 0.483 AD-1810446.1 54.281 1.793 AD-1808814.1 22.480 1.096 AD-
1810415.1 44.403 2.813 AD-1808789.1 46.633 1.426 AD-1810398.1 49.302 2.611 AD-1808773.1 61.336 2.458 AD-
1810363.1 48.303 0.735 AD-1808748.1 32.402 1.061 AD-1810345.1 56.469 6.579 AD-1808727.1 29.449 1.943 AD-
1810313.1 69.402 3.692 AD-1808698.1 25.513 1.463 AD-1810296.1 58.716 3.020 AD-1808678.1 41.395 0.879 AD-
1810292.1 52.975 4.214 AD-1808649.1 29.492 0.446 AD-1810277.1 49.085 2.042 AD-1808628.1 31.293 0.539 AD-
1810255.1 46.815 1.824 AD-1808598.1 28.085 0.851 AD-1810239.1 63.361 1.618 AD-1808561.1 48.147 1.805 AD-
1810221.1 56.534 2.689 AD-1808511.1 50.948 3.050 AD-1810189.1 64.800 5.754 AD-1808482.1 59.364 1.486 AD-
1810174.1 44.716 1.350 AD-1808448.1 51.937 1.799 AD-1810143.1 66.647 1.603 AD-1808433.1 37.064 0.551 AD-
1810127.1 59.790 2.418 AD-1808418.1 24.593 0.687 AD-1810068.1 59.756 0.562 AD-1808407.1 33.131 0.798 AD-
1810053.1 73.408 0.901 AD-1808383.1 10.345 0.532 AD-1810038.1 57.000 1.540 AD-1808356.1 29.735 1.068 AD-
1810023.1 59.887 3.676 AD-1808327.1 22.413 1.016 AD-1810004.1 56.400 4.003 AD-1808312.1 21.839 0.800 AD-
1809954.1 48.343 3.077 AD-1808273.1 44.109 1.721 AD-1809937.1 38.981 1.083 AD-1808258.1 25.762 1.062 AD-
1809919.1 51.413 1.534 AD-1808238.1 30.415 2.284 AD-1809895.1 60.898 2.216 AD-1808157.1 50.280 1.509 AD-
1809839.1 56.989 2.001 AD-1808139.1 22.362 0.865 AD-1809822.1 46.487 4.020 AD-1808098.1 19.558 0.482 AD-
1809773.1 64.240 1.399 AD-1808063.1 40.421 1.032 AD-1809757.1 54.610 1.245 AD-1807955.1 31.513 1.425 AD-
1809737.1 46.954 2.502 AD-1807919.1 42.415 1.307 AD-1809719.1 51.801 1.864 AD-1807904.1 31.449 0.822 AD-
1809702.1 60.105 2.076 AD-1807889.1 54.480 2.939 AD-1809670.1 54.809 1.835 AD-1807888.1 38.152 1.731 AD-
1809633.1 53.875 5.306 AD-1807870.1 65.949 3.009 AD-1809617.1 46.580 3.490 AD-1807844.1 68.255 2.054 AD-
1809596.1 49.445 1.087 AD-1807804.1 29.939 1.630 AD-1809585.1 44.319 2.153 AD-1807780.1 23.409 0.426 AD-
1809561.1 68.525 1.535 AD-1807764.1 21.667 1.047 AD-1809550.1 62.777 3.584 AD-1807725.1 18.939 0.695 AD-
1809535.1 58.258 2.222 AD-1807707.1 32.573 1.716 AD-1809511.1 57.067 2.201 AD-1807659.1 40.382 1.804 AD-
1809498.1 83.585 2.024 AD-1807646.1 9.393 0.687 AD-1809472.1 59.655 2.843 AD-1807631.1 8.050 0.376 AD-
1809456.1 48.165 0.977 AD-1807616.1 12.937 0.426 AD-1809414.1 43.623 3.988 AD-1807591.1 13.957 0.555 AD-
1809395.1 40.931 1.346 AD-1807578.1 10.386 0.234 AD-1809369.1 57.180 1.004 AD-1807561.1 19.352 0.706 AD-
1809354.1 43.339 1.502 AD-1807539.1 24.797 1.176 AD-1809317.1 43.226 0.758 AD-1807481.1 21.057 1.421 Positive
0.805 0.152 AD-1807455.1 23.271 0.848 Control

TABLE-US-00007 TABLE 6 Hotspot Regions in ABLIM3 mRNA Duplex IDs That Target % Message
SEQ ID mRNA Target Sequence Hotspot Region Remaining Start End NO: (NM_001301015.3) AD-1809702,
AD-1809719 <50 3042 3081 683 CCTGGATTGTAACCTCTCCCT TGTCCAAATCTAGGATTCC AD-1810038, AD-
1810053 <50 3420 3457 684 TCATCCTCCAGAGAGAAAATA GGCCGTGTCTCAAAGAA AD-1810676, AD-
1810692 <50 4223 4261 685 CCAGAGGGAAGTCTGCTGTGT TCTCATGTAGGATGTCAG AD-1807904, AD-
1807919 <45 726 763 686 GTGACCTTCAGCGGTAAAGAA TGTGTGTGCCAAACGTG AD-1808238, AD- <45
1080 1137 687 TGCCACCAGATGTTACCGAA 1808258, AD-1808273 GGAGAGGAAATGTACCTCAC
AGGTTCCGAGGTTTGGC AD-1808418, AD-1808433 <40 1342 1379 688 TCATTTCTATGAGCCTCATTC
CAGATACATGTCCGAC AD-1808842, AD-1808865 <40 1910 1955 689 GCACAAGCTCCAAAGTGGA
TTGGCCGGCTGATTCTGAAGG AAGAA AD-1809051, AD-1809066 <40 2167 2204 690
ACCTCTTCCACTACGACAGCA TGAACGCAGTCAACTGG AD-1807707, AD-1807725 <35 480 520 691
ACAACCACTTCCACATCAGA TGCTTCACCTGTCAAGTATG AD-1808238, AD-1808258 <35 1080 1122 692

GTCCACCATGATGTTACCGAA GGAGAGGAAATGTACATGCAC AG AD-1808628, AD-1808748 <35 1665
693 CTCCACTCCAACCTCTTACCA GGCTCCCAAGCACTTTCACAT CC AD-1808727, AD-1808748 <35 1744
1787 694 AAACAAGTGAAGACATCAGC CAGACCTCCAAGTACAGTCCC ATC AD-1809187, AD-1809205 <35
2323 2363 695 AGTTCTACCAAGTCTTTGGCA TGACCATCTCTGAGTTTGAC AD-1807764, AD-1807780 <25
540 578 696 TCAGGCTTCTTCTTCAAGAAC CAGGAGTACATCTGCACC AD-1808312, AD-1808327 <25 1175
1212 697 GTTAAAGCATAGACGGACATC TGAACCTCCATCTCAC AD-1807561, AD-1807578 <20 274 313
698 CCCAGGTGATGAGTGAGGTTT GAAGAACGGAAGATTAAAA AD-1807616, AD- <15 349 401 699
AGTGCAAAGACATCACCATG 1807631, AD-1807646 AACACTAGCATTCCTTATCAG CAGAATCCTTAC AD-
1809354, AD-1809369 <10 2518 2555 700 CCTGCCATGAGACTTGCTTTT CTGTACTGTCAGGCAAG
TABLE-US-00008 ABLIM3 SEQUENCES SEQ ID NO: 1 >NM_001301015.3 *Homo sapiens* actin
binding LIM protein family member 3 (ABLIM3), transcript variant 1, mRNA
ACACCCCCTTTGGGCTCCGGAGCCGCTGGGCAGGAATAGTCCGGGGTGC GCGCGAGCAGGCGCTGCCAACCCCACT
TCTGCCCGGGATTCCAATCTGAGGAGCAGGAGGACCGGGGCGCCGGTGTCTCTGCCGCTCCTTCTCCTTGCTCTCA
CCTGCGCCTATTAGTCCACGCGCCTTCAAGGCCAGGGGCTACAGCCCAGACAGAGAGGGGACAGCAGAGGGGAGAGA
GAGCACCTGAGGATACAGAGCTGGCACTGGACTGCCTTTTACACCCCCAGGTGATGAGTGAGGTTTCAAGAACGGA
AGATTTAAAAAGCAGCCGGGGCCTCCGTATTGAATGAAAGACCCAGTGCAAAGACATCACCATGAACACTAGCATT
CCTTATCAGCAGAATCCTTACAATCCACGGGGCAGCTCCAATGTCATCCAGTGCTACCGCTGTGGAGACACCTGCA
AAGGGGAAGTGGTCCGCGTGCACAACAACCACTTCCACATCAGATGCTTACCTGTCAAGTATGTGGCTGTGGCCT
GGCCCAGTCAGGCTTCTTCTTCAAGAACCAGGAGTACATCTGCACCCAGGACTACCAGCAACTCTATGGCACCCGCG
TGTGACAGCTGCCGGGACTTCATCACAGGCGAAGTCATCTCGGCCCTGGGCCGCACTTACCACCCCAAGTGCTTCG
TGTGCAGCTTGTGCAGGAAGCCTTTCCCCATTGGAGACAAGGTGACCTTCAGCGGTAAAGAATGTGTGTGCCAAAC
GTGCTCCAGTCCATGGCCAGCAGTAAGCCCATCAAGATTTCGTGGACCAAGCCACTGTGCCGGGTGCAAGGAGGAG
ATCAAGCACGGCCAGTCACTCCTGGCTCTGGACAAGCAGTGGCACGTCAGCTGCTTCAAGTGCCAGACCTGCAGCG
TCATCCTCACCGGGGAGTATATCAGCAAGGATGGTGTTCATACTGTGAGTCCGACTACCATGCCCAGTTTGGCAT
TAAATGTGAGACTTGTGACCGATACATCAGTGGCAGAGTCTTGGAGGCAGGAGGGAAGCACTACCACCCAACTGT
GCCAGGTGTGTACGCTGCCACCAGATGTTACCCGAAGGAGAGGAAATGTACCTCACAGGTTCCGAGGTTTGGCACC
CCATCTGCAAACAGGCAGCCCGGGCAGAGAAGAAGTTAAAGCATAGACGGACATCTGAAACCTCCATCTCACCCCC
TGGATCCAGCATTGGGTCACCCAAACCGAGTCATCTGCGCTAAAGTGGATAATGAGATCCTTAATTACAAAGACCTG
GCGGCTCTCCCCAAGGTTAAGTCTATCTACGAGGTACAACGCCCCGACCTCATTTCCCTATGAGCCTCATTCCAGAT
ACATGTCCGACGAGATGCTGGAGAGATGTGGCTATGGAGAGTCGCTGGGAACATTATCTCCCTACTCCCAGGACAT
CTACGAGAACCTGGACCTCCGGCAGAGACGGGCCTCCAGCCCCGGGTACATAGACTCCCCACCTACAGCCGGCAG
GGCATGTCCCCCACCTTCTCCCGCTCACCTCACCACTACTACCGCTCTGGGCCCGAGAGTGGCCGGAGCTCTCCAT
ACCATAGCCAGTTAGATGTGAGGTCTTCCACTCCAACCTCTTACCAGGCTCCCAAGCACTTTTACATCCCAGCTGG
AGACAGTAACATCTACCGGAAACCCCCGATCTACAAACGGCATGGTGAATTTGTCTACAGCAACCAAGAGCAAAACA
AGTGAAGACATCAGCCAGACCTCCAAGTACAGTCCCATCTACTCGCCAGACCCCTACTATGCTTCGGAGTCTGAGT
ACTGGACCTACCATGGGTCCCCCAAAGTGCCCCGAGCCAGAAGGTTCTCGTCTGGAGGAGAGGAGGATGATTTTGA
CCGCAGCATGCACAAGCTCCAAAGTGGAATTGGCCGGCTGATTCTGAAGGAAGAAATGAAGGCCCGGTGAGCTCC
TATGCAGATCCCTGGACCCCTCCCCGGAGCTCCACCAGCAGCCGGGAAGCCCTGCACACAGCTGGCTATGAGATGT
CCCTCAATGGCTCCCCTCGGTGCGACTACCTGGCTGACAGTGATCCTCTCATCTCCAAATCTGCCTCCCTGCCTGC
CTACCGAAGAAATGGGCTGCACAGGACACCCAGCGCAGACCTCTTCCACTACGACAGCATGAACGCAGTCAACTGG
GGCATGCGAGAGTACAAGATCTACCCTTATGAACTGCTGCTGGTGACTACAAGAGGAAGAAACCGACTGCCCAAGG
ATGTAGACAGGACCCGTTTAGAGCGCCACCTGTCCCAGGAAGAGTTTACCAAGTCTTTGGCATGACCATCTCTGA
GTTTGACCGGCTGGCCCTCTGGAAGAGGAATGAACTGAAGAAGCAAGCCCGGCTGTTCTAGGCAGAGGCTCTATAA
ATATATATGCATTTATATAAAGATATATGTAAAATCTCTCTACTGAAGCTCGGTATAATCCTCTCTTGTGTAATGG
GACACACTGCCTGCCATGAGACTTGCTTTTCTGTACTGTCAGGCAAGCCCACGTCATCGAGATATTTTATGCTCC
TTACTTTTCTTTTTCTAAGTGCTGTGGGATCTGGGAAGGGATTTGAGGGGACTCTGTCCTTTTATTGGGGATCCTT
TTTATACTGAAACATCTGTCTAACTTGAGTGCCCCAAGGTCCAACCTCTCTTTCTTAAAGAAGGTGCCTGAAGAAG
TCTCTCTTCTCTCTGCTTCGTGGCCCCCTTTCTTAAATTTCTAGGGCTGATGCTGACCATGTGGTTTTCCACACCTTA
TTGGCCCCAGAGGGGGCCCTCCCATGGGAAGATCTGCAGCAGTCTCCCCAAATCAGTGAGCACCTTTGAGCGCCAC
GAAGAACTTTCTCAACACCCCCAATTAGGAGCTCAGTGCTCTCTTGGGGCAATGCAGTTAAAAGGGTGAGCCTCAA
ATCTAGTCATTACACCAGTCAACAGAAGTGAGCAGGGCCTAGGCCTCTCCTCAGCTCCTTAACCCTCCTCCTTCTG
CCCTGGATTGTAACCTCTCCCTTGTCCAAATCTAGGATTCTGCTAGGAAAAGGAAAAGGCCCTTCCCTTCCCTCC
ACCACTTCCAACCTGGCCCCCTTTGCCTGACCTGGACTTGAGAAACCAGAGGAAAAGAGAGGGAGCGGAAGTGGGAG
TGGAGCAGGGCACCTGTTAGAATCAGAGCTGCAGGATTTCTTGGGACCCCTCCTCTCTCCCTCACTGCTCCCAGCAC
CTCCTGACCCTTCCCTCTTTCAAGGAGAAGCCCATGATTGCAGCTTGATTCTTTAGCCTTATTACAATCTATGTG
CCTGACAACTCAACACACCGCAGGGCTAATGTTCCCACCAGAGCTCCAACCTGAACAACCAGACAGACAACTCTCAT
CATCCTCCAGAGAGAAAATAGGCCGTGTCTCAAAGAAAGGTTCTTGGTCTATGCCTCTGGTCTGTGGGCTGGCAGG
GCAACCATAACCATAACCCCGCCAGTCTCGGCTCCTGCTGCAAAGTTGGCCATGTTTCACAGGGAAACTTTTGGA
GAGTGGCTGCTTATGAGATTCCAAAATGAAGTGTGGCCAACACCGCTCATGGCCATCCTGGATTTTCCAGTGGC
TTCCCTTCTGCTCGCCTCCCTGAACAGGGGAGAAAGCTTAACCTCTCTTCTCCTCTCCAAACCTTTACCTTGAA
TGGGTAATGTTTGGTGGGGGCTGTTCTTCTTGGAGAAGCCTTGAGTCGGACCATTTTGAGATCATGGAGGAAGGA
TGAAGAAGTGAAAATGACAATAATGACTCTCAAGAGGCTGGCGATGTGACATGGCAAATGTAGAACTGACTTAAAT

TGAACCAACACCTACTGACCTACTCTGATGTTGAGCAACCTGCTGAATACTGACACTGAATGAGGGGGAGGGG
GAGCACGGGGTGAGTCAACCTGGGACTCGGTCTCAGGGATATGCCTACCAATAGCGGGTATCGTAAGGCATGTACC
CAAACATAACGGATGTAAGGCAGAAAGTGATCGGAGAAGGAATGAGAAAGTGTGCGTGATGTTAATGAAAAGTCAT
ATGCAGCTAGAGCAGACCCAGGAAAGCTTTCTGGAAGAGATTGCATCTGAGGAAATTCAGGAAGGATCTTTGTAGA
TTGGGGGGAGATTCTAAATTGAAGGGGTGATGGGGTGAGGGGCCAGAGGGAAGTCTGCTGTGTTCTCATGTAGGAT
GTCAGCCCTCCCTGCAACTTCTCTTTTTGGCCAATGTCTTTTCACTTTCCTGACCCTTTAGAATCATCCCCAGCCA
GACGCAATCATGGAAGTTGCCTTATTGTCACTGGTTAAGAACTTGGCGAGATTGAAGGGCTTTTGTTATTGTTGTT
GGATATTTTGTGTTCCCATAAAAGCACATCATTTCAACCCTAAA SEQ ID NO: 2>Reverse Complement
of SEQ ID NO: 1

TTTAGGGTTGAAATGATGTGCTTTTATGGGAAACAAAAATATCCAACAACAATAACAAAAGCCCTTCAATCTCGCC
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CATGCCGTTTGTAGATCGGGGGTTTCCGGTAGATGTTACTGTCTCCAGCTGGGATGTGAAAGTGCTTGGGAGCCTG
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GGCGCCCCGGTCTCTGCTCCTCAGATTGGAATCCCGGGCAGAAGTGGGGTTGGCAGCGCCTGCTCGCGCGCACC
CCGGACTATTCTGCCCAGCGGCTCCGGAGCCCCAAAGGGGGTGT

Claims

1.-2. (canceled)

3. A double stranded ribonucleic acid (dsRNA) agent for inhibiting expression of ABLIM3, wherein the dsRNA agent comprises a sense strand and an antisense strand forming a double stranded region, wherein the antisense strand comprises a region of complementarity to an mRNA encoding Ablim3, and wherein the region of complementarity comprises at least 15 contiguous nucleotides differing by no more than 3 nucleotides from any one of the antisense nucleotide sequences in Tables 3 or 4.

4. The dsRNA agent of claim 3, wherein (a) the sense strand comprises at least 15 contiguous nucleotides differing by no more than three nucleotides from any one of the nucleotide sequence of nucleotides 82-102, 134-154, 232-252, 276-296, 293-313, 326-346, 351-371, 366-386, 381-401, 414-434, 482-502, 500-520, 542-562, 558-578, 582-602, 622-642, 674-694, 692-712, 713-733, 728-748, 743-763, 779-799, 887-907, 942-962, 983-1003, 1001-1021, 1082-1102, 1102-1122, 1117-1137, 1177-1197, 1192-1212, 1242-1262, 1269-1289, 1313-1333, 1344-1364, 1359-1379, 1374-1394, 1408-1428, 1437-1457, 1550-1570, 1594-1614, 1624-1644, 1645-1665, 1674-1694, 1715-1735, 1746-1766, 1767-1787, 1812-1832, 1859-1879, 1884-1904, 1912-1932, 1935-1955, 1971-1991, 2039-2059, 2096-2116, 2127-2147, 2169-2189, 2184-2204, 2218-2238, 2245-2265, 2273-2293, 2325-2345, 2343-2363, 2385-2405, 2422-2442, 2459-2479, 2483-2503, 2520-2540, 2535-2555, 2561-2581, 2585-2605, 2667-2687, 2703-2723, 2729-2749, 2764-2784, 2788-2808, 2830-2850, 2861-2881, 2885-2905, 2937-2957, 2958-2978, 2974-2994, 3011-3031, 3044-3064, 3061-3081, 3079-3099, 3110-3130, 3146-3166, 3203-3223, 3220-3240, 3279-3299, 3303-3323, 3321-3341, 3338-3358, 3388-3408, 3407-3427, 3422-3442, 3437-3457, 3452-3472, 3532-3552, 3548-3568, 3579-3599, 3594-3614, 3626-3646, 3644-3664, 3682-3702, 3706-3726, 3721-3741, 3746-3766, 3763-3783, 3796-3816, 3814-3834, 3849-3869, 3866-3886, 3897-3917, 3919-3939, 3986-4006, 4018-4038, 4037-4057, 4058-4078, 4084-4104, 4099-4119, 4131-4151, 4159-4179, 4225-4245, 4241-4261, 4276-4296, 4291-4311, 4306-4326, 4336-4356, 4354-4374, 4369-4389, 4387-4407, 4404-4424, or 4421-4441 of SEQ ID NO: 1, and the antisense strand comprises at least 15 contiguous nucleotides from the corresponding nucleotide sequence of SEQ ID NO: 2; (b) the antisense strand comprises at least 15 contiguous nucleotides differing by no more than three nucleotides from any one of the antisense strand nucleotide sequences of a duplex selected from the group consisting of AD-1807455, AD-1807481, AD-1807539, AD-1807561, AD-1807578, AD-1807591, AD-1807616, AD-1807631, AD-1807646, AD-1807659, AD-1807707, AD-1807725, AD-1807764, AD-1807780, AD-1807804, AD-1807844, AD-1807870, AD-1807888, AD-1807889, AD-1807904, AD-1807919, AD-1807955, AD-1808063, AD-1808098, AD-1808139, AD-1808157, AD-1808238, AD-1808258, AD-1808273, AD-1808312, AD-1808327, AD-1808356, AD-1808383, AD-1808407, AD-1808418, AD-1808433, AD-1808448, AD-1808482, AD-1808511, AD-1808561, AD-1808598, AD-1808628, AD-1808649, AD-1808678, AD-1808698, AD-1808727, AD-1808748, AD-1808773, AD-1808789, AD-1808814, AD-1808842, AD-1808865, AD-1808901, AD-1808941, AD-1808978, AD-1809009, AD-1809051, AD-1809066, AD-1809080, AD-1809107, AD-1809135, AD-1809187, AD-1809205, AD-1809246, AD-1809283, AD-1809293, AD-1809317, AD-1809354, AD-1809369, AD-1809395, AD-1809414, AD-1809456, AD-1809472, AD-1809498, AD-1809511, AD-1809535, AD-1809550, AD-1809561, AD-1809585, AD-1809596, AD-1809617, AD-1809633, AD-1809670, AD-1809702, AD-1809719, AD-1809737, AD-1809757, AD-1809773, AD-1809822, AD-1809839, AD-1809895, AD-1809919, AD-1809937, AD-1809954, AD-1810004, AD-1810023, AD-1810038, AD-1810053, AD-1810068, AD-1810127, AD-1810143, AD-1810174, AD-1810189, AD-1810221, AD-1810239, AD-1810255, AD-1810277, AD-1810292, AD-1810296, AD-1810313, AD-1810345, AD-1810363, AD-1810398, AD-1810415, AD-1810446, AD-1810468, AD-1810494, AD-1810526, AD-1810545, AD-1810566, AD-1810591, AD-1810606, AD-1810638, AD-1810666, AD-1810676, AD-1810692, AD-1810727, AD-1810742, AD-1810757, AD-1810767, AD-1810785, AD-1810800, AD-1810818, AD-1810822, and AD-1810839; (c) the nucleotide sequence of the sense and antisense strand comprise any one of the sense and antisense strand nucleotide sequences in Tables 3 or 4; (d) the base pair at the 1 position of the 5'-end of the antisense strand of the duplex is an AU base pair; and/or (e) the sense strand has a total of 21 nucleotides and the antisense strand has a total of 23 nucleotides.

5.-6. (canceled)

7. The dsRNA agent of claim 3, wherein the sense strand, the antisense strand, or both the sense strand and the antisense strand is conjugated to one or more lipophilic moieties.

8. The dsRNA agent of claim 7, wherein (a) the one or more lipophilic moieties are conjugated to one or more internal positions in the double stranded region of the dsRNA agent; (b) the one or more lipophilic moieties are conjugated via a linker or carrier; (c) lipophilicity of the one or more lipophilic moieties, measured by log Kow, exceeds 0; (d) hydrophobicity of the double-stranded RNA agent, measured by the unbound fraction in a plasma protein binding assay of the double-stranded RNA agent, exceeds 0.2; (e) hydrophobicity of the double-stranded RNA agent, measured by the

fraction in a plasma protein binding assay of the double-stranded RNA agent, exceeds 0.2, and wherein the plasma protein binding assay is an electrophoretic mobility shift assay using human serum albumin protein; and/or (f) the one or more lipophilic moieties are conjugated to one or more internal positions on at least one strand.

9.-12. (canceled)

13. The dsRNA agent of claim 3, wherein the dsRNA agent comprises at least one modified nucleotide.

14. The dsRNA agent of claim 13, wherein (a) no more than five of the sense strand nucleotides and no more than five of the nucleotides of the antisense strand are unmodified nucleotides, (b) all of the nucleotides of the sense strand and all of the nucleotides of the antisense strand are modified nucleotides; (c) the at least one modified nucleotide is selected from the group consisting of a deoxy-nucleotide, a 3'-terminal deoxythymidine (dT) nucleotide, a 2'-O-methyl modified nucleotide, a 2'-fluoro modified nucleotide, a 2'-deoxy-modified nucleotide, a locked nucleotide, an unlocked nucleotide, a conformationally restricted nucleotide, a constrained ethyl nucleotide, an abasic nucleotide, a 2'-amino-modified nucleotide, a 2'-O-allyl-modified nucleotide, 2'-C-alkyl-modified nucleotide, 2'-hydroxyl-modified nucleotide, a 2'-methoxyethyl modified nucleotide, a 2'-O-alkyl-modified nucleotide, a morpholino nucleotide, a phosphoramidate, a non-natural nucleotide base, a tetrahydropyran modified nucleotide, a 1,5-anhydrohexitol modified nucleotide, a cyclohexenyl modified nucleotide, a nucleotide comprising a 5'-phosphorothioate group, a nucleotide comprising a 5'-methylphosphonate group, a nucleotide comprising a 5' phosphate or 5' phosphate mimic, a nucleotide comprising vinyl phosphonate, a glycol nucleic acid (GNA), an S-glycol nucleic acid (S-GNA), a nucleotide comprising 2-hydroxymethyl-tetrahydrofuran-5-phosphate, a nucleotide comprising 2'-deoxythymidine-3' phosphate, a nucleotide comprising 2'-deoxyguanosine-3'-phosphate, a nucleotide comprising a 2' phosphate, and a terminal nucleotide linked to a cholesteryl derivative and a dodecanoic acid bisdecylamide group; and combinations thereof; (d) the at least one modified nucleotide is selected from the group consisting of a 2'-deoxy-2'-fluoro modified nucleotide, a 2'-deoxy-modified nucleotide, 3'-terminal deoxythymidine nucleotides (dT), a locked nucleotide, an abasic nucleotide, a 2'-amino-modified nucleotide, a 2'-alkyl-modified nucleotide, a morpholino nucleotide, a phosphoramidate, and a non-natural nucleotide base; (e) the at least one modified nucleotide comprises a short sequence of 3'-terminal deoxythymidine nucleotides (dT); and/or (f) the at least one modified nucleotide is a 2'-O-methyl, a GNA and/or a 2' fluoro modification.

15.-19. (canceled)

20. The dsRNA agent of claim 3, wherein (a) the dsRNA comprises at least one phosphorothioate internucleotide linkage; (b) the dsRNA agent comprises 6-8 phosphorothioate internucleotide linkages; (c) each of the sense strand and the antisense strand is no more than 30 nucleotides in length; (d) at least one of the sense strand or the antisense strand comprises a 3' overhang of at least 1 nucleotide; (e) at least one of the sense strand or the antisense strand comprises a 3' overhang of at least 2 nucleotides; (f) the double stranded region is 15-30, 17-23, 17-25, 23-37, 19-21, or 21-23 nucleotide pairs in length; and/or (g) each of the sense strand and antisense strand has 19-30, 19-23, or 21-23 nucleotides.

21.-34. (canceled)

35. The dsRNA agent of claim 8, wherein (a) the one or more lipophilic moieties are conjugated to the one or more internal positions on the at least one strand via a linker or carrier; (b) the one or more internal positions include all positions except the terminal two positions from each end of the at least one strand; (c) the one or more internal positions include all positions except the terminal three positions from each end of the at least one strand; (d) the one or more internal positions exclude a cleavage site region of the sense strand; (e) the one or more internal positions include all positions except positions 9-12, counting from the 5'-end of the sense strand; (f) the one or more internal positions include all positions except positions 11-13, counting from the 3'-end of the sense strand; (g) the one or more internal positions exclude a cleavage site region of the antisense strand; (h) the one or more internal positions include all positions except positions 12-14, counting from the 5'-end of the antisense strand; (i) the one or more internal positions include all positions except positions 11-13 on the sense strand, counting from the 3'-end, and positions 12-14 on the antisense strand, counting from the 5'-end; (j) the one or more lipophilic moieties are conjugated to one or more of the internal positions selected from the group consisting of positions 4-8 and 13-18 on the sense strand, and positions 6-10 and 15-18 on the antisense strand, counting from the 5' end of each strand; (k) the one or more lipophilic moieties are conjugated to one or more of the internal positions selected from the group consisting of positions 5, 6, 7, 15, and 17 on the sense strand, and positions 15 and 17 on the antisense strand, counting from the 5'-end of each strand; (l) the one or more lipophilic moieties are conjugated to one or more of the internal positions selected from the group consisting of positions 5, 6, 7, 15, and 17 on the sense strand, and positions 15 and 17 on the antisense strand, counting from the 5'-end of each strand; and/or (m) the one or more internal positions in the double stranded region exclude a cleavage site region of the sense strand.

36.-46. (canceled)

47. The dsRNA agent of claim 7, wherein (a) the sense strand is 21 nucleotides in length, the antisense strand is 23 nucleotides in length, and the lipophilic moiety is conjugated to position 21, position 20, position 15, position 1, position 7, position 6, or position 2 of the sense strand or position 16 of the antisense strand; (b) the one or more lipophilic moieties are conjugated to position 21, position 20, position 15, position 1, or position 7 of the sense strand; (c) the one or more lipophilic moieties are conjugated to position 21, position 20, or position 15 of the sense strand; (d) the one or more lipophilic moieties are conjugated to position 20 or position 15 of the sense strand; or (e) the one or more lipophilic moieties are conjugated to position 16 of the antisense strand.

48.-51. (canceled)

52. The dsRNA agent of claim 7, wherein (a) the one or more lipophilic moieties are an aliphatic, alicyclic, or polyalicyclic compound, (b) the one or more lipophilic moieties are selected from the group consisting of lipid, cholesterol, retinoic

acid, cholic acid, adamantic acid, acetic acid, 1-pyrene butyric acid, dihydrotestosterone, 1,3-bis-O(hexadecyl)glycerol, geranyloxyhexanol, hexadecylglycerol, borneol, menthol, 1,3-propanediol, heptadecyl group, palmitic acid, myristic acid, 03-(oleoyl) lithocholic acid, 03-(oleoyl)cholenic acid, dimethoxytrityl, and phenoxazine; (c) the one or more lipophilic moieties contain a saturated or unsaturated C4-C30 hydrocarbon chain, and an optional functional group selected from the group consisting of hydroxyl, amine, carboxylic acid, sulfonate, phosphate, thiol, azide, and alkyne; (d) the one or more lipophilic moieties contain a saturated or unsaturated C6-C18 hydrocarbon chain; (e) the one or more lipophilic moieties contain a saturated or unsaturated C16 hydrocarbon chain; (f) the one or more lipophilic moieties contain a saturated or unsaturated C16 hydrocarbon chain, wherein the saturated or unsaturated C16 hydrocarbon chain is conjugated to position 6, counting from the 5'-end of the strand; (g) the one or more lipophilic moieties are conjugated via a carrier that replaces one or more nucleotide(s) in the internal position(s) or the double stranded region; (h) the one or more lipophilic moieties are conjugated via a carrier that replaces one or more nucleotide(s) in the internal position(s) or the double stranded region, wherein the carrier is a cyclic group selected from the group consisting of pyrrolidinyl, pyrazolinyl, pyrazolidinyl, imidazolinyl, imidazolidinyl, piperidinyl, piperazinyl, [1,3] dioxolanyl, oxazolidinyl, isoxazolidinyl, morpholinyl, thiazolidinyl, isothiazolidinyl, quinoxalinyl, pyridazinonyl, tetrahydrofuranyl, and decalinyl; or is an acyclic moiety based on a serinol backbone or a diethanolamine backbone; (i) the one or more lipophilic moieties are conjugated to the double-stranded iRNA agent via a linker containing an ether, thioether, urea, carbonate, amine, amide, maleimide-thioether, disulfide, phosphodiester, sulfonamide linkage, a product of a click reaction, or carbamate; (j) the one or more lipophilic moieties are conjugated to a nucleobase, sugar moiety, or internucleosidic linkage; (k) the one or more lipophilic moieties are conjugated via a bio-cleavable linker selected from the group consisting of DNA, RNA, disulfide, amide, functionalized monosaccharides or oligosaccharides of galactosamine, glucosamine, glucose, galactose, mannose, and combinations thereof; and/or (l) the 3' end of the sense strand is protected via an end cap which is a cyclic group having an amine, said cyclic group being selected from the group consisting of pyrrolidinyl, pyrazolinyl, pyrazolidinyl, imidazolinyl, imidazolidinyl, piperidinyl, piperazinyl, [1,3] dioxolanyl, oxazolidinyl, isoxazolidinyl, morpholinyl, thiazolidinyl, isothiazolidinyl, quinoxalinyl, pyridazinonyl, tetrahydrofuranyl, and decalinyl.

53.-63. (canceled)

64. The dsRNA agent of claim 7, wherein the dsRNA agent further comprises (a) a targeting ligand that targets a neuronal cell; (b) a targeting ligand that is a GalNAc conjugate; (c) a terminal, chiral modification occurring at the first internucleotide linkage at the 3' end of the antisense strand, having the linkage phosphorus atom in Sp configuration, a terminal, chiral modification occurring at the first internucleotide linkage at the 5' end of the antisense strand, having the linkage phosphorus atom in Rp configuration, and a terminal, chiral modification occurring at the first internucleotide linkage at the 5' end of the sense strand, having the linkage phosphorus atom in either Rp configuration or Sp configuration; (d) a terminal, chiral modification occurring at the first and second internucleotide linkages at the 3' end of the antisense strand, having the linkage phosphorus atom in Sp configuration, a terminal, chiral modification occurring at the first internucleotide linkage at the 5' end of the antisense strand, having the linkage phosphorus atom in Rp configuration, and a terminal, chiral modification occurring at the first internucleotide linkage at the 5' end of the sense strand, having the linkage phosphorus atom in either Rp or Sp configuration; (e) a terminal, chiral modification occurring at the first, second and third internucleotide linkages at the 3' end of the antisense strand, having the linkage phosphorus atom in Sp configuration, a terminal, chiral modification occurring at the first internucleotide linkage at the 5' end of the antisense strand, having the linkage phosphorus atom in Rp configuration, and a terminal, chiral modification occurring at the first internucleotide linkage at the 5' end of the sense strand, having the linkage phosphorus atom in either Rp or Sp configuration; (f) a terminal, chiral modification occurring at the first, and second internucleotide linkages at the 3' end of the antisense strand, having the linkage phosphorus atom in Sp configuration, a terminal, chiral modification occurring at the third internucleotide linkages at the 3' end of the antisense strand, having the linkage phosphorus atom in Rp configuration, a terminal, chiral modification occurring at the first internucleotide linkage at the 5' end of the antisense strand, having the linkage phosphorus atom in Rp configuration, and a terminal, chiral modification occurring at the first internucleotide linkage at the 5' end of the sense strand, having the linkage phosphorus atom in either Rp or Sp configuration; (g) a terminal, chiral modification occurring at the first, and second internucleotide linkages at the 3' end of the antisense strand, having the linkage phosphorus atom in Sp configuration, a terminal, chiral modification occurring at the first, and second internucleotide linkages at the 5' end of the antisense strand, having the linkage phosphorus atom in Rp configuration, and a terminal, chiral modification occurring at the first internucleotide linkage at the 5' end of the sense strand, having the linkage phosphorus atom in either Rp or Sp configuration; (h) a phosphate or phosphate mimic at the 5'-end of the antisense strand; and/or (i) a phosphate or phosphate mimic at the 5'-end of the antisense strand, wherein the phosphate mimic is a 5'-vinyl phosphonate (VP).

65.-74. (canceled)

75. The dsRNA of claim 3, wherein (a) the dsRNA agent targets a hotspot region of an mRNA encoding ABLIM3; (b) the dsRNA agent targets a hotspot region of an mRNA encoding ABLIM3, wherein the hotspot region comprises nucleotides 3042-3081, 3420-3457, 4223-4261, 726-763, 1080-1137, 1342-1379, 1910-1955, 2167-2204, 480-520, 1080-1122, 1622-1665, 1744-1787, 2323-2363, 540-578, 1175-1212, 274-313, 349-401, or 2518-2555 of SEQ ID NO: 1; or (c) the dsRNA agent targets a hotspot region of an mRNA encoding ABLIM3, wherein the dsRNA agent is selected from the group consisting of AD-1809702, AD-1809719, AD-1810038, AD-1810053, AD-1810676, AD-1810692, AD-1807904, AD-1807919, AD-1808238, AD-1808258, AD-1808273, AD-1808418, AD-1808433, AD-1808842, AD-1808865, AD-1809051, AD-1809066, AD-1807707, AD-1807725, AD-1808628, AD-1808649, AD-1808727, AD-1808748, AD-

1809187, AD-1809205, AD-1807764, AD-1807780, AD-1808312, AD-1808327, AD-1807561, AD-1807578, AD-1807616, AD-1807631, AD-1807646, AD-1809354, and AD-1809369.

76.-78. (canceled)

79. A cell containing the dsRNA agent of claim 3.

80. A pharmaceutical composition for inhibiting expression of a gene encoding ABLIM3, comprising (a) the dsRNA agent of claim 3; or (b) the dsRNA agent of claim 3 and a lipid formulation.

81. (canceled)

82. The pharmaceutical composition of claim 80, wherein (a) the dsRNA agent is in an unbuffered solution, (b) the dsRNA agent is in an unbuffered solution, wherein the unbuffered solution is saline or water; (c) the dsRNA agent is in a buffer solution; (d) the dsRNA agent is in a buffer solution, wherein the buffer solution comprises acetate, citrate, prolamine, carbonate, or phosphate or any combination thereof; or (e) the dsRNA agent is in a buffer solution, wherein the buffer solution is phosphate buffered saline (PBS).

83.-86. (canceled)

87. A method of inhibiting expression of an ABLIM3 gene in a cell, the method comprising introducing into the cell the dsRNA agent of claim 3, thereby inhibiting expression of the ABLIM3 gene in the cell.

88. The method of claim 87, wherein (a) the cell is within a subject, (b) the cell is within a human subject; (c) the cell is within a human subject having an ABLIM3-associated disorder; (d) the cell is within a human subject having an ABLIM3-associated disorder, wherein the ABLIM3-associated disorder is PTSD, or age-related memory loss; (e) the dsRNA agent inhibits the expression of ABLIM3 by at least 30%; (f) introducing the dsRNA agent into the cell inhibits the expression of ABLIM3 by at least 30%; and/or (g) the cell is within a subject, wherein inhibiting expression of ABLIM3 decreases ABLIM3 protein level in serum of the subject by at least 30%.

89.-94. (canceled)

95. A method of (a) treating a subject having a disorder that would benefit from reduction in ABLIM3 expression, comprising administering to the subject a therapeutically effective amount of the dsRNA agent of claim 3, thereby treating the subject having the disorder that would benefit from reduction in ABLIM3 expression; and/or (b) preventing at least one symptom or sign in a subject having a disorder that would benefit from reduction in ABLIM3 expression, comprising administering to the subject a prophylactically effective amount of the dsRNA agent of claim 3, thereby preventing at least one symptom or sign in the subject having the disorder that would benefit from reduction in ABLIM3 expression.

96. (canceled)

97. The method of claim 95, wherein (a) the disorder is an ABLIM3-associated disorder; (b) the disorder is an ABLIM3-associated disorder, wherein the ABLIM3-associated disorder is PTSD or age-related memory loss; (c) the subject is human; (d) the administration of the agent to the subject causes improved memory, or a decrease in anxiety, depression, fear, agitation, hostility, and/or distress; (e) the dsRNA agent is administered to the subject at a dose of about 0.01 mg/kg to about 50 mg/kg; (f) the dsRNA agent is administered to the subject intrathecally; (g) the method further comprises determining the level of ABLIM3 in a sample(s) from the subject; (h) the method further comprises determining the level of ABLIM3 in a sample(s) from the subject, wherein the level of ABLIM3 in the subject sample(s) is an ABLIM3 protein level in a blood, serum, or cerebrospinal fluid sample(s); and/or (i) the method further comprises administering to the subject an additional therapeutic agent.

98.-106. (canceled)

107. A kit, a vial, a syringe, or an intrathecal pump comprising the dsRNA agent of claim 3.

108-110. (canceled)
